ENT ATLAS

CLINICAL

# Ear

34 articles · imaging key facts · November 2026

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CLINICAL · EAR

# Otitis media: acute and with effusion

The same small space behind the drum produces two very different patients: a feverish child in pain with a bulging drum, and a quiet child with fluid who simply does not hear well. The first needs pain relief and judgement about antibiotics; the second needs patience, a hearing test, and sometimes a grommet.

**AT A GLANCE**

- Acute disease is about pain and infection; effusion is about hearing and development.
- Pneumatic otoscopy makes the diagnosis in both.
- Analgesia for all; antibiotics by age and severity, or watch at 48–72 hours.
- Watch effusion for 3 months; offer tubes for lasting loss.
- In an adult with one-sided effusion, look at the nasopharynx.
- NICE: watch non-severe AOM; prophylaxis halves attacks in recurrent AOM; grommets or hearing aids after 3 months of bilateral OME with ≥25–30 dB loss.

## 1 CHAPTER ONE Overview

Two diseases of one cleft.

### 1.1 Definitions

#### 1.1.1 Acute otitis media

Rapid onset, within 48 hours, of middle ear inflammation with an effusion: pain, fever and a red, bulging drum.

#### 1.1.2 Recurrent acute otitis media

Three or more episodes in six months, or four or more in a year, with the child well in between.

#### 1.1.3 Otitis media with effusion

Fluid in the middle ear behind an intact drum, without signs of acute infection. Also called *secretory otitis media* or *glue ear*. It is *chronic* once it has lasted three months.

#### 1.1.4 Acute necrotising otitis media

> A rare, destructive form, usually from beta-haemolytic streptococcus after measles or scarlet fever, which rapidly destroys the drum, annulus and ossicles.

### 1.2 Epidemiology

Acute otitis media is one of the commonest reasons a young child sees a doctor, peaking between six months and two years. Effusion follows most acute attacks and is the commonest cause of hearing loss in childhood, with peaks around two and again around five years.

### 1.3 Why it matters

Acute disease is about pain and the rare but serious complication. Effusion is about hearing, speech, behaviour and school, in a child who often does not complain at all.

## 2 CHAPTER TWO Pathogenesis

Why children, and why after a cold.

### 2.1 The Eustachian tube in a child

The child's tube is shorter, wider and more horizontal than the adult's, and its muscles open it less efficiently. Secretions from the nasopharynx reach the middle ear easily, and the middle ear drains poorly.

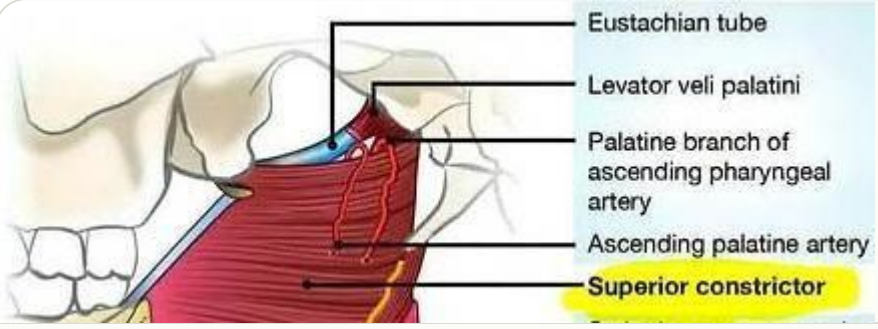



**FIGURE 1** The Eustachian tube with tensor and levator veli palatini, the muscles that open it.
Anatomy atlas · ENT Steps p. 204

## 2.2 The sequence

A viral cold inflames the tube and the adenoid. The tube blocks, gas in the middle ear is absorbed, negative pressure develops and a transudate collects. Bacteria from the nasopharynx, where the adenoid acts as a reservoir, then infect the fluid.

## 2.3 Organisms

### 2.3.1 Bacteria

*Streptococcus pneumoniae* leads, followed by non-typeable *Haemophilus influenzae* and *Moraxella catarrhalis*. About half of the *H. influenzae* and almost all *Moraxella* produce beta-lactamase.

### 2.3.2 Viruses

RSV, rhinovirus and adenovirus are found in many aspirates, alone or with bacteria.

## 2.4 Risk factors

**WHO GETS OTITIS MEDIA**

- ENVIRONMENT **Daycare, siblings** smoke exposure, dummy use, bottle feeding lying flat, no breastfeeding
- ANATOMY **Cleft palate** Down syndrome, craniofacial syndromes
- HOST **Prematurity** immunodeficiency, low socioeconomic status
- ADENOID **Bacterial reservoir** and tubal obstruction

# 3 CHAPTER THREE Acute otitis media

Pain, fever and a bulging drum.

## 3.1 Stages

Dhingra describes acute suppurative otitis media as a sequence, and the drum changes at each stage.

### 3.1.1 Tubal occlusion

The drum is retracted with a foreshortened handle of malleus; there is deafness and fullness, and little fever.

### 3.1.2 Presuppuration

Hyperaemia spreads from the handle of malleus and the periphery in a *cart-wheel* pattern. Pain is throbbing, and a child may have a high fever.

### 3.1.3 Suppuration

Pus collects under pressure: the drum is red and bulging, landmarks vanish, and a yellow spot may mark where it will rupture. Pain is severe.

### 3.1.4 Resolution

The drum ruptures, usually with a small anteroinferior perforation, discharge appears and pain and fever settle. Most perforations heal.

### 3.1.5 Complication

If infection spreads beyond the cleft: mastoiditis, facial palsy, labyrinthitis or intracranial disease.

## 3.2 Clinical picture

Otalgia, fever, irritability, ear pulling and poor feeding, often after a cold. These symptoms are non-specific on their own; the diagnosis rests on the drum.

## 3.3 Diagnosis

*Pneumatic otoscopy* is the essential examination. A bulging drum with poor mobility makes the diagnosis. A red drum in a crying child is not enough, because crying flushes the drum. Tympanometry helps when the view is poor.

**LOOK FOR A COMPLICATION FIRST**

- Facial weakness, vertigo, mastoid tenderness or swelling, neck stiffness, drowsiness.
- A child with a cochlear implant, immunodeficiency or a septic appearance is treated, not observed.

## 3.4 Management

### 3.4.1 Pain relief for everyone

Paracetamol or ibuprofen, whatever is decided about antibiotics.

### 3.4.2 Observation or antibiotics

Many children recover without antibiotics. Watchful waiting for 48 to 72 hours, with a safety net, suits children over two with mild symptoms or an uncertain diagnosis, provided the parents can return. It does not raise the rate of mastoiditis.

**WHEN TO GIVE ANTIBIOTICS**

| AGE | SEVERE | NON-SEVERE, BOTH EARS | NON-SEVERE, ONE EAR |
|---|---|---|---|
| 6–23 months | Antibiotic | Antibiotic | Antibiotic or observe |
| 2 years and over | Antibiotic | Antibiotic or observe | Antibiotic or observe |

Severe means fever over 39°C or moderate to severe pain for more than 48 hours.

### 3.4.3 Choice of antibiotic

- **First line:** high-dose amoxicillin.
- **Amoxicillin-clavulanate:** if amoxicillin was given in the last 30 days, with purulent conjunctivitis, or after recurrent failure.
- **Failure at 48 to 72 hours:** step up to amoxicillin-clavulanate, then intramuscular ceftriaxone for three days, then tympanocentesis for culture.

WHERE THE BOOKS DIFFER · DOSE AND DURATION OF AMOXICILLIN

| Dhingra | Lee |
|---|---|
| 40 mg/kg/day for at least 10 days. | 90 mg/kg/day (high dose, to overcome resistant pneumococci), with duration by age. |

Current guidelines follow the high-dose regimen; know both for examinations.

### 3.4.4 Myringotomy

Indicated for a bulging drum with severe pain, incomplete resolution with persisting hearing loss, or effusion beyond 12 weeks, and in the child who fails second-line treatment, when the fluid is sent for culture.

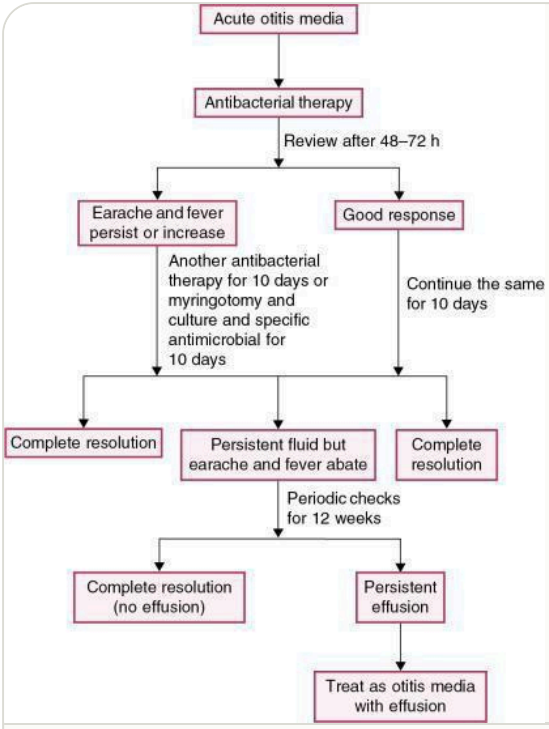



**FIGURE 2** Treatment of acute otitis media, matched to its stage.
Dhingra, fig. 10.1

## 3.5 Recurrent acute otitis media

Prophylactic antibiotics are no longer recommended; grommets reduce the number of episodes, and adenoidectomy is added in older children or for repeat surgery.

WHERE THE BOOKS DIFFER · RECURRENT OTITIS MEDIA

| Dhingra | Lee |
|---|---|

| | |
|---|---|
| Mentions amoxicillin prophylaxis, while noting its drawbacks and that many prefer grommets; grommets for 4 episodes in 6 months or 6 in a year. | Does not recommend prophylaxis; defines recurrence as 3 in 6 months or 4 in 12 months. |

**ACUTE AND RECURRENT OTITIS MEDIA: GUIDANCE AND EVIDENCE (OXFORD TEXTBOOK)**

**Antibiotics.** NICE (2018) advises analgesia and self-care for non-severe acute otitis media, including unilateral disease from 6 to 23 months, because most cases resolve and the number of children treated to prevent one case of mastoiditis is about 4,800. Immediate antibiotics are reserved for children under two with bilateral disease, those with otorrhoea, the systemically unwell, those under six months and the immunocompromised: amoxicillin first, a macrolide in penicillin allergy, and high-dose co-amoxiclav if symptoms persist. The American Academy of Pediatrics allows shared decision-making with a 48 to 72 hour wait-and-see option.

**Recurrent acute otitis media** is three or more episodes in six months or four or more in a year. Risk factors include day care, parental smoking and dummies; breastfeeding protects, and pneumococcal and Hib vaccination reduce episodes. Children who keep having infections despite vaccination may have poor antibody responses and benefit from a booster with the 23-valent vaccine. A meta-analysis of 14 trials found that long-term **antibiotic prophylaxis** (amoxicillin at half dose twice daily, or azithromycin 10 mg/kg three times a week, for two to three months) cut episodes from about three to 1.5 a year, with about five children treated to prevent one child having an attack, although the effect stops with the drug and resistance is a concern. **Grommets** reduce episodes by about one in the following six months, with little effect by a year, and allow infections to be treated with quinolone drops; they are the usual choice when effusion persists between attacks. Direct comparisons of prophylaxis and grommets are inconclusive.

**Complications** now occur in only 0.05 to 0.25% of episodes. Acute mastoiditis follows fewer than 0.4%, mostly under two years; intravenous antibiotics alone succeed in about 73%, with myringotomy, grommet, drainage of a subperiosteal abscess or cortical mastoidectomy for the rest.

CHAPTER FOUR

# 4 Otitis media with effusion

The quiet ear that cannot hear.

## 4.1 Pathogenesis

Tubal dysfunction leaves negative pressure and fluid behind; the mucosa responds with more goblet cells and mucous glands, and the fluid becomes thick and tenacious, the classic *glue*. Persistent adenoid infection, allergy and unresolved acute otitis media all contribute.

## 4.2 Clinical picture

### 4.2.1 Symptoms

Often none: a parent or teacher notices inattention, poor speech, or a child turning up the television. Older children and adults describe fullness and muffled hearing.

### 4.2.2 Signs

The drum is dull and opaque, sometimes amber, grey or bluish, with fine radial vessels rather than redness. It is often retracted, and may show an air-fluid level or bubbles. Mobility on pneumatic otoscopy is reduced.



**FIGURE 3** Otitis media with effusion: bubbles appear behind the drum on Valsalva.
Dhingra, fig. 10.2

## 4.3 Investigations

### 4.3.1 Tympanometry

A flat (type B) curve, or one shifted far to the negative side (type C).

### 4.3.2 Audiometry

Usually a 20 to 40 dB conductive loss. An audiogram is needed when the effusion has lasted three months, or sooner if there is a speech or developmental concern.

### 4.3.3 The nasopharynx in adults

**UNILATERAL EFFUSION IN AN ADULT**

Always examine the nasopharynx: nasopharyngeal carcinoma can first present this way, especially in smokers and people from endemic regions.

**WHERE THE BOOKS DIFFER · SCOPING THE NASOPHARYNX**

**Dhingra**

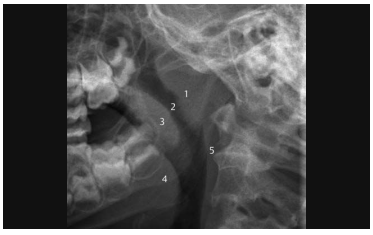
Always exclude a nasopharyngeal tumour in an adult with one-sided effusion.

**Lee**

The value of endoscopy is clearest in high-risk patients.

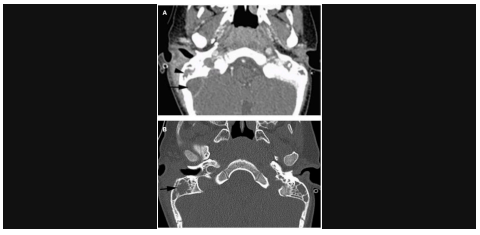
**IMAGING IN THE APP** 2 cases

Adenoid enlargement, and otomastoiditis with its complications.



**Adenoid Hypertrophy**

Infection & inflammation · 1 image



**Otomastoiditis and its complications**

Infection & inflammation · 9 images

## 4.4 Natural history

Most effusions clear on their own: about 70% within three months, and around 90%

after treated acute otitis media.

## 4.5 Management

### 4.5.1 Watchful waiting

Review every three months with a hearing test. Antibiotics, steroids, antihistamines and decongestants have no lasting benefit and are not recommended. Autoinflation with a nasal balloon helps some children.

**WHERE THE BOOKS DIFFER · MEDICAL TREATMENT OF EFFUSION**

| Dhingra | Lee |
|---|---|
| Lists decongestants, antihistamines and antibiotics among the medical options. | Guidelines find no role for them. |

Follow current guidelines in practice.

### 4.5.2 Grommets

For effusion lasting three months or more with hearing loss, or in a child at risk (speech delay, learning or sensory difficulty, cleft palate, Down syndrome). Thick glue may need two incisions, one to let air in behind the fluid.

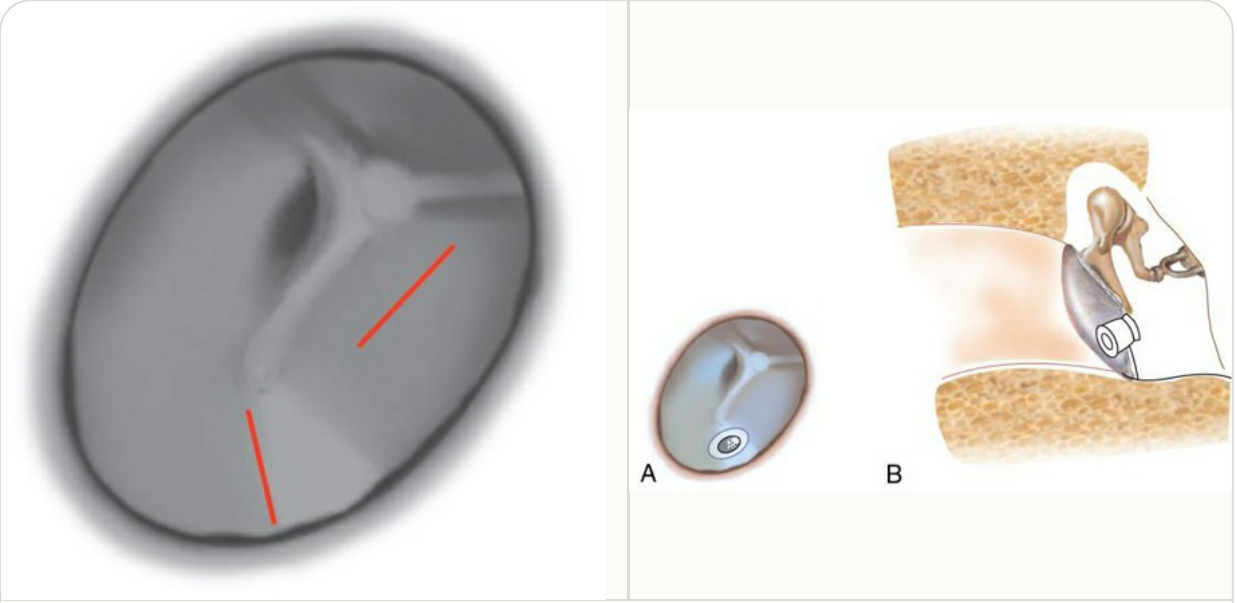



**FIGURE 4** First, two incisions in the drum let thick mucus be aspirated. Second, a grommet in place.
Dhingra, figs 10.3 and 10.4

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation**
Tubes: OME with hearing loss or at-risk child, or recurrent AOM
**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation in the Pediatric Population**
The same operation in children
**Adenoidectomy**
Adenoidectomy: age over 4 with OME, repeat tubes, or nasal or sleep symptoms

### 4.5.3 Adenoidectomy

Added for children over about four, for repeat tube insertion, or when there are nasal or sleep symptoms. It reduces the chance of needing tubes again.

### 4.5.4 Hearing aids

An alternative to surgery, especially in Down syndrome where the canals are narrow and effusion persists.

**GUIDELINES AND NUMBERS (OXFORD TEXTBOOK)**

Effusion peaks at about two years, when around one in five screened children have it, with a second peak at five; up to 90% of children have at least one episode before school. It affects 60 to 85% of children with Down syndrome or cleft palate. Screen-detected effusions clear in about 63% by three months and 88% by a year, but once a **bilateral** effusion has lasted three months, only about a quarter clear by six months and half persist three years later.

**NICE** advises three months of watchful waiting, then ventilation tubes for persistent bilateral effusion with a hearing level in the better ear of 25 to 30 dB HL or worse (averaged over 0.5, 1, 2 and 4 kHz), or with significant developmental, educational or behavioural effects; **hearing aids** are an equal alternative, and are preferred in Down syndrome. NICE does not recommend adenoidectomy with the first set of tubes unless there are persistent or frequent upper respiratory symptoms, whereas the American guideline offers tubes alone under four years and tubes, adenoidectomy or both from four years. Otoscopy by an experienced examiner, pneumatic otoscopy and tympanometry each detect over 90% of effusions; tympanometry is the least specific.

**Tube sequelae:** otorrhoea in about 16% soon after surgery and 26% later; blockage up to 7%; granulation 5%; tympanosclerosis about 32% and focal atrophy 25%; persistent perforation about 2%, rising to about 17% with long-stay tubes. Otorrhoea is best treated with topical antibiotic or antibiotic–steroid drops rather than oral antibiotics, and water precautions make little difference for most children.

**PEARLS**

- Almost every child with a cleft palate has effusion: offer earlier tubes, and keep watching after palate repair.
- Grommet otorrhoea is treated with topical drops, not oral antibiotics.
- Propping an infant up during feeds reduces reflux up the tube.

# 5 CHAPTER FIVE Complications and sequelae

Acute and long term.

## 5.1 Of acute otitis media

- Perforation in about 5%; around 90% heal.
- Acute mastoiditis and subperiosteal abscess.
- Facial palsy and labyrinthitis.
- Meningitis and other intracranial complications.

## 5.2 Of chronic effusion

- Atrophic, retracted drum and atelectasis.
- Necrosis of the long process of the incus.
- Tympanosclerosis.
- Retraction pockets and cholesteatoma.
- Cholesterol granuloma.

## 5.3 Masked disease

Inadequate antibiotic courses can leave a smouldering ear that becomes a persistent effusion or masked mastoiditis.

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 On call

**A CHILD WITH A PAINFUL EAR**

1. Look for complications first: face, balance, mastoid, neck and conscious level.
2. Give analgesia to every child.
3. Perform pneumatic otoscopy; a red drum alone is not enough.
4. Decide antibiotics or observation by age, severity and laterality.
5. Safety-net: return if worse at 48–72 hours, with neck swelling, weakness or drowsiness.

## 6.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Stages of acute suppurative otitis media** and the drum at each stage.
- **Why children:** the Eustachian tube and the adenoid.
- **The three bacteria,** and beta-lactamase production.
- **Effusion:** drum appearance, type B tympanogram, 20–40 dB conductive loss.
- **Indications for grommets,** and when to add adenoidectomy.
- **Unilateral effusion in an adult:** examine the nasopharynx.

## 6.3 Common pitfalls

**COMMON PITFALLS**

- Diagnosing otitis media from a red drum in a crying child.
- Withholding analgesia while deciding about antibiotics.

- Treating effusion with decongestants and antihistamines.
- Missing nasopharyngeal carcinoma in an adult with one-sided glue.

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Written for this app from Oxford Textbook of Otolaryngology ch. 84, 88, Oxford Specialist Handbook ch. 16, 23, Oxford Handbook of ENT ch. 6, 17, Dhingra ch. 10 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Acute, coalescent and masked mastoiditis

In acute mastoiditis the infection has broken out of the middle ear into the bone behind it. The typical patient is a small, unwell child with a red, tender swelling behind an ear that is pushed forward; the dangerous one is the patient on antibiotics whose ear has gone quiet while the bone is still being destroyed.

**AT A GLANCE**

- Otitis media plus a tender, swollen mastoid and a displaced ear.
- Admit, culture, IV antibiotics, myringotomy.
- Opaque mastoid isn't enough: look for lost septa or abscess.
- Operate for abscess, coalescence, failure or complication.
- Repeated antibiotics can hide it.

## 1 CHAPTER ONE Overview

When otitis media escapes the middle ear.

### 1.1 Definition

Every acute otitis media inflames the mastoid air cells to some degree, because the antrum is continuous with the middle ear; a CT in plain otitis media often shows an opaque mastoid. *Acute mastoiditis* means clinically evident infection of the mastoid process, and *coalescent mastoiditis* means the bony septa between air cells have been destroyed, turning the cells into a single cavity full of pus.

### 1.2 Who gets it

Half of children with acute mastoiditis are four years old or younger, and only about 30% have a history of recurrent otitis media; often it is the first severe ear infection.

### 1.3 Why it matters

Mastoiditis is the gateway to the other complications of otitis media: neck and facial abscesses, facial palsy, labyrinthitis and intracranial infection. Up to one in ten develop an intracranial complication, even on intravenous antibiotics.

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CHAPTER TWO

# 2 Pathology

From inflamed mucosa to destroyed bone.

### 2.1 Organisms

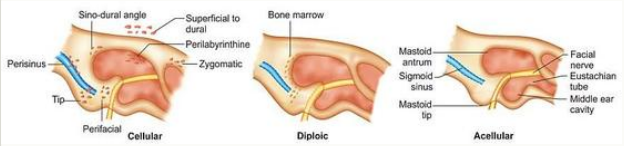
*Streptococcus pneumoniae* leads, with *Streptococcus pyogenes*, *Staphylococcus aureus* and *Haemophilus influenzae*. Pneumococcal serotype 19A is both resistant and virulent. Cultures are negative in many children who have already had antibiotics.

WHERE THE BOOKS DIFFER · COMMONEST ORGANISM

| Dhingra | Lee |
| --- | --- |
| Beta-haemolytic streptococcus. | Streptococcus pneumoniae. |
| Lee reflects more recent series. | |

### 2.2 How the bone is destroyed

Two processes run together. Pus under pressure in the cells causes ischaemic necrosis of the thin bony septa, and inflamed mucosa activates osteoclasts that resorb bone (*hyperaemic decalcification*). The cells coalesce into an abscess cavity, which then looks for a way out.



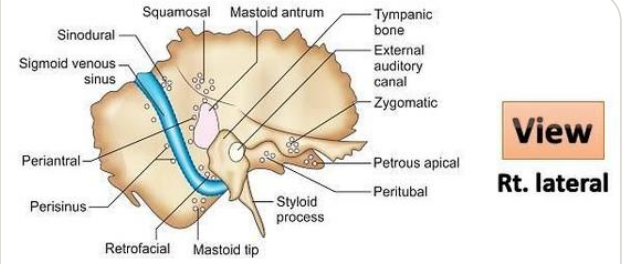




**FIGURE 1** Types of mastoid (cellular, diploic, acellular) and the groups of air cells through which infection spreads.

Anatomy atlas · ENT Steps p. 59, ENT Steps p. 60

## 2.3 Classification

**MASTOIDITIS**

- ACUTE **Acute mastoiditis** otitis media with mastoid inflammation
- BONE LOSS **Coalescent** septa destroyed, pus in a single cavity
- AFTER ANTIBIOTICS **Masked (latent)** slow bone destruction, few symptoms
- LONG-STANDING **Chronic** with perforation, cholesteatoma or grommet

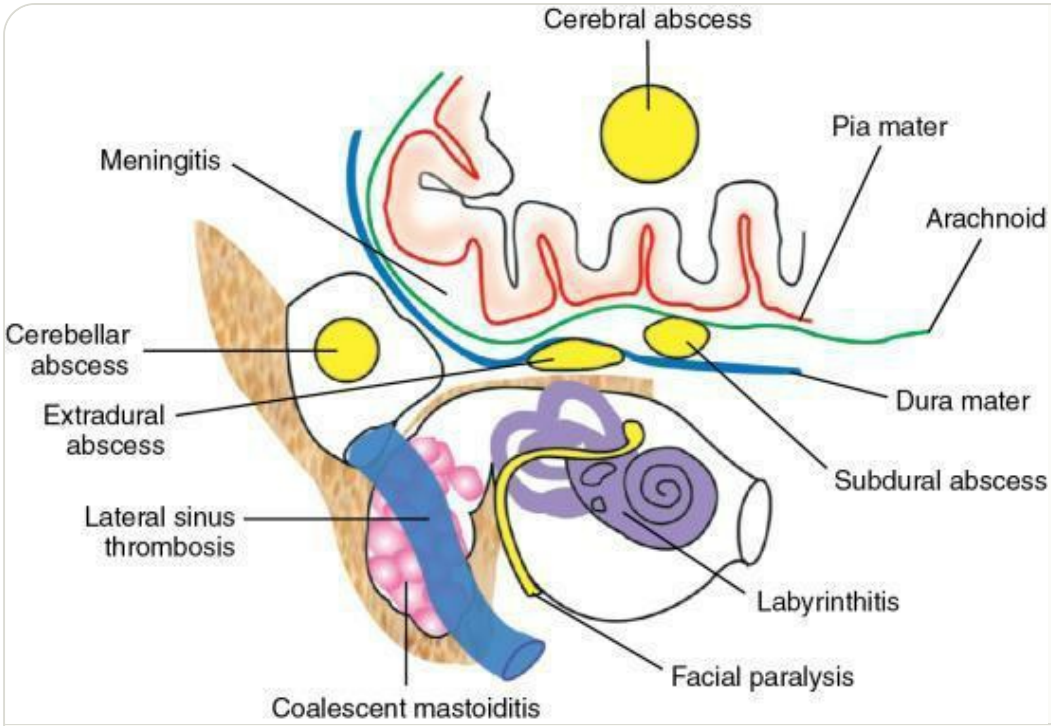



**FIGURE 2** Routes by which suppurative otitis media spreads to cause its complications.
Dhingra, fig. 12.1

# 3 CHAPTER THREE Clinical features

The child, the ear and the bone.

## 3.1 Symptoms

### 3.1.1 Pain

Persistent pain behind the ear, often after an apparently improving otitis media.

### 3.1.2 Fever

Fever that persists or returns despite treatment.

### 3.1.3 Discharge

Ear discharge that becomes profuse and creamy, or persists beyond three weeks. Mucoid discharge must come from the middle ear, because the canal has no mucous glands.

**THE CHANGE IS WHAT MATTERS**

Pain, fever or discharge that continues or comes back during an acute otitis media points to mastoiditis.

## 3.2 Signs

### 3.2.1 The mastoid

Tenderness is most telling over the tip and posterior border, rather than over the suprameatal (MacEwen's) triangle, which is tender in plain otitis media too. Always compare with the healthy side. Later the area becomes red, boggy and fluctuant.

### 3.2.2 The pinna

A subperiosteal abscess pushes the auricle forwards and downwards, and the postauricular sulcus is lost.

### 3.2.3 The canal and drum

- **Sagging** of the posterosuperior canal wall, from periostitis over the antrum.
- **Reservoir sign:** the canal refills with pus straight after mopping.
- A pars tensa perforation with pulsatile discharge, sometimes with a *light-house* reflex.
- Hearing loss is conductive.



**FIGURE 3** Acute mastoiditis: the pinna is pushed forward and downward by the swelling behind it.

Dhingra, fig. 12.2

## 3.3 Masked mastoiditis

Repeated or inadequate antibiotic courses can sterilise the middle ear while bone destruction continues. The patient has little pain and no discharge, or only a mild ache or persistent otalgia. The drum is dull and thickened with lost landmarks, and hearing is reduced. It may hide an extradural or perisinus abscess.

**WHERE THE BOOKS DIFFER · SYMPTOMS OF MASKED MASTOIDITIS**

| Dhingra | Lee |
| --- | --- |
| Painless, without discharge; mild ache at most. | Persistent, chronic otalgia after antibiotics. |

Either way, a patient whose ear never quite settles after antibiotics needs a CT.

# 4 CHAPTER FOUR Abscesses

Where the pus comes out.

## 4.1 Subperiosteal (postauricular)

The commonest. Pus breaks through the cribriform area into the space under the periosteum over the mastoid, displacing the pinna.

## 4.2 Zygomatic

Infection of the zygomatic air cells produces swelling in front of and above the ear, with oedema of the upper eyelid.

## 4.3 Bezold abscess

In an adult or older child with a pneumatised tip, pus breaks through the thin medial side of the tip and tracks under the sternocleidomastoid, presenting as a tender upper neck mass.

### 4.3.1 Differential

Upper jugular lymphadenitis, a parotid mass, an infected branchial cyst, parapharyngeal abscess and jugular vein thrombosis.

### 4.3.2 Treatment

Cortical mastoidectomy, finding the fistula at the tip, and drainage of the neck through a separate incision.

## 4.4 Meatal (Luc) abscess

Pus from the antrum tracks between the posterior canal wall and the periosteum, presenting as a swelling in the deep canal.

## 4.5 Citelli abscess

Pus from the tip tracks along the posterior belly of digastric to form an abscess behind the mastoid or in the digastric triangle.

## 4.6 Deep neck

Parapharyngeal and retropharyngeal abscesses via the tip cells or petrous cells.

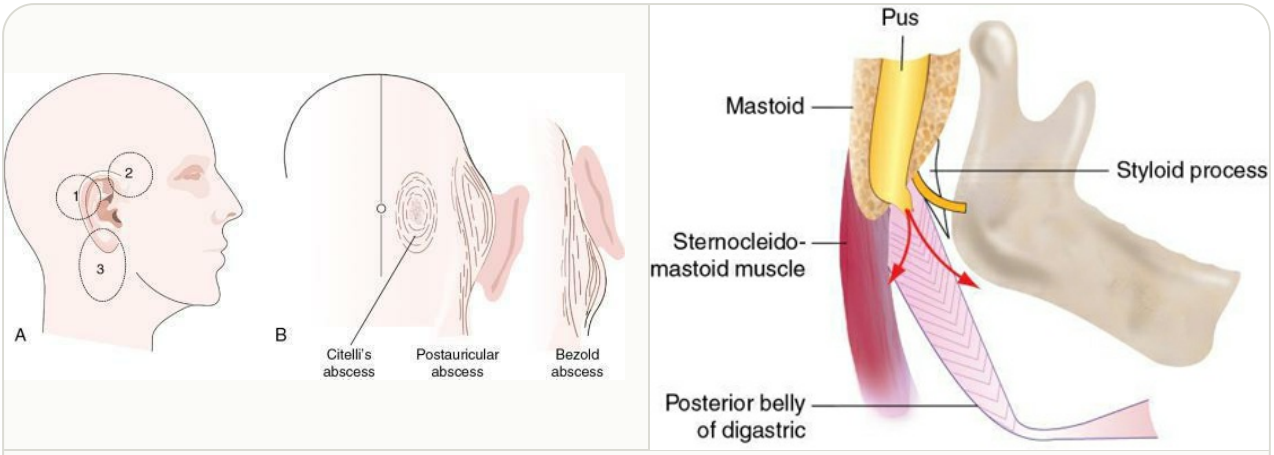



**FIGURE 4** First, abscesses around the mastoid: postauricular, zygomatic and Bezold, with Citelli's abscess. Second, a Bezold abscess, where pus bursts through the medial side of the tip into the neck.

Dhingra, figs 12.4 and 12.5




**FIGURE 5** A mastoid abscess that has burst through the skin.

Dhingra, fig. 12.3

# 5 CHAPTER FIVE Diagnosis

Mainly clinical, confirmed by imaging.

## 5.1 Blood tests

Full blood count with raised white cells, inflammatory markers, and blood cultures. Pus from the ear or myringotomy is sent for culture.

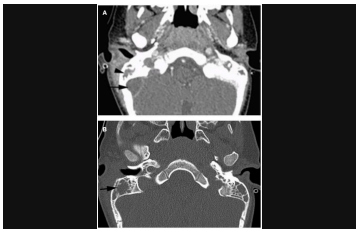
## 5.2 Imaging

### 5.2.1 CT temporal bone with contrast

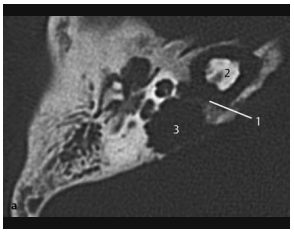
Shows coalescence (loss of the septa), cortical erosion and abscesses, and looks for intracranial spread. An opaque mastoid alone is not mastoiditis.

**IMAGING IN THE APP** 3 cases

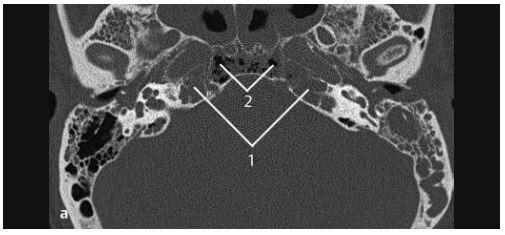
Otomastoiditis, petrositis and skull base infection on CT and MRI.



**Otomastoiditis and its complications**
Infection & inflammation · 9 images




**Petrositis**
Infection & inflammation · 3 images




**Infection of the Skull Base and Gradenigo Syndrome**
Infection & inflammation · 3 images

### 5.2.2 MRI and MR venography

When an intracranial complication or sinus thrombosis is suspected.

## 5.3 Differential diagnosis

### SWELLING BEHIND THE EAR

| | DISTINGUISHING POINTS |
|---|---|
| **Furuncle of the canal** | No preceding otitis media, painful tragus and pinna movement, normal drum, no mucoid discharge |
| **Suppurating postauricular node** | No ear history, superficial abscess, mastoid not tender |
| **Otitis externa with nodes** | Canal swollen and tender, drum normal |
| **Infected sebaceous cyst** | Superficial, punctum, ear normal |

# 6 CHAPTER SIX Management

Admit, drain, and decide on surgery.

## 6.1 First steps

**ON CALL: ACUTE MASTOIDITIS**

1. Admit; take blood and ear cultures.
2. Start intravenous third-generation cephalosporin, with anti-pseudomonal cover in chronic ears.
3. Examine the facial nerve, balance, neck and neurology.
4. Request contrast CT of the temporal bones and brain; MRI and MRV if intracranial spread is possible.
5. Perform a wide myringotomy with a grommet, and send the pus for culture.
6. Call your senior for an abscess, coalescence, or no response at 24–48 hours.

## 6.2 Medical treatment

Intravenous antibiotics, changed according to culture, with oral step-down to complete three to six weeks in total. Uncomplicated acute mastoiditis without coalescence often settles with antibiotics and myringotomy alone.

## 6.3 Surgery

### 6.3.1 Indications for cortical mastoidectomy

- Subperiosteal abscess.
- Coalescence on CT.
- Sagging of the posterosuperior canal wall or a positive reservoir sign.
- No improvement after 48 hours of adequate treatment.
- Any complication, or cholesteatoma.

### 6.3.2 Small children

A small subperiosteal abscess may be aspirated or incised, but cortical mastoidectomy is generally advised. The infant mastoid is small and vascular, landmarks are poor, and the facial nerve exits superficially.

### 6.3.3 The operation

Cortical (Schwartze) mastoidectomy removes the diseased cells and opens the antrum to the middle ear, leaving the canal wall and ossicles intact.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Mastoid Surgery**
Cortical mastoidectomy for coalescence, abscess or failure

**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation**
Myringotomy and tube

**Intracranial Complications of Otitis Media**
Intracranial complications

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Coalescence:** what it means and how it is shown on CT.
- **Signs:** forward-displaced pinna, sagging canal wall, reservoir sign.
- **The abscesses:** subperiosteal, zygomatic, Bezold, Luc and Citelli, and where each points.
- **Indications for cortical mastoidectomy.**
- **Masked mastoiditis:** antibiotics without cure.

## 7.2 Common pitfalls

**COMMON PITFALLS**

- Treating an opaque mastoid on CT as mastoiditis without clinical signs.
- Relying on suprameatal tenderness alone.
- Missing masked mastoiditis in a child whose otalgia never settled.
- Forgetting the superficial facial nerve when incising behind an infant's ear.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 12 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Chronic suppurative otitis media without cholesteatoma

Chronic suppurative otitis media of the mucosal, or safe, type is a wet ear discharging through a hole that will not heal. It rarely threatens life, but it wears down hearing and patience. The aim is a dry, safe ear: clean it properly, treat it topically for long enough, prove there is no cholesteatoma, and then decide whether repair is worth it.

**AT A GLANCE**

- A permanent central perforation with recurring mucoid discharge.
- Clean it first, then topical quinolone for weeks.
- Keep water out and treat the nose and throat.
- Prove there's no cholesteatoma.
- Repair the drum once dry.

## 1 CHAPTER ONE Overview

A persistent hole and a discharging ear.

### 1.1 Definition

Chronic suppurative otitis media is long-standing infection of part or all of the middle ear cleft, with ear discharge and a permanent perforation. It is classically divided into two types: the *tubotympanic* or *safe* type, which involves the mucosa and the anteroinferior cleft, and the *atticoantral* or *unsafe* type, which involves bone and usually cholesteatoma. This chapter covers the safe, mucosal type.

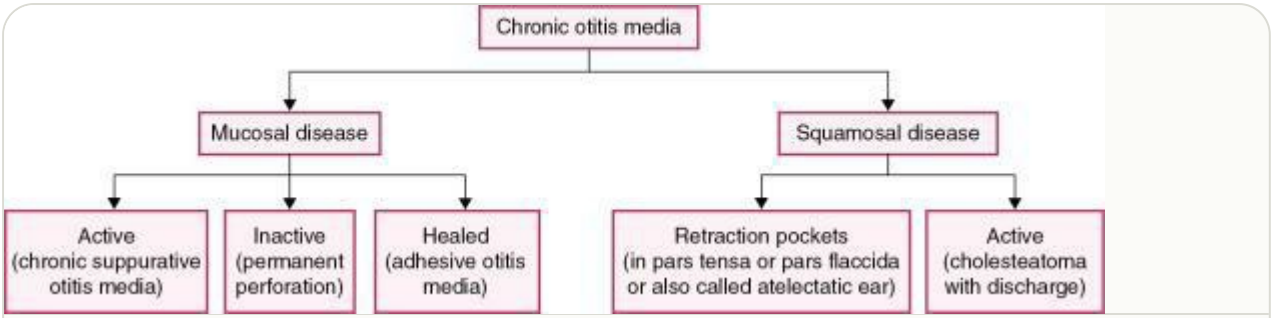



**FIGURE 1** Classification of chronic otitis media into tubotympanic (mucosal) and atticoantral (squamous) types.

Dhingra, fig. 11.7

## 1.2 Clinical stages

### 1.2.1 Active mucosal disease

Discharge through a pars tensa perforation.

### 1.2.2 Inactive mucosal disease

A dry, permanent perforation that may become active with a cold or water.

### 1.2.3 Healed chronic otitis media

An atrophic, healed drum, often with tympanosclerosis and some conductive loss.

## 1.3 Why it matters

It is a common cause of preventable hearing loss worldwide, especially where crowding, poverty and limited access to care allow acute infections to go untreated.

# 2 CHAPTER TWO Pathogenesis

Why the hole does not heal and the ear keeps running.

## 2.1 Origins

Most cases follow repeated acute otitis media in childhood, or a perforation from acute necrotising otitis or trauma. Chronic effusion and Eustachian tube

dysfunction keep the mucosa inflamed.

## 2.2 The permanent perforation

Once the squamous epithelium of the drum grows round the edge of the hole and meets the mucosa on the inside, the perforation behaves like a lined fistula and will not close on its own.

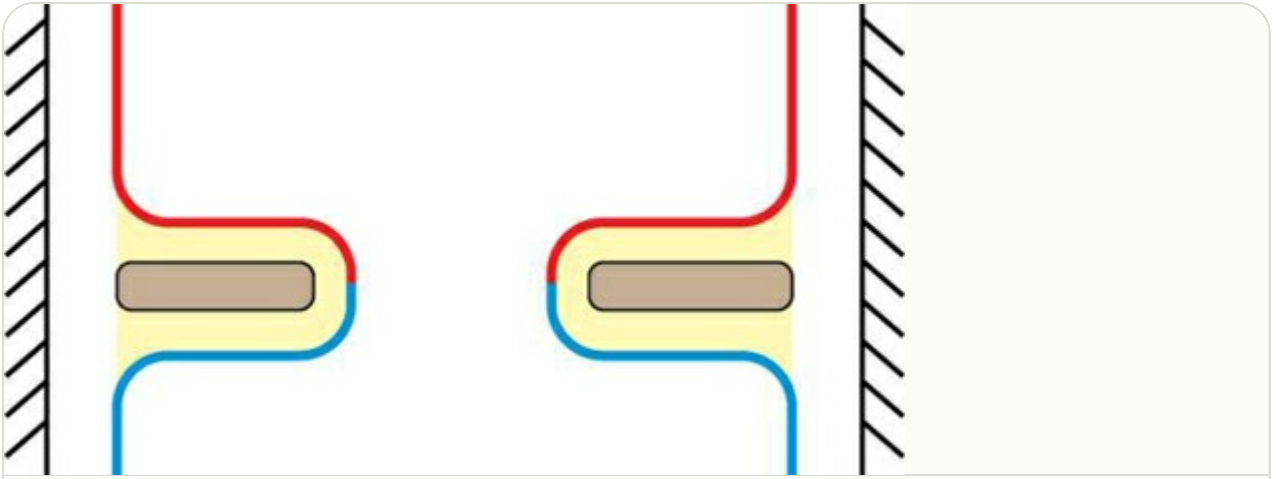


**FIGURE 2** A permanent perforation of the pars tensa: the epidermal and mucosal layers meet across its edge, so it behaves like a non-healing fistula.
Dhingra, fig. 11.5

## 2.3 Why the discharge recurs

### 2.3.1 Reinfection through the tube

Infection ascends from the nose, sinuses, adenoids and tonsils, especially in children, and nose blowing pushes secretions into the middle ear.

### 2.3.2 Reinfection through the canal

Water entering through the perforation during bathing or swimming.

### 2.3.3 Biofilms

Bacteria, particularly *Pseudomonas aeruginosa*, form biofilms on the mucosa that shield them from antibiotics and immune cells, explaining the relapses after apparently successful treatment.

### 2.4 Organisms

*Pseudomonas aeruginosa*, *Proteus*, *Escherichia coli* and *Staphylococcus aureus* are the commonest aerobes, often with anaerobes such as *Bacteroides*. Fungi may follow repeated antibiotic drops.

---

# 3 CHAPTER THREE Clinical features

The safe ear.

### 3.1 Symptoms

#### 3.1.1 Discharge

Profuse, mucoid or mucopurulent, without a foul smell, and intermittent: it flares with colds and with water in the ear and dries between episodes.

#### 3.1.2 Hearing loss

Conductive, of variable degree, and rarely over 50 dB. Some patients hear better when the ear is wet, because the discharge shields the round window and restores a phase difference between the windows.

---

### 3.2 Signs

#### 3.2.1 The perforation

Central, with a rim of drum all round it. It may lie anterior, posterior or inferior to the handle of malleus, or be subtotal. A large perforation may expose the ossicles and the promontory.

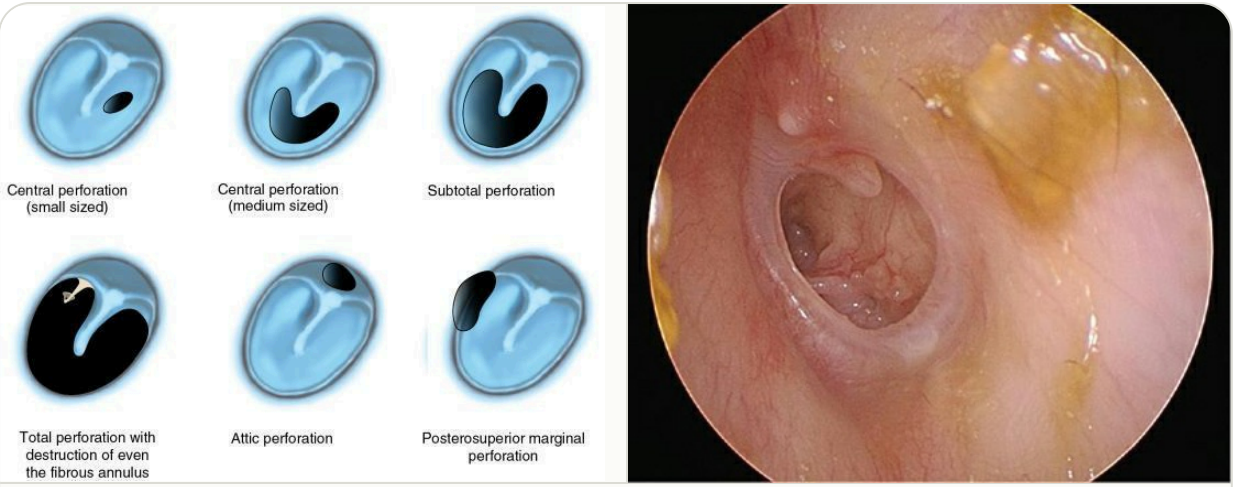



**FIGURE 3** First, the types of perforation seen in the pars tensa. Second, a large central perforation.

Dhingra, figs 11.10 and 11.11

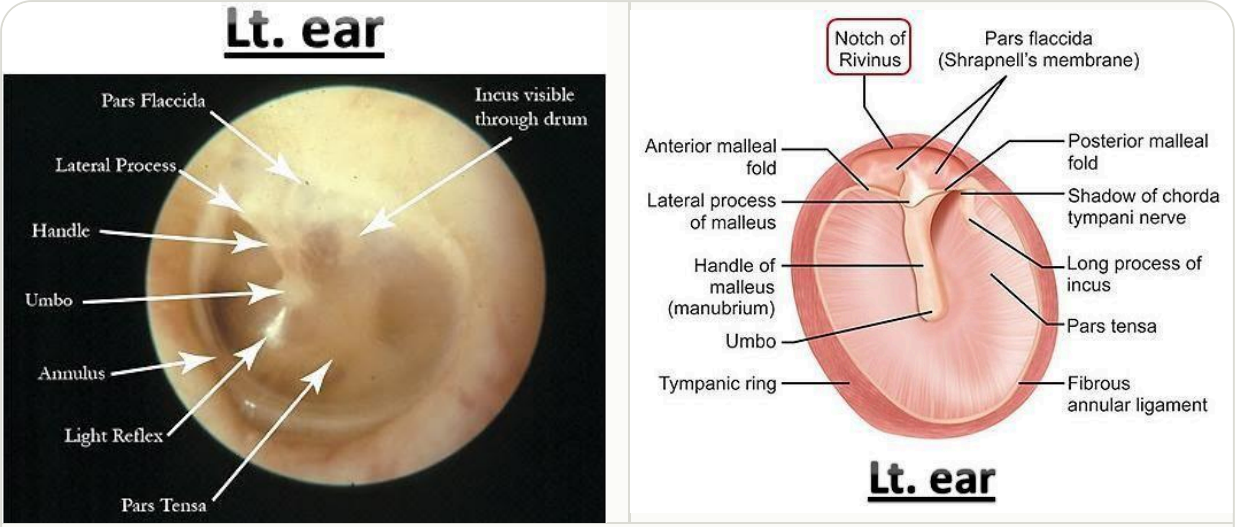



**FIGURE 4** A normal tympanic membrane and its parts, for describing the site of a perforation.

Anatomy atlas · ENT Steps p. 36

### 3.2.2 The middle ear mucosa

Normal and pale when inactive, red and oedematous when active. A pale polyp may fill the perforation.

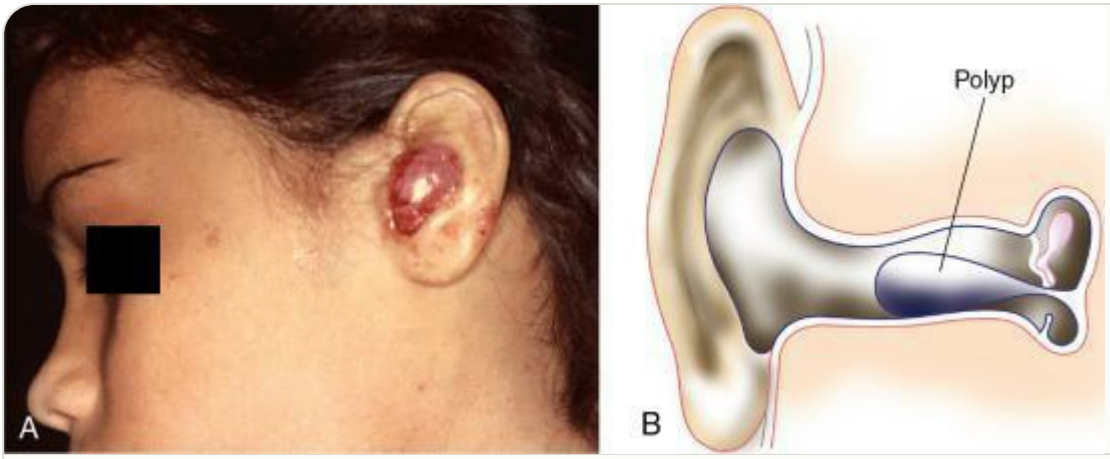



**FIGURE 5** An aural polyp in the canal, and how a polyp from the promontory can present through the perforation.

Dhingra, fig. 11.9

**NEVER AVULSE A POLYP**

A polyp may arise from the stapes, the facial nerve or the lateral semicircular canal. Pulling it out risks facial palsy or a dead ear; remove it under the microscope or at surgery, and send it for histology.

## 3.3 Safe or unsafe?

### MUCOSAL VERSUS SQUAMOUS CHRONIC OTITIS MEDIA

| | MUCOSAL (SAFE) | SQUAMOUS (UNSAFE) |
|---|---|---|
| **Discharge** | Profuse, mucoid, not foul | Scanty, foul-smelling |
| **Perforation** | Central | Attic or posterosuperior marginal |
| **Polyp** | Pale | Red, fleshy |
| **Cholesteatoma** | Absent | Present |
| **Complications** | Rare | More common |
| **Hearing** | Mild to moderate conductive | Conductive or mixed |

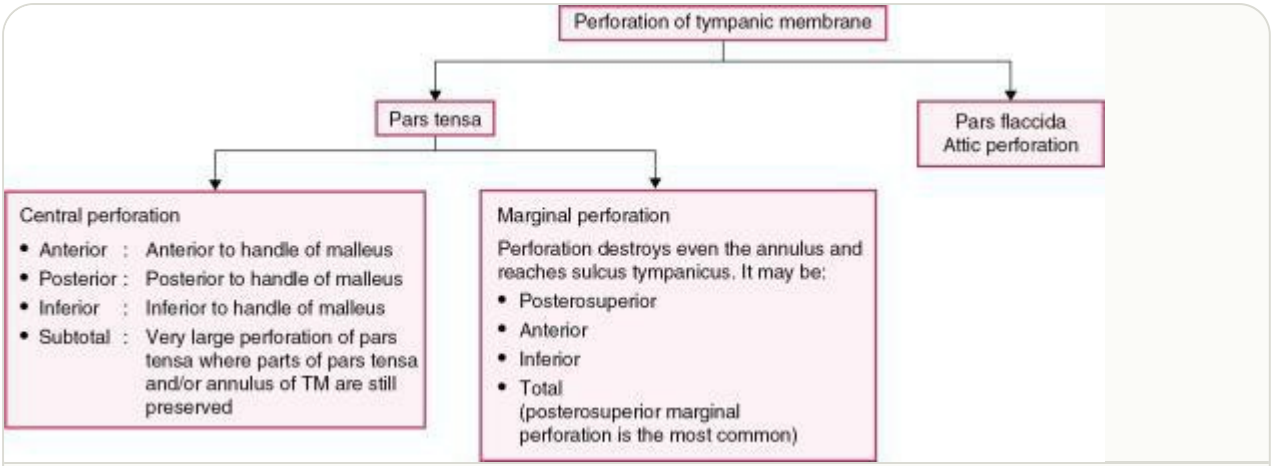



**FIGURE 6** Sites of perforation: central perforations belong to the safe type; attic and posterosuperior marginal perforations to the dangerous type.

Dhingra, fig. 11.8

# 4 CHAPTER FOUR Investigations

Look, test, and image when needed.

## 4.1 Examination under the microscope

Every chronic ear should be cleaned and examined with the microscope. An apparently dry ear can hide discharge, and rarely cholesteatoma coexists with a central perforation.



**FIGURE 7** Examination of the ear under the operating microscope.
Dhingra, fig. 11.12

## 4.2 Swab for culture

Guides the choice of drops when the ear does not respond.

## 4.3 Tuning forks and audiogram

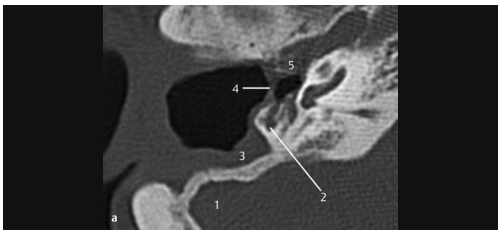
Measure the conductive loss and bone conduction. An air-bone gap over about 40 dB suggests ossicular discontinuity, often erosion of the long process of the incus.

## 4.4 Imaging

CT is not routine for mucosal disease, where it shows no bone erosion. It is used when cholesteatoma or a complication is suspected, and before revision surgery. MRI is reserved for intracranial concerns.

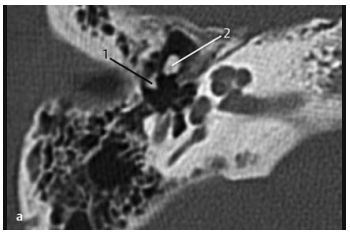
**IMAGING IN THE APP** 3 cases

A discharging modified radical cavity, and ossicular luxation and fixation.



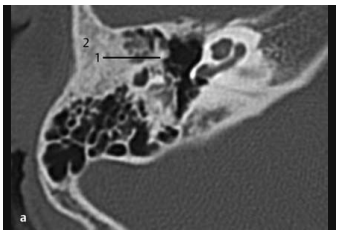

Modified Radical Cavity, Persistent Infections

After surgery · 2 images




Luxation of the Ossicular Chain

Trauma · 10 images




Fixation of the Ossicular Chain

Bone disease · 3 images

## 4.5 Biopsy

For granulation that persists despite treatment, to exclude tuberculosis and malignancy.

## 4.6 Differential diagnosis

- **Cholesteatoma:** scanty foul discharge, attic or marginal defect, white flakes.
- **Tuberculous otitis media:** painless thin discharge, pale granulation, multiple perforations, early facial palsy.
- **Granular myringitis:** granulation on the drum itself, without a perforation.
- **Carcinoma of the temporal bone:** pain, bleeding, and granulation that does not settle.

# 5 CHAPTER FIVE Management

Dry the ear, then decide on repair.

## 5.1 Aural toilet

Drops cannot work through pus. Clean the ear under the microscope by suction, dry mopping or gentle saline irrigation; never use forceful syringing.

## 5.2 Topical treatment

### 5.2.1 Drops

Quinolone drops are preferred when the drum is perforated. Continue for four to six weeks after debridement when the ear is persistently active.

### 5.2.2 Technique

Tragal pumping after instilling the drops pushes them through the perforation into the middle ear.

### 5.2.3 Acidifying drops

Acetic acid 1.5% helps against *Pseudomonas*.

WHERE THE BOOKS DIFFER · OTOTOXIC DROPS

| Dhingra | Lee |
| --- | --- |
| Advises avoiding aminoglycoside and polymyxin drops with a perforation. | Lists neomycin and polymyxin preparations among the options. |

Quinolone drops are the safer choice with an open middle ear.

## 5.3 Systemic antibiotics

Mainly for acute flares, chosen by culture.

## 5.4 Treating the source

Treat infected tonsils, adenoids and sinuses, and allergy; avoid hard nose blowing.

## 5.5 Keep water out

Cotton wool with petroleum jelly, or a moulded plug, during bathing; no swimming while the ear is active.

## 5.6 Surgery

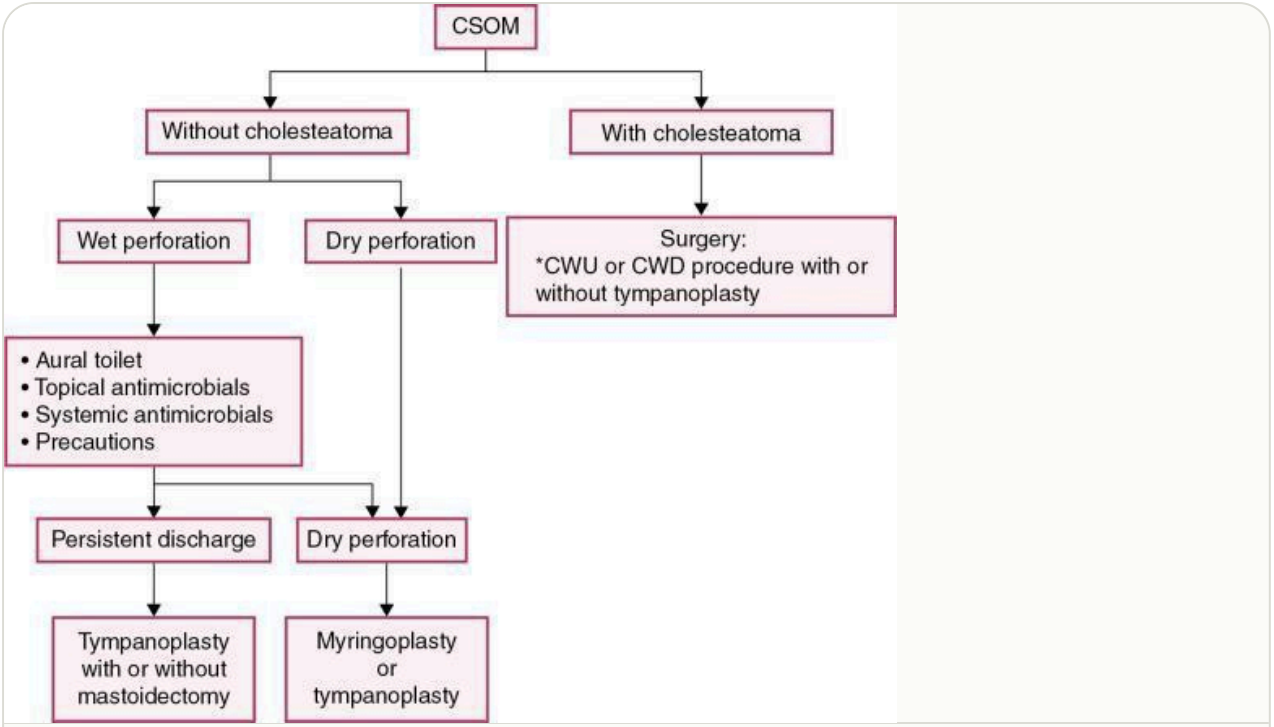



**FIGURE 8** A management pathway for chronic suppurative otitis media, from medical treatment to tympanoplasty and mastoid surgery.

Dhingra, fig. 11.14

### 5.6.1 Tympanoplasty

Once the ear has been dry for some weeks, repair of the perforation (myringoplasty), with ossicular reconstruction if needed, restores a closed, protected middle ear. Success is 60 to 90%, higher when air passes through the perforation on Valsalva, a sign of a working tube.

### 5.6.2 Mastoidectomy

For discharge that persists despite proper medical treatment. Cortical mastoidectomy may be combined with tympanoplasty; on its own it does not improve the rate of drum closure.

### 5.6.3 Hearing aids

For patients unfit for or unwilling to have surgery, ideally once the ear is dry.

**PEARLS**

- The other ear predicts how well the Eustachian tube works.

- Think of tuberculosis when an ear fails to respond, with pale granulation, multiple holes and facial weakness.
- No treatment of the Eustachian tube has proven benefit.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Myringoplasty and Tympanoplasty**
Myringoplasty or tympanoplasty

**Ossicular Chain Reconstruction**
Ossicular reconstruction

**Mastoid Surgery**
Tympanomastoidectomy for refractory disease

# 6 CHAPTER SIX Complications

Uncommon in the safe ear, but real.

## 6.1 Ossicular erosion

The long process of the incus is a vascular watershed and erodes first, giving a larger conductive loss.

## 6.2 Sensorineural hearing loss

Toxins crossing the round window can damage the basal turn of the cochlea, producing a high-frequency loss.

## 6.3 Tympanosclerosis

Chalky hyaline plaques in the drum and middle ear, which may fix the ossicles.

## 6.4 Others

Rarely, mastoiditis, facial palsy, labyrinthitis or intracranial infection in an acute flare.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Safe versus unsafe:** the six-row comparison.
- **Why the perforation does not heal,** and why the discharge recurs.
- **Aural toilet,** quinolone drops and water precautions.
- **Hearing better when the ear is wet.**
- **Never avulse a polyp.**
- **Indications for tympanoplasty,** and what predicts success.

## 7.2 Common pitfalls

**COMMON PITFALLS**

- Prescribing drops into an ear full of pus.
- Using aminoglycoside drops through a perforation without need.
- Calling a chronic ear safe without examining it under the microscope.
- Missing tuberculosis or malignancy in granulation that will not settle.

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CLINICAL · EAR

# Cholesteatoma

A cholesteatoma is skin in the wrong place: keratinising squamous epithelium growing inside the middle ear and mastoid, shedding keratin it cannot clear, expanding and eroding bone as it goes. Drops will not cure it. The operation aims for a safe, dry ear first and hearing second, and the follow-up lasts for years.

**AT A GLANCE**

- Keratinising skin in the middle ear that erodes bone.
- Scanty foul discharge with an attic or marginal defect.
- CT maps the disease; non-EPI MRI finds recurrence.
- Surgery is the treatment: safe ear first, hearing second.
- Assume a lateral canal fistula until shown otherwise.
- EAONO/JOS staging; endoscopic ear surgery and mastoid obliteration are growing; DWI misses disease under about 3 mm.

## 1 CHAPTER ONE Overview

Skin in the wrong place.

### 1.1 Definition

A cholesteatoma is a three-dimensional collection of keratinising stratified squamous epithelium in the middle ear cleft, with accumulated keratin debris in its centre. Despite the name it contains no cholesterol and is not a tumour; *keratoma* would be a better word.

### 1.2 Structure

#### 1.2.1 Matrix

The living wall: keratinising squamous epithelium resting on a layer of inflamed connective tissue, the *perimatrix*.

#### 1.2.2 Contents

White, flaky or pearly keratin debris, often infected and foul smelling.

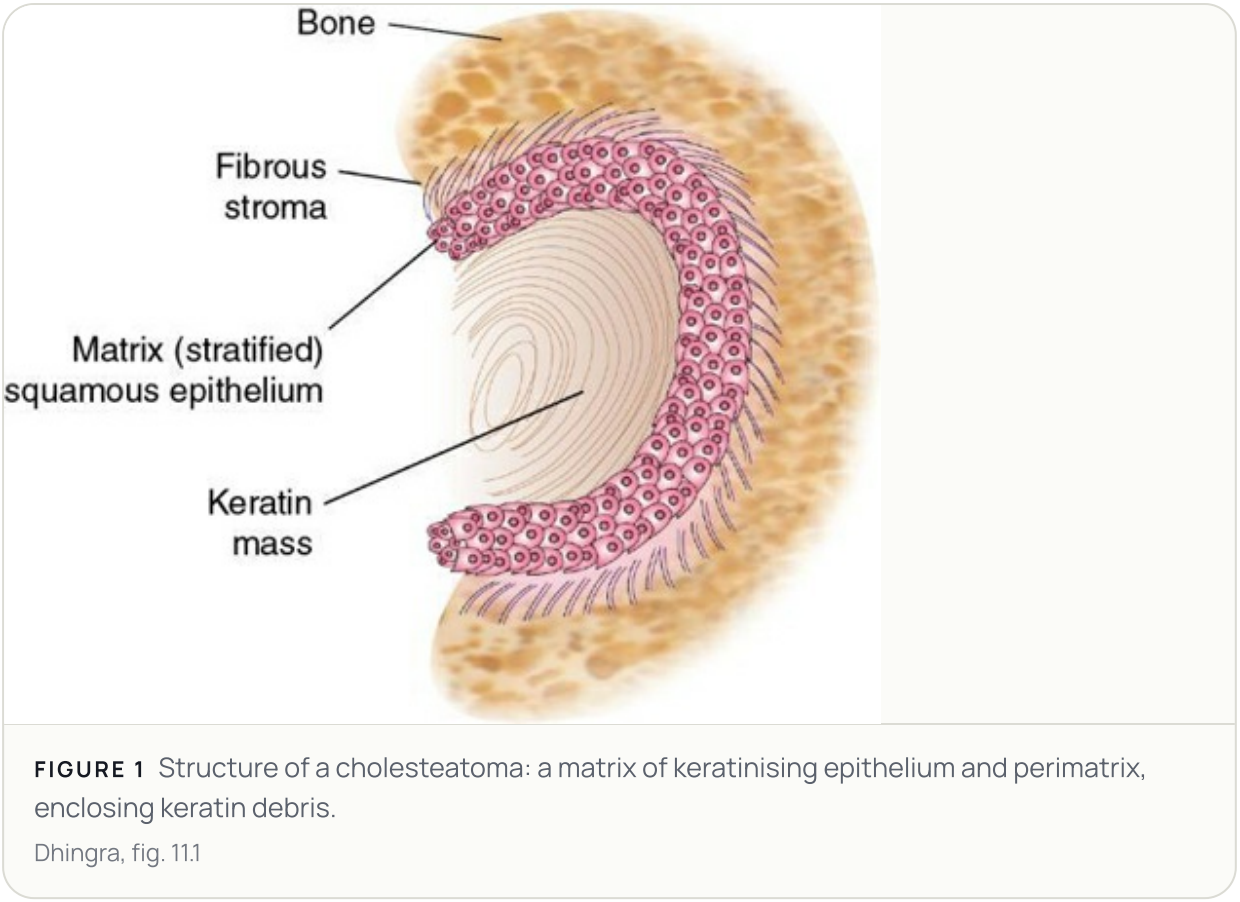



**FIGURE 1** Structure of a cholesteatoma: a matrix of keratinising epithelium and perimatrix, enclosing keratin debris.

Dhingra, fig. 11.1

### 1.3 Why it matters

It is the *unsafe* chronic ear. Cholesteatoma destroys the ossicles, the labyrinth, the facial canal and the bony plates over the dura and sigmoid sinus, and it causes most of the serious complications of chronic otitis media.

## 2 CHAPTER TWO Classification and genesis

Where it comes from.

### 2.1 Congenital cholesteatoma

A white mass behind an intact drum, usually anterosuperior, in a child with no history of otitis media. It arises from epithelial rests left in the temporal bone, and may also occur in the petrous apex and the cerebellopontine angle.

## 2.2 Primary acquired cholesteatoma

The commonest type. A retraction pocket of the pars flaccida, drawn in by negative middle ear pressure, deepens into the attic; keratin accumulates in it, the neck narrows, and it becomes a self-perpetuating sac.

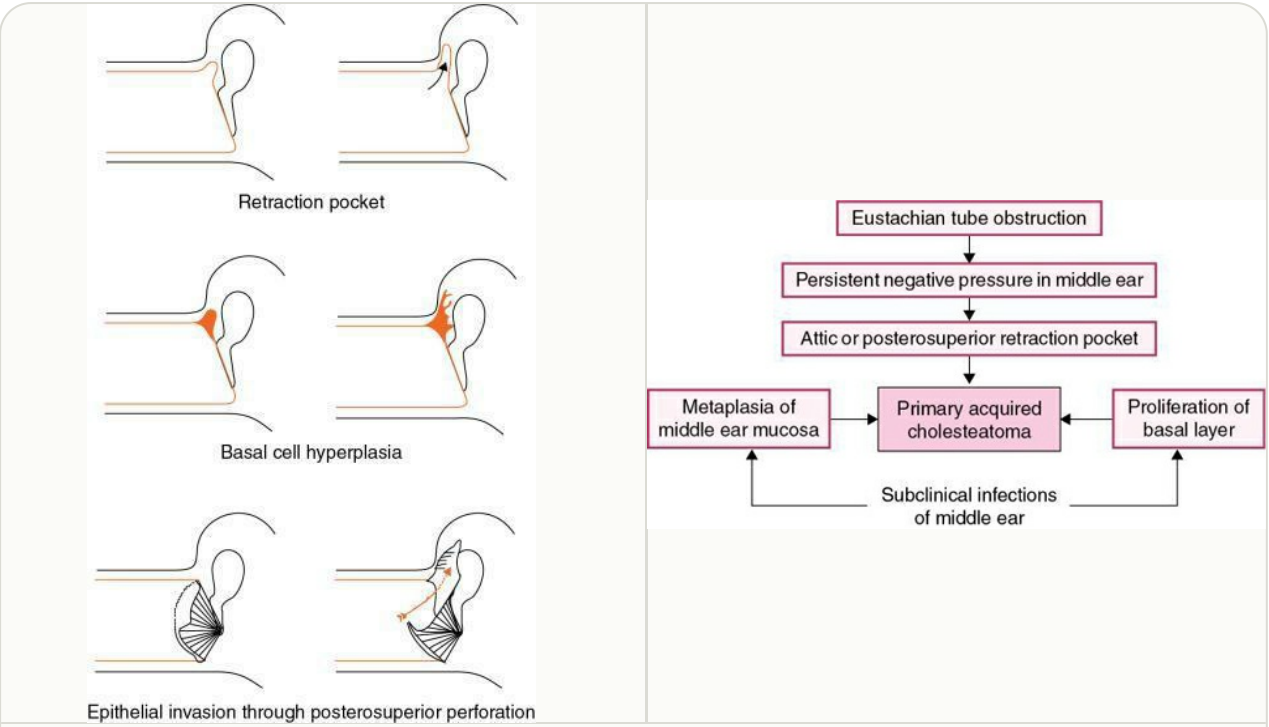



**FIGURE 2** First, the theories of genesis: a retraction pocket, basal cell hyperplasia and epithelial invasion through a posterosuperior perforation. Second, the sequence leading to primary acquired cholesteatoma.

Dhingra, figs 11.2 and 11.3

## 2.3 Secondary acquired cholesteatoma

Squamous epithelium grows into the middle ear through a marginal or posterosuperior perforation, or is implanted by trauma or surgery.

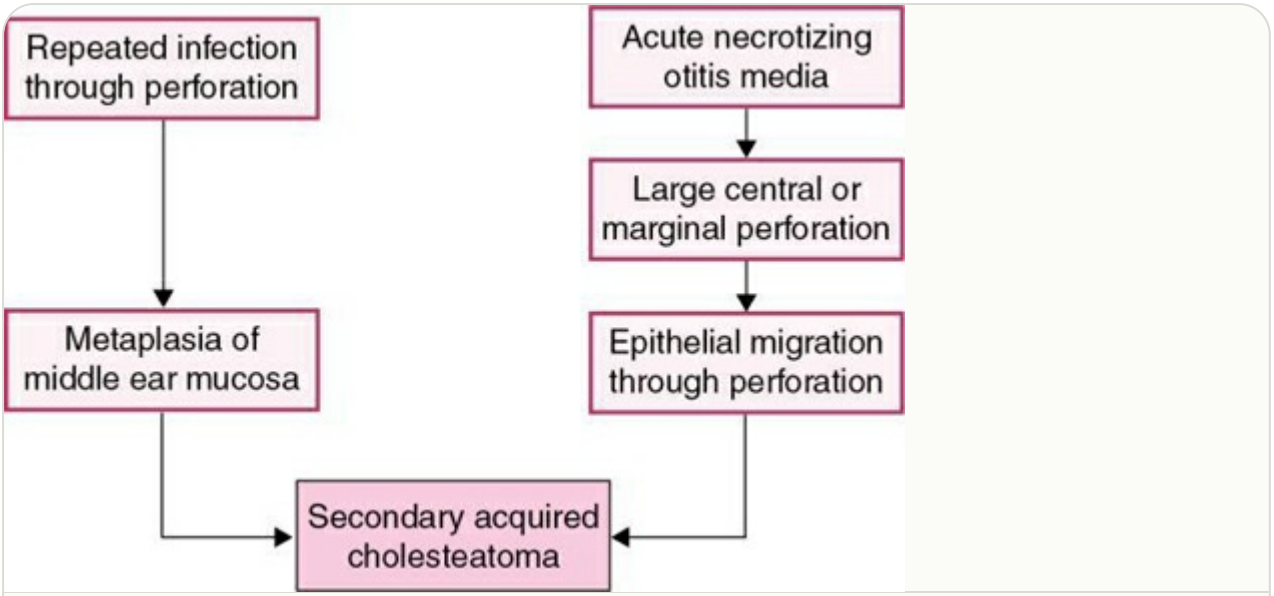



**FIGURE 3** Pathways to secondary acquired cholesteatoma: metaplasia of the middle ear mucosa, and epithelial migration through a large central or marginal perforation.

Dhingra, fig. 11.4

## 2.4 Theories of genesis

- **Cell rests:** for congenital cholesteatoma.
- **Invagination or retraction (Wittmaack):** for primary acquired.
- **Basal cell hyperplasia (Ruedi):** prickle cells of the pars flaccida invade the subepithelial tissue.
- **Epithelial invasion or migration (Habermann):** through a marginal perforation.
- **Squamous metaplasia (Sade):** of the middle ear mucosa after chronic inflammation.

## 2.5 Retraction pockets

Retraction of the pars tensa is staged by how deep it goes.

- **Stage I:** retracted, not touching the incus.
- **Stage II:** touching the incus.
- **Stage III:** lying on the promontory (atelectasis), but can be lifted off with suction.
- **Stage IV:** adherent to the promontory (adhesive otitis), with no middle ear space.

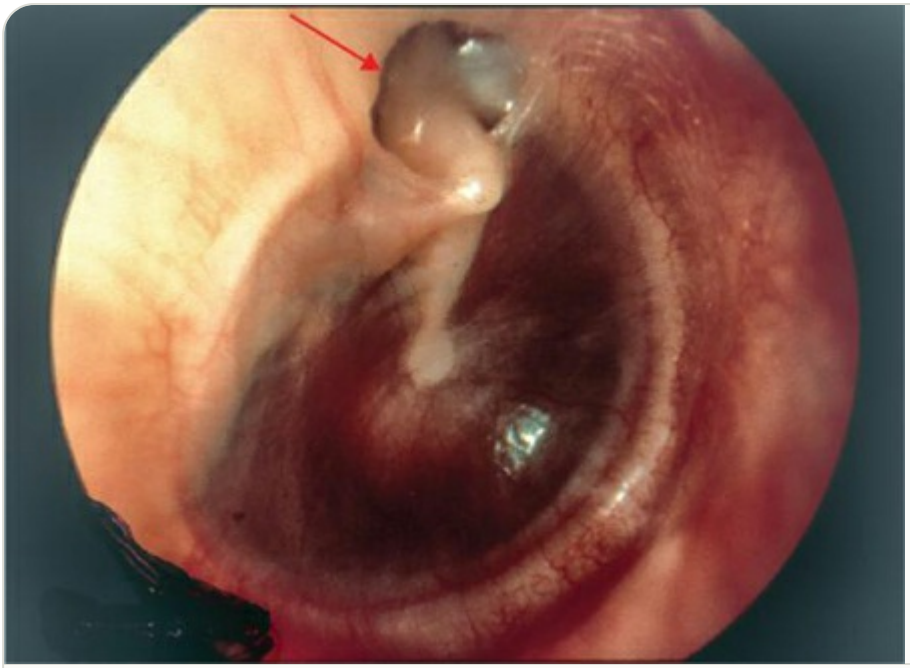


**FIGURE 4** A retracted drum with an attic retraction pocket (arrow).
Dhingra, fig. 11.6

**VENTILATION OF THE ATTIC**

The attic is ventilated through the narrow anterior and posterior tympanic isthmus. When the isthmus is blocked by swollen mucosa, the attic and Prussak's space become isolated and the pars flaccida retracts, which is why attic cholesteatoma can coexist with a normally ventilated mesotympanum.

# 3 CHAPTER THREE Pathology

How it destroys bone.

## 3.1 Bone erosion

Collagenases and other enzymes from the matrix and perimatrix, osteoclast activation by inflammatory cytokines, and infection all contribute. The mastoid is usually sclerotic.

**WHERE THE BOOKS DIFFER · HOW BONE IS ERODED**

| Dhingra | Lee |
| --- | --- |

| | |
|---|---|
| Attributes erosion to enzymes from osteoclasts and inflammatory cells, and rejects pressure necrosis. | Includes pressure along with the inflammatory mediators. |

## 3.2 Routes of spread

The cholesteatoma follows the path of least resistance: from Prussak's space into the attic, then through the aditus to the antrum and mastoid; or posteriorly into the sinus tympani and facial recess, the sites hardest to clear at surgery.

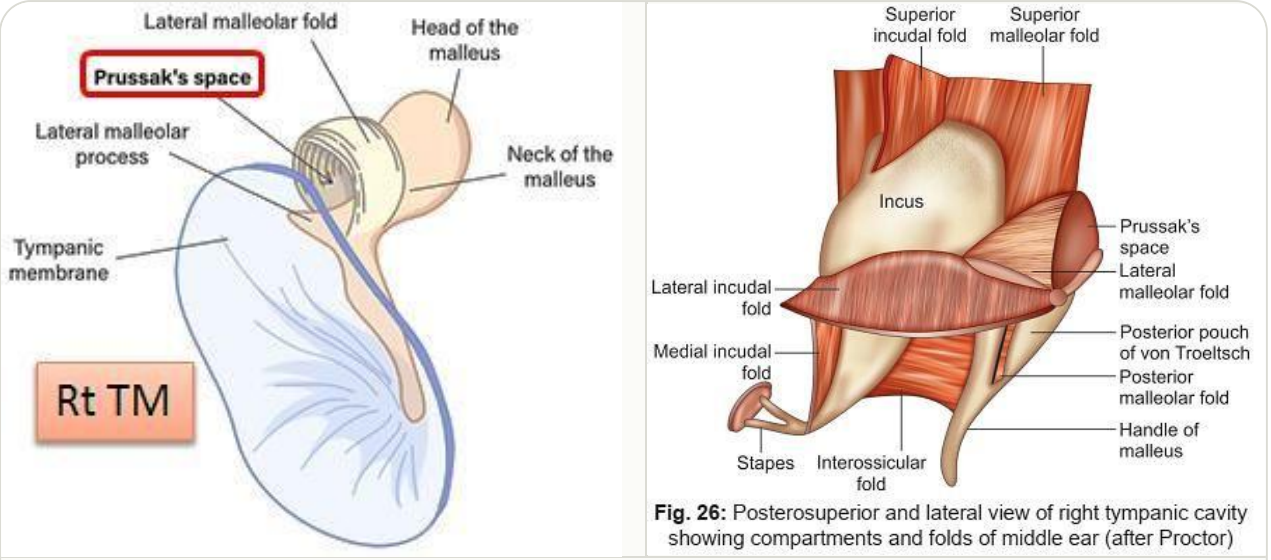



**FIGURE 5** Prussak's space, where attic cholesteatoma begins, and the folds and compartments that guide its spread.

Anatomy atlas · ENT Steps p. 54

## 3.3 Hidden sites for residual disease

Sinus tympani, facial recess, the undersurface of the scutum, and the anterior epitympanic recess.

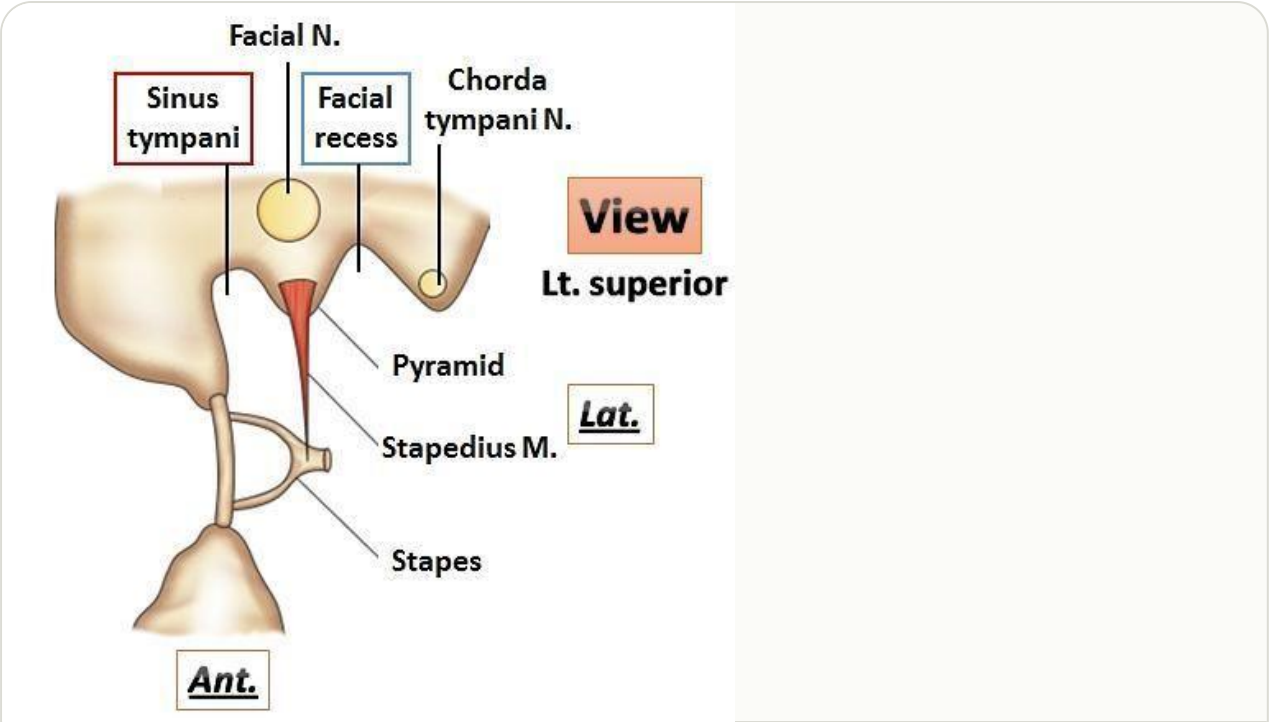



**FIGURE 6** The posterior wall of the middle ear: the facial recess and sinus tympani, the commonest hiding places for residual disease.

Anatomy atlas · ENT Steps p. 46

# 4 CHAPTER FOUR Clinical features

The scanty, foul, unsafe ear.

## 4.1 Symptoms

### 4.1.1 Discharge

Scanty, foul smelling and sometimes bloodstained, from infected keratin and granulation.

**A DISCHARGE THAT SUDDENLY STOPS**

Pus may have been blocked and may be tracking inwards. Ask about headache, fever, vertigo and facial weakness.

### 4.1.2 Hearing loss

Conductive, or mixed when the cochlea is involved. A cholesteatoma that bridges an eroded chain can transmit sound well: the *cholesteatoma hearer*, who must be

warned that hearing may worsen after surgery.

#### 4.1.3 Symptoms of complications

Earache, headache, vertigo, facial weakness, fever and vomiting, neck stiffness, double vision or unsteadiness, or a listless child who will not feed.

## 4.2 Signs

- An attic perforation or posterosuperior marginal perforation.
- White keratin flakes or a pearly mass.
- Erosion of the scutum or posterosuperior canal wall.
- A red, fleshy polyp that bleeds when touched.
- A positive fistula test in some.

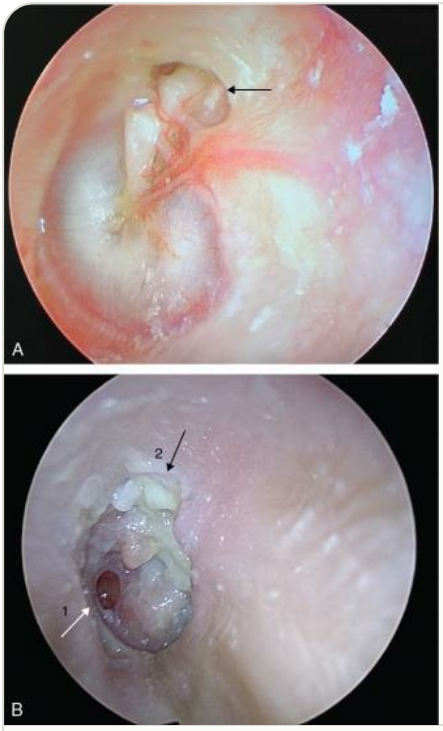



**FIGURE 7** (A) An attic retraction pocket. (B) A double perforation, in the pars tensa and in the attic.
Dhingra, fig. 11.13

**A BLUE DRUM**

A blue drum with dark fluid behind it suggests cholesterol granuloma, not cholesteatoma.

# 5 CHAPTER FIVE Investigations

Map the disease before you operate.

## 5.1 Examination under the microscope

Clean with suction; look for keratin, the extent of the defect and polyps.

## 5.2 Fistula test

Pressing on the tragus or using a Siegle speculum causes vertigo and nystagmus if the lateral canal is eroded. It is specific but insensitive, positive in only 30 to 50% of fistulae. Do it in every case.

## 5.3 Audiogram

Measures conductive and sensorineural components before surgery.

## 5.4 High-resolution CT

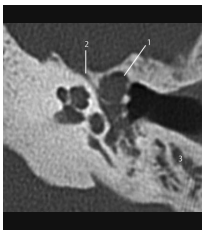
Shows the extent of disease and the state of the scutum, ossicles, tegmen, lateral semicircular canal, facial canal and sigmoid plate, and the anatomy for surgery.

## 5.5 Non-echo-planar diffusion-weighted MRI

Keratin restricts diffusion and shows bright. It detects residual or recurrent disease after canal wall up surgery, often replacing a second look, and helps when an intracranial complication is suspected.

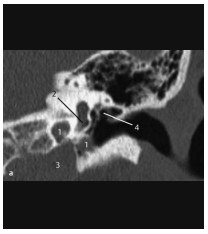
**IMAGING IN THE APP** 5 cases

Acquired and congenital cholesteatoma, a labyrinthine fistula, intracranial extension and canal cholesteatoma.



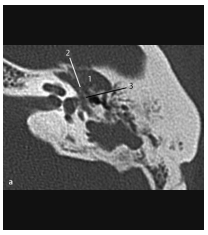
**Cholesteatoma of the Middle Ear**

Infection & inflammation · 12 images



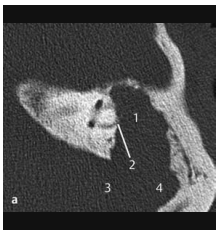
**Congenital Cholesteatoma**

Congenital · 11 images



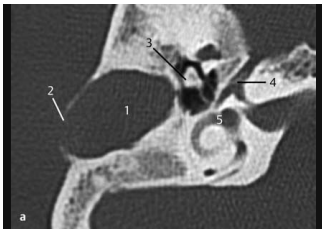
**Labyrinthine Fistula due to Cholesteatoma**

Infection & inflammation · 11 images



**Cholesteatoma with Intracranial Involvement**

Infection & inflammation · 7 images



**Inclusion Cholesteatoma and Atresia of the External Auditory Canal**

Congenital · 3 images

# 6 CHAPTER SIX Management

Surgery is the treatment.

## 6.1 Aims

In order: eradicate disease and make the ear safe; obtain a dry ear; preserve or restore hearing.

## 6.2 Conservative care

Regular microscopic cleaning suits small accessible disease, patients over 65 or unfit for surgery, or those who refuse an operation.

## 6.3 Choice of operation

### 6.3.1 Canal wall up

The posterior canal wall is kept (combined approach tympanoplasty). The ear looks normal, swimming is allowed and hearing aids are easy to fit, but residual disease is commoner, so a second look at about six months or MRI follow-up is needed.

### 6.3.2 Canal wall down

The posterior canal wall is removed and the attic, antrum and mastoid are exteriorised into one cavity (modified radical or radical mastoidectomy). Exposure is better and recurrence lower, but the cavity needs lifelong cleaning, water precautions, and aids are harder to fit.

CANAL WALL UP OR DOWN

| | CANAL WALL UP | CANAL WALL DOWN |
|---|---|---|
| **Appearance** | Normal meatus | Wide meatus and cavity |
| **Residual disease** | Higher | Lower |
| **Second look** | Often needed | Not routine |
| **Cavity care** | None | Regular cleaning |
| **Swimming** | Allowed | Limited |
| **Hearing aid** | Easy | Harder |

Choose canal wall up only when the patient can return for long-term follow-up.

### 6.3.3 Endoscopic ear surgery

Transcanal endoscopes suit limited attic and mesotympanic disease, and see around corners into the sinus tympani and anterior attic without a postauricular incision. Disease reaching the antrum or mastoid needs a combined approach. The surgeon works one-handed, and bleeding and heat from the light source are drawbacks. Residual disease found by the endoscope after apparent microscopic clearance ranges from 17 to 38%.

OPERATIVE GUIDES IN THEATRE 3 operations

**Mastoid Surgery**
Mastoidectomy

**Ossicular Chain Reconstruction**
Ossicular reconstruction

**Cholesterol Granulomas and Congenital Epidermoid Tumors of the Temporal Bone**
Congenital and petrous disease

### 6.4 Lateral canal fistula

Present in 7 to 10% of cholesteatomas, 90% of them in the lateral canal; only about 60% are seen on preoperative CT. Assume one in every case.

- Small fistula: the matrix is often removed at the end of the operation and the fistula sealed.
- Large fistula or one in an only hearing ear: leave the matrix on it and convert to canal wall down.
- If the endosteum tears, cover it immediately.
- Never strip the matrix off a cochlear fistula.

### 6.5 Hearing reconstruction

Ossiculoplasty, at the first stage or the second look.

## 7 CHAPTER SEVEN Complications

Why it is called unsafe.

### 7.1 Intratemporal

- Ossicular erosion, usually the incus first.
- Labyrinthine fistula and labyrinthitis.
- Facial palsy, usually from the tympanic segment.
- Mastoiditis and abscesses.

### 7.2 Intracranial

Extradural and subdural abscess, meningitis, brain abscess, lateral sinus thrombosis and otitic hydrocephalus, through an eroded tegmen or sinus plate.

**AT SURGERY**

Facial canal dehiscence and tegmen dehiscence often accompany a fistula. The epineurium does not need opening unless disease has invaded the nerve.

CURRENT PRACTICE (OXFORD TEXTBOOK)

**Staging.** The EAONO/JOS system, from European and Japanese otological societies, divides the middle ear and mastoid into sites, with the *sinus tympani* and *protympanum* counted as difficult-access sites where residual disease is most often left. Stage I disease occupies one site, stage II two or more, stage III has an extracranial complication (facial palsy, labyrinthine fistula, postauricular abscess, canal wall destruction), and stage IV has intracranial spread. Using it allows outcomes to be compared between surgeons and techniques.

**Endoscopes.** A rigid 0° endoscope in clinic shows the full depth of a retraction pocket and documents it photographically. *Total endoscopic ear surgery* through the canal suits disease confined to the middle ear and attic: the wide angle shows the sinus tympani, facial recess and anterior attic better than the microscope, without a postauricular incision, but it is more demanding and its long-term recidivism figures are still emerging.

**Canal wall up or down.** Neither is a gold standard, and evidence on hearing outcomes is equivocal. Canal wall up surgery is often staged, with a planned second stage some months later (the Oxford Textbook suggests 3 to 6) to remove residual disease and rebuild the ossicular chain. About 40% of open cavities discharge or are intolerant of water, which is why *mastoid obliteration* (bone pâté, muscle flaps or bioactive glass, with cartilage canal wall reconstruction) is increasingly used: it avoids a cavity and lowers recidivism.

**Follow-up imaging.** *Non-echo-planar diffusion-weighted MRI* has a sensitivity of about 90% and specificity of about 95% for residual cholesteatoma, but disease smaller than about 3 mm can be missed. It can replace a planned second look when clinical suspicion is low, but a negative scan does not cancel a second look when suspicion is high. Recidivism is either **residual** (incompletely removed) or **recurrent** (a new retraction), and both demand long-term follow-up.

# 8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

## 8.1 On call

**CHRONIC EAR WITH A NEW SYMPTOM**

1. New vertigo, facial weakness, headache, fever or neck stiffness: treat as a complication until proven otherwise.
2. Examine the facial nerve and do a fistula test.
3. Request CT, and MRI if intracranial spread is possible.
4. Start intravenous antibiotics and inform your senior.

## 8.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Definition** and structure: matrix, perimatrix, keratin.
- **Congenital, primary and secondary acquired,** and the theories of genesis.
- **Signs of the unsafe ear.**
- **Canal wall up versus down,** and who is suitable for each.
- **Fistula:** test, incidence and management.
- **Non-EPI diffusion MRI** for residual disease.

## 8.3 Common pitfalls

**COMMON PITFALLS**

- Treating an attic crust with drops and never lifting it.
- Forgetting the fistula test before surgery.
- Choosing canal wall up for a patient who cannot attend follow-up.
- Stripping matrix off a large fistula in an only hearing ear.

Written for this app from Oxford Textbook of Otolaryngology ch. 19, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 11 and K. J. Lee ch. 22, 25. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Intracranial complications of otitis media

A discharging ear with headache, fever, fits or drowsiness is an intracranial complication until proven otherwise. Meningitis is the commonest, brain abscess the next; often there is more than one. The order of work matters: resuscitate, image before any lumbar puncture, start antibiotics, involve neurosurgery, and only then deal with the ear.

**AT A GLANCE**

- Discharging ear plus fever, headache, seizures, focal signs or drowsiness.
- Meningitis is commonest; brain abscess is next.
- Contrast imaging comes before any lumbar puncture.
- IV antibiotics, neurosurgery, then treat the ear source.
- Check hearing afterwards.

## 1 CHAPTER ONE Overview

When otitis media crosses the dura.

### 1.1 Definition and scope

Infection spreading from the middle ear and mastoid into the cranial cavity: extradural abscess, subdural empyema, meningitis, brain abscess, lateral (sigmoid) sinus thrombophlebitis and otitic hydrocephalus. Perforation, ossicular erosion, tympanosclerosis and atelectasis are *sequelae* of otitis media, not complications.

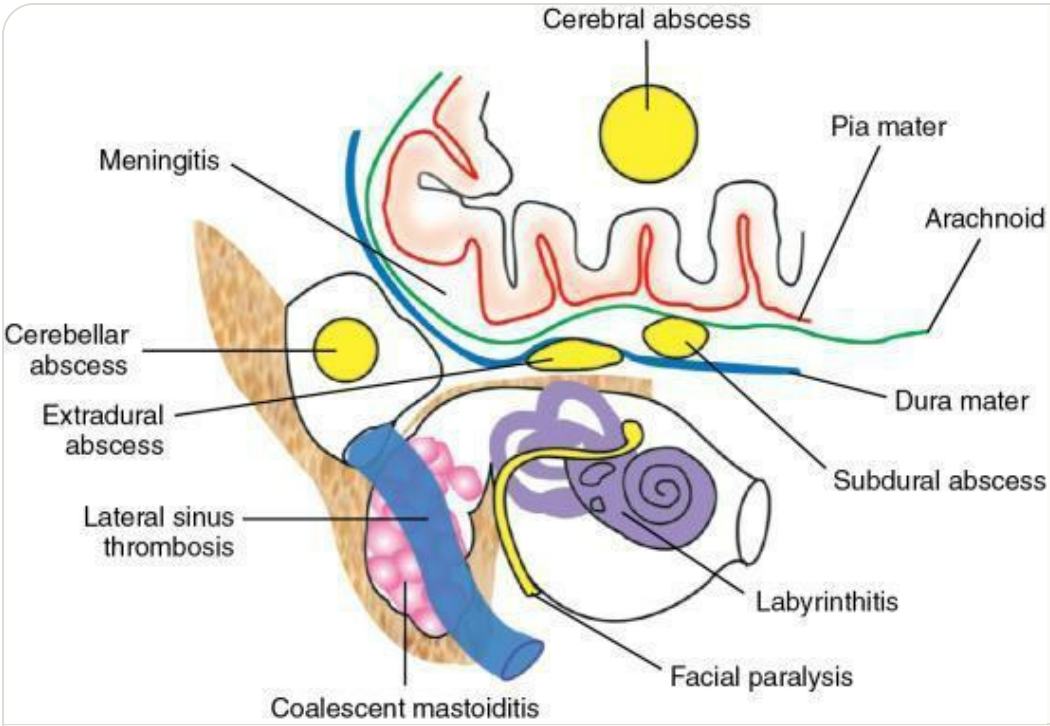



**FIGURE 1** The complications of suppurative otitis media and the routes of spread.
Dhingra, fig. 12.1

## 1.2 Epidemiology

Antibiotics have made these complications rare where care is good, but they remain important where chronic ears go untreated. Half of brain abscesses in adults and a quarter in children are otogenic, and a temporal lobe abscess is about twice as common as a cerebellar one.

## 1.3 Acute or chronic disease

Infants with acute otitis media tend to develop meningitis by blood-borne spread. Adults with chronic otitis media, especially cholesteatoma, develop abscesses and sinus thrombosis by bone erosion.

WHERE THE BOOKS DIFFER · HOW OFTEN THERE IS MORE THAN ONE

| Dhingra | Lee |
| --- | --- |
| About one-third of patients with otogenic meningitis have another intracranial lesion. | About half. |

Either way, look for a second complication in every patient.

CHAPTER TWO

# 2 Pathogenesis

Three routes into the skull.

## 2.1 Routes of spread

### 2.1.1 Direct bone erosion

Cholesteatoma, osteitis and granulation destroy the tegmen or the sinus plate, bringing infection to the dura.

### 2.1.2 Venous thrombophlebitis

Infected thrombus in small veins crosses intact bone, which is why an abscess can arise without visible erosion.

### 2.1.3 Preformed pathways

Congenital dehiscences, a patent petrosquamous suture or Hyrtl's fissure in children, inner ear malformations, fractures, previous surgery, and the round and oval windows via the labyrinth.

## 2.2 Risk factors

Extremes of age, poverty and poor access to care, virulent or resistant organisms, inadequate antibiotic courses, immunosuppression, preformed defects and cholesteatoma.

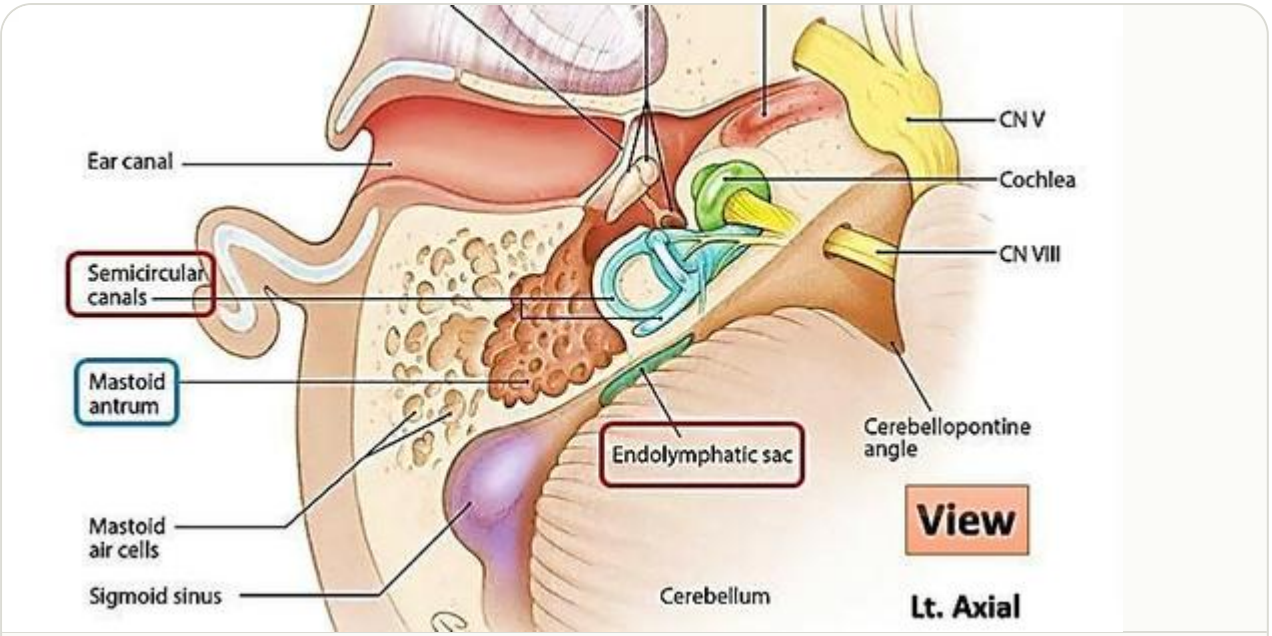



**FIGURE 2** Axial relations of the middle ear and mastoid: the dura, sigmoid sinus and labyrinth that infection can reach.

Anatomy atlas · ENT Steps p. 57

## 2.3 Organisms

Mixed: *Staphylococcus aureus*, streptococci, *Proteus*, *Pseudomonas* and *E. coli*, with anaerobes such as *Bacteroides*, particularly in abscesses.

# 3 CHAPTER THREE The complications

Each with its own picture.

## 3.1 Extradural abscess

### 3.1.1 Picture

Pus between the bone and the dura of the middle or posterior fossa. Often silent and found at mastoidectomy; sometimes a deep, one-sided headache that eases when pus flows from the ear.

### 3.1.2 Treatment

At mastoidectomy, remove bone until healthy dura is exposed. If pus is suspected behind an intact plate, take the plate down.

## 3.2 Subdural empyema

### 3.2.1 Picture

Pus in the subdural space spreads quickly over the hemisphere: severe headache, fever, meningism, seizures, aphasia and rapidly progressive hemiplegia, with raised pressure.

### 3.2.2 Treatment

A neurosurgical emergency: urgent drainage through burr holes or craniotomy, with high-dose antibiotics.

## 3.3 Meningitis

### 3.3.1 Picture

The commonest complication. Headache, fever, photophobia, vomiting, neck stiffness, and positive Kernig and Brudzinski signs; later, drowsiness, fits and coma.

### 3.3.2 CSF

Turbid, under raised pressure, with many polymorphs, high protein and low sugar.

### 3.3.3 Treatment

Intravenous antibiotics that penetrate the CSF, and steroids. Treat the ear: myringotomy for acute otitis media, mastoidectomy for chronic disease once the patient is fit.

**AFTER MENINGITIS**

Check hearing early. Labyrinthitis ossificans can close the cochlea within weeks, and a cochlear implant must go in before it does.

## 3.4 Brain abscess

### 3.4.1 Stages

WHERE THE BOOKS DIFFER · STAGES OF AN OTOGENIC BRAIN ABSCESS

| Dhingra | Lee |
| --- | --- |
| Four stages: invasion (initial encephalitis), localisation (the latent, quiet stage), enlargement, and termination by rupture into the ventricle or subarachnoid space. | Three stages: encephalitis, a quiet period, and rupture. |

The quiet stage is the trap: the patient seems better while the abscess is forming.

### 3.4.2 Raised intracranial pressure

Morning headache, projectile vomiting, papilloedema, a slow pulse and a subnormal temperature, with drowsiness.

### 3.4.3 Temporal lobe abscess

Nominal aphasia (with a dominant-side lesion), a contralateral upper quadrant field defect, contralateral weakness, and uncinate fits with hallucinations of taste and smell. An ipsilateral third nerve palsy with a dilated pupil warns of tentorial herniation.

### 3.4.4 Cerebellar abscess

Suboccipital headache, nystagmus, ipsilateral ataxia with past-pointing, intention tremor and dysdiadochokinesia, and early papilloedema.

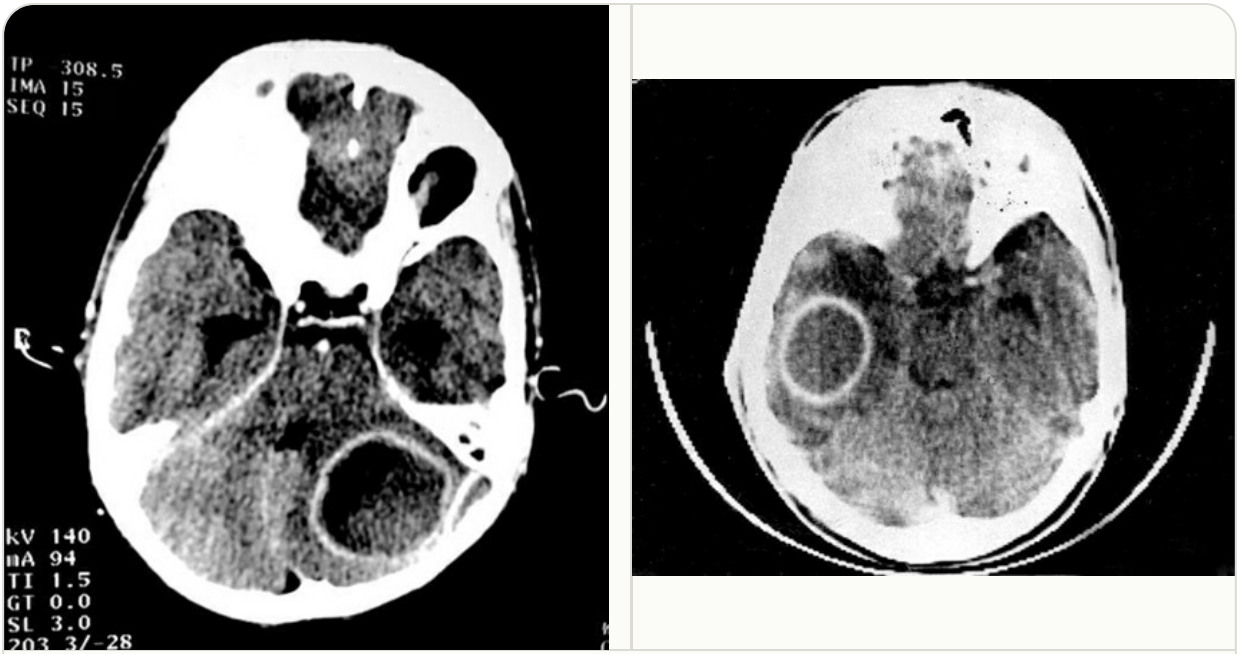



**FIGURE 3** First, CT of a left cerebellar abscess. Second, CT of a right otogenic cerebral (temporal lobe) abscess.

Dhingra, figs 12.6 and 12.7

### 3.4.5 CSF

Moderately raised cells and protein with normal sugar; but a lumbar puncture is dangerous and is avoided while there is mass effect.

### 3.4.6 Treatment

Broad-spectrum intravenous antibiotics with anaerobic cover, steroids or mannitol for raised pressure, and neurosurgical drainage by aspiration through a burr hole or excision. The ear is then made safe through a separate operative field.

**WHERE THE BOOKS DIFFER · TIMING OF EAR SURGERY WITH AN ABSCESS**

| Dhingra | Lee |
|---|---|
| Operate on the ear once the abscess is controlled. | Timing is debated; some treat both at one sitting when the patient is stable. |

## 3.5 Lateral sinus thrombophlebitis

### 3.5.1 Picture

Inflammation of the sigmoid sinus wall, mural thrombus, then an infected clot. The fever is the clue: high, swinging, *picket-fence* fever with rigors and well periods

in between, like malaria but irregular.

#### 3.5.2 Signs

- **Griesinger's sign:** oedema over the mastoid emissary vein.
- **Tobey-Ayer test:** compressing the internal jugular vein on the healthy side raises CSF pressure, but compressing the affected side does not.
- **Crowe-Beck test:** compressing the healthy jugular vein engorges the retinal veins.
- Papilloedema, more often with a right-sided thrombus as the right sinus is usually dominant.
- Tenderness along the jugular vein; septic emboli to the lungs, bones and joints.

#### 3.5.3 Investigations

Blood culture during a rigor, a smear to exclude malaria, and contrast CT (the *empty delta sign*) or MR venography.

#### 3.5.4 Treatment

Intravenous antibiotics and mastoidectomy with exposure of the sinus. If it is needle-aspirated and yields pus or no blood, it is opened between packs placed above and below, and infected clot is removed; healthy clot is left alone. Anticoagulation is debated. Ligation of the internal jugular vein is kept for septic emboli or clot spreading down the neck.

## 3.6 Otitic hydrocephalus

#### 3.6.1 Picture

Raised intracranial pressure with normal CSF and normal-sized ventricles, usually after lateral sinus thrombosis that blocks venous outflow. Headache, vomiting, blurred vision, papilloedema and a sixth nerve palsy.

#### 3.6.2 CSF

Pressure over 300 mm of water, with normal contents.

#### 3.6.3 Treatment

Treat the ear and the sinus; lower the pressure with acetazolamide, steroids and mannitol, with repeated lumbar punctures or a shunt if needed. Serial eye examinations are essential, because the danger is blindness from optic atrophy.

# 4 CHAPTER FOUR Investigations

Image before you puncture.

## 4.1 Imaging

### 4.1.1 Contrast CT of the brain and temporal bone

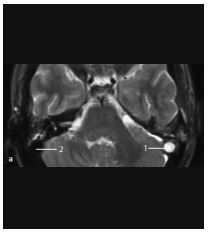
The first study: abscess, empyema, bone erosion and the delta sign.

### 4.1.2 MRI with contrast and MR venography

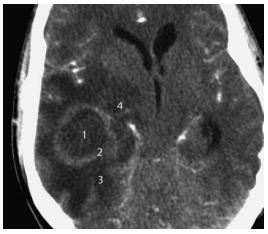
More sensitive for early cerebritis, subdural collections and sinus thrombosis.

**IMAGING IN THE APP** 5 cases

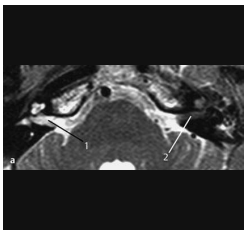
Sinus thrombosis, otogenic abscess, labyrinthitis with meningitis, and petrositis.



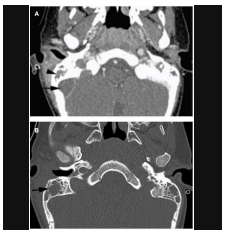
**Dural Sinus Thrombosis**
Infection & inflammation · 10 images



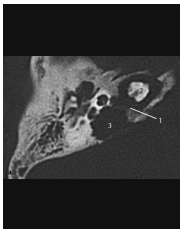
**Intracranial Abscess from an Otologic Focus**
Infection & inflammation · 1 image



**Labyrinthitis and Subclinical Meningitis**
Infection & inflammation · 5 images



**Otomastoiditis and its complications**
Infection & inflammation · 9 images



**Petrositis**
Infection & inflammation · 3 images

## 4.2 Lumbar puncture

Only when imaging has excluded mass effect and raised pressure.

**CSF BY COMPLICATION**

| COMPLICATION | PRESSURE | CELLS | SUGAR |
|---|---|---|---|
| Meningitis | Raised | Very high, polymorphs | Low |
| Brain abscess | Raised | Moderate | Normal |

| COMPLICATION | PRESSURE | CELLS | SUGAR |
|---|---|---|---|
| Sinus thrombosis | Raised | Normal | Normal |
| Otitic hydrocephalus | Over 300 mm H2O | Normal | Normal |

## 4.3 Other tests

Blood cultures, full blood count, ear swab, audiogram when possible.

### WHICH COMPLICATION?

| COMPLICATION | CLUE | NEXT STEP |
|---|---|---|
| Meningitis | Neck stiffness | CT, then lumbar puncture |
| Temporal abscess | Aphasia, field defect | MRI, drain |
| Cerebellar abscess | Ataxia, nystagmus | MRI, drain |
| Sinus thrombosis | Picket-fence fever | MR venography |
| Otitic hydrocephalus | Papilloedema, clear CSF | MRV, lower pressure |

# 5 CHAPTER FIVE Management

Patient first, ear last.

## 5.1 The sequence

**ON CALL: DISCHARGING EAR WITH FEVER AND HEADACHE**

1. Airway, breathing, circulation; treat sepsis.
2. Full neurological examination, including fundi and cranial nerves.
3. Contrast CT of brain and temporal bones, then MRI and MRV.
4. Start intravenous antibiotics with CSF penetration and anaerobic cover.
5. Call neurosurgery early.

6 No lumbar puncture until mass effect and raised pressure are excluded.

## 5.2 Treating the ear

Acute otitis media: myringotomy and grommet. Chronic disease or cholesteatoma: mastoidectomy to remove the source, once the intracranial problem is controlled and the patient is fit.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Intracranial Complications of Otitis Media**
Intracranial complications

**Mastoid Surgery**
Mastoid source control

## 5.3 Outcomes

Deafness, epilepsy, focal deficits, loss of vision and death. Rupture of an abscess into the ventricle carries a mortality of about 80%.

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Routes of spread:** erosion, thrombophlebitis, preformed pathways.
- **Stages of brain abscess,** and the danger of the quiet stage.
- **Temporal versus cerebellar abscess signs.**
- **Lateral sinus thrombosis:** picket-fence fever, Griesinger's sign, Tobey-Ayer and Crowe-Beck tests, delta sign.
- **CSF in each complication.**

- **Otitic hydrocephalus:** raised pressure, normal CSF, risk of blindness.

## 6.2 Common pitfalls

COMMON PITFALLS

- Doing a lumbar puncture before imaging.
- Being reassured by the quiet stage of an abscess.
- Treating swinging fever as malaria without examining the ear.
- Stopping at one complication and missing a second.
- Forgetting a hearing test after meningitis.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 12 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Otitis externa and skull base osteomyelitis

Most otitis externa is a painful, swollen canal that needs cleaning and the right drops, not oral antibiotics. The exception changes everything: an older diabetic with severe night pain and granulation in the canal has skull base osteomyelitis until proven otherwise.

**AT A GLANCE**

- Painful canal with tragal tenderness and a mobile drum.
- Clean it, wick it, use drops; oral antibiotics are rarely needed.
- Fungal disease itches and shows spores.
- Night pain and granulation in a diabetic is skull base osteomyelitis.
- Technetium confirms; gallium follows; treat about 6 weeks.
- Necrotising otitis externa: biopsy, MRI or PET-CT, dual antipseudomonal therapy for at least 6 weeks.

## 1 CHAPTER ONE Overview

Infection of the ear canal skin.

### 1.1 Definition

Inflammation of the skin of the external auditory canal, bacterial, fungal or reactive. The canal is a blind, warm, dark tube lined by thin skin over bone and cartilage, and anything that breaks its defences (moisture, scratching, loss of wax) lets infection in.

### 1.2 Forms

OTITIS EXTERNA

- BACTERIAL **Acute diffuse otitis externa** furuncle (localised)
- FUNGAL **Otomycosis** Aspergillus, Candida

- REACTIVE **Eczematous, seborrhoeic** contact dermatitis from drops or moulds, neurodermatitis
- CHRONIC **Chronic otitis externa** stenosis of the canal
- INVASIVE **Necrotising (malignant) otitis externa** skull base osteomyelitis

## 1.3 Applied anatomy

The outer third of the canal is cartilaginous, with hair follicles and sebaceous and ceruminous glands; the inner two-thirds is bony with thin, gland-free skin. Infection spreads from the canal through the *fissures of Santorini* in the cartilage and the tympanomastoid suture to the parotid, the skull base and the temporomandibular joint.

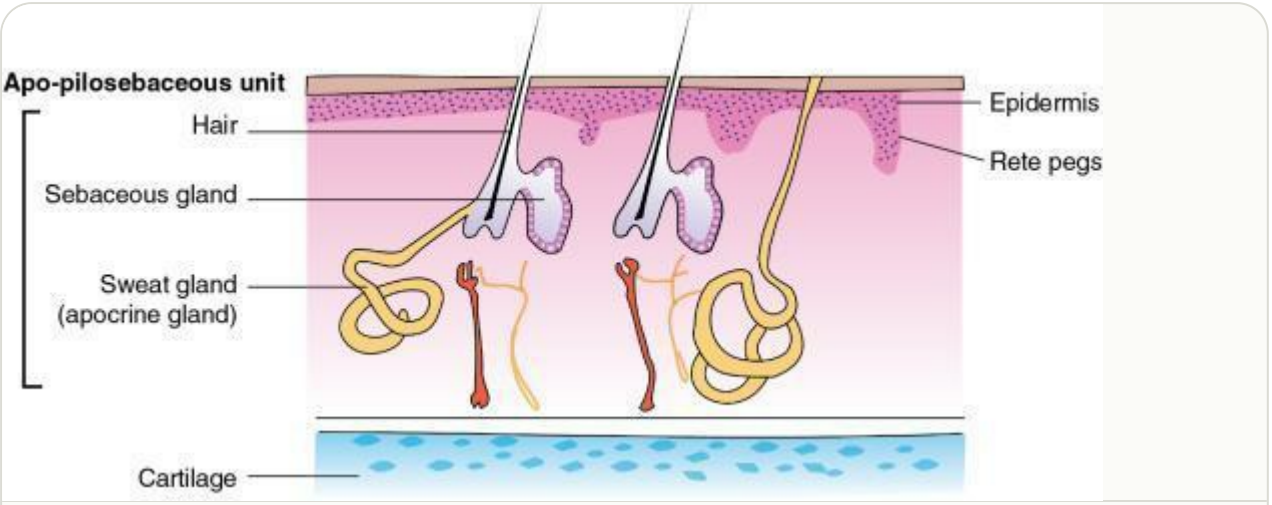



**FIGURE 1** Skin of the cartilaginous canal, with its hair follicles and sebaceous and ceruminous glands.
Dhingra, fig. 8.12

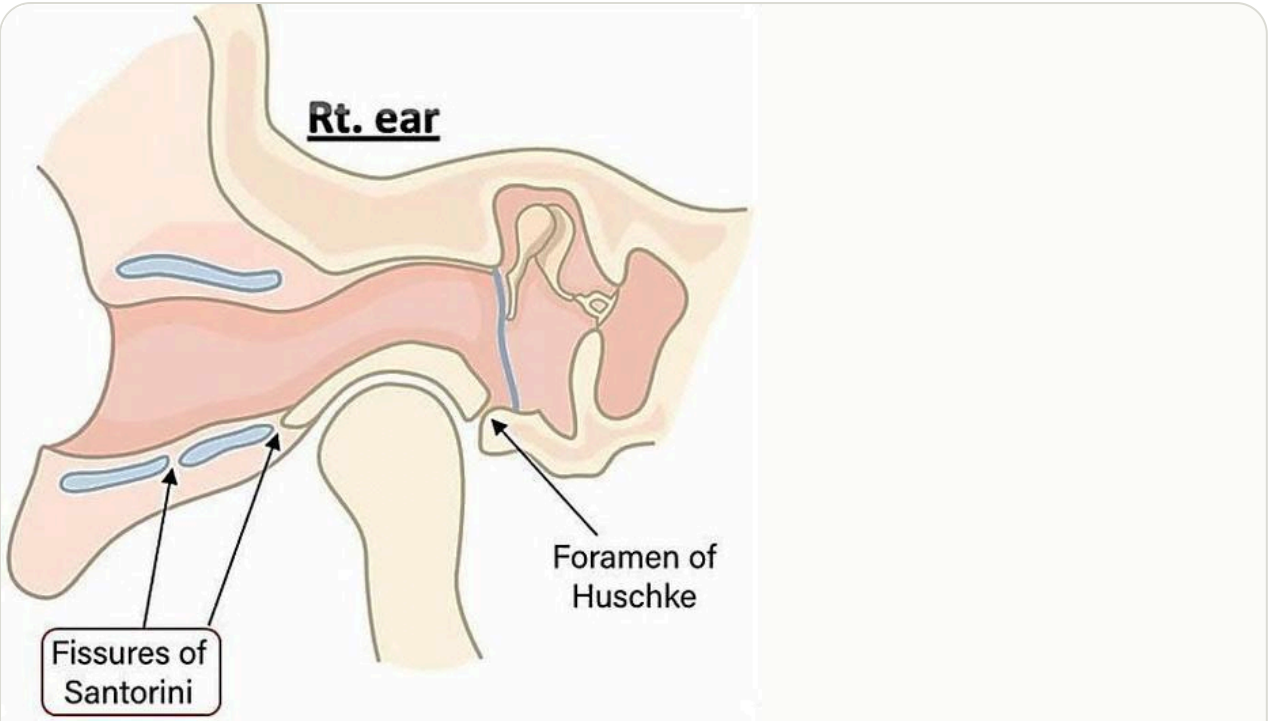



**FIGURE 2** Fissures of Santorini and the foramen of Huschke, routes from the canal to the parotid and skull base.

Anatomy atlas · ENT Steps p. 32

CHAPTER TWO

# 2 Acute diffuse otitis externa

Swimmer's ear.

## 2.1 Causes

Moisture (swimming, humid climates), trauma from cotton buds and fingernails, hearing aids and earbuds, eczema and loss of protective wax. *Pseudomonas aeruginosa* and *Staphylococcus aureus* are the usual organisms.

## 2.2 Clinical picture

### 2.2.1 Symptoms

Pain that develops over a day or two, itching, a feeling of blockage and a scanty discharge.

### 2.2.2 Signs

Pain on moving the tragus or pinna; a red, swollen canal with debris; tender nodes in front of or behind the ear. When visible, the drum moves normally on pneumatic otoscopy, unlike otitis media.

## 2.3 Management

### 2.3.1 Aural toilet

Clean the canal under vision by suction or dry mopping. Debris collects in the anteroinferior recess, so look there.

**OPERATIVE GUIDES IN THEATRE** 1 operation

**Office-Based Procedures in Otology**
Office debridement

### 2.3.2 Wick

If the canal is swollen shut, insert an expanding wick soaked in antibiotic-steroid drops, or in an astringent such as aluminium acetate 8% or silver nitrate 3%.

**WHERE THE BOOKS DIFFER · CHANGING THE WICK**

| Dhingra | Lee |
| --- | --- |
| Daily for two to three days. | Every three to five days. |

### 2.3.3 Drops

Antibiotic-steroid drops for a week to ten days. Acidifying with dilute vinegar helps mild cases and prevention.

### 2.3.4 Oral antibiotics

Not needed, unless there is spreading cellulitis or the patient is diabetic or immunocompromised.

### 2.3.5 Keep it dry

No swimming until healed; cotton wool with petroleum jelly when bathing.

**WHEN DROPS FAIL**

> Think of contact allergy to neomycin, fungal superinfection, or an underlying necrotising infection.

CHAPTER THREE

# 3 Furuncle

A boil in the canal.

## 3.1 Picture

A staphylococcal infection of a hair follicle, so it occurs only in the hair-bearing outer cartilaginous canal. Pain is out of proportion to its size and worse on chewing; it can swell the postauricular groove and mimic mastoiditis.

## 3.2 Distinguishing from mastoiditis

The drum is normal, there is no preceding otitis media and no mucoid discharge, and pinna movement is painful.

## 3.3 Treatment

Analgesia, oral antistaphylococcal antibiotics, heat, and an ichthammol-glycerine pack; incise when it points. Recurrent furuncles call for testing for diabetes and treating nasal staphylococcal carriage.

CHAPTER FOUR

# 4 Otomycosis

Itch and spores.

## 4.1 Organisms

*Aspergillus niger* (black-headed spores), *Aspergillus fumigatus* (pale blue or green) and *Candida* (creamy white). Repeated antibiotic drops, humidity and swimming

predispose.

## 4.2 Picture

Intense itching, discomfort, blockage and a watery discharge with a musty smell. The canal contains a mass like *wet newspaper*, with black, grey or white spores.

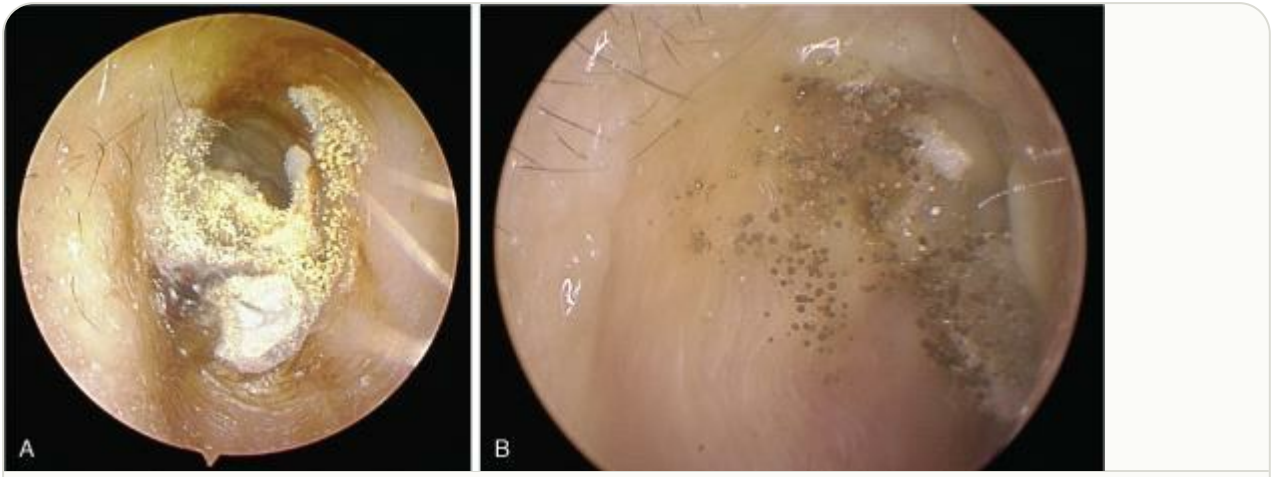



**FIGURE 3** Endoscopic view of otomycosis with yellow (A) and black (B) fungal spores.
Dhingra, fig. 8.11

## 4.3 Treatment

Meticulous cleaning, then a topical antifungal, continued for a week beyond apparent cure: clotrimazole for most, nystatin for *Candida*. Salicylic acid in alcohol and povidone iodine are alternatives.

# 5 CHAPTER FIVE Chronic and reactive otitis externa

Itch, scaling and narrowing.

## 5.1 Chronic otitis externa

Long-standing itch and irritation with thick, dry, scaling skin, which may narrow the canal. Treat underlying eczema or seborrhoeic dermatitis, use steroid drops or

ointment, and stop the patient scratching.

### 5.2 Reactive forms

Eczematous otitis externa from drops or hearing-aid moulds, seborrhoeic dermatitis with scalp involvement, and neurodermatitis from habitual scratching.

### 5.3 Otitis externa haemorrhagica

Blood blisters on the deep canal and drum, often with influenza.

### 5.4 Chronic stenosing otitis externa

Fibrous narrowing of the deep canal. Treated by excising the thick skin, widening the bony canal and grafting.

## 6 CHAPTER SIX Necrotising otitis externa

The dangerous one.

### 6.1 What it is

An invasive *Pseudomonas* infection (about 90% of cases) that spreads from the canal to the bone of the skull base: a skull base osteomyelitis. It affects older diabetics and the immunocompromised. *Aspergillus* causes a minority.

### 6.2 Clinical picture

#### 6.2.1 Symptoms

Severe, deep, unrelenting pain, worse at night and out of proportion to what is seen, with a persistent discharge despite treatment.

#### 6.2.2 Signs

Granulation tissue on the floor of the canal at the bony-cartilaginous junction. Cranial nerve palsies mark spread: the facial nerve first at the stylomastoid

> foramen, then the ninth, tenth and eleventh at the jugular foramen, and the twelfth. Multiple palsies carry a worse prognosis.

## 6.3 Investigations

### WHICH SCAN?

| TEST | USE | LIMITATION |
|---|---|---|
| Technetium-99 bone scan | Confirms early bone involvement | Stays positive for months |
| Gallium or indium scan | Follows response; repeated every few weeks | Less anatomical detail |
| CT | Bone erosion | Changes appear late |
| MRI | Soft tissue, marrow and dura | Not a guide to cure |

Add ESR or CRP (useful to follow), glucose, culture, and biopsy of the granulation to exclude carcinoma.

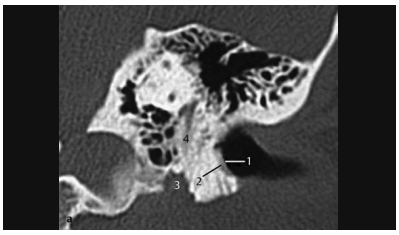
**WHERE THE BOOKS DIFFER · MONITORING AND DURATION**

**Dhingra**
Gallium repeated every three weeks; treatment for six to eight weeks, including intravenous aminoglycoside combinations.

**Lee**
Gallium every four weeks; oral fluoroquinolone for about six weeks as first line.

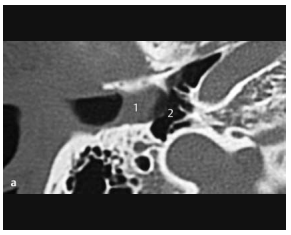
**IMAGING IN THE APP** 3 cases

Necrotising otitis externa, canal stenosis and canal carcinoma.



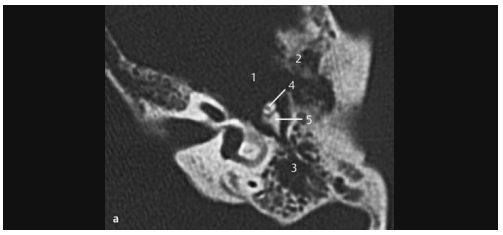
Malignant External Otitis or Necrotizing External Otitis

Tumours · 6 images



Stenosis of the External Auditory Canal

Bone disease · 2 images



Malignancy of the External Auditory Canal Squamous Cell Carcinoma

Tumours · 2 images

## 6.4 Treatment

Admit; control the diabetes; regular cleaning; high-dose oral ciprofloxacin, or intravenous anti-pseudomonal antibiotics for advanced disease; voriconazole for *Aspergillus*. Continue until the examination, inflammatory markers and scans have normalised. Surgery is limited to biopsy and removal of sequestra or abscesses. Hyperbaric oxygen is unproven.

## 6.5 Outcome

Mortality remains around 5 to 20%, from sepsis, sinus thrombosis and cranial neuropathies. Quinolone resistance is rising.

**SUSPECT IT WHEN**

- A diabetic or immunocompromised patient has severe pain and granulation in the canal.
- Otitis externa fails to respond after a week of proper treatment.
- A cranial nerve palsy appears with ear pain.

**NECROTISING OTITIS EXTERNA: CURRENT PRACTICE (OXFORD TEXTBOOK)**

Infection spreads from the canal floor through the **fissures of Santorini** into the infratemporal fossa and skull base. Between 65 and 100% of patients are diabetic, but the classic "elderly diabetic with a painful discharging ear" is not always present, especially in other forms of immunosuppression. *Pseudomonas aeruginosa* is the usual organism, about a quarter of isolates are resistant, and fungi are cultured in over 20%

of cases, more often in HIV with low CD4 counts and with a worse prognosis. Surface swabs may grow nothing, so send both a swab and a **granulation biopsy** for bacterial and fungal culture and for histology, because squamous carcinoma of the temporal bone is the key differential; if there is no tissue in the canal, deep samples may be taken at cortical mastoidectomy.

**Imaging.** CT shows erosion only after about 30% demineralisation, so a normal early CT is falsely reassuring. Contrast MRI with fat suppression and diffusion shows soft tissue and marrow disease earlier and detects intracranial spread. Gallium and technetium scans have largely been replaced by MRI, and **FDG PET-CT** is increasingly used to diagnose and to monitor response. CT and MRI changes persist for up to a year after successful treatment, so they cannot be used to decide when to stop.

**Treatment.** Admit for multidisciplinary care (otology, diabetes, microbiology, radiology, elderly care), start empirical intravenous antipseudomonal therapy at once, and give **dual therapy**, typically piperacillin–tazobactam or ceftazidime with a fluoroquinolone, adjusted to cultures. There is no agreed guideline, but consensus favours at least **six weeks** of intravenous treatment, often completed at home, guided by clinical response, inflammatory markers and imaging. Surgery is now limited to biopsy, deep sampling and drainage of abscesses; debridement and nerve decompression have no proven benefit.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 On call

**A PAINFUL, SWOLLEN EAR CANAL**

1. Clean under vision; insert a wick if the canal is closed.
2. Antibiotic-steroid drops; analgesia; keep dry.
3. Check glucose in diabetics and anyone with recurrent infection.
4. Diabetic or immunocompromised with severe pain and granulation: biopsy, ESR, imaging and admit.

## 7.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Tragal tenderness** and a mobile drum versus otitis media.
- **Furuncle** in the cartilaginous canal only.
- **Otomycosis:** organisms and appearance.
- **Necrotising otitis externa:** who, organism, signs, scans and treatment.

## 7.3 Common pitfalls

**COMMON PITFALLS**

- Prescribing oral antibiotics instead of cleaning and drops.
- Missing a necrotising infection in a diabetic.
- Neomycin contact allergy mistaken for treatment failure.
- Drops into a canal full of debris.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, 13, Oxford Specialist Handbook ch. 15, Oxford Handbook of ENT ch. 5, Dhingra ch. 8 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Tympanic membrane: trauma, retraction, barotrauma and sequelae

The drum is thin, and it records what happens around it. A traumatic hole usually heals by itself; a retracted drum is a sign of a failing Eustachian tube; and a sudden pressure injury can reach through the middle ear into the inner ear. In every case, check for vertigo, facial weakness and sensorineural hearing loss.

**AT A GLANCE**

- Most traumatic perforations heal on their own: keep dry.
- Vertigo or sensorineural loss needs urgent review.
- Retraction reflects tube failure.
- Barotrauma: decongestants, then myringotomy if needed.

## 1 CHAPTER ONE Overview

Injuries and scars of the drum.

### 1.1 The normal drum

A three-layered membrane: outer squamous epithelium, a middle fibrous layer, and inner mucosa. The fibrous layer gives it strength and shape; when it is lost after healing, the drum becomes thin and floppy. On otoscopy the normal right drum shows the handle of malleus, the lateral process and a bright cone of light at five o'clock.

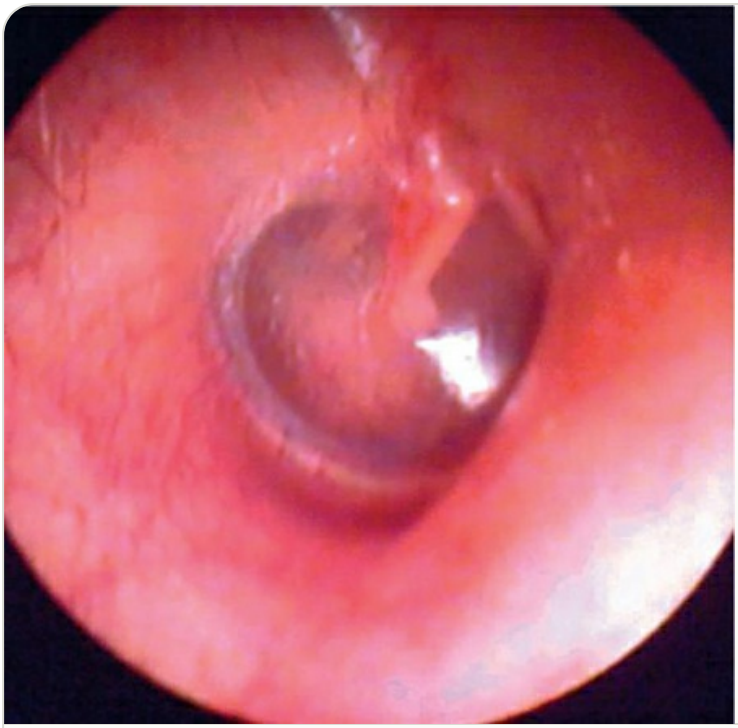


**FIGURE 1** A normal right tympanic membrane, with the cone of light at five o'clock.
Dhingra, fig. 8.17

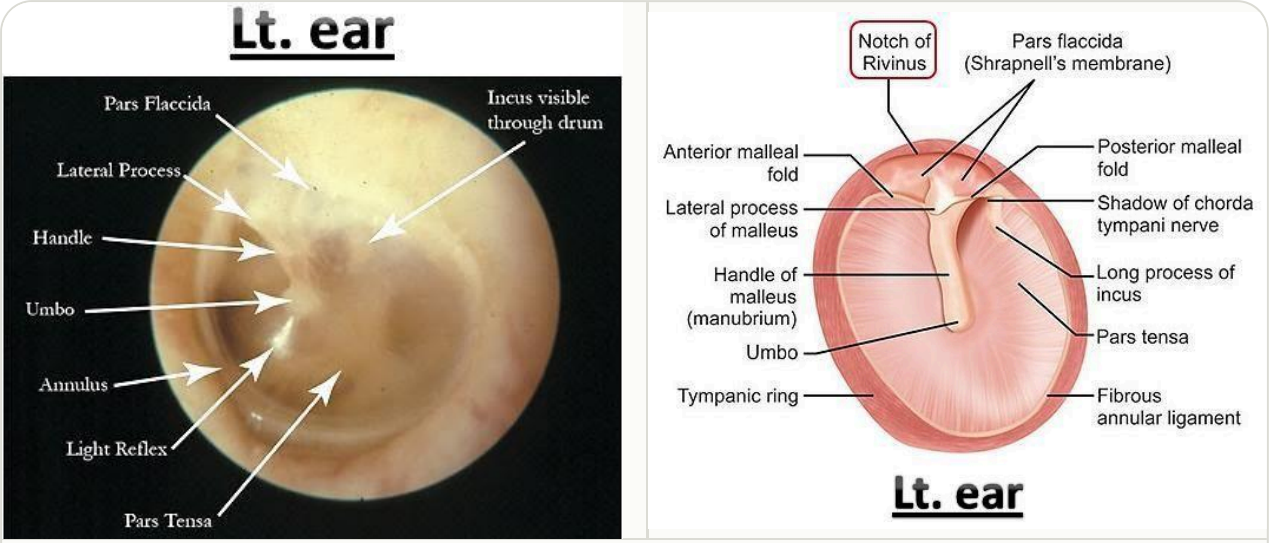



**FIGURE 2** A normal tympanic membrane seen through the otoscope, and its parts.
Anatomy atlas · ENT Steps p. 36

## 1.2 Conditions covered

- Traumatic perforation.
- Retraction, atelectasis and retraction pockets.
- Atrophic drum and tympanosclerosis.

- Barotrauma of the middle ear.

# 2 CHAPTER TWO Traumatic perforation

Most heal on their own.

## 2.1 Causes

- **Direct:** cotton buds, hairpins, matchsticks, careless syringing or instrument use.
- **Pressure:** a slap on the ear, blast injury, a fall onto water, a kiss on the ear.
- **Temporal bone fracture.**
- **Thermal:** welding sparks or hot slag, which heal poorly.

## 2.2 Clinical picture

Sudden pain, a little bleeding, deafness and tinnitus. The perforation is irregular or slit-like, often with blood clot and with its edges turned inwards.



**FIGURE 3** A traumatic perforation from blast, with ecchymosis (white arrow head) and tears of the drum (black arrows).

Dhingra, fig. 8.19

LOOK DEEPER

Vertigo, sensorineural hearing loss or facial weakness suggests stapes subluxation, perilymph fistula or a temporal bone fracture. Arrange urgent assessment and consider exploration.

## 2.3 Management

### 2.3.1 Assess

Tuning forks, nystagmus and the facial nerve; an audiogram when practical. CT if a fracture is suspected.

### 2.3.2 Keep dry, keep out

No drops, no syringing and no water in the ear. Antibiotics only if the ear becomes infected.

### 2.3.3 Repair

Most perforations heal within weeks. Large ones with in-turned edges may be repositioned and splinted with a paper patch under the microscope. Those that persist beyond about three months are repaired by myringoplasty.

OPERATIVE GUIDES IN THEATRE 3 operations

**Myringoplasty and Tympanoplasty**
Repair if it persists

**Temporal Bone Trauma**
Temporal bone trauma

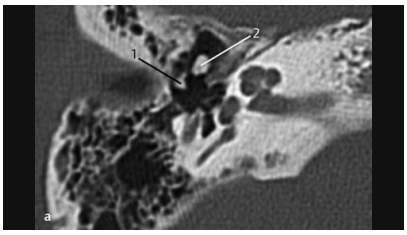
**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation**
Ventilation tube

## 2.4 Complications

Persistent perforation, infection, ossicular dislocation, perilymph fistula, and implantation cholesteatoma from squamous epithelium driven into the middle ear.

IMAGING IN THE APP 1 case

Ossicular luxation after trauma.




**Luxation of the Ossicular Chain**

Trauma · 10 images

# 3 CHAPTER THREE Retraction and atelectasis

The drum sucked in.

## 3.1 Why it happens

When the Eustachian tube fails, middle ear gas is absorbed and negative pressure draws the drum inwards. Healed or atrophic areas, which lack the fibrous layer, retract first.

## 3.2 Signs of a retracted drum

- A dull drum with a broken or absent light reflex.
- A prominent lateral process of the malleus.
- A foreshortened handle of malleus.
- Prominent anterior and posterior malleolar folds.

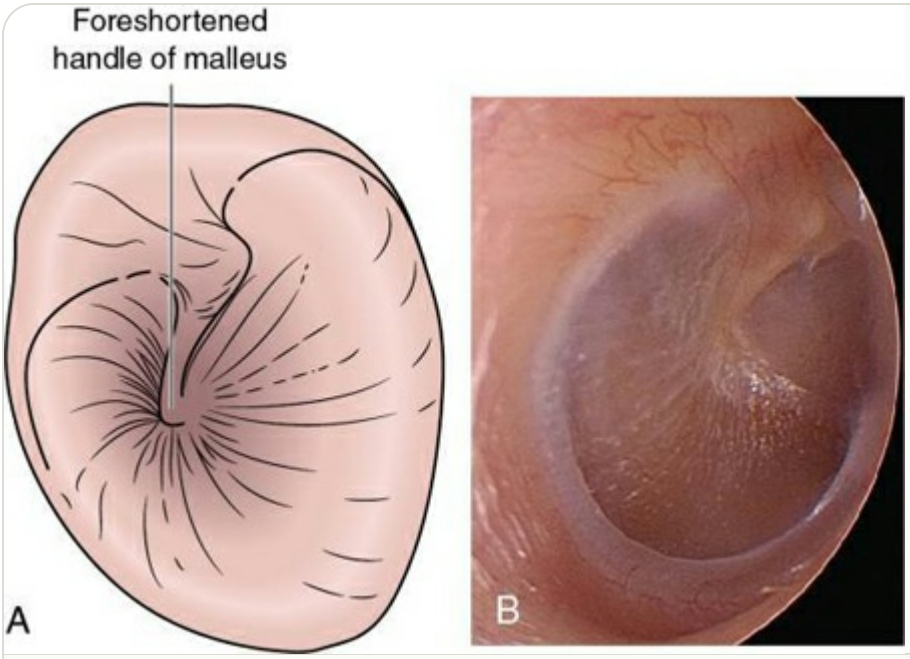



**FIGURE 4** Features of a retracted tympanic membrane.
Dhingra, fig. 8.18

## 3.3 Retraction pockets and atelectasis

### STAGES OF PARS TENSA RETRACTION

| STAGE | APPEARANCE |
|---|---|
| I | Retracted, not touching the incus |
| II | Touching the incus |
| III | Lying on the promontory (atelectasis), can still be lifted |
| IV | Adherent to the promontory (adhesive otitis), no middle ear space |

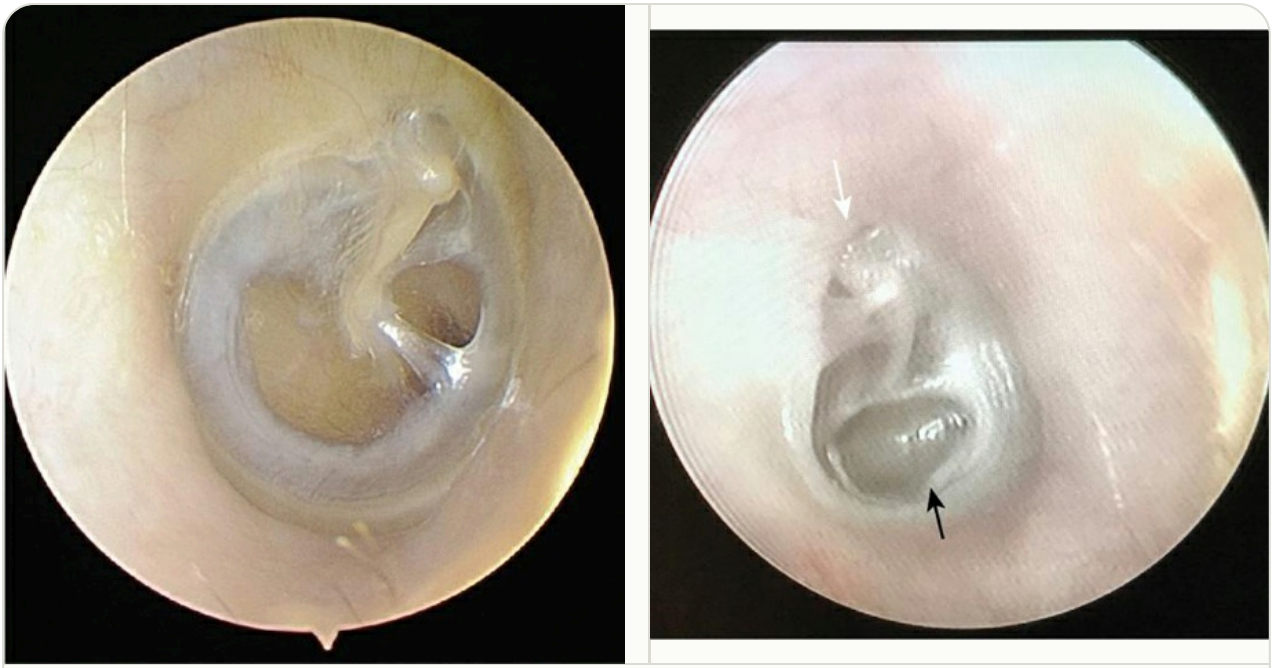


**FIGURE 5** First, an atrophic drum with a retraction pocket and an adhesion from the malleus. Second, atelectasis with retraction pockets in the attic (white arrow) and the pars tensa (black arrow).

Dhingra, figs 8.20 and 8.21

### 3.4 Management

Treat the cause of tubal failure (adenoids, sinusitis, allergy). A grommet re-aerates the middle ear while the drum can still be lifted (stage III). Deep pockets that collect keratin, or that cannot be cleaned, are excised and reinforced with cartilage before they become cholesteatoma.

## 4 CHAPTER FOUR Atrophy and tympanosclerosis

Scars of past disease.

### 4.1 Atrophic drum

A thin, transparent segment without a fibrous layer, after healed perforations or long-standing effusion. It moves excessively and retracts easily.

## 4.2 Tympanosclerosis

Chalky white hyaline and calcified plaques in the fibrous layer of the drum and sometimes in the middle ear around the ossicles, after recurrent otitis media or grommets. Plaques on the drum usually cause no symptoms; deposits around the ossicles can fix them and cause a conductive loss that is difficult to correct surgically.



**FIGURE 6** Tympanosclerosis with a central perforation: chalky plaques lie in front of and behind the hole.
Dhingra, fig. 8.22

# 5 CHAPTER FIVE Barotrauma

Aero-otitis media.

## 5.1 Mechanism

On descent in an aircraft or when diving, outside pressure rises. If the Eustachian tube does not open, middle ear pressure falls relative to it. Once the difference reaches about 90 mm Hg, the soft tissues at the pharyngeal end of the tube are

pushed together and the tube *locks*, so swallowing and Valsalva can no longer open it. A cold, allergy or sleeping through descent make this more likely.

### 5.2 Clinical picture

Severe pain, deafness, tinnitus and sometimes vertigo. The drum is retracted and congested, with haemorrhage into it, and later an effusion or blood (haemotympanum), or a perforation.

### 5.3 Inner ear barotrauma

Forceful Valsalva against a locked tube can rupture the round or oval window, causing sensorineural loss and vertigo: a perilymph fistula that may need exploration.

### 5.4 Treatment

Mild cases: decongestant drops or spray, analgesia and gentle Valsalva. Persistent effusion or failure: politzerisation or myringotomy, and a grommet for recurrent problems.

**PREVENTION**

Avoid flying with a cold, use a decongestant spray before descent, stay awake on descent, and swallow, yawn or perform Valsalva repeatedly.

## 6 CHAPTER SIX For the examination

What the examiner wants to hear.

### 6.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Traumatic perforation:** causes, keep dry, most heal.

- **Red flags:** vertigo, sensorineural loss, facial weakness.
- **Signs of a retracted drum.**
- **Retraction staging,** and when a grommet helps.
- **Barotrauma:** the locked tube at about 90 mm Hg, treatment and prevention.

## 6.2 Common pitfalls

COMMON PITFALLS

- Syringing or putting drops into an ear with a fresh traumatic perforation.
- Missing a perilymph fistula in a patient with vertigo after a slap or a dive.
- Ignoring a deep retraction pocket until it becomes a cholesteatoma.

Written for this app from Oxford Textbook of Otolaryngology ch. 21, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 8, 10 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Otosclerosis

Otosclerosis is a disease of the bony capsule of the inner ear, in which dense bone is remodelled into spongy bone that can lock the stapes in the oval window. The patient is usually a young adult with slowly worsening hearing and a perfectly normal drum; the cure, when it is needed, is one of the most satisfying operations in otology.

**AT A GLANCE**

- Young adult with progressive conductive loss and a normal drum.
- Rinne negative, As tympanogram, Carhart's notch, absent reflex.
- Choose a hearing aid or stapes surgery.
- Don't operate on an only hearing ear.
- Small risk of dead ear, about 1 in 200.
- Stapedotomy closes the gap to ≤10 dB in 80–95%; dead ear 0.4–2%; offer a hearing aid first.

## 1 CHAPTER ONE Overview

A young adult, a normal drum, and a hearing loss that keeps creeping on.

### 1.1 Definition

Otosclerosis, also called *otospongiosis*, is a primary disease of the otic capsule. Normal dense enchondral bone is replaced by vascular, spongy bone that later hardens again. When a focus reaches the oval window it fixes the stapes footplate and causes a conductive hearing loss; when it involves the cochlea it can add a sensorineural loss.

### 1.2 Epidemiology

The histological disease is far commoner than the clinical one. Lee quotes histological foci in 8 to 12% of white adults, but clinical otosclerosis in only 0.5 to 2%. It is less common in Black and Asian populations. Women are affected about

twice as often as men, and hearing loss usually begins between 20 and 30 years of age; onset before 10 or after 40 is unusual.

### 1.3 Why it matters

Otosclerosis is the classic cause of progressive conductive deafness behind an intact, normal-looking drum. It is entirely treatable, with a hearing aid or with surgery that restores hearing in nine patients out of ten. But the operation opens the inner ear, and a small number of patients lose the hearing in that ear altogether, so the decision to operate is a careful one.

## 2 CHAPTER TWO Pathology

A remodelling disease of the otic capsule.

### 2.1 The otic capsule

The bony labyrinth is formed from three layers: an outer periosteal layer, an inner endosteal layer, and a middle enchondral layer that normally stays dense and inert throughout life. Otosclerosis is a disease of that middle layer. Its favourite site is the *fissula ante fenestram*, a small cleft of fibrous tissue just in front of the oval window.

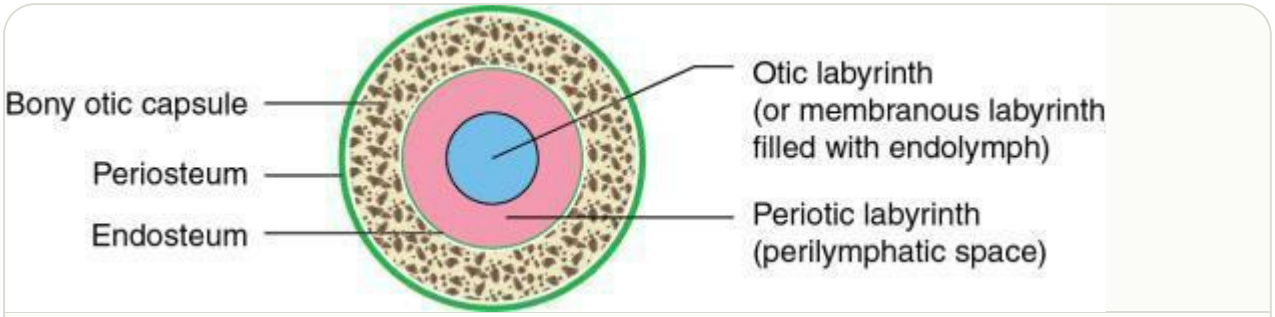



**FIGURE 1** The bony and membranous labyrinth. The otosclerotic focus usually begins just in front of the oval window, where it can fix the stapes footplate.
Dhingra, fig. 13.1

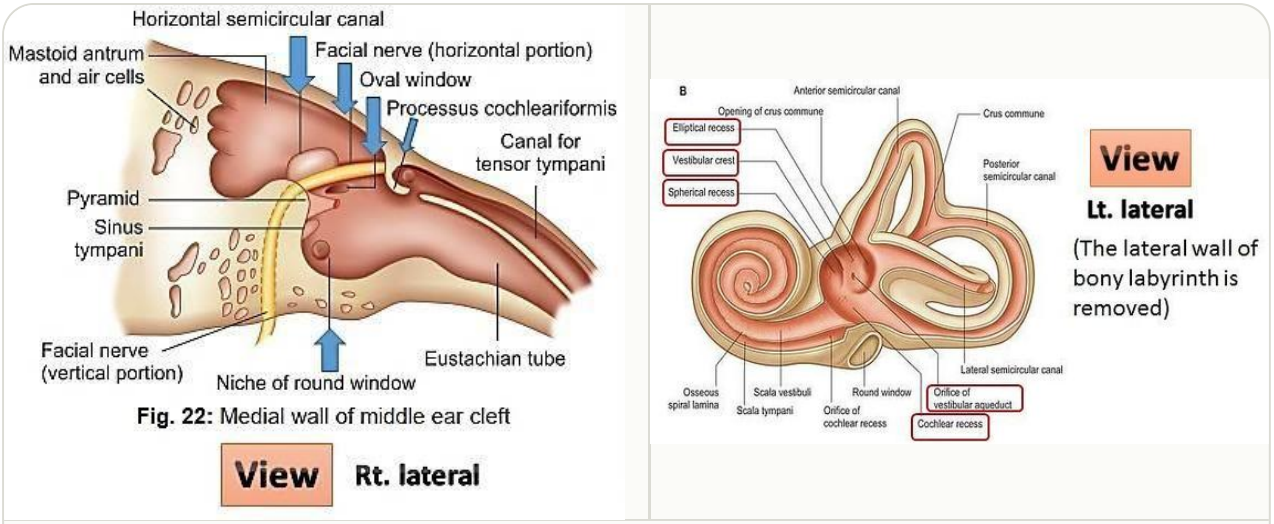



**FIGURE 2** The medial wall of the middle ear with the oval window, and the bony labyrinth: the oval window niche is where stapedial otosclerosis fixes the footplate.

Anatomy atlas · ENT Steps p. 48, ENT Steps p. 66

## 2.2 Histology: active and mature foci

Grossly, a focus may look chalky white, grey or yellow; a red, vascular focus is active and likely to progress quickly.

### 2.2.1 Active (otospongiotic) phase

Immature lesions are full of marrow and vascular spaces, with osteoclasts resorbing bone and osteoblasts laying down new bone. Cement substance around the vessels stains blue with haematoxylin and eosin, forming the *blue mantles of Manasse*, the histological mark of activity.

### 2.2.2 Mature (sclerotic) phase

Older foci are less vascular, with more laminated bone and fibrillar matrix that stains red. Hearing loss reflects where the focus sits, not how active it is.

## 2.3 Types of otosclerosis

### 2.3.1 Stapedial otosclerosis

The commonest clinical form, causing stapes fixation and conductive loss. The focus may be:

- **Anterior:** at the fissula ante fenestram, the site of predilection.
- **Posterior:** behind the oval window.

- **Circumferential:** around the margin of the footplate.
- **Biscuit type:** within a thickened footplate, the annular ligament still free.
- **Obliterative:** filling the whole oval window niche.

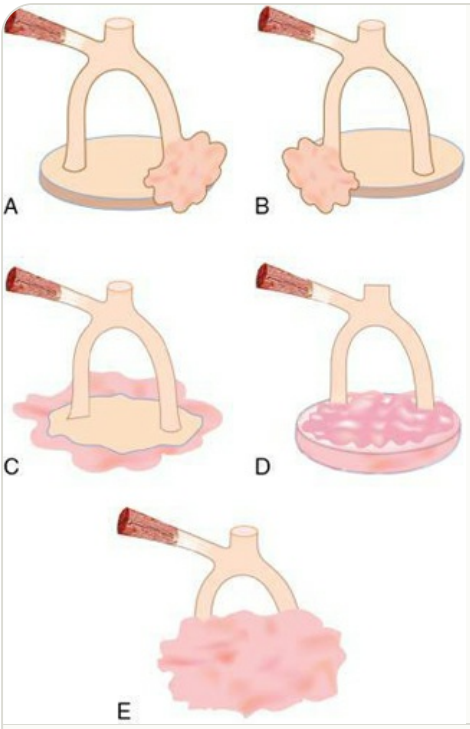



**FIGURE 3** Types of stapedial otosclerosis: (A) anterior focus, (B) posterior focus, (C) circumferential, (D) biscuit type with a thick footplate, (E) obliterative.
Dhingra, fig. 13.2

### 2.3.2 Cochlear otosclerosis

Foci around the round window or elsewhere in the cochlear capsule cause sensorineural loss, probably from enzymes and toxic products released into the inner ear fluids. It is often combined with stapedial disease, giving a mixed loss.

### 2.3.3 Histological otosclerosis

Foci that touch neither the stapes nor the cochlea remain silent and are found only at post-mortem.

# 3 CHAPTER THREE Aetiology

Genes, hormones, and perhaps a virus.

## 3.1 Heredity

About half of patients have a family history. Inheritance is autosomal dominant with incomplete penetrance, and several genetic loci have been described.

## 3.2 Sex and hormones

Women are affected about twice as often, and the hearing loss may begin or accelerate during pregnancy, around the menopause, or after an accident or major operation.

## 3.3 Associations

### 3.3.1 Osteogenesis imperfecta

Brittle bones, blue sclerae and conductive deafness form *van der Hoeve syndrome*. The otic capsule lesions look identical to otosclerosis under the microscope, and both involve genes for type I collagen.

### 3.3.2 Measles virus

Measles RNA and antigens have been found in active foci, and some argue for a viral cause, as has been proposed for Paget's disease.

# 4 CHAPTER FOUR Clinical features

The history usually tells you; the drum does not.

## 4.1 Symptoms

### 4.1.1 Hearing loss

Painless, insidious and progressive, beginning in the twenties and usually affecting both ears, often one before the other.

### 4.1.2 Paracusis Willisii

The patient hears speech better in a noisy room. Background noise makes normal-hearing speakers raise their voices, while it barely reaches the otosclerotic ear.

#### 4.1.3 Tinnitus and vertigo

Tinnitus is common, especially with cochlear involvement and active disease. Vertigo is uncommon.

#### 4.1.4 Speech

Because the patient hears their own voice loudly by bone conduction, they speak softly, in a monotonous, well-modulated voice.

## 4.2 Signs

The tympanic membrane is normal and mobile, and Eustachian tube function is normal. In about one in ten, a pink blush shows through the drum over the promontory: *Schwartze sign*, which reflects increased vascularity in an active focus.

**CLINICAL PEARL**

A conductive loss with a normal drum has a short differential: otosclerosis, ossicular discontinuity, malleus head fixation, and superior canal dehiscence. The stapedial reflex separates the last of these (see
5.4
).

# 5 CHAPTER FIVE Investigations

Forks, audiogram and impedance tell the story.

## 5.1 Tuning forks

Rinne turns negative first at 256 Hz, then at 512 Hz and finally at 1024 Hz as fixation progresses, so the forks give a rough measure of the air-bone gap. Weber lateralises to the ear with the greater conductive loss. Absolute bone conduction is normal, unless the cochlea is involved.

**WHERE THE BOOKS DIFFER · AIR-BONE GAP THAT TURNS EACH FORK NEGATIVE**

| Dhingra | Lee |
| --- | --- |
| | 15, 25 and 35 dB respectively. |

256 Hz at about 15 dB, 512 Hz at about 30 dB, 1024 Hz at about 45 dB.

Both describe the same sequence: negative Rinne at 1024 Hz means a large gap.

## 5.2 Pure-tone audiometry

Air conduction is reduced, most at the lower frequencies early on. Bone conduction is normal, but often shows a dip at 2000 Hz: *Carhart's notch*, about 5 dB at 500 Hz, 10 dB at 1000 Hz, 15 dB at 2000 Hz and 5 dB at 4000 Hz. The notch is a mechanical artefact of stapes fixation, not a cochlear loss, and it disappears after successful surgery. A pure conductive loss rarely exceeds about 60 dB; anything worse means cochlear involvement.

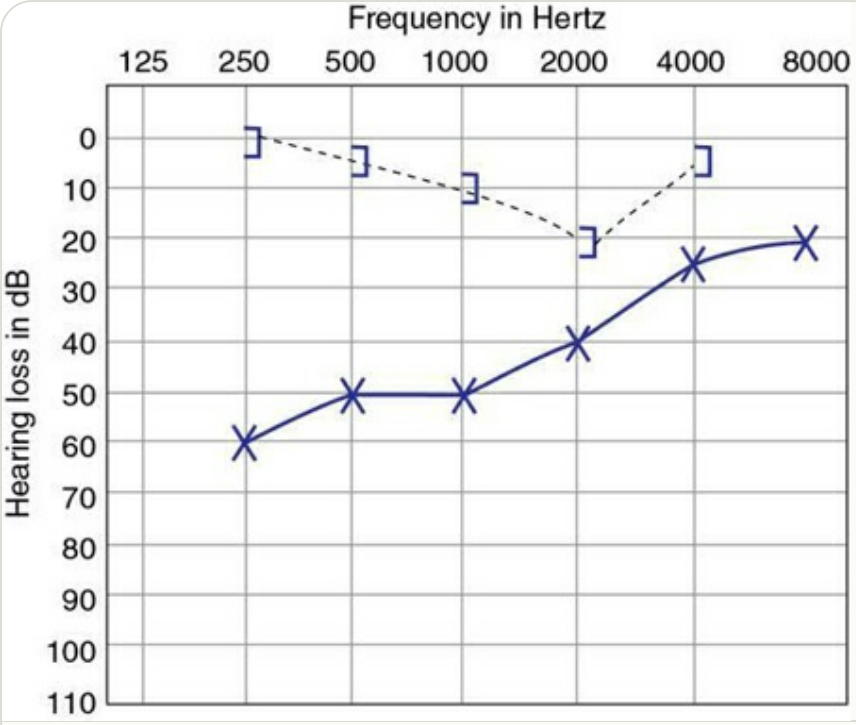



**FIGURE 4** Audiogram of a left ear with otosclerosis. Air conduction is reduced with an air-bone gap; bone conduction dips at 2000 Hz (Carhart's notch).
Dhingra, fig. 13.3

## 5.3 Speech audiometry

Discrimination is good, unless there is cochlear disease. This matters for surgical selection.

## 5.4 Impedance audiometry

### 5.4.1 Tympanometry

Normal early; later a shallow, stiff As curve.

### 5.4.2 Stapedial reflex

Absent once the stapes is fixed. In early disease a characteristic biphasic (on-off) pattern may appear.

**PEARL**

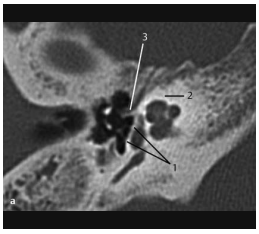
A conductive loss with a normal drum and a present stapedial reflex is not otosclerosis. Think of superior canal dehiscence, which produces an inner-ear conductive loss with intact reflexes.

## 5.5 Imaging

High-resolution CT is not needed routinely. It is useful when the loss progresses rapidly, when cochlear otosclerosis is suspected (a pericochlear halo, the *double ring sign*), when the gap is doubtful, before revision surgery, and when there are vestibular symptoms.

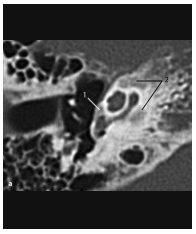
**IMAGING IN THE APP** 6 cases

CT signs of otosclerosis, a piston after stapedectomy, a gusher ear and osteogenesis imperfecta.



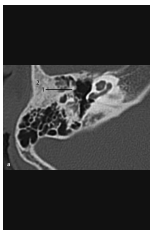
**Otosclerosis, Radiologic Signs around the Oval Niche**

Bone disease · 10 images



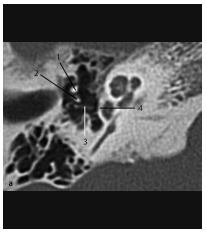
**Otosclerosis, Retrofenestral**

Bone disease · 3 images



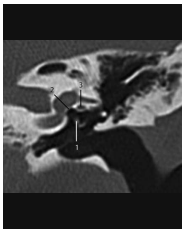
**Fixation of the Ossicular Chain**

Bone disease · 3 images



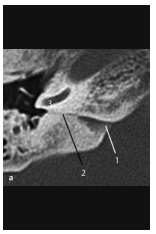
**Stapedectomy, Control of Piston Position**

After surgery · 4 images



**Osteogenesis Imperfecta**

Bone disease · 3 images



**Gusher**

Infection & inflammation · 3 images

# 6 CHAPTER SIX Differential diagnosis

Other causes of a conductive loss.

## 6.1 Conductive loss with a normal drum

### HOW TO TELL THEM APART

| | CLUE | KEY TEST |
|---|---|---|
| **Otosclerosis** | Young adult, progressive, family history | Absent reflex, Carhart's notch |
| **Superior canal dehiscence** | Autophony, sound- or pressure-induced vertigo | Reflex present, low VEMP threshold, CT |
| **Malleus head fixation** | Similar history | Malleus immobile on palpation at surgery, CT |
| **Ossicular discontinuity** | Trauma or previous infection | Hypermobile drum, Ad tympanogram, gap often over 50 dB |
| **Congenital stapes fixation** | Childhood onset, non-progressive | Flat 50–60 dB loss, no Carhart's notch |

### 6.2 Conductive loss with an abnormal drum

Otitis media with effusion, adhesive otitis media and tympanosclerosis are separated by otoscopy and tympanometry.

## 7 CHAPTER SEVEN Management

Watch, aid, or operate.

### 7.1 The three options

**OTOSCLEROSIS**

- MILD LOSS **Observation** review hearing yearly
- ANY LOSS **Hearing aid** effective, no risk; the choice for an only hearing ear
- SUITABLE GAP **Stapes surgery** closes the gap in 90–95%
- FAR ADVANCED **Cochlear implant** when aids no longer help

Each is legitimate. The patient chooses after an honest account of the benefits and the small risk of losing the hearing in the operated ear.

### 7.2 Medical treatment

No drug reverses fixation. *Sodium fluoride* has been used to hasten maturation of active foci and slow sensorineural progression in cochlear otosclerosis.

**WHERE THE BOOKS DIFFER · SODIUM FLUORIDE**

| Dhingra | Lee |
|---|---|
| Controversial, and not generally recommended. | Widely accepted for progressive cochlear disease, though not FDA approved. |

In practice it is considered for active disease with progressive sensorineural loss, a positive Schwartze sign or a halo on CT.

## 7.3 Hearing aid

An excellent alternative for patients who do not want surgery or are unfit for it, and the treatment of choice for an only hearing ear. Conductive losses aid particularly well.

## 7.4 Stapes surgery

The fixed stapes is removed, wholly or in part, and a prosthesis is placed between the long process of the incus and the inner ear.

### 7.4.1 Selection

The patient should feel the handicap (air conduction around 30 dB or worse), have a real air-bone gap with Rinne negative at 256 and 512 Hz, and keep good speech discrimination.

WHERE THE BOOKS DIFFER · CRITERIA FOR SURGERY

| Dhingra | Lee |
|---|---|
| Air-bone gap at least 15 dB, discrimination 60% or more, ideal age 20–30. | Air-bone gap 20 dB or more (25–30 dB in the rehabilitation chapter), discrimination 70% or better, typical age 30–50. |

Operate on the poorer-hearing ear first.

### 7.4.2 Contraindications

- **Only or better-hearing ear:** use a hearing aid.
- **Active Ménière's disease:** a hydropic ear is at higher risk of sensorineural loss.
- **Active middle ear disease or perforation:** treat first.
- **Children:** recurrent Eustachian tube problems and fast regrowth of the focus.
- **Occupations:** divers, pilots and frequent flyers, roofers and athletes (balance), and noisy workplaces, since the stapedius reflex is lost.
- **Relative:** otitis externa, exostoses and pregnancy; defer until settled.

### 7.4.3 Stapedectomy or stapedotomy

In *stapedectomy* all or part of the footplate is removed and the window sealed with a graft; in *stapedotomy* only a small fenestra (about 0.6–0.8 mm) is made in the footplate. Stapedotomy disturbs the inner ear less, gives better high-frequency results and has largely replaced stapedectomy.

### 7.4.4 Steps

The operation is usually done under local anaesthesia through the ear canal.

- Meatal incision and elevation of the tympanomeatal flap.
- Exposure of the oval window, removing the posterosuperior bony overhang with curette or drill, preserving the chorda tympani.
- Confirmation of stapes fixation and a mobile malleus and incus.
- Measurement from the incus to the footplate.
- Division of the stapedius tendon and removal of the stapes superstructure.
- A hole in the footplate (stapedotomy) or partial removal (stapedectomy).
- Placement of the prosthesis on the incus and into the fenestra, and crimping.
- Replacement of the flap.

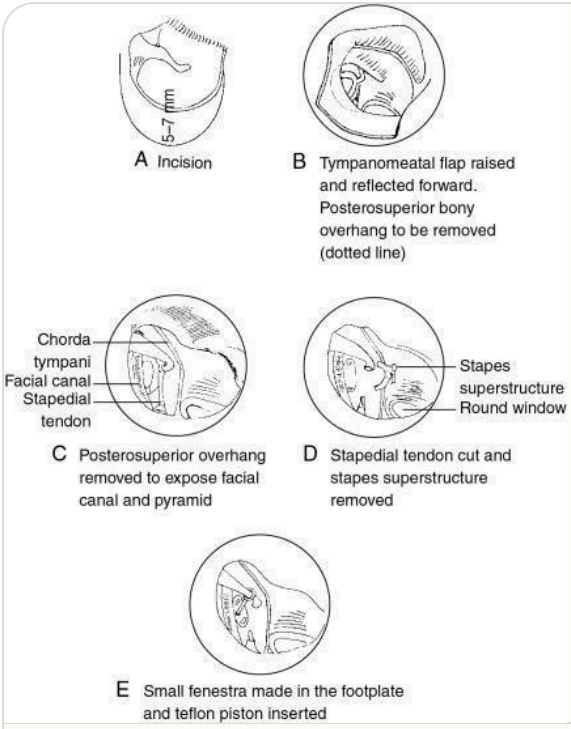



**FIGURE 5** Steps of stapedectomy, from elevation of the tympanomeatal flap to placement of the prosthesis.

Dhingra, fig. 13.6

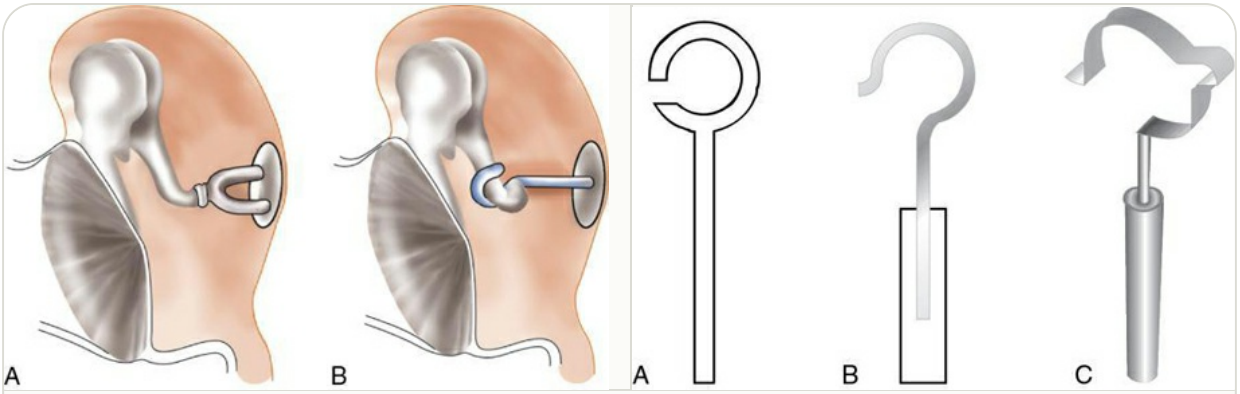



**FIGURE 6** First, the fixed stapes before removal, and a Teflon piston in its place afterwards. Second, stapes prostheses: Teflon, platinum–Teflon and titanium–Teflon pistons.
Dhingra, figs 13.4 and 13.5

#### 7.4.5 Results

The air-bone gap closes to within 10 dB in about 90 to 95% of experienced hands. Carhart's notch disappears with it.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Otosclerosis**
Stapedotomy
**Bone-Anchored Hearing Devices**
Bone-anchored devices
**Cochlear Implantation in Adults**
Implant for far-advanced disease

## 7.5 Far-advanced otosclerosis

When combined loss is so severe that aids no longer help, a cochlear implant gives good results. Stapes surgery plus a powerful aid is an alternative in selected patients. Otosclerotic bone can cause facial nerve stimulation by the implant.

## 7.6 Historical operations

Stapes mobilisation gave only temporary benefit, as the stapes refixed. Lempert's fenestration created a new window in the lateral semicircular canal; it left a mastoid cavity and an inherent loss of about 25 dB, and is now obsolete.

# 8 CHAPTER EIGHT Complications of stapes surgery

Rare, but the patient must hear about them beforehand.

## 8.1 During the operation

- **Tear of the tympanomeatal flap,** which can leave a perforation.
- **Chorda tympani injury,** causing a metallic taste in 5–10%, worse if the opposite chorda was already damaged.
- **Incus dislocation.**
- **Floating footplate,** which drops into the vestibule if pushed.
- **Perilymph gusher,** about 1 in 300, more likely with a wide cochlear aqueduct, especially in X-linked deafness.
- **Facial nerve injury,** more likely with a dehiscent or overhanging nerve.

## 8.2 Early after surgery

### 8.2.1 Vertigo

Usually from surgical trauma, serous labyrinthitis or a prosthesis that is too long. Persistent severe vertigo needs urgent review.

### 8.2.2 Reparative granuloma

Rising sensorineural loss and vertigo at one to two weeks; needs urgent re-exploration.

## 8.3 Late complications

### 8.3.1 Conductive loss returning

A short, loose or displaced prosthesis, or erosion of the incus (the commonest finding at revision, in about 5%).

### 8.3.2 Perilymph fistula

Fluctuating hearing and vertigo, often after straining or barotrauma.

### 8.3.3 Sensorineural loss

About 2% have some sensorineural loss; about 1 in 200 lose all hearing in the operated ear (a *dead ear*). A slow high-frequency loss may appear over the years.

### AFTER STAPES SURGERY

1. Lie the patient with the operated ear up for the first hours; expect mild unsteadiness.
2. Vertigo with nystagmus, sudden hearing loss or facial weakness needs same-day senior review.
3. Advise against straining, nose blowing, heavy lifting and flying for about two to four weeks.
4. Avoid water in the ear until the flap has healed.
5. Book an audiogram at about six weeks.

### NUMBERS AND CURRENT PRACTICE (OXFORD TEXTBOOK)

Clinical otosclerosis affects about 2% of adults, rising to 3% over 60, although histological foci are found in about 3.4% of temporal bones; the familial form (25 to 50%) is autosomal dominant with incomplete penetrance, and the measles link remains unproven. Population data show similar prevalence in men and women, with women more severely affected, which may explain the 2:1 female excess in surgical series. Untreated, the air–bone gap widens by about 1 dB a year, with a further 1 dB a year of sensorineural decline.

**Hearing aids** give excellent results and should be offered to every patient before surgery; a bone-conduction device suits those who cannot wear an aid and decline surgery. **Fluoride** is not used in the UK because its evidence is weak, and bisphosphonates remain experimental.

**Small fenestra stapedotomy** closes the air–bone gap to within 10 dB in 80% to over 95% of ears. Total sensorineural loss occurs in about 0.4 to 2%, taste disturbance from chorda tympani handling in up to 30% (usually temporary), and vertigo is now uncommon. Pre-existing tinnitus improves in about three-quarters and worsens in 5%. A delayed facial palsy after surgery usually recovers fully within six weeks on oral steroids.

**Far-advanced otosclerosis** (thresholds worse than 85 dB with unmeasurable bone conduction) is treated first by stapedotomy with a hearing aid, then by cochlear implantation if that fails; an active middle ear implant coupled to a stapedotomy piston suits patients too good for an implant.

# 9 CHAPTER NINE For the examination

What the examiner wants to hear.

## 9.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Site of predilection:** the fissula ante fenestram.
- **Paracusis Willisii,** and why it happens.
- **Schwartze sign:** active, vascular focus.
- **Carhart's notch:** 2000 Hz dip in bone conduction, mechanical, disappears after surgery.
- **Selection criteria and contraindications** for stapes surgery, especially the only hearing ear.
- **van der Hoeve syndrome:** osteogenesis imperfecta, blue sclerae, deafness.
- **Dead ear** risk of about 1 in 200, and gusher in X-linked deafness.

## 9.2 Common pitfalls

**COMMON PITFALLS**

- Operating on the only or better-hearing ear.
- Missing superior canal dehiscence: the reflex is present.
- Reading Carhart's notch as a true cochlear loss and turning a good candidate away.
- Forgetting to ask about occupation, flying and diving before offering surgery.

Written for this app from Oxford Textbook of Otolaryngology ch. 20, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 13 and K. J. Lee ch. 19, 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Sudden sensorineural hearing loss

A patient who wakes with one ear suddenly dull, and a drum that looks completely normal, has an otological emergency until the tuning forks and an audiogram say otherwise. Treat early with steroids, then look for a cause: most cases are idiopathic, but a few hide a tumour, a stroke or an infection.

**AT A GLANCE**

- 30 dB over three frequencies within 3 days.
- Forks first; don't assume it's wax.
- Steroids early, intratympanic if needed.
- MRI for a retrocochlear cause.
- Half recover by 2 weeks, mostly with mild loss and no vertigo.

## 1 CHAPTER ONE Overview

Deafness overnight.

### 1.1 Definition

A sensorineural hearing loss of 30 dB or more over at least three contiguous frequencies, developing within three days.

### 1.2 Epidemiology

About 5 to 20 per 100,000 people a year, most often in middle age (median 40 to 54), equally in men and women. It is almost always one-sided.

### 1.3 Why it matters

Time matters: treatment works best in the first two weeks. And a minority have a serious underlying cause, including vestibular schwannoma and posterior circulation stroke.

# 2 CHAPTER TWO Causes

Usually idiopathic, but exclude the rest.

## 2.1 Idiopathic

Most cases have no identifiable cause. Three theories compete, and each may be right for some patients:

- **Viral:** cochleitis or neuritis, supported by seasonal clusters and viral serology.
- **Vascular:** thrombosis, embolism, spasm or haemorrhage in the labyrinthine artery, an end artery.
- **Membrane rupture:** of the cochlear membranes or the round window, letting perilymph leak.

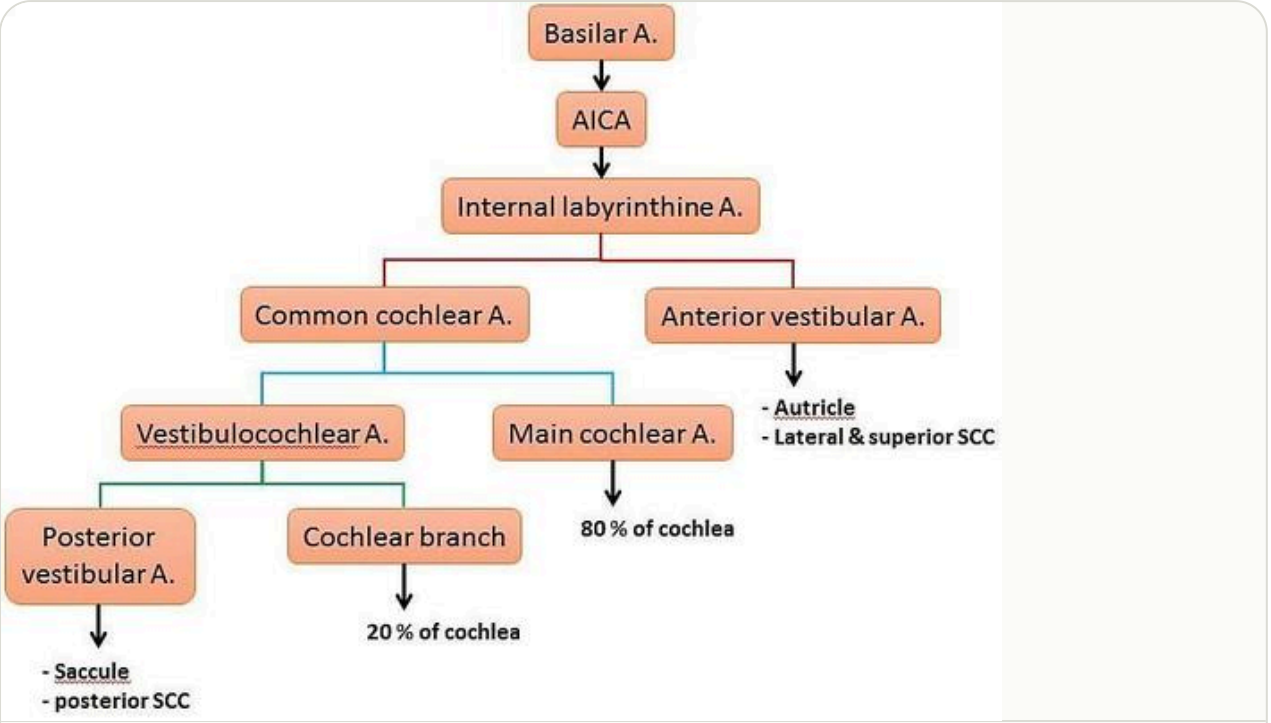



**FIGURE 1** Branches of the labyrinthine artery, an end artery, which explains the vascular theory.
Anatomy atlas · ENT Steps p. 82

## 2.2 Identifiable causes

**CAUSES TO EXCLUDE**

- INFECTION **Mumps, measles, herpes zoster** meningitis, syphilis, acute otitis media, HIV
- TRAUMA **Head injury, ear surgery** blast, barotrauma, perilymph fistula
- VASCULAR **Stroke (AICA)** leukaemia, sickle cell, polycythaemia, embolism, bypass surgery
- OTOLOGICAL **Ménière's disease** Cogan syndrome, enlarged vestibular aqueduct
- TUMOUR **Vestibular schwannoma** other cerebellopontine angle tumours, metastases
- OTHER **Ototoxic drugs** multiple sclerosis, hypothyroidism, sarcoidosis, autoimmune disease, non-organic loss

**PERILYMPH FISTULA**

A leak at the oval or round window may follow stapes surgery, barotrauma, heavy lifting or straining. Spontaneous fistula is possible but probably over-diagnosed.

# 3 CHAPTER THREE Clinical features

One ear, often on waking.

## 3.1 Symptoms

### 3.1.1 Hearing loss

Sudden, one-sided, often noticed on waking or over a few hours. Patients frequently describe it as a blocked ear.

### 3.1.2 Tinnitus and fullness

Often precede or accompany the loss.

### 3.1.3 Vertigo

> Present in some, and a sign of a worse prognosis.

### 3.2 Signs

The canal and drum are normal. Tuning forks are the key bedside test: Weber lateralises to the *good* ear, and Rinne remains positive in the affected ear.

**NOT WAX UNTIL PROVEN OTHERWISE**

A blocked ear after a cold, a swim or a flight can be sudden sensorineural loss. If you assume wax or fluid without doing forks, the window for treatment can close.

## 4 CHAPTER FOUR Investigations

Confirm, then search.

### 4.1 Audiometry

Pure-tone audiogram with masking, and speech discrimination. The pattern (low-frequency, high-frequency, flat or total) helps prognosis.

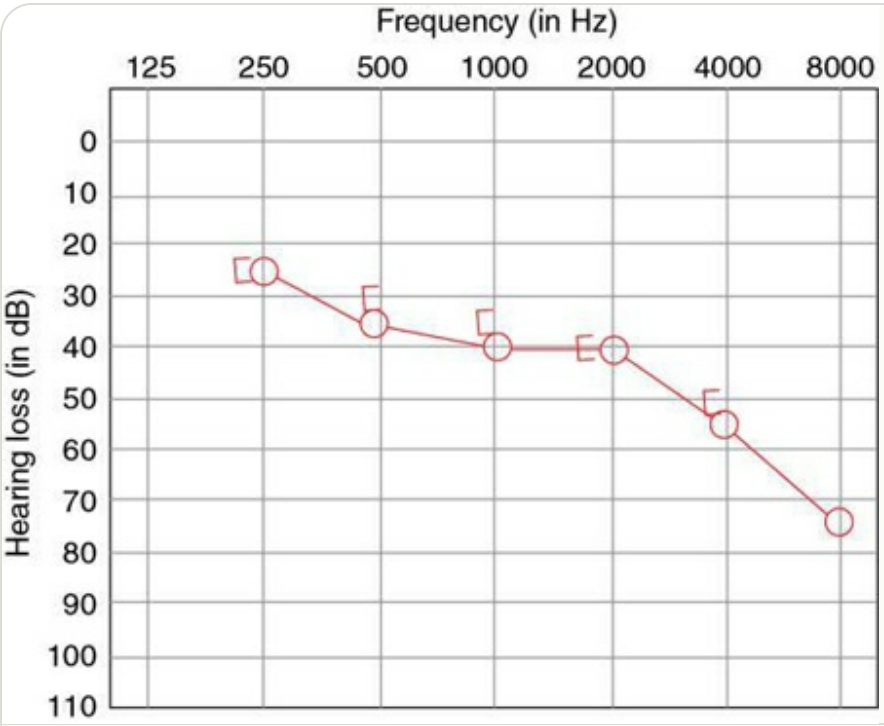



**FIGURE 2** Audiogram of a right ear with sensorineural loss: air and bone conduction fall together, with no air-bone gap.

Dhingra, fig. 5.6

## 4.2 Tympanometry

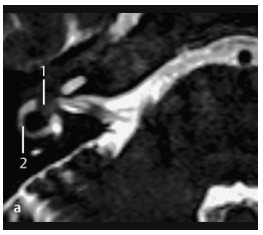
Normal, which excludes middle ear fluid.

## 4.3 MRI

MRI of the internal auditory canals and cerebellopontine angles with gadolinium, to exclude vestibular schwannoma and other retrocochlear lesions. If MRI is impossible, auditory brainstem response.

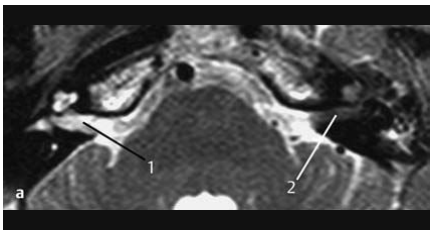
**IMAGING IN THE APP** 3 cases

Retrocochlear and labyrinthine causes: schwannoma, labyrinthitis and otosyphilis.



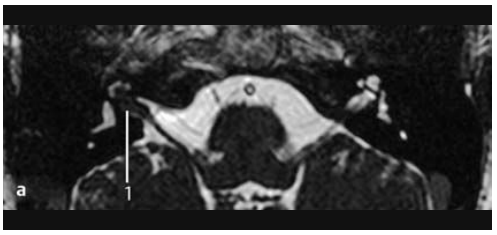

**Vestibular Schwannoma**

Tumours · 13 images




**Labyrinthitis and Subclinical Meningitis**

Infection & inflammation · 5 images




**Otosyphilis and Intracranial Complications**

Infection & inflammation · 6 images

## 4.4 Blood tests by history

Not routine for everyone. Full blood count, ESR, glucose, lipids, thyroid function and syphilis serology when the history suggests them; autoimmune screens for bilateral or fluctuating loss.

## 4.5 Exploration

Exploratory tympanotomy only when a fistula is strongly suspected, after trauma or straining.

## 4.6 Differential diagnosis

- Wax or middle ear effusion: forks and tympanometry.
- Ménière's disease: fluctuation, low-frequency loss, attacks of vertigo.
- Vestibular schwannoma: MRI.
- Anterior inferior cerebellar artery stroke: vertigo, ataxia, facial weakness.
- Non-organic hearing loss: inconsistent results, objective tests normal.

# 5 CHAPTER FIVE Management

Early steroids, then rehabilitation.

## 5.1 On the day

**SUDDEN HEARING LOSS IN CLINIC OR THE EMERGENCY DEPARTMENT**

1. Do the forks: Weber to the good ear, Rinne positive in the bad ear.
2. Look for wax; don't accept a blocked ear without testing.
3. Arrange an audiogram the same or the next day.
4. Offer steroids early.
5. Facial weakness, vertigo with ataxia, or vascular risk factors: treat as a posterior circulation stroke.

## 5.2 Steroids

### 5.2.1 Oral

High-dose prednisolone, for example 1 mg/kg (up to about 60 mg) a day for a week, then tapered. Dhingra gives prednisolone 40 to 60 mg each morning for a week, tapered over three weeks.

### 5.2.2 Intratympanic

Dexamethasone or methylprednisolone injected through the drum. Used first-line when systemic steroids are unsafe (poorly controlled diabetes, for example), or as salvage when oral treatment has failed.

## 5.3 Other treatments

Antivirals have no proven benefit. Older adjuncts, such as bed rest, carbogen inhalation, vasodilators, dextran, a low-salt diet with diuretics and hyperbaric oxygen, have not clearly beaten natural recovery. Dextran is contraindicated in heart failure and bleeding disorders.

**WHERE THE BOOKS DIFFER · ADJUNCT TREATMENTS**

**Dhingra**
Lists carbogen, low salt and hyperbaric oxygen among the options, while noting that no treatment has proven better than spontaneous recovery.

**Lee**
Recommends steroids, orally or intratympanically; finds no evidence for antivirals.

## 5.4 Rehabilitation

If recovery is poor: a hearing aid for a partial loss, or a CROS aid, bone-conduction device or cochlear implant for single-sided deafness. Tinnitus counselling as needed.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Cochlear Implantation in Adults**
Implant for lasting profound loss

**Acoustic Neuroma**
If a schwannoma is found

# 6 CHAPTER SIX Prognosis

What predicts recovery.

About half recover by two weeks; recovery after a month is unusual.

**PROGNOSIS IN SUDDEN HEARING LOSS**

| FACTOR | BETTER | WORSE |
|---|---|---|
| Severity | Mild | Severe or total |
| Frequencies | Low or mid | High or flat |
| Vertigo | Absent | Present |
| Age | Younger | Older |
| Treatment | Early | Late |
| Early recovery | Within two weeks | None by two weeks |

### 6.1 Follow-up

Repeat the audiogram during treatment and at the end. Counsel the patient to protect the other ear from noise and ototoxic drugs.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

### 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **The definition:** 30 dB, three frequencies, three days.
- **Tuning forks:** Weber to the good ear, Rinne positive.
- **Theories:** viral, vascular, membrane rupture.
- **MRI** to exclude a schwannoma.
- **Steroids,** oral and intratympanic.
- **Prognostic factors.**

### 7.2 Common pitfalls

**COMMON PITFALLS**

- Calling it wax or a cold without doing forks.
- Delaying treatment while waiting for an audiogram slot.
- Forgetting MRI once the acute phase has passed.
- Missing a stroke in a patient with vertigo and ataxia.

Written for this app from Oxford Textbook of Otolaryngology ch. 11, Oxford Specialist Handbook ch. 18, Oxford Handbook of ENT ch. 7, Dhingra ch. 5 and K. J. Lee ch. 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Ménière's disease

Ménière's disease is a disorder of the inner ear fluids: attacks of spinning vertigo lasting hours, with fluctuating low-tone hearing loss, roaring tinnitus and a feeling of fullness. It is a diagnosis made by criteria after other causes are excluded, and most patients are controlled without ever needing an operation.

**AT A GLANCE**

- Episodic vertigo, fluctuating low-frequency loss, tinnitus and fullness.
- Diagnosis by criteria, after excluding others.
- Salt reduction and medicines control most.
- Intratympanic gentamicin next; labyrinthectomy if hearing is useless.
- Bárány 2015: definite needs attacks of 20 min–12 h with documented low-frequency loss; betahistine no better than placebo (BEMED).

## 1 CHAPTER ONE Overview

Episodes of vertigo, an ear that fills and fades.

### 1.1 Definition

An idiopathic disorder of the inner ear characterised by *endolymphatic hydrops*: distension of the endolymphatic system. It produces the classic tetrad of episodic vertigo, fluctuating sensorineural hearing loss, tinnitus and aural fullness.

### 1.2 Epidemiology

Onset is usually between 35 and 60 years. It begins in one ear, but the other ear becomes involved in a substantial minority over the years.

### 1.3 Ménière's disease or Ménière's syndrome

*Ménière's disease* is idiopathic hydrops. *Ménière's syndrome*, or secondary hydrops, has an identifiable cause: head injury, viral labyrinthitis, syphilis, Cogan syndrome,

autoimmune disease, or *delayed hydrops* years after a profound loss in that ear.

# 2 CHAPTER TWO Pathology

Too much endolymph.

## 2.1 Hydrops

The cochlear duct and saccule distend, bowing Reissner's membrane into the scala vestibuli. Small ruptures let potassium-rich endolymph mix with perilymph, poisoning hair cells and nerve fibres and producing an attack; as the membrane heals, function partly recovers.

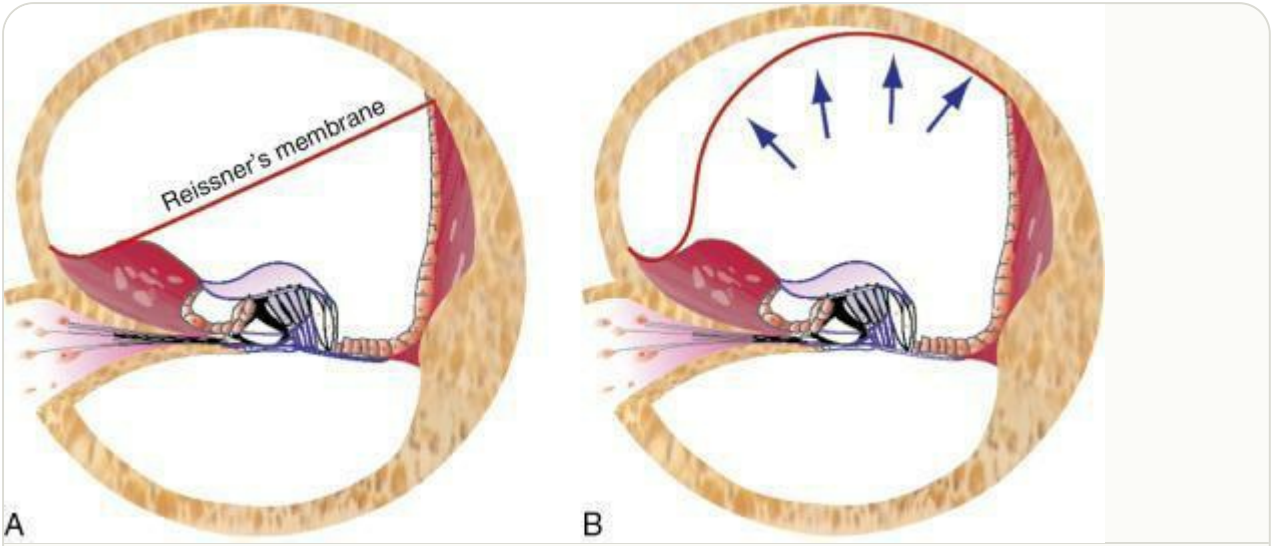



**FIGURE 1** (A) The normal cochlear duct. (B) In hydrops the duct is distended with endolymph and Reissner's membrane bulges into the scala vestibuli.
Dhingra, fig. 15.1

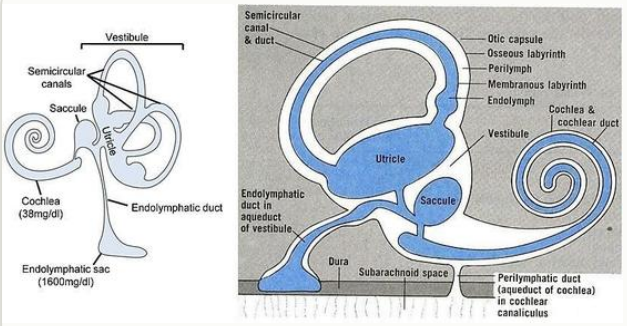



**TABLE 2** Differences between the composition of endolymph, perilymph and CSF**

| | Endolymph | Perilymph | CSF |
|---|---|---|---|
| $Na^+$ (mEq/L) | 3 | 150 | 152 |
| $K^+$ (mEq/L) | 150 | 3–5 | 4 |
| $Cl^-$ (mEq/L) | 130 | 125 | |
| Protein (mg/dL) | 126 | 200–400 | 20–50 |
| Glucose (mg/dL) | 10–40 | 85 | 70 |

**Values vary from the site of collection such as cochlea, saccule and endolymphatic sac in cases of endolymph and scala tympani and vestibuli in cases of perilymph.

**FIGURE 2** The membranous labyrinth with the endolymphatic sac, and the composition of endolymph and perilymph.

Anatomy atlas · ENT Steps p. 74, ENT Steps p. 75

## 2.2 Why it happens

The cause is unknown; several theories may each be true in some patients.

- **Defective absorption** by the endolymphatic sac, or a small, poorly pneumatised sac and duct.
- **Vasomotor disturbance** with spasm of the internal auditory artery.
- **Allergy** and **autoimmunity.**
- **Salt and water retention.**
- **Hypothyroidism.**
- **Viral infection.**

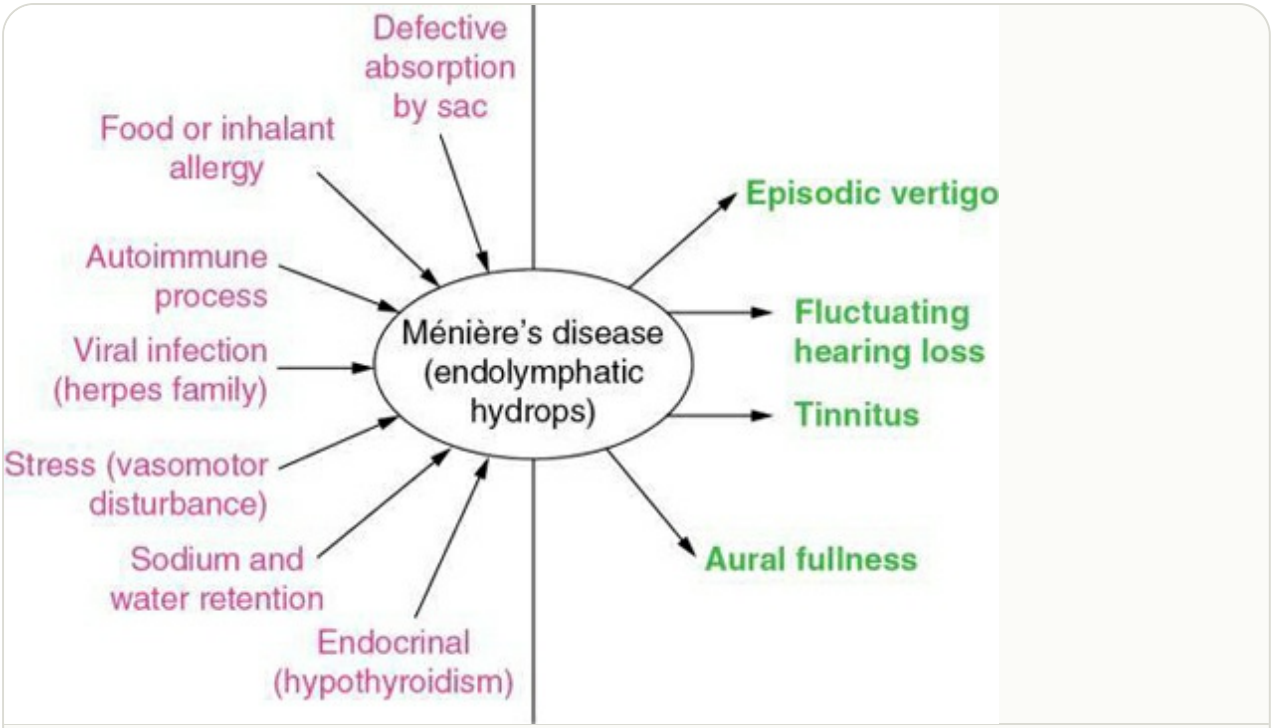



**FIGURE 3** Aetiological factors and symptoms of endolymphatic hydrops.
Dhingra, fig. 15.2

# 3 CHAPTER THREE Clinical features

The attack and the interval.

## 3.1 Vertigo

Sudden, rotatory vertigo lasting from 20 minutes to several hours, often with nausea, vomiting, sweating and pallor. Attacks come in clusters with remissions of weeks or months. The patient lies still with the bad ear up or down; there is no loss of consciousness.

## 3.2 Hearing loss

Fluctuating and sensorineural, worse during an attack and recovering afterwards, at first in the low frequencies. Over the years the loss becomes permanent and flat. Distortion, *diplacusis* (hearing the same tone at a different pitch in each ear) and intolerance of loud sound are common.

## 3.3 Tinnitus

Low-pitched and roaring, louder before and during attacks.

## 3.4 Fullness

Pressure in the ear, which may change or worsen before an attack and act as a warning.

## 3.5 Signs

The drum is normal. During an attack there is horizontal-rotatory nystagmus.

WHERE THE BOOKS DIFFER · DIRECTION OF NYSTAGMUS IN AN ATTACK

| Dhingra | Lee |
| --- | --- |
| Fast phase towards the healthy ear. | Towards the affected ear at the start (irritative), away from it later (paretic), and back towards it during recovery. |

**TULLIO PHENOMENON**

Vertigo provoked by loud sound may occur when the distended saccule lies against the stapes footplate.

## 3.6 Variants

### 3.6.1 Cochlear hydrops

Hearing symptoms without vertigo.

### 3.6.2 Vestibular hydrops

Vertigo without hearing symptoms.

### 3.6.3 Lermoyez syndrome

Hearing gets worse before an attack and improves with the vertigo, the reverse of the usual pattern.

### 3.6.4 Tumarkin's otolithic crisis

> Sudden drop attacks without warning or loss of consciousness, from abrupt otolith stimulation. Dangerous falls make it an indication for definitive treatment.

# 4 CHAPTER FOUR Diagnosis

Criteria, audiogram, and exclusion.

## 4.1 Diagnostic criteria

**AAO-HNS DIAGNOSTIC CATEGORIES**

| CATEGORY | REQUIREMENTS |
|---|---|
| Certain | Definite disease plus histological confirmation |
| Definite | Two or more spontaneous episodes of vertigo of 20 minutes or longer; documented hearing loss on at least one occasion; tinnitus or fullness; other causes excluded |
| Probable | One definite episode, documented hearing loss, tinnitus or fullness |
| Possible | Episodic vertigo without documented loss, or fluctuating loss with imbalance |

**BÁRÁNY SOCIETY CRITERIA (2015)**

The criteria now used internationally, and by the Oxford Textbook, recognise only two categories. **Definite** Ménière's disease needs two or more spontaneous episodes of vertigo each lasting 20 minutes to 12 hours; audiometrically documented low- to medium-frequency sensorineural loss in the affected ear on at least one occasion before, during or after an attack; fluctuating aural symptoms (hearing, tinnitus or fullness) in that ear; and no better explanation. **Probable** disease needs two or more episodes of 20 minutes to 24 hours with fluctuating aural symptoms, without the audiometric requirement. The upper limit of 12 hours helps separate Ménière's disease from vestibular migraine, which is about ten times more common, overlaps with it in many patients, and is the commonest misdiagnosis.

## 4.2 Staging

By the average of thresholds at 0.5, 1, 2 and 3 kHz in the worst audiogram of the previous six months: stage 1 up to 25 dB, stage 2 26 to 40 dB, stage 3 41 to 70 dB,

stage 4 over 70 dB.

## 4.3 Audiometry

### 4.3.1 Pure-tone audiogram

Sensorineural loss affecting the low frequencies, giving a rising curve; later the curve flattens.

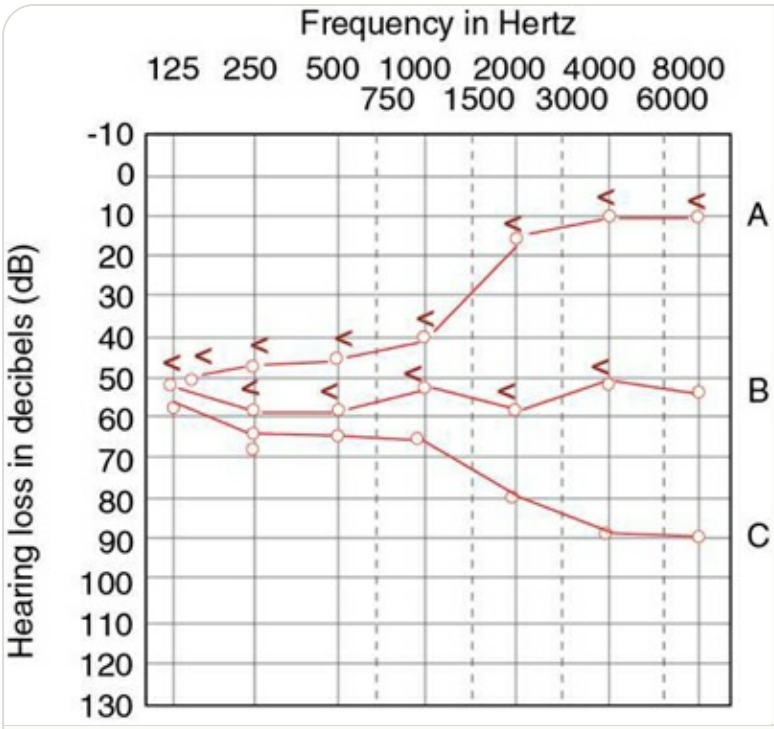



**FIGURE 4** Audiogram in early Ménière's disease: sensorineural loss, worse in the low frequencies (a rising curve).

Dhingra, fig. 15.3

### 4.3.2 Speech and recruitment

Discrimination is moderately reduced, around 55 to 85% between attacks, and recruitment is present, marking a cochlear lesion.

## 4.4 Electrocochleography

The summating potential grows relative to the action potential in a hydropic ear.

WHERE THE BOOKS DIFFER · RAISED SP/AP RATIO

| Dhingra | Lee |
| --- | --- |
| Above 30%. | Above 0.35. |

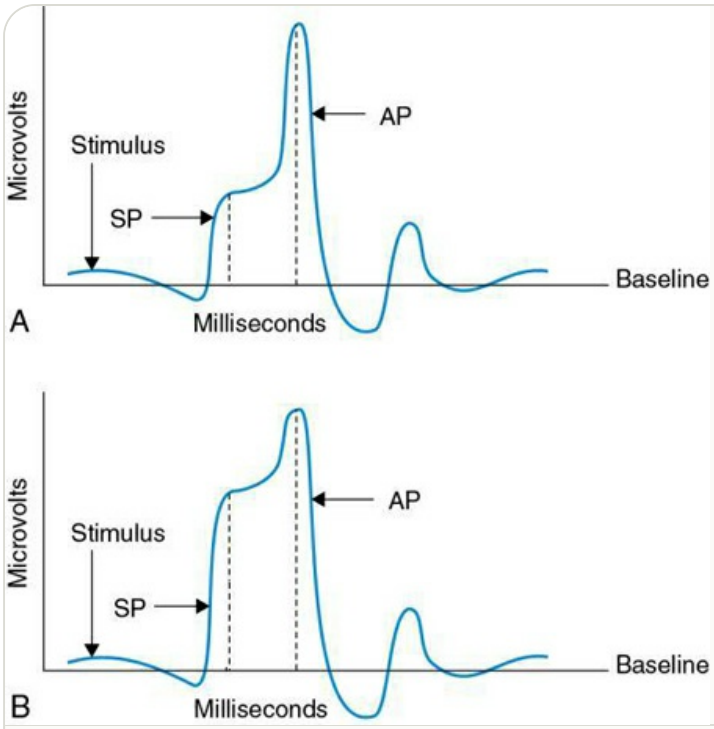



**FIGURE 5** Electrocochleography: (A) normal ear, (B) Ménière's disease with an enlarged summating potential relative to the action potential.

Dhingra, fig. 15.4

## 4.5 Glycerol test

An osmotic diuretic, 1.5 mL/kg, reduces hydrops temporarily. An improvement of 10 dB at two adjacent frequencies, or 10% in discrimination, is positive.

## 4.6 Caloric test

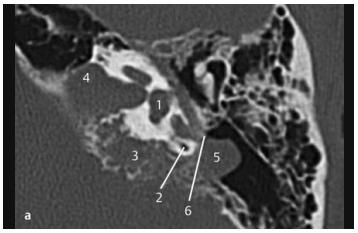
Reduced response on the affected side in about three-quarters.

## 4.7 MRI with contrast

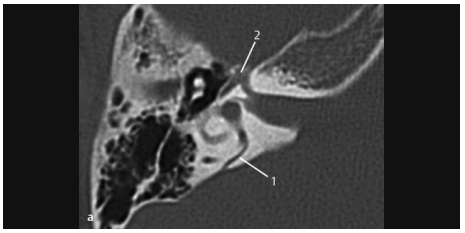
Essential to exclude a vestibular schwannoma or other retrocochlear lesion. Delayed contrast MRI can now show hydrops directly in specialist centres.

**IMAGING IN THE APP** 2 cases

An endolymphatic sac tumour and an enlarged vestibular aqueduct, two causes of secondary hydrops.




**Endolymphatic Sac Tumour**
Tumours · 8 images




**Vestibular Aqueduct, Enlargements**
Congenital · 11 images

## 4.8 Differential diagnosis

- Vestibular schwannoma.
- Vestibular migraine.
- Recurrent vestibulopathy.
- Superior canal dehiscence.
- Autoimmune inner ear disease and otosyphilis.
- Transient ischaemic attacks in the vertebrobasilar territory.

# 5 CHAPTER FIVE Management

Most are controlled medically.

## 5.1 The acute attack

**AN ACUTE ATTACK**

1. Reassure; let the patient lie still with the head supported.
2. Check the neurology and the head impulse test: new neurological signs mean stroke imaging, not sedatives.
3. Give an antiemetic or vestibular sedative: prochlorperazine, dimenhydrinate, or diazepam.

4 Intravenous fluids if vomiting.

5 Keep sedatives short: they slow compensation.

## 5.2 Long-term medical treatment

### 5.2.1 Lifestyle

Salt restriction to about 1.5 to 2 g a day, adequate water, limiting caffeine and alcohol, stopping smoking, and managing stress. Avoid working at heights or driving during active periods.

### 5.2.2 Medicines

Betahistine (with caution in asthma) and a diuretic. Treat allergy and hypothyroidism. About 80% are controlled.

WHERE THE BOOKS DIFFER · BETAHISTINE

| Dhingra | Oxford |
| --- | --- |
| Betahistine and a diuretic as the standard long-term medical treatment. | A large randomised trial (BEMED, 2016) found no benefit of betahistine over placebo in reducing attacks, even at high dose. Many units still offer it as a low-risk first step, but patients with frequent attacks should not be held on it for long before moving to intratympanic steroids. Trials of intratympanic gentamicin and steroids show similar rates of hearing loss with either. The other ear's vestibular function must be checked before any ablation, because up to half of patients develop disease in the second ear over twenty years. |

## 5.3 When medical treatment fails

The choice depends on whether hearing in the ear is still useful.

INTRACTABLE VERTIGO

HEARING USEFUL **Preserve the labyrinth** intratympanic steroid, endolymphatic sac surgery, pressure-pulse device

HEARING USEFUL **Destroy vestibular function** intratympanic gentamicin, vestibular neurectomy

HEARING LOST **Labyrinthectomy** removes all inner ear function

### 5.3.1 Intratympanic steroid

Repeated injections of dexamethasone or methylprednisolone; no risk to hearing.

### 5.3.2 Intratympanic gentamicin

Chemical ablation of the vestibular hair cells, given in small doses through a grommet or by injection. Vertigo is controlled in 60 to 80%, at the price of some hearing loss in 4 to 30%.

### 5.3.3 Pressure-pulse therapy

A device (Meniett) delivers low-pressure pulses to the middle ear through a grommet.

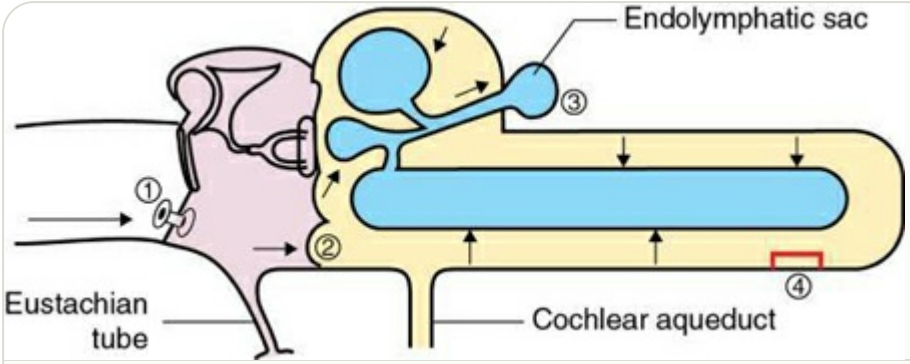



**FIGURE 6** The mechanism of intermittent low-pressure pulse therapy.
Dhingra, fig. 15.6

### 5.3.4 Endolymphatic sac surgery

Decompression of the sac, or a shunt to the mastoid, preserves hearing. Its benefit over placebo is debated.

WHERE THE BOOKS DIFFER · SAC SURGERY

| Dhingra | Lee |
|---|---|
| Listed among the operations that preserve hearing. | Controversial. |

### 5.3.5 Vestibular neurectomy

Section of the vestibular nerve through the middle or posterior cranial fossa: high vertigo control with hearing preserved, at the price of a craniotomy.

#### 5.3.6 Labyrinthectomy

For an ear with no useful hearing: removal of the membranous labyrinth through the canal or the mastoid. A cochlear implant can follow in the same or another sitting.

**PEARLS**

- Recruitment makes hearing aids hard to fit.
- Sacculotomy and the Cody tack are historical.
- Before destroying one ear, make sure the other ear is healthy.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Surgery for Vertigo**
Surgery for vertigo

**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation**
Tube for intratympanic or Meniett therapy

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **The tetrad,** and the pathology of hydrops.
- **AAO-HNS criteria** and staging.
- **Audiogram:** low-frequency, fluctuating, rising curve.
- **Electrocochleography** and the glycerol test.
- **Variants:** Lermoyez and Tumarkin.
- **Ladder of treatment** by hearing status.

## 6.2 Common pitfalls

COMMON PITFALLS

- Labelling every dizzy patient as Ménière's without documented hearing loss.
- Forgetting MRI to exclude a schwannoma.
- Keeping patients on vestibular sedatives for months.
- Using gentamicin when the other ear may also be affected.

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Written for this app from Oxford Textbook of Otolaryngology ch. 14, Oxford Specialist Handbook ch. 18, Oxford Handbook of ENT ch. 7, Dhingra ch. 7, 15 and K. J. Lee ch. 16. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Vestibular schwannoma and CP angle tumours

A vestibular schwannoma is a slow, benign tumour of the eighth nerve sheath that announces itself through the ear: one-sided hearing loss and tinnitus, with speech understanding worse than the audiogram predicts. An MRI makes the diagnosis; the choice to watch, irradiate or remove depends on size, hearing, age and growth.

**AT A GLANCE**

- Asymmetric sensorineural loss with poor speech understanding.
- Gadolinium MRI is the gold standard.
- Watch, radiate or remove by size, hearing and age.
- The route depends on tumour size and hearing.
- Bilateral tumours mean NF2.
- Most small tumours are watched (17–29% grow over 4 years); radiosurgery or surgery chosen by size and hearing.

## 1 CHAPTER ONE Overview

The ear is the first to know.

### 1.1 Definition

A benign tumour of Schwann cells, usually arising from the vestibular division of the eighth nerve in the internal auditory canal. *Acoustic neuroma* is the traditional, inaccurate name: it is rarely acoustic and never a neuroma.

### 1.2 Epidemiology

It accounts for 8 to 10% of intracranial tumours and about 80% of cerebellopontine angle tumours, usually presenting between 40 and 60. Most are sporadic and one-sided. Bilateral tumours mean *neurofibromatosis type 2*.

### 1.3 Why it matters

It is the commonest serious cause of asymmetric sensorineural hearing loss, and every asymmetric loss is screened for it. Untreated large tumours compress the brainstem and cause hydrocephalus.

---

# 2 CHAPTER TWO Pathology

Slow growth from inside the canal.

### 2.1 Origin and growth

The tumour arises near the glial-Schwann cell junction in the internal auditory canal, widening the canal before it grows into the cerebellopontine angle. Growth is slow, around 1 to 2 mm a year, and some tumours never grow at all.

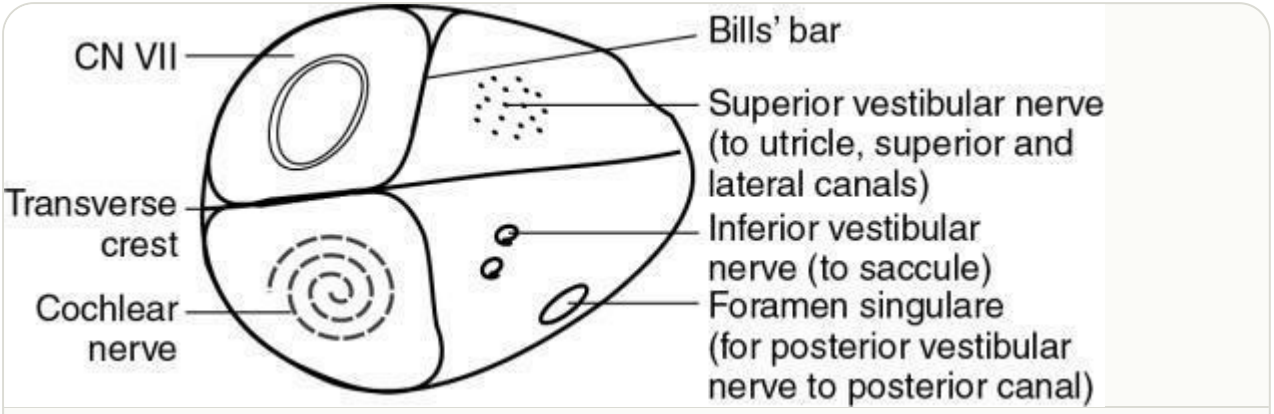



**FIGURE 1** The fundus of the internal auditory canal, showing the position of the facial, cochlear and superior and inferior vestibular nerves.

Dhingra, fig. 18.1

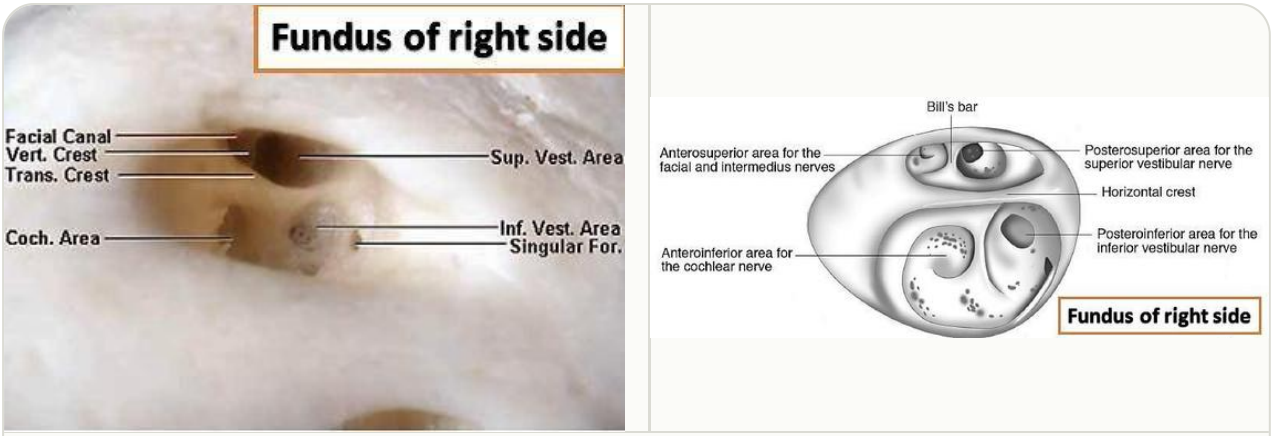



**FIGURE 2** The fundus of the internal auditory canal, with Bill's bar and the transverse crest separating the facial, cochlear and vestibular nerves.

Anatomy atlas · ENT Steps p. 83

## 2.2 Histology

Two patterns: *Antoni A*, compact spindle cells with palisading nuclei (Verocay bodies), and *Antoni B*, loose, cystic tissue.

## 2.3 Neurofibromatosis type 2

Autosomal dominant, from loss of the NF2 gene on chromosome 22. Bilateral vestibular schwannomas, often with meningiomas and other schwannomas, present in young adults.

## 2.4 Size

SIZE AND SPREAD

| STAGE | SIZE (DHINGRA) | WHAT IT AFFECTS |
|---|---|---|
| Intracanalicular | In the canal | Eighth nerve |
| Small | Up to 1.5 cm | Into the angle |
| Medium | 1.5 to 4 cm | Fifth nerve, cerebellum |
| Large | Over 4 cm | Brainstem, fourth ventricle, hydrocephalus |

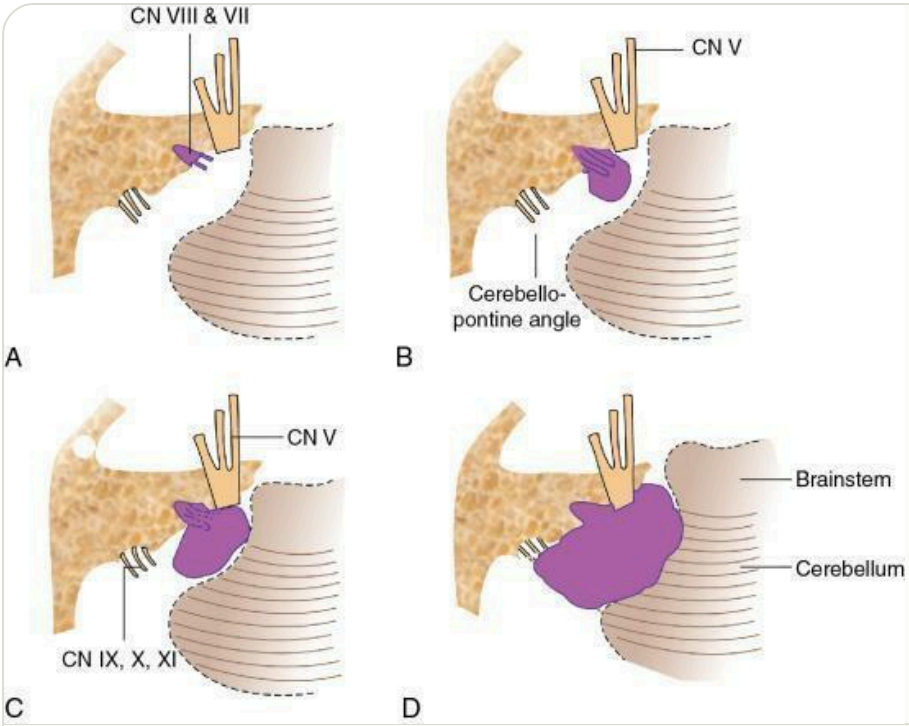



**FIGURE 3** Expansion of an acoustic neuroma: (A) intracanalicular, (B) into the cerebellopontine angle, (C) pressing on the fifth nerve, (D) onwards to the brainstem and lower cranial nerves.
Dhingra, fig. 18.2

CHAPTER THREE

# 3 Clinical features

A stepwise spread.

## 3.1 Eighth nerve

### 3.1.1 Hearing loss

Progressive, one-sided sensorineural loss with poor speech discrimination, often far worse than the pure-tone thresholds suggest. About one in ten presents as sudden hearing loss.

### 3.1.2 Tinnitus

One-sided and persistent.

### 3.1.3 Balance

Unsteadiness rather than true vertigo, because the brain compensates as slowly as the tumour grows.

### 3.2 Fifth nerve

At around 2.5 cm: reduced corneal reflex, often the earliest sign, then facial numbness.

### 3.3 Seventh nerve

The sensory fibres fail first: numbness of the posterior meatal wall (*Hitselberger's sign*), loss of taste and reduced lacrimation. Motor weakness of the face comes late.

### 3.4 Lower cranial nerves

Ninth and tenth: dysphagia, hoarseness and a reduced gag.

### 3.5 Cerebellum and brainstem

Ataxia, nystagmus, and finally raised intracranial pressure with headache, vomiting and papilloedema from obstructed CSF flow.

> **URGENT**
> Headache, vomiting, ataxia or papilloedema with a large tumour means hydrocephalus or brainstem compression: urgent neurosurgical review.

## 4 CHAPTER FOUR Investigations

MRI first.

### 4.1 Audiometry

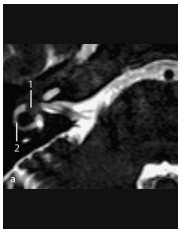
Asymmetric sensorineural loss, poor discrimination, *roll-over* (discrimination falls as loudness rises), absent or decaying stapedial reflex, and abnormal tone decay: the signs of a retrocochlear lesion.

## 4.2 MRI with gadolinium

The gold standard: finds tumours of a few millimetres. A brightly enhancing mass centred on the internal auditory canal, with a cone into the angle (*ice-cream cone*).

**IMAGING IN THE APP** 9 cases

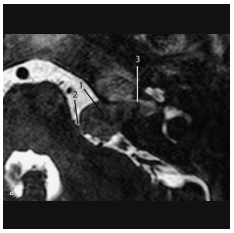
Intracanalicular and angle schwannomas, cystic change, radiotherapy effects, NF2, and the mimics: meningioma, epidermoid and arachnoid cyst.



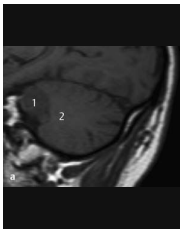
**Vestibular Schwannoma**
Tumours · 13 images



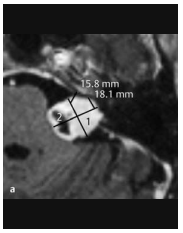
**Schwannoma of the Internal Auditory Canal**
Tumours · 3 images



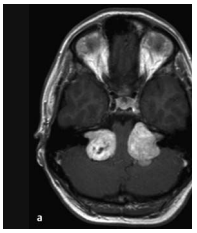
**Schwannoma of the Cerebellopontine Angle**
Tumours · 7 images



**Cystic Degeneration in CPA Schwannoma**
Tumours · 4 images




**Effects of Radiotherapy on CPA Schwannoma**
After surgery · 2 images



**Neurofibromat II (NF II)**
Tumours · 6 imag

## 4.3 Auditory brainstem response

Used when MRI is not possible. A prolonged inter-aural wave V latency or I–V interval suggests a lesion, but small tumours are missed.

**WHERE THE BOOKS DIFFER · ABNORMAL WAVE V DIFFERENCE**

| Dhingra | Lee |
|---|---|
| More than 0.2 ms. | 0.4 to 0.5 ms. |

## 4.4 Caloric test

Reduced on the affected side in most, normal in very small tumours.

### 4.5 CT

If MRI cannot be done, and to show the bony canal and approach anatomy.

### 4.6 Avoid lumbar puncture

CSF protein is raised, but a puncture with a large tumour risks herniation.

### 4.7 Differential diagnosis

OTHER CEREBELLOPONTINE ANGLE LESIONS

| LESION | DISTINGUISHING FEATURES |
|---|---|
| Meningioma | Broad dural base, dural tail, eccentric to the canal, calcification, hyperostosis |
| Epidermoid | Does not enhance; bright on diffusion-weighted MRI |
| Facial schwannoma | Facial palsy, enlarges the geniculate or labyrinthine segment |
| Lipoma | Bright on T1 before contrast; suppresses with fat saturation |
| Haemangioma | Ground-glass erosion; palsy out of proportion to size |
| Arachnoid cyst | Follows CSF on all sequences |

## 5 CHAPTER FIVE Management

Watch, radiate, or remove.

### 5.1 Choosing

The decision rests on tumour size and growth, hearing in both ears, age and health, and the patient's wishes.

WHERE THE BOOKS DIFFER · FIRST CHOICE

| Dhingra | Lee |
| --- | --- |
| Favours surgical removal. | Presents observation, radiosurgery and microsurgery as equal options, chosen by patient and tumour. |

Modern practice follows Lee: many small tumours are simply watched.

## 5.2 Observation

For small tumours, older or unfit patients, and those with good hearing: serial MRI at 6 to 12 months, then less often. Treat if it grows.

## 5.3 Stereotactic radiosurgery

A single high dose (around 12 to 13 Gy at the margin), usually for tumours under 2.5 to 3 cm. Growth is controlled in 90 to 95%. Hearing is not reliably preserved; the facial nerve risk is about 1%; malignant change is very rare, about 1 in 1000. Conventional external beam radiotherapy has no role.

## 5.4 Microsurgery

SURGICAL APPROACHES

| APPROACH | BEST FOR | HEARING | MAIN RISK |
| --- | --- | --- | --- |
| Translabyrinthine | Any size | Lost | CSF leak |
| Retrosigmoid (suboccipital) | Medial, small or large tumours | May be preserved | Headache, cerebellar retraction |
| Middle fossa | Canal tumours under 1.5 cm | Preserved in 60–70% | Temporal lobe retraction, facial nerve |

### 5.4.1 Translabyrinthine

Through the mastoid and labyrinth: no brain retraction and the best view of the facial nerve, but hearing is sacrificed.

### 5.4.2 Retrosigmoid

Through the posterior fossa behind the sigmoid sinus: hearing can be preserved for tumours that spare the lateral canal; postoperative headache is commoner.

#### 5.4.3 Middle fossa

Above the temporal bone: for small intracanalicular tumours with good hearing. The facial nerve lies over the tumour and is at greater risk.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Acoustic Neuroma**
Acoustic neuroma surgery
**Stereotactic Radiosurgery**
Stereotactic radiosurgery
**Bone-Anchored Hearing Devices**
Rehabilitating single-sided deafness

## 5.5 Subtotal removal

Leaving a little tumour on the facial nerve preserves facial function; the remnant often stays stable and can be watched or irradiated.

## 5.6 NF2

Bevacizumab may shrink tumours and improve hearing. An auditory brainstem implant or cochlear implant can restore some hearing.

# 6 CHAPTER SIX Complications

Of the tumour and of treatment.

- Facial palsy.
- Hearing loss.
- CSF leak and meningitis.
- Chronic headache, especially after the retrosigmoid route.
- Hydrocephalus and brainstem compression with large tumours.

**CHOOSING BETWEEN WAITING, RADIOSURGERY AND SURGERY (OXFORD TEXTBOOK)**

**Observation first** is now common for small tumours. In a large series followed for four years, only 17% of intracanalicular and 29% of extrameatal tumours grew, and growing tumours usually enlarge by 1 to 2 mm a year, so waiting rarely makes later treatment harder. The first repeat MRI is at six months, to catch a fast-growing tumour or a wrong diagnosis, then yearly while stable. The cost of waiting is hearing, which may decline even when the tumour does not grow.

**Stereotactic radiosurgery**, usually a single session of 11 to 14 Gy at the margin, controls most small and medium tumours. Surgery after failed radiosurgery carries a worse facial nerve outcome than primary surgery.

**Surgical approach** follows size and hearing. The *middle fossa* approach suits intracanalicular tumours up to about 1.5 cm with serviceable hearing. The *retrosigmoid* approach suits tumours under about 2 cm, not filling more than two-thirds of the internal meatus, with serviceable hearing; in a large review, class A or B hearing was kept in 57% of intracanalicular tumours, 37% under 2 cm and 12% over 2 cm. The *translabyrinthine* approach sacrifices hearing but gives direct access without brain retraction and can remove a tumour of any size; facial nerve outcomes at House–Brackmann I or II are high for tumours under about 3.5 cm. Guidelines find no approach clearly superior for facial function. Presenting features are mostly asymmetric hearing loss and tinnitus; facial numbness occurs in 7 to 9% and facial weakness in only about 5%, even with large tumours.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Origin:** the vestibular nerve in the internal auditory canal.
- **Presenting symptoms,** and why vertigo is uncommon.
- **Order of cranial nerve involvement:** VIII, V (corneal reflex), VII (Hitselberger's sign), IX and X.
- **Retrocochlear audiology:** discrimination, roll-over, reflex decay.
- **MRI with gadolinium** as the gold standard.
- **The three surgical approaches,** with hearing outcome for each.

- **NF2:** bilateral tumours, chromosome 22.

## 7.2 Common pitfalls

COMMON PITFALLS

- Not requesting MRI for an asymmetric loss.
- Attributing sudden hearing loss to a virus without imaging.
- Doing a lumbar puncture with a large tumour.
- Promising hearing preservation after radiosurgery.

Written for this app from Oxford Textbook of Otolaryngology ch. 18, Oxford Specialist Handbook ch. 19, Oxford Handbook of ENT ch. 8, Dhingra ch. 18 and K. J. Lee ch. 24. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Facial nerve palsy

A face that will not move is frightening for the patient and a test of method for the doctor. First decide whether the forehead is spared; then remember that Bell's palsy is a diagnosis of exclusion, examine the ear, parotid, neck and other cranial nerves, and protect the eye today.

**AT A GLANCE**

- Forehead spared means central.
- Bell's is a diagnosis of exclusion; check ear, parotid and neck.
- Protect the eye and give early steroids.
- ENoG and EMG guide decompression and repair.
- Slow or recurrent palsy needs imaging.

## 1 CHAPTER ONE Overview

One nerve, a long course, many causes.

### 1.1 Definition

Facial nerve palsy is weakness of the muscles of facial expression from a lesion anywhere between the motor cortex and the muscles. The approach depends on where the lesion is, how complete it is, and whether the nerve is likely to recover.

### 1.2 Why it matters

The eye is at risk within hours, the face carries a person's expression and identity, and a palsy may be the first sign of a tumour, a cholesteatoma or a fracture.

## 2 CHAPTER TWO Applied anatomy

A long and narrow course.

## 2.1 Central connections

The part of the facial nucleus that supplies the forehead receives fibres from both cerebral hemispheres; the part that supplies the lower face is controlled only from the opposite side. An upper motor neurone (*supranuclear*) lesion therefore spares the forehead. Emotional movement travels by separate pathways, so a smile may survive a stroke that abolishes voluntary movement.

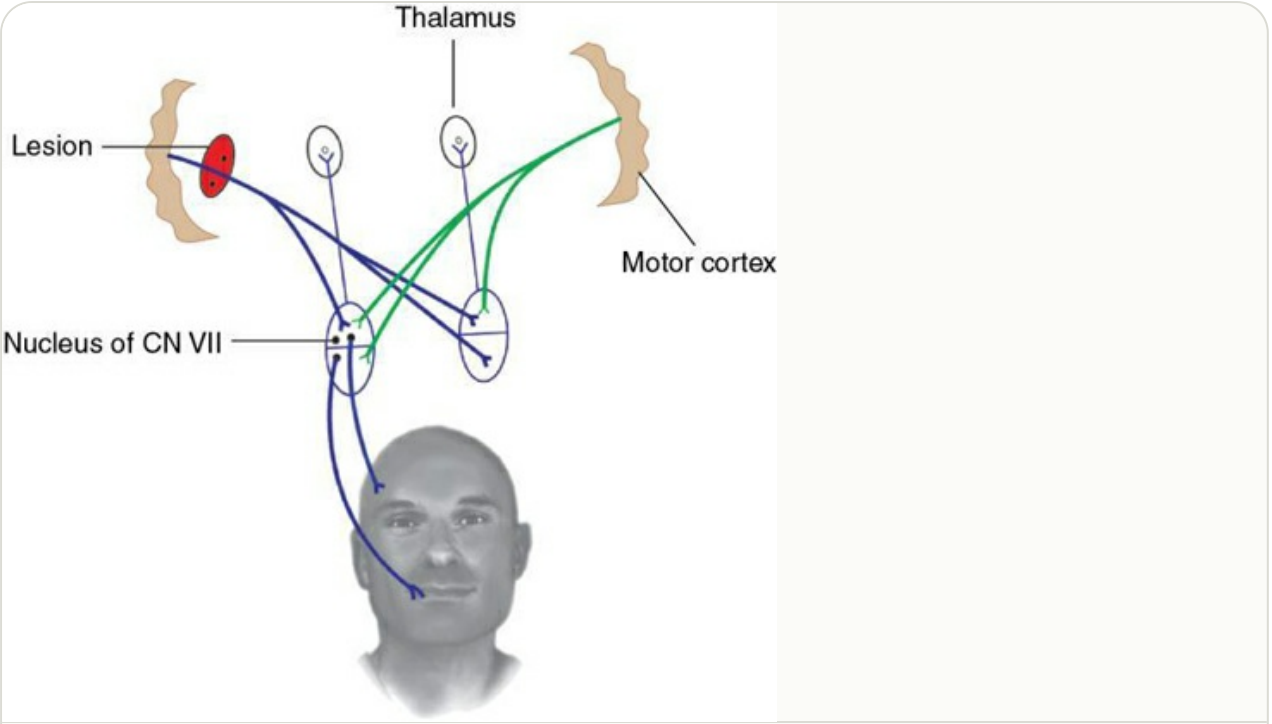



**FIGURE 1** The forehead receives bilateral cortical innervation and is spared in supranuclear paralysis; emotional movements travel by separate pathways.
Dhingra, fig. 14.1

## 2.2 The course of the nerve

### 2.2.1 Intracranial

From the pons across the cerebellopontine angle with the eighth nerve.

### 2.2.2 Intratemporal

- **Meatal segment:** in the internal acoustic meatus, anterosuperior.
- **Labyrinthine segment:** about 4 mm long, the narrowest part of the canal, ending at the geniculate ganglion. Swelling here is thought to cause Bell's palsy.
- **Tympanic segment:** along the medial wall of the middle ear above the oval window; the commonest site of bony dehiscence.

- **Mastoid segment:** vertically down to the stylomastoid foramen.

#### 2.2.3 Extratemporal

Through the parotid gland, dividing into temporal, zygomatic, buccal, marginal mandibular and cervical branches.

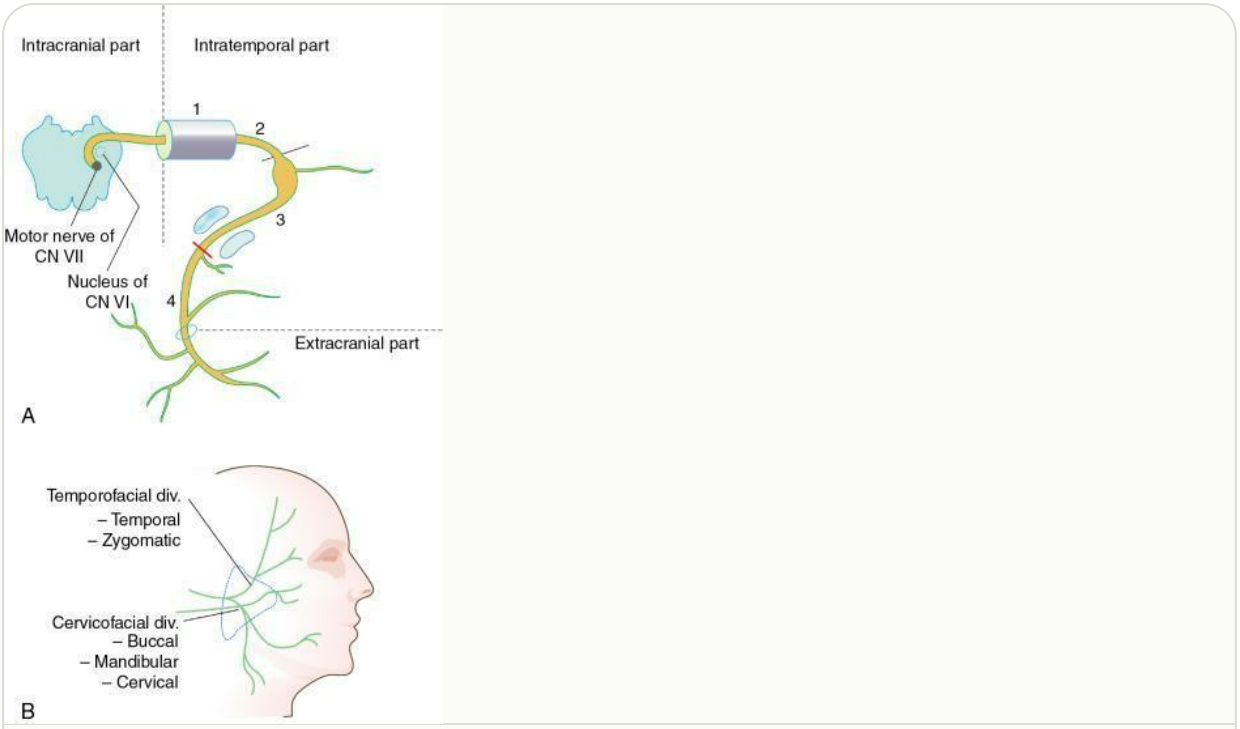



**FIGURE 2** Course of the facial nerve: meatal, labyrinthine, tympanic and mastoid segments, and its branches.

Dhingra, fig. 14.2

## 2.3 Branches inside the temporal bone

- **Greater superficial petrosal nerve:** lacrimation.
- **Nerve to stapedius:** the stapedial reflex.
- **Chorda tympani:** taste from the anterior two-thirds of the tongue and submandibular and sublingual secretion.

These allow a lesion to be localised (see 4.3 ).

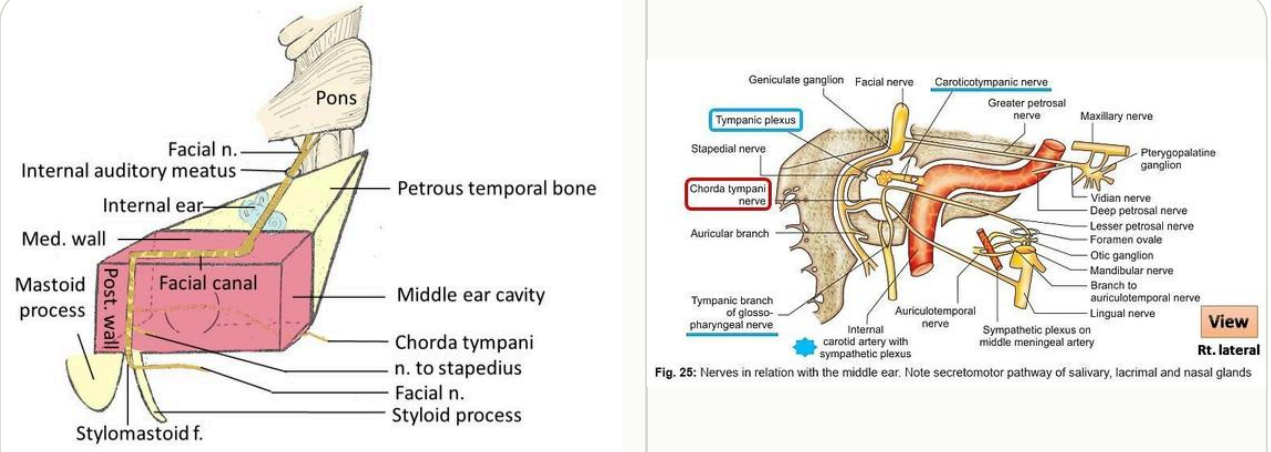



**FIGURE 3** The course of the facial nerve through the temporal bone, and the petrosal nerves and tympanic plexus of the middle ear.

Anatomy atlas · Doss p. 149, ENT Steps p. 53

## 2.4 Surgical landmarks

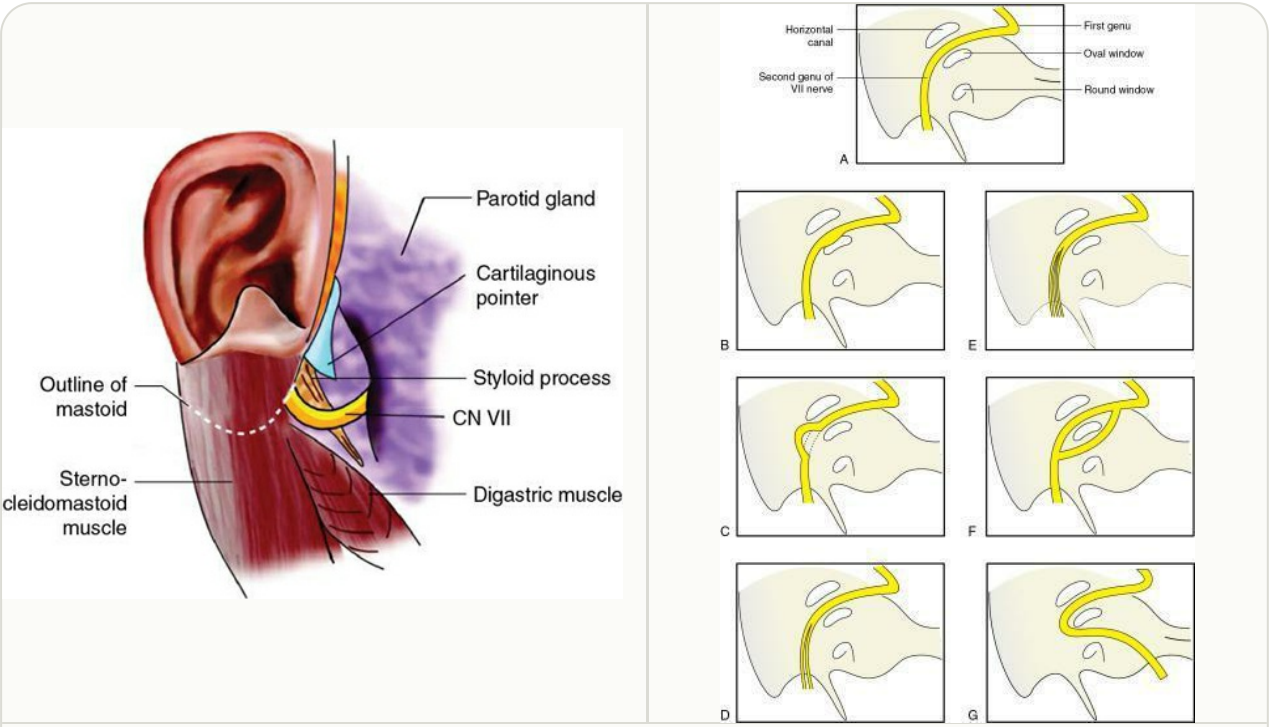



**FIGURE 4** First, landmarks for finding the nerve in parotid surgery. Second, variations in its course, including bony dehiscence.

Dhingra, figs 14.4 and 14.5

## 2.5 Nerve injury

### SUNDERLAND CLASSIFICATION

| DEGREE | WHAT IS DAMAGED | OUTCOME |
|---|---|---|
| 1 | Conduction block (neurapraxia) | Full recovery |
| 2 | Axons, tubes intact (axonotmesis) | Full recovery, slower |
| 3 | Endoneurium | Incomplete, synkinesis likely |
| 4 | Perineurium | Poor without repair |
| 5 | Whole nerve (neurotmesis) | No recovery without repair |

# 3 CHAPTER THREE Causes

Think beyond Bell's palsy.

**CAUSES OF FACIAL PALSY**

- CENTRAL **Stroke, tumour** forehead spared
- INFECTION **Bell's palsy, Ramsay Hunt** otitis media, cholesteatoma, necrotising otitis externa, Lyme disease
- TRAUMA **Temporal bone fracture** surgery, birth (forceps), facial wounds
- TUMOUR **Parotid cancer** facial schwannoma, haemangioma, glomus, vestibular schwannoma, temporal bone cancer
- SYSTEMIC **Diabetes** sarcoid (Heerfordt), Guillain-Barré, leukaemia, granulomatosis
- CONGENITAL **Möbius syndrome** birth injury

## 3.1 Bell's palsy

### 3.1.1 What it is

An acute idiopathic lower motor neurone palsy, 60 to 75% of all facial palsies. Reactivation of herpes simplex virus is the likely cause; the nerve swells inside the tight labyrinthine segment. It is commoner in diabetes and pregnancy.

### 3.1.2 Course

Weakness comes on over hours and peaks within 72 hours, often after a day of pain behind the ear. About 85 to 90% recover fully, and 95% if the palsy never becomes complete.

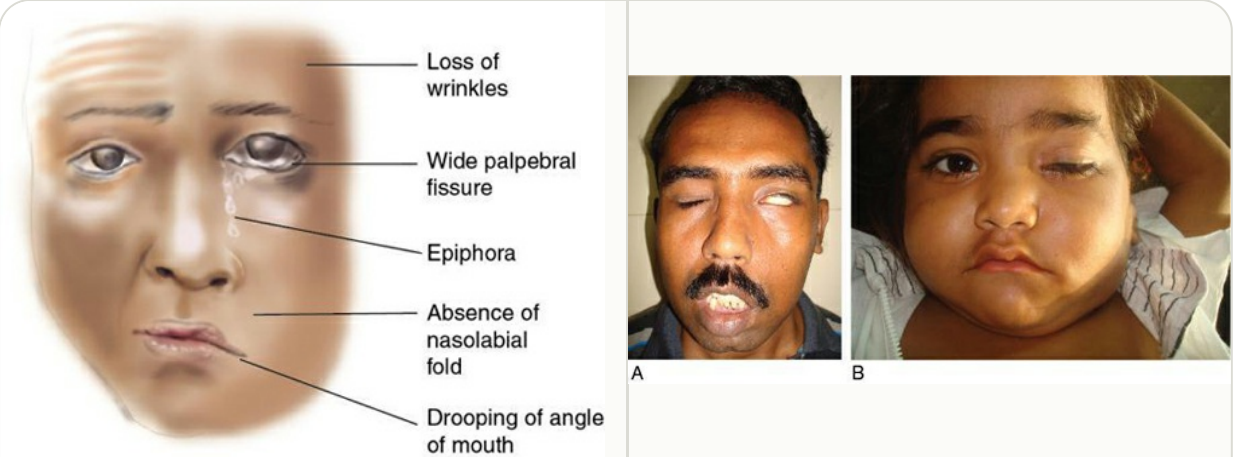



**FIGURE 5** First, the features of a left-sided facial palsy. Second, Bell's palsy on the left in (A) an adult and (B) a child.

Dhingra, fig. 14.7

## 3.2 Ramsay Hunt syndrome

Herpes zoster oticus: vesicles in the concha, canal or palate, severe ear pain, facial palsy, and often sensorineural hearing loss and vertigo. The outlook for the face is worse than in Bell's palsy.



**FIGURE 6** Ramsay Hunt syndrome: facial palsy with vesicles in the concha.
Dhingra, fig. 14.9

## 3.3 Otitis media and cholesteatoma

Acute otitis media can inflame a dehiscent nerve; cholesteatoma erodes the canal, usually in the tympanic segment. Both need urgent otological care.

## 3.4 Trauma

### TEMPORAL BONE FRACTURES AND THE FACIAL NERVE

| | LONGITUDINAL | TRANSVERSE |
|---|---|---|
| **Frequency** | About 80% | About 20% |
| **Blow** | Side of the head | Back of the head |
| **Bleeding from the ear** | Common | Usually absent |
| **Hearing** | Conductive | Sensorineural |
| **Vertigo** | Less often | Severe |
| **Facial palsy** | About 20%, often delayed | About 50%, often immediate |

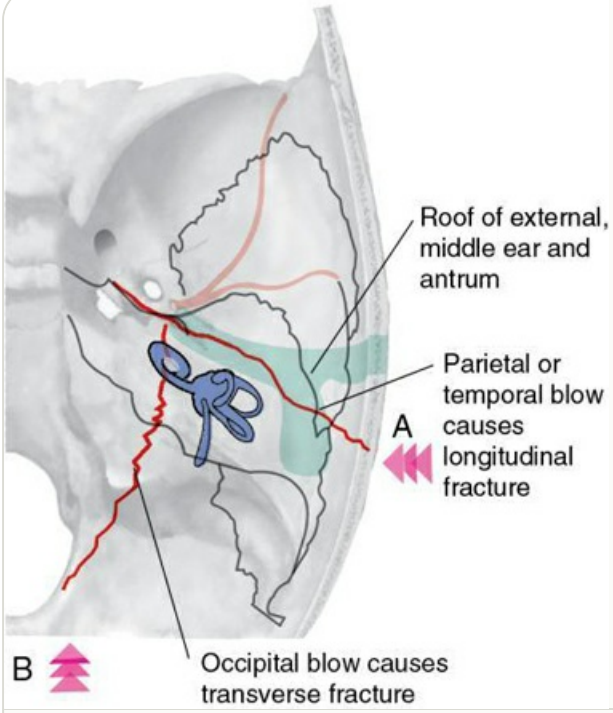



**FIGURE 7** Longitudinal and transverse fractures of the temporal bone.
Dhingra, fig. 14.10

## 3.5 Tumours

A slowly progressive palsy, a recurrent palsy on the same side, or palsy with twitching suggests a tumour. A facial palsy with a parotid lump is almost always malignant.

## 3.6 Bilateral or recurrent palsy

Bilateral: Lyme disease, Guillain-Barré, sarcoidosis, leukaemia, Möbius syndrome, HIV, neurofibromatosis 2 and bilateral fractures. Recurrent on the same side: a tumour in about a third; *Melkersson-Rosenthal syndrome* combines recurrent palsy, lip swelling and a fissured tongue.

# 4 CHAPTER FOUR Assessment

Where, how bad, and why.

## 4.1 History

Speed of onset, pain, vesicles, hearing loss and vertigo, ear discharge, trauma or surgery, previous episodes, and systemic illness.

## 4.2 Examination

### 4.2.1 The face

Ask the patient to raise the eyebrows, close the eyes tightly, smile and puff the cheeks. On attempted eye closure the eye rolls up and out (*Bell's phenomenon*). Note drooling and tearing.

### 4.2.2 Grading

HOUSE-BRACKMANN GRADING

| GRADE | DESCRIPTION |
|---|---|
| I | Normal |
| II | Slight weakness; complete eye closure with minimal effort |
| III | Obvious weakness, complete eye closure with effort, synkinesis may appear |
| IV | Obvious disfiguring weakness, incomplete eye closure |
| V | Barely perceptible movement |
| VI | No movement |

### 4.2.3 Everything else

Otoscopy, the parotid and neck, the other cranial nerves, and hearing. A sixth nerve palsy with it points to the brainstem; fifth, eighth or lower nerve involvement points to the cerebellopontine angle; a motor-only palsy points to the parotid.

## 4.3 Topodiagnosis

LOCALISING THE LESION

| LOST FUNCTION | LESION LIES |
|---|---|
| Lacrimation (Schirmer test) | At or above the geniculate ganglion |

| LOST FUNCTION | LESION LIES |
|---|---|
| Stapedial reflex | Above the nerve to stapedius |
| Taste and submandibular flow | Above the chorda tympani |
| Motor only | Below the chorda, in the mastoid or parotid |

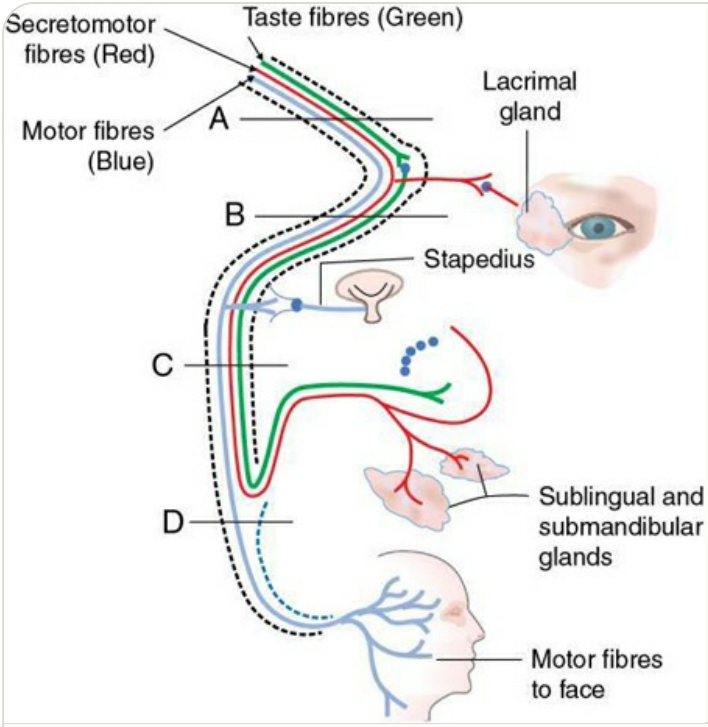



**FIGURE 8** Topographical localisation of facial nerve lesions.
Dhingra, fig. 14.12

## 4.4 Electrical tests

### 4.4.1 Electroneuronography (ENoG)

Compares the compound muscle response on the two sides. Degeneration of 90% or more within the first three weeks predicts a poor recovery.

### 4.4.2 Electromyography (EMG)

Fibrillation potentials at 14 to 21 days mean denervation; polyphasic potentials appear 6 to 12 weeks before visible recovery; electrical silence means the motor end plates are lost.

WHERE THE BOOKS DIFFER · THE ENOG WINDOW

| Dhingra | Lee |
|---|---|

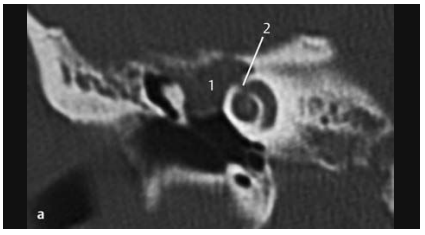
| | |
|---|---|
| From day 4 to day 21. | Days 3 to 12 when deciding on decompression. |

## 4.5 Imaging

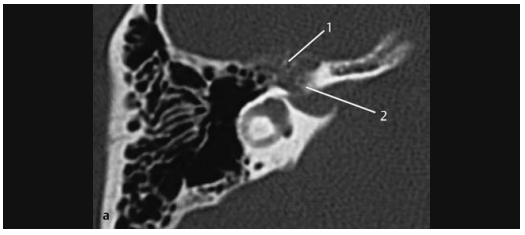
Contrast MRI of the whole course, and CT of the temporal bone for fractures and cholesteatoma. Indicated for an atypical, slow, recurrent or bilateral palsy, or no recovery by three to four months.

**IMAGING IN THE APP** 2 cases

Facial nerve schwannoma and geniculate haemangioma, the tumours behind a slow or recurrent palsy.




**Schwannoma of the Facial Nerve**
Tumours · 3 images




**Hemangioma of the Geniculate Ganglion**
Tumours · 3 images

# 5 CHAPTER FIVE Management

Eye first, then the nerve.

## 5.1 Eye protection

Lubricating drops by day, ointment and taping at night, sunglasses or a moisture chamber. If closure stays poor, a gold or platinum weight in the upper lid, a lateral tarsal strip or tarsorrhaphy.

## 5.2 Bell's palsy

Oral steroids started within 72 hours, for example prednisolone 60 mg daily for five days followed by a five-day taper. An antiviral may add a small benefit in severe palsy.

WHERE THE BOOKS DIFFER · STEROIDS IN BELL'S PALSY

| Dhingra | Lee |
| --- | --- |
| Prednisolone 1 mg/kg/day, extended if the palsy stays complete; notes that benefit is not proved beyond doubt. | Recommends steroids within 48 to 72 hours. |
| Current practice gives early steroids. | |

Be cautious with steroids in diabetes, hypertension, peptic ulcer, tuberculosis, glaucoma and pregnancy.

## 5.3 Ramsay Hunt syndrome

High-dose steroids with an antiviral such as aciclovir or valaciclovir, and analgesia.

## 5.4 Otitis media and cholesteatoma

Acute otitis media: antibiotics with myringotomy. Chronic disease: urgent exploration, clearing disease from the nerve; granulation that has invaded the nerve is left alone.

## 5.5 Trauma

An immediate, complete palsy with a fracture through the canal is explored and decompressed or repaired. A delayed palsy is managed conservatively, with serial ENoG.

WHERE THE BOOKS DIFFER · DELAYED TRAUMATIC PALSY

| Dhingra | Lee |
| --- | --- |
| Conservative treatment. | Serial ENoG; decompress if the response falls below 10% of normal. |

## 5.6 Palsy after ear surgery

Remove the packing and any tight dressing first. If the nerve was not seen and the palsy is complete, explore unless continuity is certain.

## 5.7 Decompression

Opening the bony canal, and sometimes the sheath, at the site of compression. For Bell's palsy it is controversial and, if done, must include the labyrinthine segment within two weeks.

## 5.8 Repair and reanimation

### 5.8.1 Nerve repair

Tension-free primary anastomosis, or a cable graft from the great auricular (not when there are neck metastases), sural or medial antebrachial cutaneous nerve.

### 5.8.2 Nerve transfer

When the distal nerve and muscle survive but the proximal stump is unavailable, ideally within 18 months: the masseteric nerve is most used; hypoglossal transfer costs some tongue function; a cross-face graft can give a spontaneous smile.

### 5.8.3 Muscle transfer

For long-standing palsy with atrophied muscle: temporalis transfer, or a free gracilis transfer powered by the masseteric nerve (one stage, about 92% success, not spontaneous) or by a cross-face graft (two stages, about 80%, spontaneous).

### 5.8.4 Static procedures

Brow lift, fascia lata sling and lid procedures.

**OPERATIVE GUIDES IN THEATRE** 7 operations

**Facial Nerve Decompression**
Decompression
**Tumors of the Facial Nerve**
Facial nerve tumours
**Temporal Bone Trauma**
Temporal bone trauma
**Nerve Graft Techniques**
Nerve grafts
**Eyelid Reanimation in Patients With Facial Paralysis**
Eyelid reanimation
**Facial Reanimation**
Facial reanimation
**Mastoid Surgery**
Mastoid surgery when ear disease is the cause

CHAPTER SIX
# 6 Complications

When recovery is incomplete.

- **Exposure keratitis.**
- **Synkinesis:** the eye closes when the patient smiles, from aberrant regeneration; treated with physiotherapy and botulinum toxin.
- **Contracture and hemifacial spasm.**
- **Crocodile tears (Bogorad syndrome):** tearing while eating, as salivary fibres regrow into the lacrimal pathway.
- **Psychological distress.**



**FIGURE 9** Hemifacial spasm: all the facial muscles and platysma contract together.
Dhingra, fig. 14.13

### 6.1 Hemifacial spasm

Usually from a vascular loop compressing the nerve at the brainstem. Treated with botulinum toxin, or microvascular decompression.

## 7 CHAPTER SEVEN
# For the examination

What the examiner wants to hear.

### 7.1 On call

**ACUTE FACIAL WEAKNESS**

1. Forehead spared with emotional movement kept: central; follow the stroke pathway if acute.
2. Protect the eye now: drops by day, ointment and tape at night.
3. Look in the ear: vesicles mean Ramsay Hunt, so add an antiviral; a foul ear or cholesteatoma needs urgent otology review.

4 Examine the parotid, neck and other cranial nerves.

5 After trauma, record whether the palsy was immediate or delayed.

6 Start steroids within 72 hours for Bell's palsy.

## 7.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Upper versus lower motor neurone** palsy, and why the forehead is spared.
- **Segments of the nerve,** and the narrow labyrinthine segment.
- **Topodiagnosis:** Schirmer, stapedial reflex, taste.
- **House-Brackmann grading.**
- **ENoG and EMG:** timing and meaning.
- **Longitudinal versus transverse fractures.**
- **Bell's palsy is a diagnosis of exclusion.**

## 7.3 Common pitfalls

**COMMON PITFALLS**

- Calling it Bell's palsy without looking in the ear or feeling the parotid.
- Forgetting eye care.
- Missing a slowly progressive or recurrent palsy that needs imaging.
- Not recording the timing of a post-traumatic palsy.

Written for this app from Oxford Textbook of Otolaryngology ch. 10, Oxford Specialist Handbook ch. 17, Dhingra ch. 14 and K. J. Lee ch. 21. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Temporal bone fracture and CSF otorrhoea

A temporal bone fracture comes with a head injury, so trauma care comes first and the ear second. Once the patient is safe, three questions decide what happens next: is the face moving, can the patient hear, and is there CSF leaking?

**AT A GLANCE**

- Trauma care first.
- Record facial function and hearing early.
- Capsule-violating fractures do worst.
- Most CSF leaks stop; delayed palsy usually recovers.
- Classify as otic capsule-sparing or -violating; capsule-violating fractures risk late meningitis.

## 1 CHAPTER ONE Overview

A fracture through the hearing bone.

### 1.1 Epidemiology

Most fractures follow road traffic accidents, then assaults and falls. Young adult men in their second to fourth decades are affected about three times as often as women. A fracture is found in a substantial share of patients with significant head injury.

### 1.2 Why it matters

The temporal bone contains the cochlea, the labyrinth, the facial nerve, the ossicles, the carotid artery and the dura. A fracture can damage any of them, and early documentation of facial function and hearing shapes every later decision.

# 2 CHAPTER TWO Classification

Two systems, both still taught.

## 2.1 Traditional: by the direction of the fracture line

### 2.1.1 Longitudinal fracture

About 80%. Usually from a blow to the side of the head. The line runs along the long axis of the petrous bone, through the squamous part, the posterosuperior canal wall, the tegmen and the middle ear, and tends to spare the otic capsule.

### 2.1.2 Transverse fracture

About 20%. Usually from a blow to the back of the head. The line runs across the petrous bone, through the vestibule, cochlea or internal auditory canal.

### 2.1.3 Mixed

Elements of both, common in severe injuries.

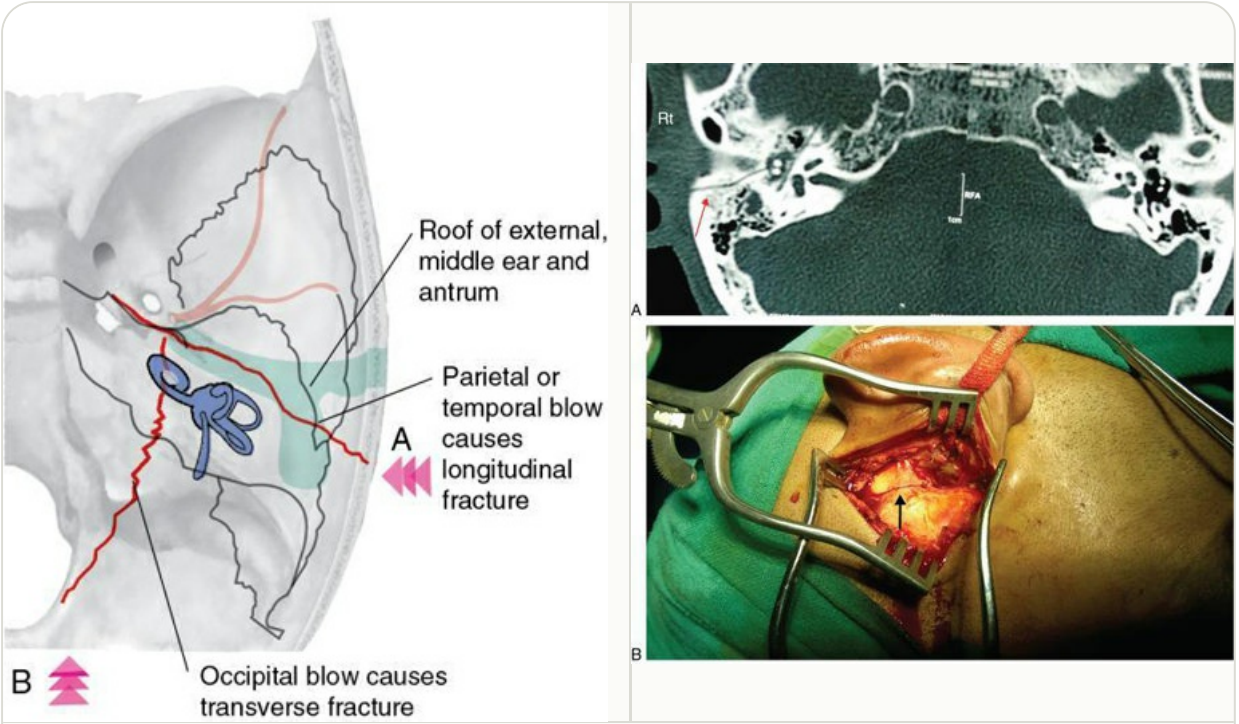



**FIGURE 1** First, the courses of longitudinal and transverse fractures of the temporal bone. Second, a right longitudinal fracture on CT and as seen at operation.
Dhingra, figs 14.10 and 14.11

LONGITUDINAL AND TRANSVERSE FRACTURES

| | LONGITUDINAL | TRANSVERSE |
|---|---|---|
| **Frequency** | About 80% | About 20% |
| **Blow** | Side of the head | Back of the head |
| **Drum** | Torn, bleeding from the ear | Intact, haemotympanum |
| **Hearing** | Conductive | Sensorineural |
| **Vertigo** | Less common | Severe |
| **Facial palsy** | About 20%, often delayed | About 50%, often immediate |

## 2.2 Modern: does it involve the otic capsule?

*Otic capsule sparing* fractures are by far the commonest; *otic capsule violating* fractures, only about 6%, are much more likely to cause facial palsy, sensorineural hearing loss, CSF leak and meningitis. This system predicts outcome better.

WHERE THE BOOKS DIFFER · LONGITUDINAL OR TRANSVERSE

**Dhingra**
Teaches conductive loss with longitudinal and sensorineural loss with transverse fractures.

**Lee**
Modern series do not fit the old categories neatly; the otic capsule system predicts outcome better.

**Know both**
the old table is still examined.

# 3 CHAPTER THREE Clinical features

Look, listen, test the face.

## 3.1 General

Signs of head injury and of a base of skull fracture: *Battle's sign* (bruising over the mastoid, appearing after a day or two), periorbital bruising, and altered

consciousness.

## 3.2 The ear

Bleeding from the ear with a torn drum or a step in the canal wall (longitudinal), or a blue-black drum from blood in the middle ear (*haemotympanum*) behind an intact drum (transverse).

## 3.3 Hearing

### 3.3.1 Conductive loss

From haemotympanum (the commonest early cause), a perforation, or ossicular injury: incudostapedial joint separation first, then dislocation of the incus, then fracture of the stapes crura.

### 3.3.2 Sensorineural loss

From fracture through the cochlea or a labyrinthine concussion; often permanent. Mixed losses are common.

## 3.4 Vertigo

Severe vertigo with spontaneous nystagmus after a transverse fracture through the labyrinth; milder, positional vertigo after concussion or BPPV.

## 3.5 Facial nerve

Record whether the palsy was immediate or delayed, and complete or incomplete. An immediate complete palsy suggests transection or a bone spicule in the nerve; a delayed palsy suggests oedema and usually recovers.

## 3.6 CSF leak

Clear fluid from the ear (CSF otorrhoea) through a torn drum, or from the nose (CSF otorhinorrhoea) via the Eustachian tube behind an intact drum.

# 4 CHAPTER FOUR Investigations

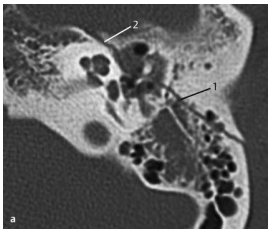
Image the bone and confirm the fluid.

## 4.1 High-resolution CT of the temporal bones

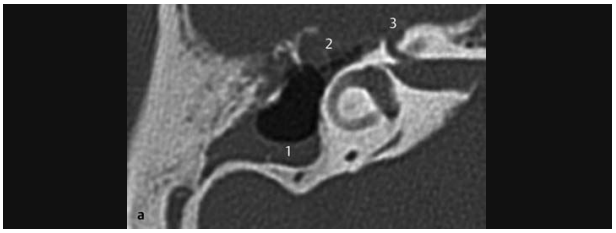
Shows the fracture line, the otic capsule, the ossicles, the facial canal and the tegmen. CT angiography if the fracture crosses the carotid canal.

**IMAGING IN THE APP** 2 cases

Skull base fractures, and dural prolapse with CSF leak.




**Skull Base Fractures**
Trauma · 13 images




**Dural Prolapse and Cerebrospinal Fluid Leakage**
After surgery · 2 images

## 4.2 Beta-2 transferrin

Confirms that the fluid is CSF.

## 4.3 Audiogram

As soon as practical, and repeated at six to eight weeks when blood has cleared from the middle ear.

## 4.4 Electroneuronography

For a complete facial palsy, to guide decisions on exploration.

# 5 CHAPTER FIVE Management

Trauma care first.

## 5.1 The first assessment

**HEAD INJURY WITH EAR SIGNS**

1. Follow trauma protocols and complete a neurological assessment.
2. Document facial function as early as possible, before sedation if you can.
3. Look for haemotympanum, a canal step, CSF, vertigo and nystagmus.
4. Do not syringe, probe or pack the ear deeply.
5. Arrange HRCT of the temporal bones, and CT angiography if the carotid canal is involved.

## 5.2 CSF leak

Most stop within a week or two with the head raised, bed rest, stool softeners and avoiding nose blowing and straining. Routine prophylactic antibiotics are not recommended. A persistent leak, or meningitis, needs a lumbar drain or surgical repair.

## 5.3 Facial palsy

Immediate complete palsy with evidence of nerve injury is explored and decompressed or repaired. Delayed palsy is managed with steroids and observation, with serial ENoG (see the facial nerve topic).

## 5.4 Hearing

Haemotympanum clears in weeks. A conductive loss persisting at six to eight weeks suggests ossicular injury and is explored and reconstructed. Sensorineural loss is rehabilitated with aids or, if profound, a cochlear implant.

**OPERATIVE GUIDES IN THEATRE** 4 operations

**Temporal Bone Trauma**
Temporal bone trauma
**Cerebrospinal Fluid, Otorrhea and Encephalocele**
CSF otorrhoea repair
**Facial Nerve Decompression**
Facial decompression
**Ossicular Chain Reconstruction**
Ossicular repair

## 5.5 Vertigo

Short-term vestibular sedatives, then vestibular rehabilitation; treat post-traumatic BPPV with repositioning manoeuvres.

# 6 CHAPTER SIX Complications

Early and late.

- **Meningitis,** early or years later, through an unhealed fracture.
- **Permanent sensorineural deafness.**
- **Facial palsy.**
- **Late cholesteatoma,** from skin trapped in the fracture line.
- **Chronic vertigo** and perilymph fistula.
- **Carotid injury** and dissection.

**FIGURES AND LATE PROBLEMS (OXFORD TEXTBOOK)**

The temporal bone is involved in 30 to 70% of head injuries, but only 14 to 22% of these produce a true fracture, mostly from road traffic accidents and falls. The old longitudinal–transverse scheme fails to describe the many mixed fractures, which is why *otic capsule-sparing* versus *otic capsule-violating* is now preferred: it predicts sensorineural loss, facial palsy and CSF leak better. Ossicular injury most often dislocates the incudostapedial joint (about 80%) and gives a conductive loss of at least 30 dB. About a quarter of fractures cause facial palsy (half of transverse and a tenth of longitudinal), and in about three-quarters of these the onset is delayed by one to sixteen days, which predicts a good recovery. A CSF leak complicates about 17%.

Bone of the otic capsule heals only by fibrous union, so capsule-violating fractures leave a weak barrier: CSF leaks can recur with coughing or straining, meningitis can occur years later, and meningoceles or encephaloceles may present long after the injury with a conductive loss, an effusion or meningitis. Patients with these fractures need pneumococcal vaccination and advice to seek help for fever and headache. Audiometry is best done two to three days after injury (after the patient wakes) and repeated; the Sunnybrook scale records synkinesis better than House–Brackmann during recovery.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Longitudinal versus transverse:** the six-row table.
- **Otic capsule sparing versus violating,** and why it predicts outcome.
- **Immediate versus delayed facial palsy,** and how each is managed.
- **Beta-2 transferrin** for CSF.
- **Ossicular injuries** in order of frequency.

## 7.2 Common pitfalls

COMMON PITFALLS

- Forgetting to record facial function before the patient is sedated.
- Syringing or packing a bleeding ear after head injury.
- Assuming a conductive loss is permanent before the blood has cleared.
- Missing late meningitis or cholesteatoma years after the injury.

Written for this app from Oxford Textbook of Otolaryngology ch. 23, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 25, Dhingra ch. 14 and K. J. Lee ch. 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Middle ear and temporal bone tumours

Tumours of the middle ear and temporal bone are rare, and two of them must never be missed. The first is a red, pulsating mass behind the drum, a glomus tumour that bleeds torrentially if biopsied in clinic. The second is the chronic, painful, bleeding ear that is really a squamous cell carcinoma.

**AT A GLANCE**

- Never biopsy a pulsatile red mass.
- CT and MRI separate glomus from carotid and jugular anomalies.
- Check catecholamines; embolise before surgery.
- A chronic bloody painful ear needs biopsy.
- Facial schwannoma: operate only for weakness worse than House–Brackmann III.

## 1 CHAPTER ONE Overview

Rare, but two are not to be missed.

### 1.1 The tumours

**TUMOURS OF THE MIDDLE EAR AND TEMPORAL BONE**

- BENIGN **Paraganglioma (glomus)** tympanicum, jugulare, vagale
- BENIGN **Others** adenomatous neuroendocrine tumour, facial schwannoma, haemangioma, choristoma, endolymphatic sac tumour
- MALIGNANT **Squamous cell carcinoma** the commonest
- MALIGNANT **Others** adenoid cystic carcinoma, adenocarcinoma, rhabdomyosarcoma (children), metastases
- SKULL BASE **Chordoma, chondrosarcoma** petrous apex and clivus

### 1.2 Why it matters

Both key tumours present to the ear clinic looking like something common: pulsatile tinnitus or a polyp, and a discharging ear.

## 2 CHAPTER TWO Paraganglioma

The glomus tumour.

### 2.1 Origin

Paragangliomas arise from neural-crest *glomus bodies* along the tympanic branch of the glossopharyngeal nerve (Jacobson's nerve) and the auricular branch of the vagus (Arnold's nerve), and on the dome of the jugular bulb. They are benign, slow-growing and highly vascular.

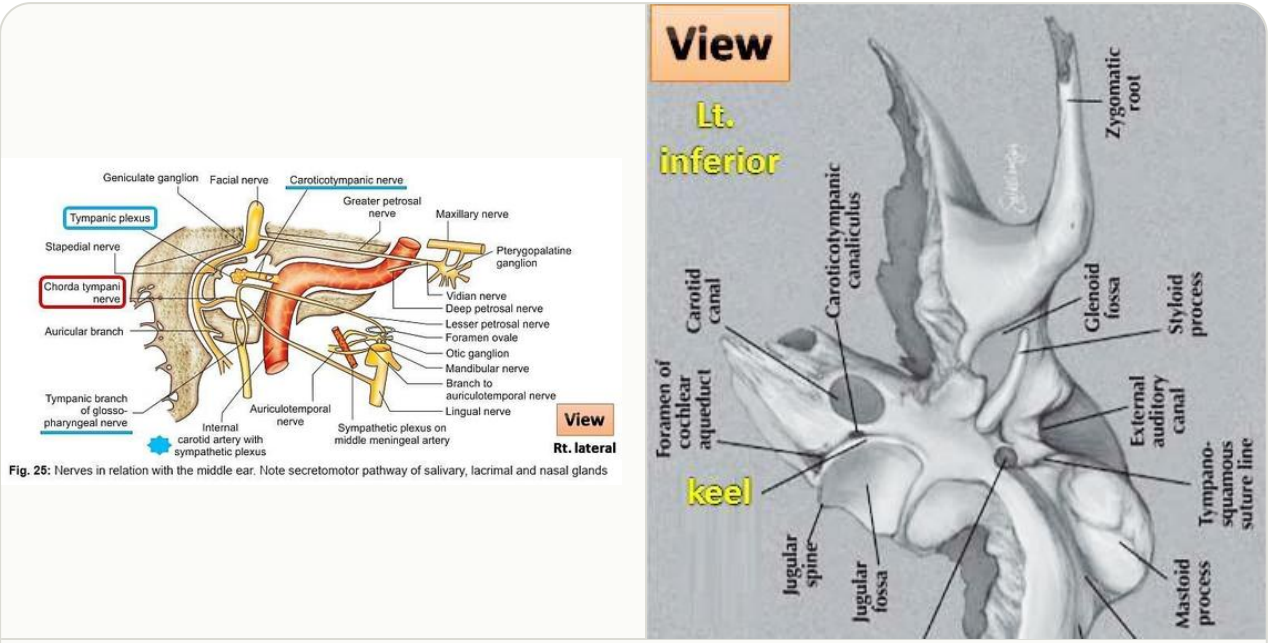



Fig. 25: Nerves in relation with the middle ear. Note secretomotor pathway of salivary, lacrimal and nasal glands

**FIGURE 1** The tympanic plexus on the promontory (Jacobson's nerve), and the keel of bone between the carotid canal and jugular fossa.

Anatomy atlas · ENT Steps p. 53, ENT Steps p. 24

## 2.2 Who gets them

Usually in the fourth and fifth decades, with a strong female predominance (about 5 to 1).

## 2.3 Pathology

Nests of chief cells (*zellballen*) surrounded by sustentacular cells in a vascular stroma.

## 2.4 Genetics

Some are familial, from mutations in the succinate dehydrogenase genes. SDHD mutations cause multiple tumours and are expressed only when inherited from the father; SDHB mutations carry a higher risk of malignancy.

WHERE THE BOOKS DIFFER · HOW OFTEN

| Dhingra | Lee |
| --- | --- |
| About 10% familial, about 3% secrete catecholamines, about 4% metastasise. | Up to 30% familial, 5% or fewer secrete, 2–4% metastasise. |

## 2.5 Types

### 2.5.1 Glomus tympanicum

Arises on the promontory and fills the middle ear.

### 2.5.2 Glomus jugulare

Arises in the jugular bulb, eroding upwards into the middle ear and medially to the jugular foramen and skull base.

### 2.5.3 Glomus vagale

Arises on the vagus below the skull base, presenting in the neck.

## 2.6 Clinical features

### 2.6.1 Symptoms

Pulsatile tinnitus (about 80%), conductive hearing loss (about 60%), fullness, and later bleeding from the ear. Jugulare tumours add lower cranial nerve palsies (ninth to twelfth: hoarseness, dysphagia, wasting of the tongue and trapezius) and facial weakness.

### 2.6.2 Signs

A red mass behind the drum, the *rising sun* when it emerges from the floor of the middle ear. Pneumatic pressure on the drum makes the mass pulsate, then blanch (*Brown's sign*). A bruit may be heard over the mastoid. In advanced cases a bleeding polyp fills the canal.



**FIGURE 2** Glomus jugulare: a red mass behind the drum.
Dhingra, fig. 17.1

**NEVER BIOPSY IN CLINIC**

A red, pulsatile mass behind the drum, or a polyp with pulsatile tinnitus, is a glomus tumour until imaged. Biopsy can cause catastrophic bleeding.

### 2.6.3 Secreting tumours

Hypertension, headache, palpitations and sweating mean catecholamine secretion: measure urinary or plasma metanephrines before any anaesthetic.

## 2.7 Investigations

### 2.7.1 CT

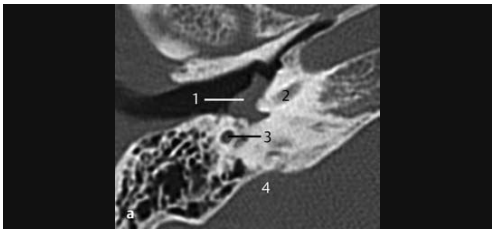
Erosion of the *caroticojugular spine* separates a jugulare tumour from a tympanicum. CT also excludes an aberrant carotid artery and a high jugular bulb.

### 2.7.2 MRI

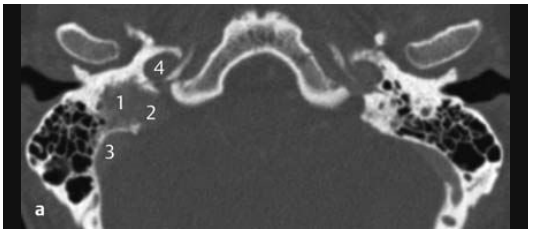
A *salt-and-pepper* appearance from flow voids; defines intracranial and neck extension.

**IMAGING IN THE APP** 2 cases

Glomus tympanicum and glomus jugulare on CT and MRI.




Glomus Tympanicum (Paraganglioma)
Tumours · 11 images




Glomus Jugulare Tumor (Paraganglioma)
Tumours · 11 images

### 2.7.3 Angiography

Shows the feeding vessels (mainly the ascending pharyngeal artery), looks for multiple tumours, and allows embolisation. Balloon occlusion testing if the carotid may need to be sacrificed.

## 2.8 Staging

GLASSCOCK-JACKSON CLASSIFICATION

| STAGE | GLOMUS TYMPANICUM | GLOMUS JUGULARE |
|---|---|---|
| I | Limited to the promontory | Jugular bulb, middle ear and mastoid |
| II | Fills the middle ear | Extends under the internal auditory canal |
| III | Into the mastoid | Into the petrous apex |

| STAGE | GLOMUS TYMPANICUM | GLOMUS JUGULARE |
|---|---|---|
| IV | Through the drum into the canal, or beyond | Beyond the apex into the clivus or infratemporal fossa |

The *Fisch* classification grades A (middle ear), B (tympanomastoid), C (infralabyrinthine and apex, C1 to C4 by carotid involvement) and D (intracranial).

## 2.9 Differential diagnosis

### RED, WHITE OR BLUE BEHIND THE DRUM

| COLOUR | THINK OF | KEY TEST |
|---|---|---|
| Red, pulsatile | Glomus tumour | CT and MRI |
| Pink, anterior | Aberrant internal carotid artery | CT, MR angiography |
| Blue, inferior | High jugular bulb | CT, MRI |
| Blue with dark fluid | Cholesterol granuloma | CT, MRI |
| White | Congenital cholesteatoma | CT |

## 2.10 Treatment

### 2.10.1 Surgery

The treatment of choice for most, after embolisation 24 to 48 hours before and alpha-blockade if the tumour secretes. Small tympanicum tumours whose whole margin can be seen are removed through the canal; larger tumours need hypotympanic, extended facial recess or mastoid-neck approaches; jugulare tumours need the Fisch infratemporal fossa approach.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Glomus Tumors**
Glomus tumours

**Resection of the Temporal Bone**
Temporal bone resection

**Tumors of the Facial Nerve**
Tumours of the facial nerve

#### 2.10.2 Radiotherapy

Stereotactic radiosurgery or fractionated radiotherapy arrests growth in about 90 to 94%, though the tumour remains. For the elderly, the unfit, residual disease and tumours where surgery would cost lower cranial nerves.

#### 2.10.3 Observation

Slow-growing tumours in elderly patients may simply be watched.

**PEARLS**

- A glomus tympanicum is only certain when its whole rim is visible; otherwise image.
- Pain is more typical of carcinoma than of a glomus tumour.

# 3 CHAPTER THREE Squamous cell carcinoma

The chronic ear that bleeds.

## 3.1 Who gets it

Adults of 40 to 60, about three-quarters with a long history of chronic ear discharge.

## 3.2 Clinical features

Bloodstained, foul discharge; deep, boring pain, especially at night; friable granulation or a polyp that bleeds on touch; and later facial palsy, vertigo, trismus and a neck or parotid mass. It is easily mistaken for chronic otitis media or otitis externa.

## 3.3 Investigations

Biopsy of any non-healing granulation in the canal or middle ear. CT of the temporal bone and neck, MRI for soft tissue and dura, and chest imaging.

### 3.4 Treatment

- **Limited to the canal without bone involvement:** sleeve resection of the canal.
- **Bony canal involved:** lateral temporal bone resection.
- **Middle ear involved:** subtotal temporal bone resection.
- **Parotidectomy and neck dissection** according to spread.
- **Postoperative radiotherapy** for most stages beyond the earliest, with chemotherapy for advanced disease.

Outlook is poor once the middle ear and beyond are involved. The dura usually resists invasion.

---

CHAPTER FOUR

# 4 Other tumours

Rarer, each with a clue.

### 4.1 Rhabdomyosarcoma

The commonest malignant ear tumour in children: a polyp in a child's ear with early facial palsy. Treated with chemoradiation, with resection when feasible.



**FIGURE 3** A four-year-old child with rhabdomyosarcoma of the right middle ear and mastoid, with facial palsy on the same side.

Dhingra, fig. 17.2

## 4.2 Endolymphatic sac tumour

A locally aggressive papillary tumour of the sac, sporadic or with von Hippel-Lindau disease (about 11% of patients, often bilateral). It mimics Ménière's disease, or causes sudden hearing loss from bleeding into the inner ear. Removed with the dura around the sac; VHL patients are screened by MRI every one to three years.

## 4.3 Facial nerve schwannoma and haemangioma

A schwannoma causes slowly progressive facial palsy or hearing loss; a haemangioma near the geniculate ganglion causes a palsy out of proportion to its small size and should be removed early. Facial schwannomas arise most often near the geniculate ganglion, and many patients keep good facial function for years. Because removal means sacrificing and grafting the nerve, and the best result after a graft is House–Brackmann III, surgery is reserved for weakness worse than grade III; enlarging tumours with good function are treated by radiosurgery (control in

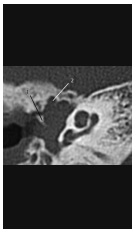
about 90% at five years, with function preserved in 90%) or by bony decompression of the canal.

## 4.4 Adenomatous neuroendocrine tumour

A benign middle ear tumour presenting as a mass behind an intact drum with conductive loss.

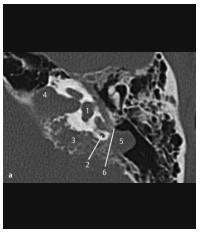
**IMAGING IN THE APP** 5 cases

Middle ear adenoma, endolymphatic sac tumour, canal carcinoma, fibrous dysplasia and apical pseudotumours.



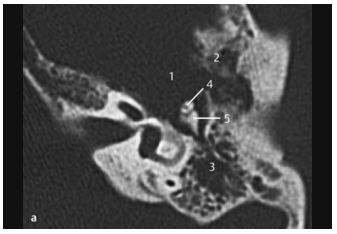
**Adenoma of the Middle Ear**

Tumours · 1 image



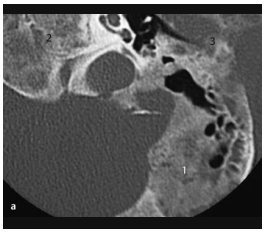
**Endolymphatic Sac Tumour**

Tumours · 8 images



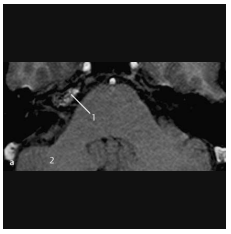
**Malignancy of the External Auditory Canal Squamous Cell Carcinoma**

Tumours · 2 images



**Fibrous Dysplasia of the Temporal Bone and Skull Base**

Bone disease · 7 images



**Pseudotumors in the Apex of the Petrous Bone**

Tumours · 12 images

## 4.5 Chordoma and chondrosarcoma

Chordoma arises in the midline clivus in older patients, with headache and sixth nerve palsy; it is bright on T2 MRI and contains physaliferous cells. Chondrosarcoma arises off the midline at the petroclival junction. Both are treated by maximal resection followed by high-dose or proton radiotherapy.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 On call

**A MASS OR GRANULATION IN THE EAR**

1. Pulsatile red mass: do not biopsy; arrange CT and MRI.
2. Non-healing, bleeding granulation in a chronic ear: biopsy it.
3. Hypertension, sweating or palpitations with a glomus tumour: check catecholamines before any anaesthetic.

## 5.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Origin of glomus tumours:** Jacobson's and Arnold's nerves, jugular bulb.
- **Signs:** rising sun, Brown's sign, pulsatile tinnitus.
- **CT:** caroticojugular spine; aberrant carotid and high bulb.
- **Staging** and treatment options.
- **Squamous carcinoma:** who, symptoms, and the types of temporal bone resection.

## 5.3 Common pitfalls

**COMMON PITFALLS**

- Biopsying a red mass behind the drum in clinic.
- Treating a bleeding, painful ear as chronic otitis media for months.
- Forgetting catecholamines before surgery.

Written for this app from Oxford Textbook of Otolaryngology ch. 18, Oxford Specialist Handbook ch. 19, Oxford Handbook of ENT ch. 8, Dhingra ch. 16, 17 and K. J. Lee ch. 24. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Labyrinthine fistula and labyrinthitis

A dizzy patient with an infected ear raises one question: has the labyrinth been breached, or inflamed? A fistula can be sealed, and serous labyrinthitis recovers; suppurative labyrinthitis destroys the hearing for good. After meningitis, the cochlea can turn to bone within weeks, and the window for a cochlear implant can close.

**AT A GLANCE**

- Vertigo in an infected ear means the labyrinth is breached or inflamed.
- Serous disease can recover; suppurative disease causes a dead ear.
- Fistula test is specific but insensitive.
- Treat the source; add steroids and follow hearing after meningitis.
- Early implant if ossification threatens.

## 1 CHAPTER ONE Overview

When infection reaches the inner ear.

### 1.1 Definitions

#### 1.1.1 Labyrinthine fistula

Erosion of the bony labyrinth, usually the lateral semicircular canal, exposing the membranous labyrinth without infecting it. Dhingra calls it *circumscribed labyrinthitis*.

#### 1.1.2 Serous labyrinthitis

Diffuse irritation of the labyrinth by bacterial toxins through the windows or a fistula, without bacteria in the inner ear. Reversible.

#### 1.1.3 Suppurative labyrinthitis

Bacteria and pus in the labyrinth: total, permanent loss of hearing and vestibular function.

#### 1.1.4 Perilymph fistula

> A leak of perilymph through the oval or round window, after stapes surgery, barotrauma, diving, straining or trauma.

### 1.2 Other forms

Viral labyrinthitis (mumps, measles, rubella, cytomegalovirus) and syphilitic labyrinthitis.

## 2 CHAPTER TWO Pathways

How infection gets in.

### 2.1 From the middle ear

Cholesteatoma eroding the lateral canal (the commonest), through the oval or round window in acute or chronic otitis media, and after surgery or trauma.

### 2.2 From the meninges

Bacteria spread from the subarachnoid space through the cochlear aqueduct or the internal auditory canal. *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Neisseria meningitidis* are the organisms; pneumococcus carries the highest risk of deafness, around 20%.

### 2.3 Blood-borne

Viral and syphilitic infections.

## 3 CHAPTER THREE Labyrinthine fistula

The eroded lateral canal.

## 3.1 Causes

Cholesteatoma in most cases, then tumours, trauma and surgery. About 7 to 10% of cholesteatomas have a fistula, 90% of them in the lateral canal.

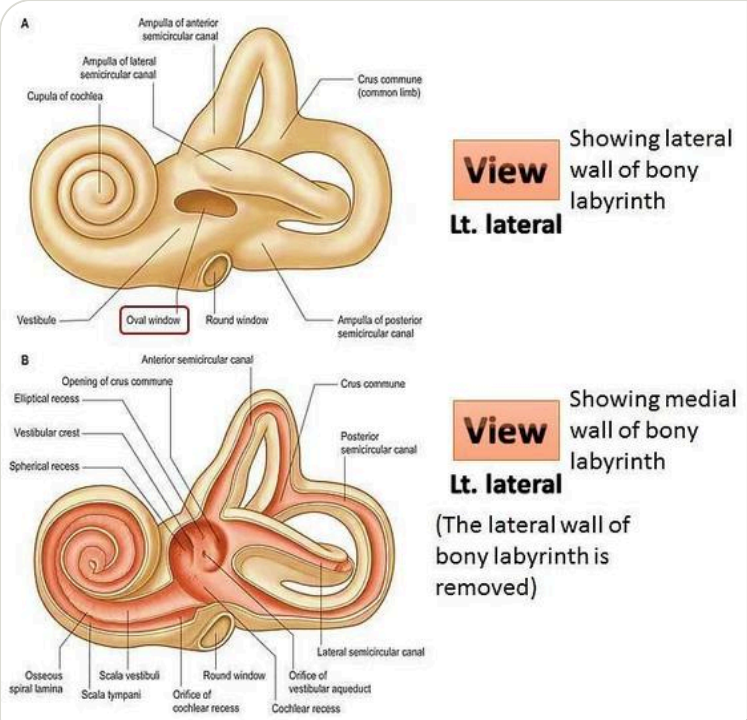



**FIGURE 1** The bony labyrinth: the lateral semicircular canal bulges into the medial wall of the attic and antrum, where cholesteatoma erodes it.

Anatomy atlas · ENT Steps p. 64

## 3.2 Clinical picture

Brief vertigo provoked by pressure on the ear: pressing the tragus, cleaning the ear, or nose blowing. Hearing is often preserved.

## 3.3 Fistula test

Pressure applied to the canal (tragal pressure or a Siegle speculum) moves the endolymph through the fistula and causes vertigo and nystagmus. With positive pressure the fast phase usually beats towards the tested ear; with negative pressure, away from it.

**THE FISTULA TEST**

It is specific but insensitive: positive in only 30 to 50% of fistulae. A negative test does not exclude one; it may also be negative in a dead labyrinth or if cholesteatoma seals

the fistula. Assume a fistula in every cholesteatoma.

## 3.4 Other positive fistula tests

Ménière's disease (Hennebert's sign), superior canal dehiscence, perilymph fistula and congenital syphilis.

## 3.5 Investigations

HRCT shows only about 60% of fistulae before surgery.

## 3.6 Management

Surgery for the cholesteatoma; the matrix over a small fistula may be removed carefully at the end of the operation and the defect sealed with fascia or bone pâté; over a large fistula, or in an only hearing ear, the matrix is left and the ear exteriorised.

# 4 CHAPTER FOUR Serous and suppurative labyrinthitis

Irritation or destruction.

## 4.1 Serous labyrinthitis

### 4.1.1 Clinical picture

Vertigo with nausea, nystagmus towards the affected ear (an irritative lesion), and some sensorineural hearing loss that recovers if treated.

### 4.1.2 Treatment

Bed rest with the affected ear uppermost, full-dose antibiotics, labyrinthine sedatives, and myringotomy for a bulging drum. Mastoidectomy treats a chronic source once medical treatment has begun.

## 4.2 Suppurative labyrinthitis

### 4.2.1 Clinical picture

Severe vertigo with nausea and vomiting, nystagmus beating towards the healthy ear (the labyrinth has failed), total deafness in the ear, and a toxic patient. The vertigo settles over three to six weeks as the brain compensates.

### 4.2.2 Danger

Infection can spread from the labyrinth to the meninges along the internal auditory canal.

### 4.2.3 Treatment

Intravenous antibiotics, removal of the middle ear source, and rarely drainage of the labyrinth if it is feeding an intracranial infection.

SEROUS OR SUPPURATIVE?

| | SEROUS | SUPPURATIVE |
|---|---|---|
| **Cause** | Toxins | Bacteria |
| **Nystagmus** | Towards the affected ear | Towards the healthy ear |
| **Hearing** | Partial loss, recovers | Total, permanent loss |
| **Patient** | Unwell | Toxic, may have meningitis |
| **Outcome** | Recovery | Dead ear, ossification |

## 4.3 Labyrinthitis ossificans

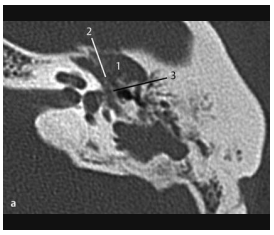
After suppurative labyrinthitis, especially meningitic, fibrous tissue and new bone fill the cochlea, starting in the basal turn near the round window, within weeks to months.

DEAFNESS AFTER MENINGITIS

- Test hearing in every patient after bacterial meningitis.
- Bilateral profound loss: image urgently (MRI and CT) for ossification and refer for early cochlear implantation.
- Steroids reduce hearing loss in *H. influenzae* and pneumococcal meningitis.

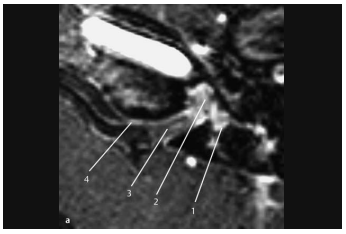
**IMAGING IN THE APP** 4 cases

Labyrinthine fistula, labyrinthitis ossificans, labyrinthitis with meningitis, and otosyphilis.



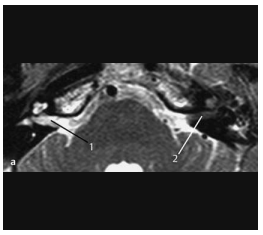
**Labyrinthine Fistula due to Cholesteatoma**

Infection & inflammation · 11 images



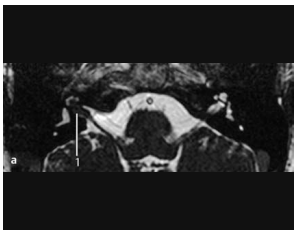
**Labyrinthitis Ossificans (Ossification after Meningitis)**

Bone disease · 14 images



**Labyrinthitis and Subclinical Meningitis**

Infection & inflammation · 5 images



**Otosyphilis and Intracranial Complications**

Infection & inflammation · 6 images

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Mastoid Surgery**
Remove source

**Cochlear Implantation in Adults**
Implant before ossification

**Pediatric Cochlear Implantation**
Paediatric implant

CHAPTER FIVE

# 5 Perilymph fistula

A leak at the windows.

## 5.1 Causes

Stapes surgery, barotrauma, diving, heavy lifting, straining or coughing, and head injury. A leak in an unoperated ear usually sits around the annular ligament or the round window membrane. Spontaneous fistula is controversial.

### 5.2 Clinical picture

Fluctuating or sudden hearing loss, roaring tinnitus, disequilibrium or vertigo, often after the precipitating event, and a positive fistula test (Hennebert's sign). A penetrating ear injury with vertigo and hearing loss suggests stapes subluxation into the vestibule.

### 5.3 Management

Bed rest for 7 to 10 days with the head raised, no straining, and stool softeners. Explore and patch the window if hearing is worsening or symptoms persist.

## 6 CHAPTER SIX
# Viral and syphilitic labyrinthitis

The other causes.

### 6.1 Viral

Mumps (usually one-sided profound loss), measles, rubella and cytomegalovirus. CMV is the commonest non-syndromic cause of congenital sensorineural hearing loss, and the loss may progress after a normal newborn screen.

### 6.2 Syphilis

Congenital or acquired. Late congenital syphilis may first appear at 8 to 20 years as bilateral sudden or fluctuating loss with vertigo, and Hennebert's sign. Test with VDRL or RPR and FTA-ABS; treat with penicillin and steroids.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 On call

**VERTIGO IN AN INFECTED EAR**

1. Examine for nystagmus and its direction; do a gentle fistula test.
2. Audiogram and HRCT.
3. Intravenous antibiotics; myringotomy for acute otitis media.
4. Plan surgery for cholesteatoma.
5. Hearing loss with meningitis: treat urgently and image for ossification.

## 7.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Three types:** circumscribed, serous, suppurative.
- **The fistula test,** its direction, and why it can be negative.
- **Nystagmus direction** in serous versus suppurative labyrinthitis.
- **Labyrinthitis ossificans** and early cochlear implantation.
- **Perilymph fistula:** causes and treatment.

## 7.3 Common pitfalls

**COMMON PITFALLS**

- Being reassured by a negative fistula test.
- Forgetting a hearing test after meningitis.

- Missing late congenital syphilis in a young adult with fluctuating bilateral loss.

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CLINICAL · EAR

# Petrous apicitis

Petrous apicitis is infection that has tracked from the middle ear deep into the tip of the petrous bone. The classic picture is Gradenigo's triad: a discharging ear, deep pain behind the eye and a sixth nerve palsy. It is rare now, but it can appear just as the ear seems to be improving.

**AT A GLANCE**

- Spread of infection into a pneumatised apex.
- Gradenigo triad: discharge, deep eye pain, VI palsy.
- Needs CT and MRI together.
- Long IV antibiotics first; surgery for abscess or failure.
- Cholesterol granuloma is bright on T1 and T2; cholesteatoma restricts diffusion.

## 1 CHAPTER ONE Overview

Infection at the tip of the temporal bone.

### 1.1 Definition

Inflammation, and sometimes abscess, of the air cells of the petrous apex, spreading from acute or chronic otitis media and mastoiditis.

### 1.2 Why only some people get it

Only about 30% of petrous apices are pneumatised; the rest are diploic (marrow-filled) or sclerotic. Infection can only track into an aerated apex.

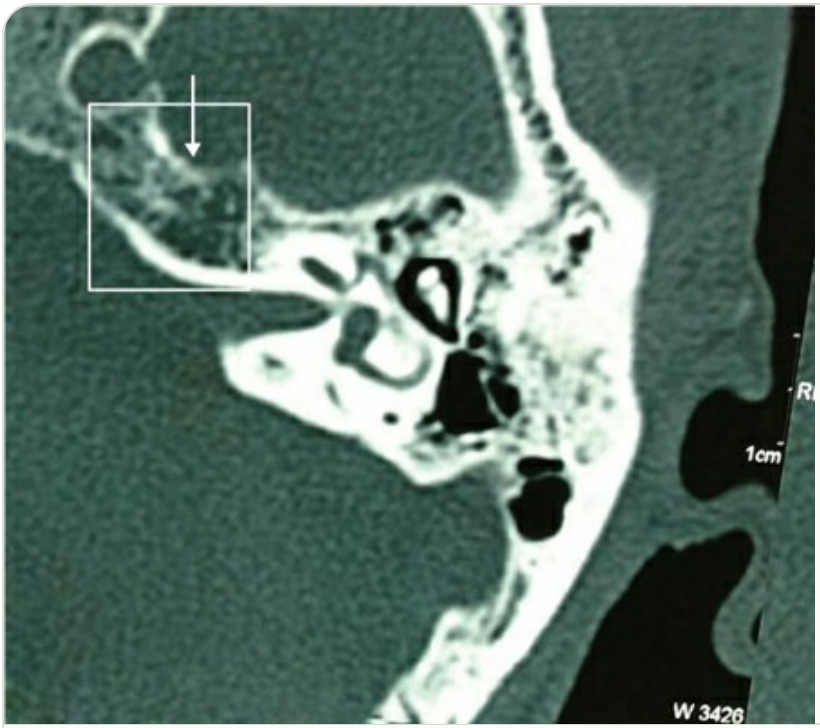



**FIGURE 1** A pneumatised petrous apex.
Dhingra, fig. 1.15

### 1.3 Why it matters

The apex lies against the sixth nerve, the trigeminal ganglion, the internal carotid artery and the meninges. Infection there can cause meningitis, carotid injury and skull base osteomyelitis.

## 2 CHAPTER TWO Applied anatomy

Tracts into the apex.

### 2.1 Air cell tracts

The middle ear and mastoid connect with the apex by perilabyrinthine cell tracts:

- **Posterosuperior:** from the mastoid and antrum, above and behind the labyrinth, through *Trautmann's triangle* and the subarcuate region.
- **Anteroinferior:** from the hypotympanum and the region around the Eustachian tube, below the cochlea.

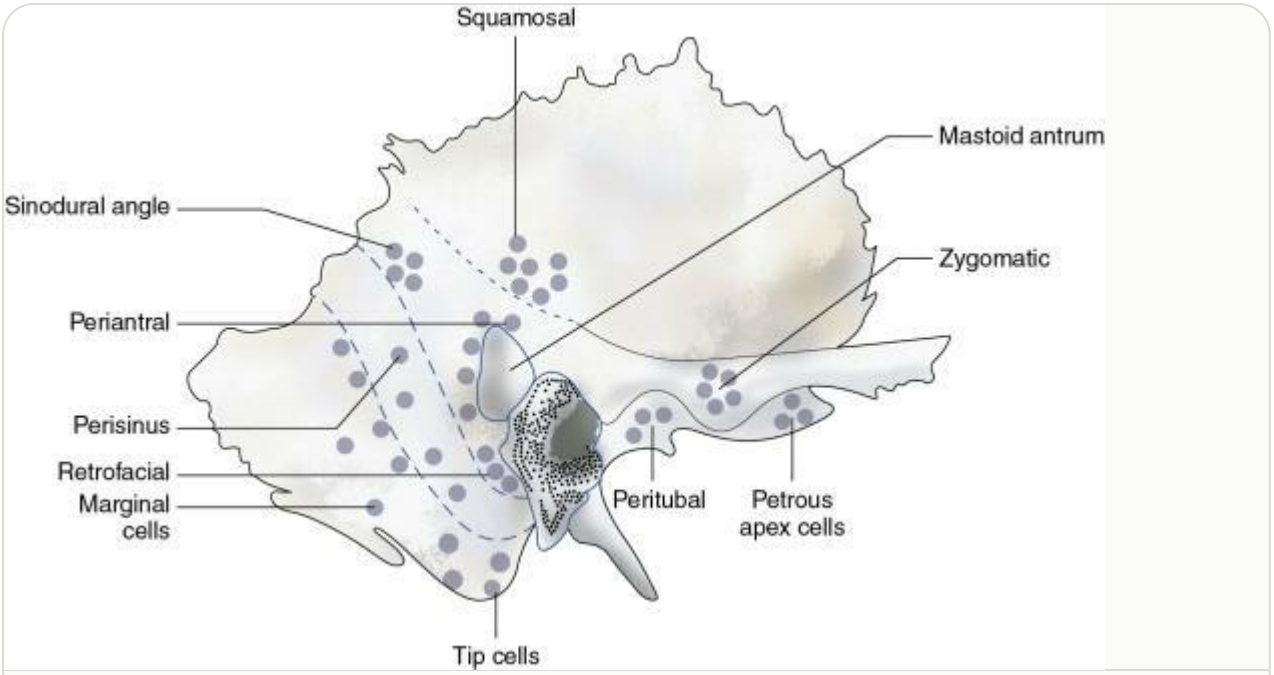



**FIGURE 2** The air cell tracts of the temporal bone.
Dhingra, fig. 1.13

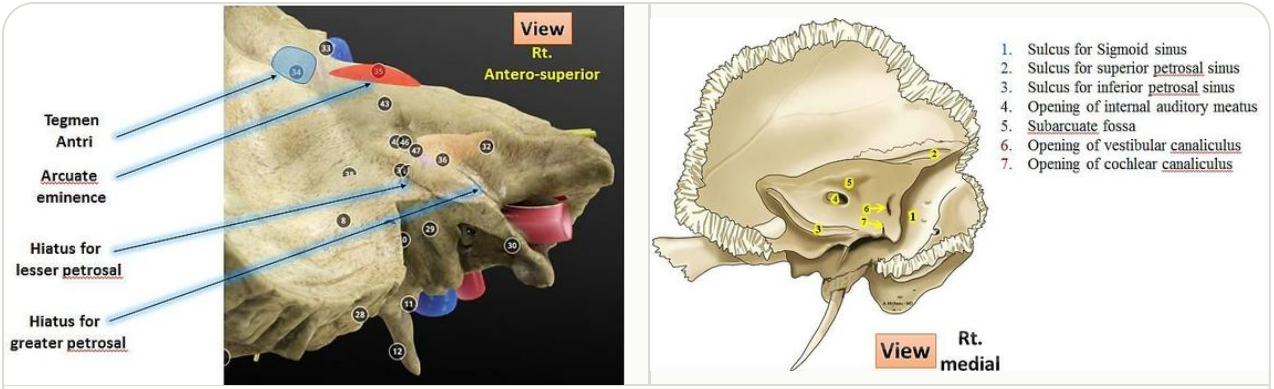



**FIGURE 3** The anterior and posterior surfaces of the petrous bone.
Anatomy atlas · ENT Steps p. 21, ENT Steps p. 23

## 2.2 Neighbours

### 2.2.1 Dorello's canal

The sixth nerve passes under the petroclinoid (Gruber's) ligament, between the petrous tip and the sphenoid; inflammation here causes a lateral rectus palsy and double vision.

### 2.2.2 Meckel's cave

> The trigeminal ganglion sits in a depression on the apex; its irritation causes deep pain behind the eye.

# 3 CHAPTER THREE Clinical features

Gradenigo's triad.

## 3.1 The triad

- **Persistent ear discharge.**
- **Deep retro-orbital or facial pain,** from the trigeminal ganglion.
- **Diplopia from a sixth nerve palsy.**

The full triad is uncommon today; often only one or two features are present.

## 3.2 Other features

Fever, headache, neck stiffness and vomiting suggest meningeal spread; the seventh and eighth nerves are occasionally involved.

**WHEN TO SUSPECT IT**

Persistent discharge or deep pain despite an adequate mastoidectomy, or new pain after the ear has apparently settled.

# 4 CHAPTER FOUR Investigations

You need both CT and MRI.

## 4.1 High-resolution CT

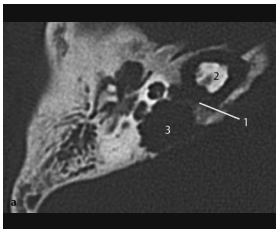
Shows the pneumatisation of the apex, breakdown of the cell septa and erosion of the bone.

## 4.2 MRI with contrast

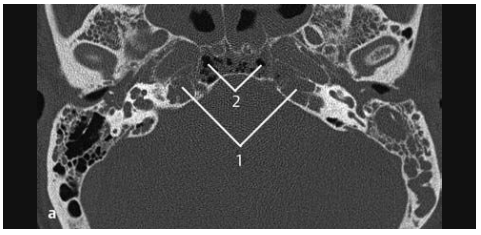
Separates marrow from fluid or pus, shows dural enhancement and intracranial spread, and assesses the carotid and cavernous sinus.

**IMAGING IN THE APP** 3 cases

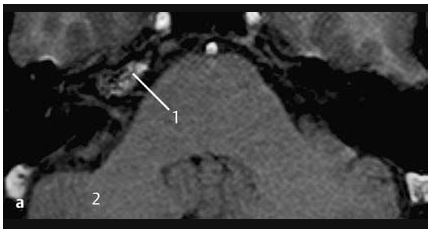
Petrositis, Gradenigo syndrome and apical pseudotumours.



**Petrositis**
Infection & inflammation · 3 images



**Infection of the Skull Base and Gradenigo Syndrome**
Infection & inflammation · 3 images



**Pseudotumors in the Apex of the Petrous Bone**
Tumours · 12 images

## 4.3 Differential diagnosis

- **Apical effusion:** fluid in an aerated apex, usually asymptomatic.
- **Asymmetric marrow:** a non-pneumatised apex full of fat that looks abnormal on MRI; a common false alarm.
- **Cholesterol granuloma:** bright on both T1 and T2, expanding slowly.
- **Cholesteatoma** of the apex.
- **Nasopharyngeal carcinoma** and other skull base tumours.

**PETROUS APEX LESIONS ON MRI (OXFORD TEXTBOOK)**

| LESION | T1 | T2 | DIFFUSION AND ENHANCEMENT | CT |
|---|---|---|---|---|
| Asymmetric marrow (fat) | Bright, suppresses with fat saturation | Intermediate | No restriction | Normal bone, non-pneumatised apex |
| Trapped fluid or effusion | Dark | Bright | No restriction, no enhancement | Intact septa |

| LESION | T1 | T2 | DIFFUSION AND ENHANCEMENT | CT |
|---|---|---|---|---|
| Cholesterol granuloma | Bright | Bright | No restriction; rim only | Smooth expansion |
| Cholesteatoma or epidermoid | Dark | Bright | Restricted diffusion | Smooth erosion |
| Apicitis | Dark | Bright | Enhancing, with dural reaction | Destroyed septa |
| Chordoma or chondrosarcoma | Intermediate | Very bright | Heterogeneous enhancement | Bone destruction; clivus or petroclival fissure |

Cholesterol granuloma is the commonest primary apex lesion, about ten times commoner than apical cholesteatoma, and forms when bleeding fills an obstructed air cell. Asymptomatic lesions are watched with serial imaging; symptomatic ones are drained, preserving hearing where possible, through an infracochlear or infralabyrinthine route, or endoscopically through the sphenoid when the lesion abuts it. Lower cranial nerve palsies with an apex lesion raise concern for malignancy.

# 5 CHAPTER FIVE Management

Antibiotics first.

## 5.1 Medical treatment

Admission, prolonged intravenous antibiotics that cover the organisms of otitis media and penetrate bone, and treatment of the middle ear source with myringotomy or mastoidectomy. Most acute cases now settle on antibiotics alone.

### 5.2 Surgery

For an abscess, bone necrosis, or failure of medical treatment. The route follows the cell tracts and the state of the hearing:

- **Hearing preserved:** through the mastoid via Trautmann's triangle or the subarcuate cells, infralabyrinthine or infracochlear routes, or through the middle cranial fossa.
- **Dead ear:** translabyrinthine or transcochlear routes, which give the widest access.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Cholesterol Granulomas and Congenital Epidermoid Tumors of the Temporal Bone**
Petrous apex lesions

**Mastoid Surgery**
Mastoid approach

## 6 CHAPTER SIX Complications

Neighbours at risk.

- Meningitis and other intracranial infection.
- Injury to or thrombosis of the internal carotid artery.
- Skull base osteomyelitis.
- Cavernous sinus thrombosis.

## 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

### 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Gradenigo's triad,** and the anatomy behind each part: Meckel's cave and Dorello's canal.
- **Pneumatisation** of the apex in about 30%.

- **CT and MRI together.**
- **Surgical routes,** chosen by the hearing.

## 7.2 Common pitfalls

COMMON PITFALLS

- Reading asymmetric apical marrow as infection.
- Missing the diagnosis because the ear seemed to be getting better.
- Waiting for the full triad.

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CLINICAL · EAR

# Perichondritis, zoster and pinna infections

A red, hot, swollen pinna that spares the lobule is a disease of the cartilage, because the lobule has none. Perichondritis needs anti-pseudomonal cover and early drainage of any pus; otherwise the cartilage dies and the ear crumples. Vesicles in the concha with facial weakness are a different emergency: Ramsay Hunt syndrome.

**AT A GLANCE**

- Red, hot pinna that spares the lobule means cartilage.
- Cover Pseudomonas and drain early.
- Zoster: vesicles, pain, and often facial palsy.
- Steroids plus antivirals for Ramsay Hunt.

## 1 CHAPTER ONE Overview

Infections and inflammation of the pinna.

### 1.1 The conditions

**RED, PAINFUL PINNA**

- SKIN **Cellulitis, erysipelas** staphylococci and streptococci
- CARTILAGE **Perichondritis and chondritis** Pseudomonas
- VIRUS **Herpes zoster oticus** Ramsay Hunt syndrome
- AUTOIMMUNE **Relapsing polychondritis** also the nose, larynx and trachea
- DEGENERATIVE **Chondrodermatitis nodularis helicis** painful nodule on the helix

### 1.2 Applied anatomy

The cartilage of the pinna has no blood supply of its own and lives on its perichondrium. When pus or blood separates the two, the cartilage starves and necroses. The lobule has no cartilage, so a cartilage disease spares it.

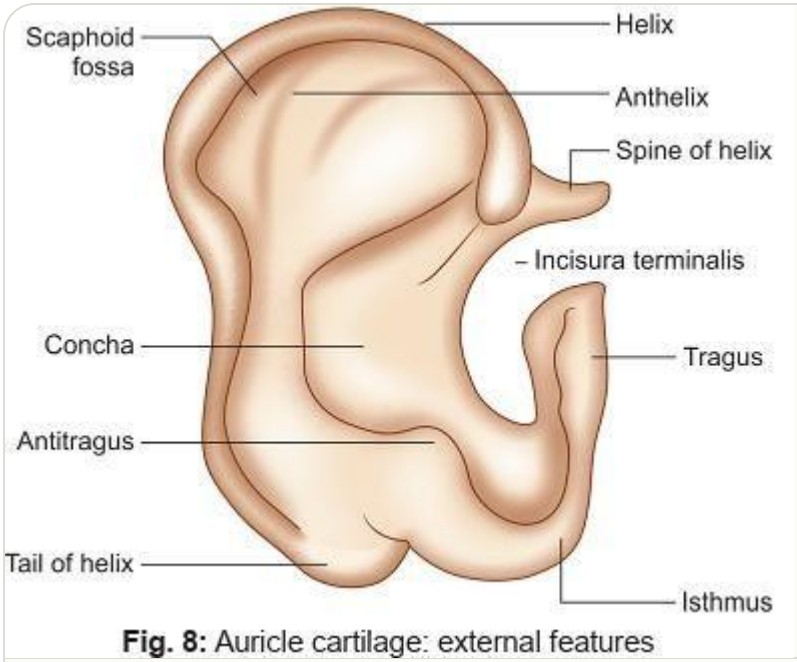



**FIGURE 1** The auricular cartilage: everything except the lobule.

Anatomy atlas · ENT Steps p. 29

# 2 CHAPTER TWO Perichondritis

Cartilage under threat.

## 2.1 Causes

Blunt trauma with an infected haematoma, lacerations and burns, high (cartilage) piercing, ear surgery, and spread from otitis externa. *Pseudomonas aeruginosa* is the usual organism, with staphylococci.

## 2.2 Clinical picture

### 2.2.1 Early

Pain and a red, hot, tender, thickened pinna, sparing the lobule.

### 2.2.2 Late

Fluctuant swelling as pus collects between perichondrium and cartilage, then necrosis.



**FIGURE 2** Perichondritis of the pinna: the whole pinna is red and swollen.
Dhingra, fig. 8.10

## 2.3 Treatment

### 2.3.1 Early, without abscess

An oral fluoroquinolone such as ciprofloxacin, or intravenous anti-pseudomonal antibiotics for severe infection; cool aluminium acetate compresses.

### 2.3.2 Abscess

Incise along a natural fold, remove dead cartilage, irrigate, and apply a bolster dressing for several days to keep the skin against the cartilage. Antibiotics continue for two to four weeks.

## 2.4 Outcome

Loss of cartilage leaves a shrivelled, deformed *cauliflower ear*.



**FIGURE 3** A cauliflower ear, the end result of cartilage necrosis after haematoma or perichondritis.

Dhingra, fig. 8.7

# 3 CHAPTER THREE Cellulitis and erysipelas

Skin infections.

## 3.1 Cellulitis

Spreading skin infection, usually staphylococcal or streptococcal, often after a scratch, piercing or otitis externa. Unlike perichondritis, it may involve the lobule.

## 3.2 Erysipelas

Streptococcal infection with a sharply defined, raised red edge, which can spread onto the face. Treated with penicillin.

# 4 CHAPTER FOUR Herpes zoster oticus

Ramsay Hunt syndrome.

## 4.1 Cause

Reactivation of varicella-zoster virus in the geniculate ganglion.

## 4.2 Clinical picture

Severe, burning ear pain, followed by vesicles in the concha, canal and sometimes the palate and tongue, then facial palsy. The eighth nerve is often involved, with tinnitus, sensorineural hearing loss and vertigo.



**FIGURE 4** Ramsay Hunt syndrome: facial palsy with small vesicles in the right concha.
Dhingra, fig. 14.9

## 4.3 Treatment

High-dose prednisolone (about 1 mg/kg for 14 days) with an antiviral such as valaciclovir, analgesia, and eye protection. Gabapentin for post-herpetic neuralgia.

Surgical decompression is not advised, because the nerve damage is diffuse.

### 4.4 Outlook

Worse than Bell's palsy: only about 60% regain normal facial function.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Eyelid Reanimation in Patients With Facial Paralysis**
Eyelid reanimation
**Facial Reanimation**
Facial reanimation

## 5 CHAPTER FIVE Inflammatory conditions

Not infections.

### 5.1 Relapsing polychondritis

An autoimmune disease of cartilage: recurrent painful red swelling of the pinna sparing the lobule, often both ears, with nasal (saddle nose), laryngeal and tracheal cartilage involvement that can threaten the airway. Treated with high-dose steroids and immunosuppression.

### 5.2 Chondrodermatitis nodularis helicis

A painful, tender nodule on the rim of the helix, usually in older men who sleep on that side. Relieve pressure; excise if persistent, and send for histology to exclude skin cancer.

### 5.3 Differential diagnosis

Gout (tophi on the helix), lymphoma and skin tumours.

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 On call

**A RED, SWOLLEN EAR**

1. Does it spare the lobule? Then it is cartilage.
2. Fluctuant: drain under anaesthesia and bolster.
3. Start anti-pseudomonal antibiotics.
4. Vesicles with facial weakness: steroids, antivirals and eye protection.
5. Eye symptoms: check the cornea.

## 6.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Why the lobule is spared.**
- **Pseudomonas** as the usual organism.
- **Drainage and bolster,** and why cartilage dies.
- **Ramsay Hunt:** vesicles, palsy, eighth nerve involvement, treatment and prognosis.
- **Relapsing polychondritis** and the airway.

## 6.3 Common pitfalls

**COMMON PITFALLS**

- Using flucloxacillin alone for perichondritis.
- Leaving pus under the perichondrium.

- Calling Ramsay Hunt syndrome Bell's palsy because nobody looked in the ear.

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CLINICAL · EAR

# Myringitis

Myringitis is inflammation of the drum itself. In the bullous form, blood-filled blisters on the drum cause sudden, intense pain that settles in a few days. In the granular form, the drum's skin is replaced by wet granulation tissue that discharges without pain, and the middle ear behind it is healthy.

**AT A GLANCE**

- Bullous: severe pain with blisters on the drum.
- Granular: painless foul discharge with granulation.
- Rule out middle ear disease.
- Treat with analgesia and drops, or vinegar and removal.

## 1 CHAPTER ONE Overview

Disease of the drum, not the middle ear.

### 1.1 Definition

Inflammation confined to the tympanic membrane, sometimes with the adjacent deep canal skin, without primary disease of the middle ear or mastoid.

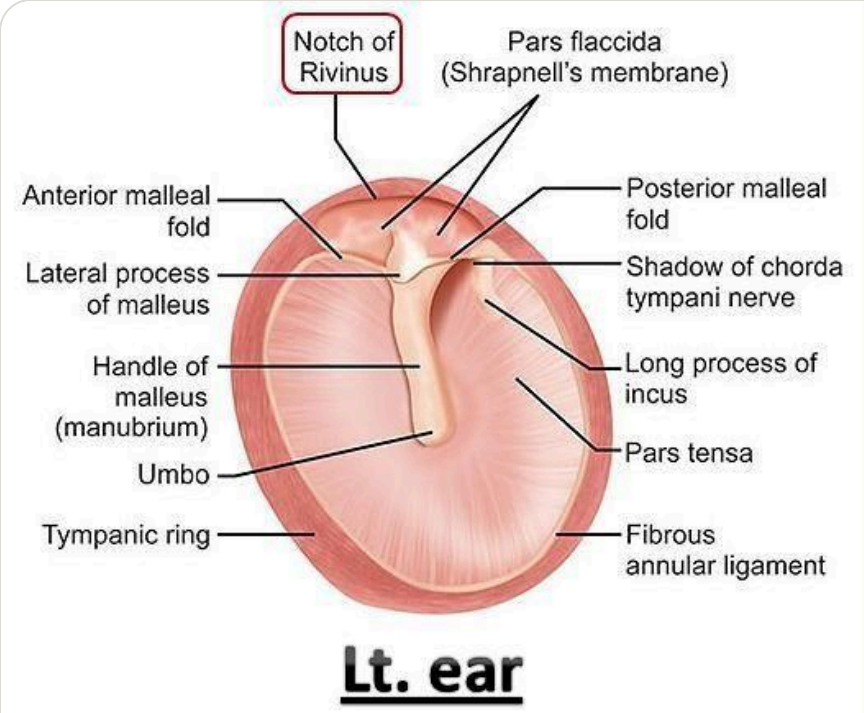



**FIGURE 1** Parts of the tympanic membrane.
Anatomy atlas · ENT Steps p. 36

## 1.2 Forms

### 1.2.1 Acute bullous myringitis

Blebs or bullae on the outer surface of the drum and the deep canal. Dhingra describes the same picture as *otitis externa haemorrhagica*.

### 1.2.2 Chronic granular myringitis

The epithelium of the drum is replaced by granulation tissue, without a perforation.

# 2 CHAPTER TWO Bullous myringitis

Blisters and pain.

## 2.1 Cause

It follows an upper respiratory infection. The organisms are those of acute otitis media: *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Moraxella*. *Mycoplasma* is

the classic association, but it has not been proven.

WHERE THE BOOKS DIFFER · MYCOPLASMA

| Dhingra | Lee |
| --- | --- |
| Lists it as a possible cause. | The link is unproven. |

## 2.2 Clinical picture

Sudden, intense ear pain for three or four days that then settles, sometimes with a little bloody discharge when the blebs burst. Haemorrhagic bullae sit on the drum and deep canal. A third to two-fifths have a middle ear effusion as well.

## 2.3 Hearing

A sensorineural or mixed loss, often in the high frequencies, is common and usually recovers. Always check hearing.

## 2.4 Treatment

Analgesia; lancing the blebs relieves pain; keep the ear dry; topical drops to prevent secondary infection; treat acute otitis media if present.

# 3 CHAPTER THREE Granular myringitis

A wet, painless drum.

## 3.1 Cause

Usually *Staphylococcus*, *Pseudomonas* or *Proteus*.

WHERE THE BOOKS DIFFER · PREDISPOSING FACTORS

| Dhingra | Lee |
| --- | --- |
| Links it to wax, foreign bodies and external ear infection. | No predisposing factor has been identified. |

## 3.2 Clinical picture

Itching, fullness and a foul, painless discharge. Granulation tissue covers part of the drum, often posterosuperiorly; the drum may perforate and heal repeatedly. Hearing is normal or slightly reduced.

## 3.3 Diagnosis

Clinical, under the microscope. CT only if chronic otitis media or cholesteatoma has to be excluded.

## 3.4 Treatment

Regular cleaning, dilute vinegar irrigation and topical drops; cauterising the granulation with silver nitrate or trichloroacetic acid, or laser; removal of the granulation. Severe or recurrent cases need tympanoplasty with canalplasty. Recurrence is common after surgery.

## 3.5 Complication

Late fibrosis can blunt the anterior angle between the drum and canal wall, causing a conductive loss.

# 4 CHAPTER FOUR Differential diagnosis

Other painful or wet drums.

- **Acute otitis media:** bulging drum with poor mobility.
- **Chronic otitis media:** a perforation and middle ear disease.

- **Herpes zoster oticus:** vesicles on the concha and canal, facial palsy.
- **Otitis externa:** canal swelling and tragal tenderness.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Bullous versus granular,** and the symptoms of each.
- **Sensorineural loss** in bullous myringitis.
- **Treatment** of each form.

## 5.2 Common pitfalls

**COMMON PITFALLS**

- Not checking hearing in bullous myringitis.
- Calling granular myringitis chronic otitis media without examining under the microscope.

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CLINICAL · EAR

# Keratosis obturans and canal cholesteatoma

A hard plug of keratin in the deep canal can be one of two conditions. In keratosis obturans the plug fills the canal all round, presses on it and widens it, painfully. In canal cholesteatoma the problem is a focal patch of eroded, dead bone on the canal floor. Both come from skin that has stopped migrating outwards.

**AT A GLANCE**

- Keratin plug: circumferential in keratosis obturans, focal in canal cholesteatoma.
- Clean regularly under the microscope.
- Biopsy granulation to exclude cancer.
- Surgery only for refractory disease.

## 1 CHAPTER ONE Overview

When the canal stops cleaning itself.

### 1.1 Normal self-cleaning

The skin of the drum and deep canal migrates outwards, carrying keratin and wax to the entrance of the canal. When this migration fails, keratin builds up in the deep canal.

### 1.2 Two conditions

**KERATOSIS OBTURANS OR CANAL CHOLESTEATOMA?**

| | KERATOSIS OBTURANS | CANAL CHOLESTEATOMA |
|---|---|---|
| **Keratin** | Plug filling the whole canal | Focal, over an area of eroded bone |
| **Bone** | Widened all round, smooth | Focal erosion and necrosis, usually posteroinferior |
| **Pain** | Acute, severe | Dull, chronic, or none |
| **Hearing** | Conductive loss | Usually normal |

| | KERATOSIS OBTURANS | CANAL CHOLESTEATOMA |
|---|---|---|
| **Sides** | Often both | Usually one |
| **Age** | Younger | Older |
| **Associations** | Bronchiectasis, chronic sinusitis | Previous surgery, trauma, radiation |

CHAPTER TWO

# 2 Keratosis obturans

The circumferential plug.

## 2.1 Cause

Failure of outward migration, possibly from inflammation of the canal skin. In young people it is associated with bronchiectasis and chronic sinusitis, which should be looked for.

## 2.2 Clinical picture

Acute, severe pain, conductive hearing loss and sometimes otorrhoea. A dense, white, laminated keratin plug fills the deep canal; once removed, the canal is widened and the drum may be thickened.

WHERE THE BOOKS DIFFER · AGE AND PAIN

**Dhingra**
Typically 5 to 20 years; pain is among the presenting symptoms.

**Lee**
Typically 30 to 60 years; severe pain is characteristic.

## 2.3 Treatment

Removal of the plug under the microscope, which may need a general anaesthetic in children, treatment of any secondary otitis externa, and regular cleaning. Salicylic

acid 2% in alcohol reduces reaccumulation. Follow-up is lifelong.

### 2.4 Complication

Progressive widening of the canal can erode into the mastoid (an *auto-mastoidectomy*) and expose the facial nerve.

## 3 CHAPTER THREE Canal cholesteatoma

Focal bone erosion.

### 3.1 Cause

Spontaneous in older patients, or after surgery, trauma, radiation or canal stenosis.

### 3.2 Clinical picture

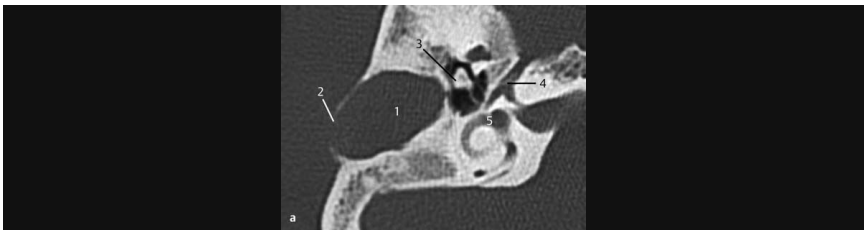
A dull, chronic ache and a foul, scanty discharge, with normal hearing. A localised ulcer on the floor or posterior wall of the bony canal, with keratin debris and exposed, necrotic bone.

### 3.3 Investigations

CT shows the extent of the erosion and its relation to the mastoid and facial nerve. Biopsy granulation tissue to exclude carcinoma.

**IMAGING IN THE APP** 1 case

Canal cholesteatoma with canal atresia.




**Inclusion Cholesteatoma and Atresia of the External Auditory Canal**
Congenital · 3 images

## 3.4 Treatment

Regular debridement and topical drops; when the disease keeps returning, surgery removes the matrix and dead bone and covers the defect with fascia or a skin graft.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Keratosis Obturans and Canal Cholesteatoma**
Keratosis obturans and canal cholesteatoma

**Osteomas and Exostoses of the External Auditory Canal**
Exostoses and osteomas

# 4 CHAPTER FOUR Differential diagnosis

What else causes a canal plug or erosion.

- Impacted wax.
- Necrotising otitis externa.
- Squamous cell carcinoma of the canal.
- Middle ear cholesteatoma eroding the canal wall.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Failure of epithelial migration** as the cause.
- **The comparison table:** circumferential versus focal.
- **Association with bronchiectasis and sinusitis.**
- **Biopsy** to exclude carcinoma.

## 5.2 Common pitfalls

**COMMON PITFALLS**

- Syringing a keratin plug repeatedly instead of removing it under the microscope.
- Missing carcinoma in canal granulation.

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CLINICAL · EAR

# Tuberculous and syphilitic otitis media

Tuberculous otitis media hides in plain sight as an ordinary chronic ear that simply will not get better. The clues are a painless, thin discharge, several holes in a thickened drum, pale granulation, and a hearing loss worse than the symptoms suggest. Send the tissue to the laboratory, because the treatment is medical, not surgical.

**AT A GLANCE**

- Painless discharge, multiple perforations and pale granulation.
- Hearing is worse than symptoms suggest.
- Confirm by PCR, histology and culture.
- Antituberculous drugs first; repair later.

## 1 CHAPTER ONE Overview

The chronic ear that refuses to heal.

### 1.1 Definition

Infection of the middle ear cleft by *Mycobacterium tuberculosis*. Syphilitic otitis media, much rarer, is included here because it shares the picture of bone destruction and a disproportionate hearing loss.

### 1.2 Epidemiology

Uncommon in low-burden countries, but important where tuberculosis is endemic, and rising with drug-resistant strains and immunosuppression, particularly HIV.

### 1.3 Routes of infection

- Through the Eustachian tube, from active pulmonary disease, which most patients have.
- Through the blood, from a distant focus.

- In infants, through the tube from infected milk.

# 2 CHAPTER TWO Clinical features

Painless, thin, and resistant.

## 2.1 Symptoms

- **Discharge:** painless, thin and cloudy, persisting despite treatment.
- **Hearing loss:** out of proportion to the symptoms, often mixed, from early ossicular and cochlear involvement.

## 2.2 Signs

- A thickened drum early on, then *multiple perforations* that merge into one large defect.
- Pale, thick mucosa and abundant pale granulation.
- Bone sequestra in the mastoid.
- Non-tender enlarged cervical and preauricular nodes.
- Facial palsy in about 10%, sometimes the presenting feature in a child.

## 2.3 Complications

Facial palsy, sequestrum, postauricular fistula, Bezold abscess and labyrinthitis.

**SUSPECT IT WHEN**

A chronic ear fails ordinary treatment, a mastoidectomy fails, granulation is pale, or facial palsy appears early in a painless ear.

# 3 CHAPTER THREE Investigations

Prove it.

## 3.1 From the ear

- **PCR of the discharge or tissue:** gives an answer within days.
- **Histology:** caseating granulomas with Langhans giant cells.
- **Acid-fast smear and culture:** a negative smear does not exclude it.

## 3.2 General

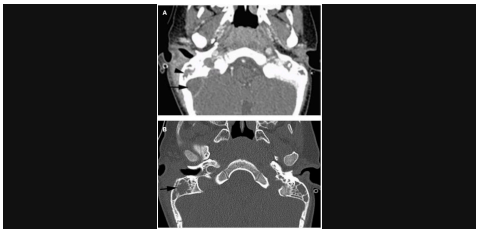
Chest imaging, tuberculin skin test or interferon-gamma release assay, and HIV testing.

## 3.3 CT

Often looks like ordinary chronic otitis media, with more bone erosion and sequestra.

IMAGING IN THE APP 1 case

Otomastoiditis on CT.



**Otomastoiditis and its complications**

Infection & inflammation · 9 images

## 3.4 Differential diagnosis

Ordinary chronic suppurative otitis media, carcinoma of the middle ear, granulomatosis with polyangiitis, and fungal infection.

# 4 CHAPTER FOUR Management

Drugs first, repair later.

## 4.1 Antituberculous chemotherapy

Standard multidrug therapy with isoniazid, rifampicin, pyrazinamide and ethambutol, guided by sensitivities, for at least six months.

## 4.2 Surgery

Limited to biopsy, removal of sequestra and drainage of abscesses or complications. Reconstruction is delayed until treatment is complete, because healing is slow and fistulae are common during active disease; afterwards, repair succeeds in most.

**OPERATIVE GUIDES IN THEATRE** 1 operation

**Myringoplasty and Tympanoplasty**
Late repair

## 4.3 Facial palsy

Usually recovers well with medical treatment.

# 5 CHAPTER FIVE Syphilitic otitis

Rare, destructive.

Congenital or acquired syphilis can cause osteitis of the temporal bone with necrosis, and labyrinthitis with profound, often fluctuating sensorineural loss and vertigo. Diagnose with treponemal tests (FTA-ABS) and treat with penicillin and steroids.

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The features:** painless discharge, multiple perforations, pale granulation, early facial palsy, disproportionate hearing loss.
- **Diagnosis** by PCR, histology and culture.
- **Treatment is medical;** surgery for complications only.

## 6.2 Common pitfalls

COMMON PITFALLS

- Operating on a chronic ear without sending the tissue for tuberculosis.
- Being reassured by a negative acid-fast smear.
- Reconstructing during active treatment.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, 24, Oxford Specialist Handbook ch. 16, Dhingra ch. 11 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Wax, foreign bodies and canal stenosis

Wax and foreign bodies are among the commonest problems in the ear clinic, and among the easiest to make worse. Ask about a previous perforation before syringing, never chase a smooth round object deeper with forceps, kill a live insect before removing it, and treat a button battery as an emergency.

**AT A GLANCE**

- Ask about perforation before syringing.
- Never grasp a smooth object with forceps.
- Kill live insects first.
- Use a microscope and anaesthesia for impacted objects.
- Exostoses are multiple and bilateral (cold water); osteomas single at suture lines; never irrigate a button battery.

## 1 CHAPTER ONE Overview

The canal and what gets stuck in it.

### 1.1 Applied anatomy

The external auditory canal is about 24 mm long. Its outer third is cartilaginous, with hairs and sebaceous and ceruminous glands; its inner two-thirds is bony, with thin, sensitive skin. The canal narrows at the *isthmus*, about 6 mm from the drum: objects pushed past it become impacted. The *anterior recess*, in front of and below the drum, collects debris.

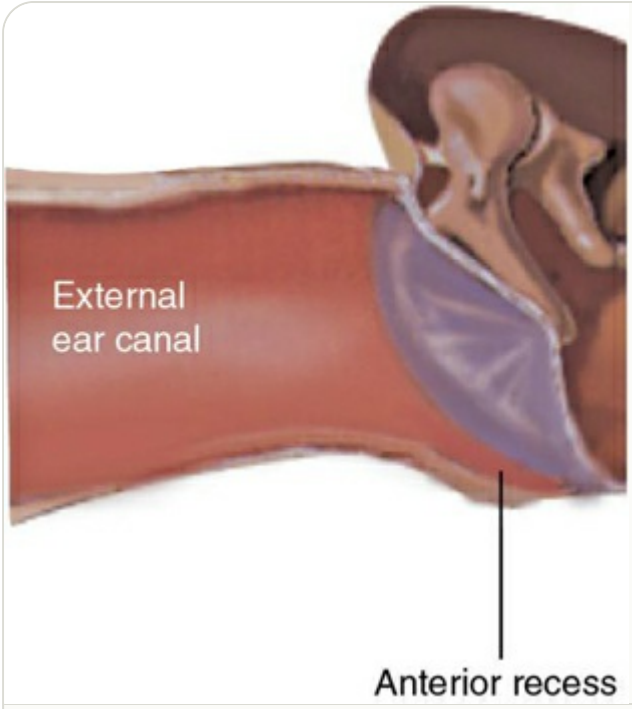



**FIGURE 1** The anterior recess of the canal, where discharge and debris collect.
Dhingra, fig. 1.2

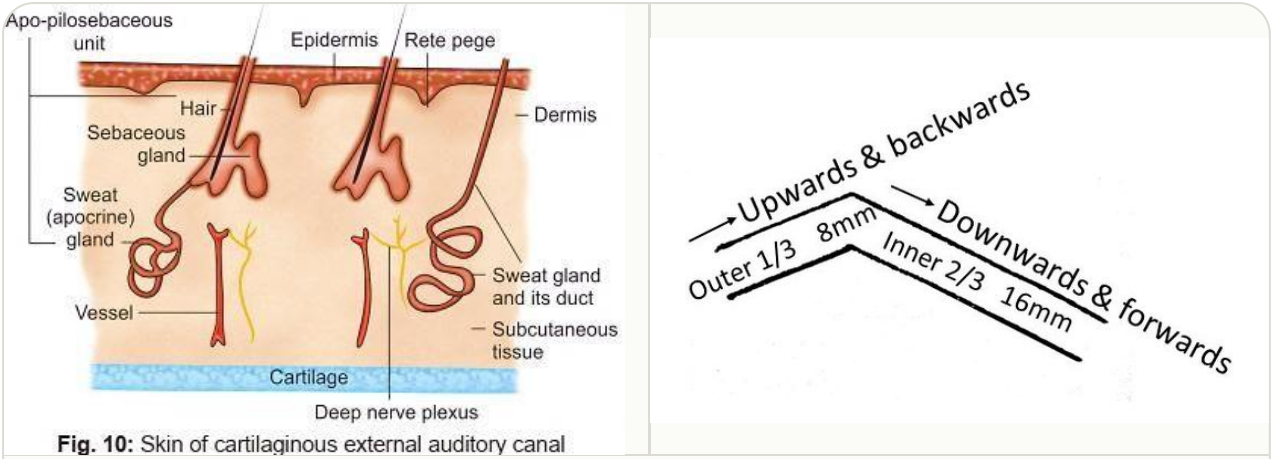



**FIGURE 2** Skin of the cartilaginous canal, and the direction of the canal.
Anatomy atlas · ENT Steps p. 31, Doss p. 184

## 1.2 Pulling the pinna

In adults, pull the pinna up and back to straighten the canal; in infants, down and back.

# 2 CHAPTER TWO Wax

A useful secretion.

## 2.1 What it is

Wax (cerumen) is a mixture of secretions of the ceruminous and sebaceous glands with shed keratin. It is acidic, water-repellent and antibacterial, and the canal normally clears it by epithelial migration.

## 2.2 Why it impacts

A narrow or tortuous canal, stiff hairs, exostoses, overactive glands, and pushing wax deeper with cotton buds or hearing aid moulds.

## 2.3 Clinical picture

A blocked ear and hearing loss, sometimes tinnitus or giddiness, and occasionally a reflex cough through the auricular branch of the vagus (*Arnold's nerve*). Sudden blockage after bathing happens when water swells dry wax.

## 2.4 Removal

### 2.4.1 Softening

Hard wax is softened first with sodium bicarbonate glycerine drops, olive or almond oil, or hydrogen peroxide for a few days.

### 2.4.2 Syringing (irrigation)

Pull the pinna up and back and direct a stream of water at body temperature along the posterosuperior canal wall, so that it flows behind the wax and pushes it out. Dry the ear and inspect the drum afterwards.

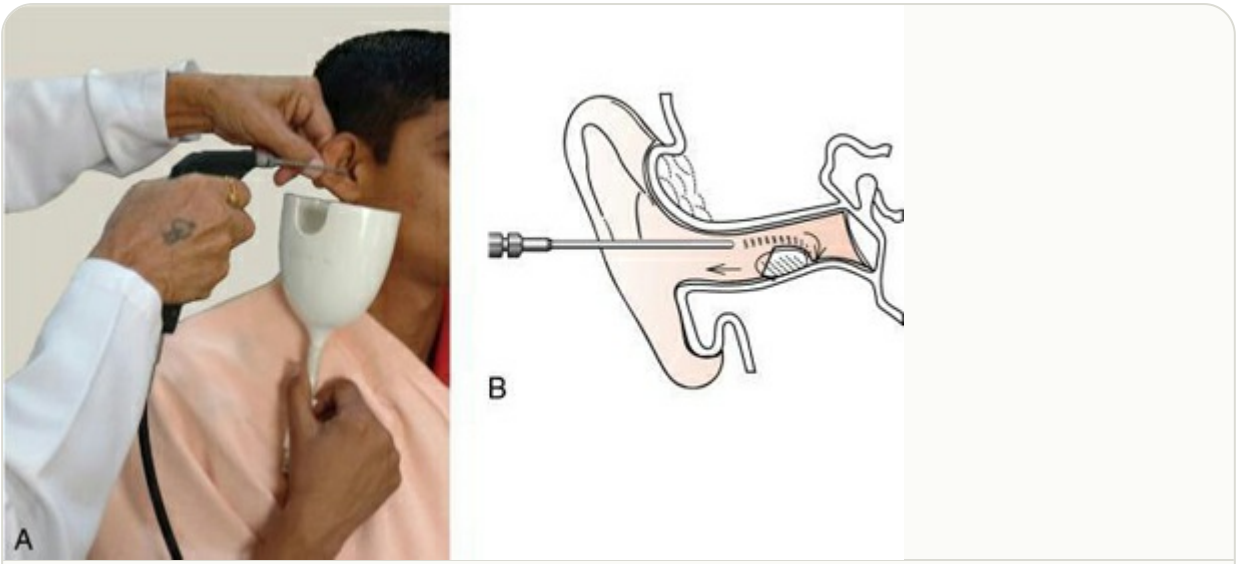



**FIGURE 3** (A) Irrigation of the ear canal. (B) The jet passes behind the wax or foreign body and pushes it out.

Dhingra, fig. 8.13

**DO NOT SYRINGE WHEN**

- There is a perforation, or a history of discharge or ear surgery.
- The patient has had a previous perforation that may be thin.
- The foreign body is vegetable matter that will swell.
- The patient has an only hearing ear (relative).

### 2.4.3 Under the microscope

Suction or instruments (a wax hook, Jobson-Horne probe or crocodile forceps) under direct vision, when syringing is unsuitable.

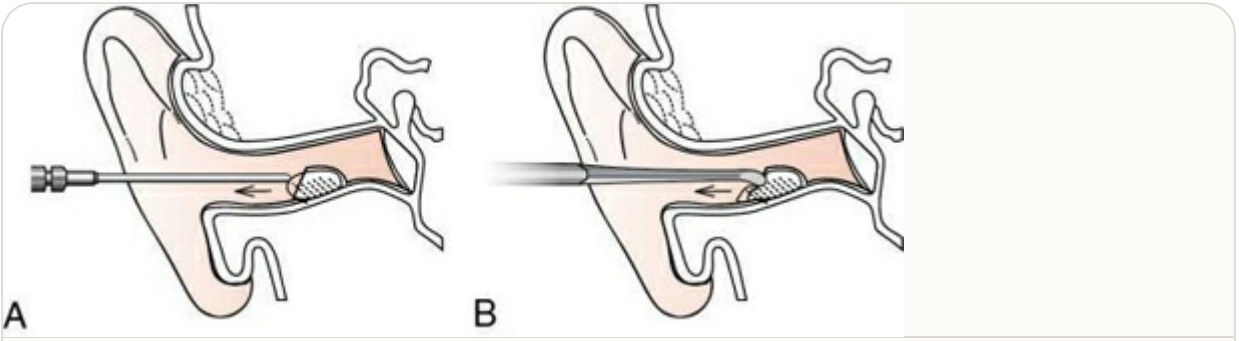



**FIGURE 4** Other methods of removal: (A) suction and (B) forceps.

Dhingra, fig. 8.14

### 2.5 Complications of removal

Perforation, canal laceration and bleeding, otitis externa, and vertigo if the water is not at body temperature.

# 3 CHAPTER THREE Foreign bodies

What children and insects leave behind.

## 3.1 Types

#### 3.1.1 Non-living

Beads, stones, paper, cotton, seeds, peas and beans, pencil tips, and button batteries. Vegetable matter swells with moisture and can cause infection.

#### 3.1.2 Living

Insects (flies, cockroaches, ants), which cause intense discomfort and noise, and maggots, from flies laying eggs in a foul-discharging ear.

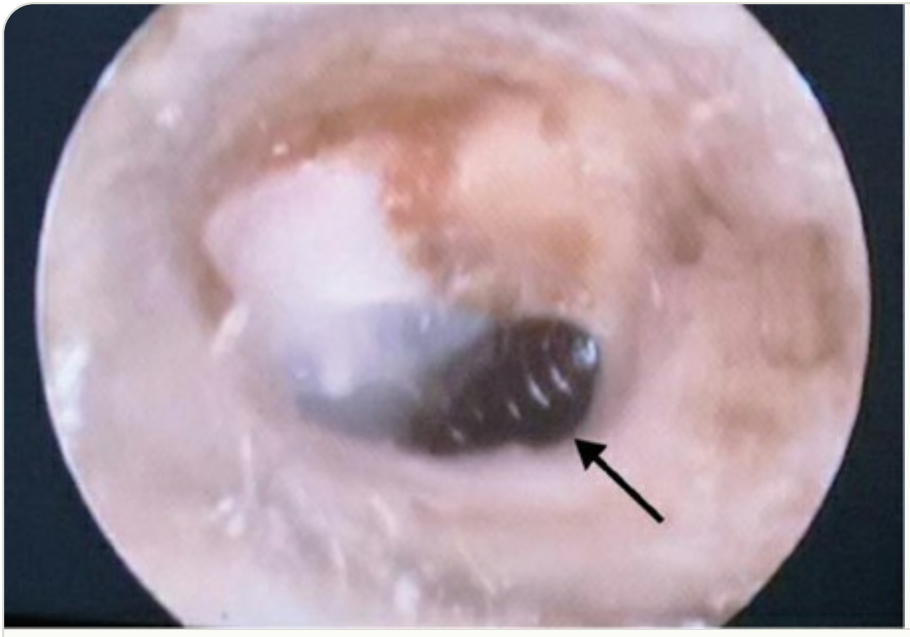


**FIGURE 5** An insect in the ear canal, seen with an endoscope (arrow).
Dhingra, fig. 8.15

## 3.2 Removal

### 3.2.1 General principles

Use good light, the microscope and the right instrument. A cooperative patient, or a general anaesthetic for children and for impacted objects or a failed first attempt.

### 3.2.2 Smooth, round objects

Do not grasp with forceps: they slip and are pushed deeper past the isthmus. Use a hook passed beyond the object, suction, or syringing (if not vegetable).

### 3.2.3 Insects

Kill first by filling the canal with oil or spirit, then remove by forceps or syringing.

### 3.2.4 Maggots

Instil chloroform water or turpentine to kill them, then remove with forceps over several sittings; do not syringe. Treat the underlying ear disease.

### 3.2.5 Button batteries

Remove urgently: they leak alkali and cause rapid tissue necrosis. Do not instil drops or water.

### 3.2.6 Impacted deep objects

A post-aural approach may be needed for objects medial to the isthmus.

**PEARLS**

- Most foreign bodies can wait for the right setting; a bad first attempt makes the second harder.
- Always inspect the drum and the other ear (and the nostrils in a child) afterwards.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Foreign Bodies of the External Auditory Canal**
Foreign bodies of the canal

**Office-Based Procedures in Otology**
Office ear procedures

**Congenital and Acquired Atresia and Stenosis of the External Auditory Canal**
Atresia and stenosis

# 4 CHAPTER FOUR Canal stenosis

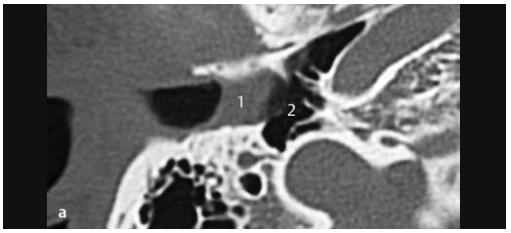
Narrowing after injury.

## 4.1 Causes

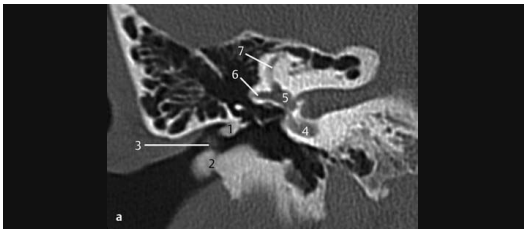
Acquired from chronic otitis externa, trauma, burns, surgery or radiation; congenital stenosis and atresia are covered separately.

**IMAGING IN THE APP** 2 cases

Canal stenosis and exostoses.




**Stenosis of the External Auditory Canal**
Bone disease · 2 images




**Exostoses of the External Auditory Canal**
Bone disease · 2 images

## 4.2 Treatment

Meatoplasty with excision of the fibrous tissue and skin flaps or grafts, and canal stenting while it heals.

**EXOSTOSES, OSTEOMAS AND FOREIGN BODIES (OXFORD TEXTBOOK)**

**Exostoses** are broad-based, usually multiple and bilateral bony swellings deep in the bony canal, caused by repeated exposure to cold water and wind ("surfer's ear"). **Osteomas** are single, pedunculated lesions arising at the tympanomastoid or tympanosquamous suture lines, unrelated to cold water. Both are usually symptomless but can trap wax and debris, causing recurrent otitis externa, a conductive loss and, rarely, canal cholesteatoma. Symptomatic patients are managed with regular microsuction and drops, and surgery is offered when that fails: a postauricular or endaural approach with a posterior meatal skin flap and removal of bone with a drill or chisel, protecting the canal skin, the drum and, posteriorly, the

facial nerve. Cold-water ear protection reduces recurrence. In a child, an apparent bony stenosis may be a mild congenital atresia, so CT is advised before operating.

**Foreign bodies** in the canal are removed under direct vision. Graspable objects are taken with forceps; round or smooth objects are hooked from behind with an angled hook, curette or suction rather than grasped, because forceps push them towards the drum. Irrigation is not recommended in the ENT clinic, and must never be used for vegetable matter, which swells, or for button batteries, which need urgent removal because they cause liquefactive necrosis.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 On call

SOMETHING STUCK IN THE EAR

1. Button battery: remove urgently, no drops.
2. Live insect: kill it first with oil or spirit, then remove.
3. Smooth round object: hook or suction, never forceps.
4. Impacted object or failed attempt: microscope, and consider general anaesthesia.
5. Inspect the drum afterwards.

## 5.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Anatomy:** cartilaginous and bony parts, the isthmus, the anterior recess.
- **Technique of syringing** and its contraindications.
- **Removal of foreign bodies,** living and non-living.

- **Arnold's nerve** and the ear cough reflex.

## 5.3 Common pitfalls

COMMON PITFALLS

- Syringing an ear with a perforation.
- Grasping a bead with forceps.
- Syringing vegetable matter.
- Delaying removal of a button battery.

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CLINICAL · EAR

# Pinna: congenital anomalies, haematoma and trauma

The pinna is all cartilage and skin, and the cartilage depends entirely on its perichondrium. A haematoma after a blow must be drained early, because blood trapped under the perichondrium becomes a cauliflower ear. Congenital anomalies are many, and most are cosmetic; the one to examine fully is microtia, which may hide a deaf ear.

**AT A GLANCE**

- Haematoma: drain and bolster early to prevent cauliflower ear.
- Repair lacerations with cartilage covered.
- Know the congenital anomalies and when to correct them.
- Rewarm frostbite and delay debridement.

## 1 CHAPTER ONE Overview

Shape, development and injury.

### 1.1 Applied anatomy

A single plate of elastic cartilage covered by skin that is tightly adherent on the lateral surface. The lobule has no cartilage. The *incisura terminalis*, between the tragus and the crus of the helix, also has no cartilage, which is why an endaural incision there avoids cutting it.

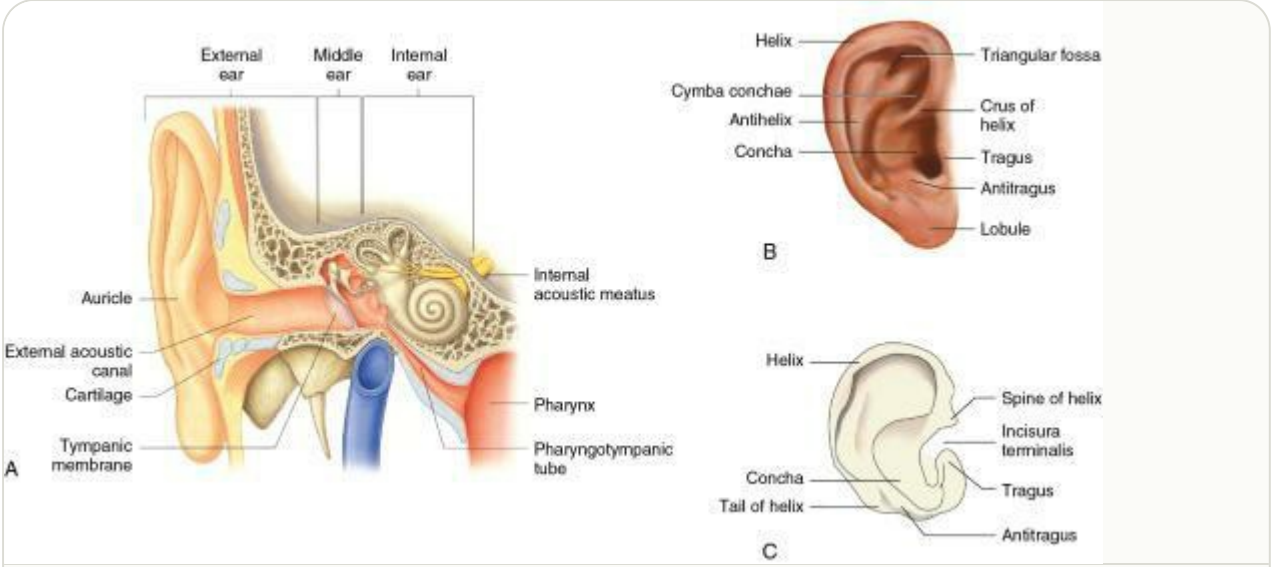



**FIGURE 1** (A) The divisions of the ear. (B) Elevations and depressions of the lateral surface of the pinna. (C) The auricular cartilage.

Dhingra, fig. 1.1

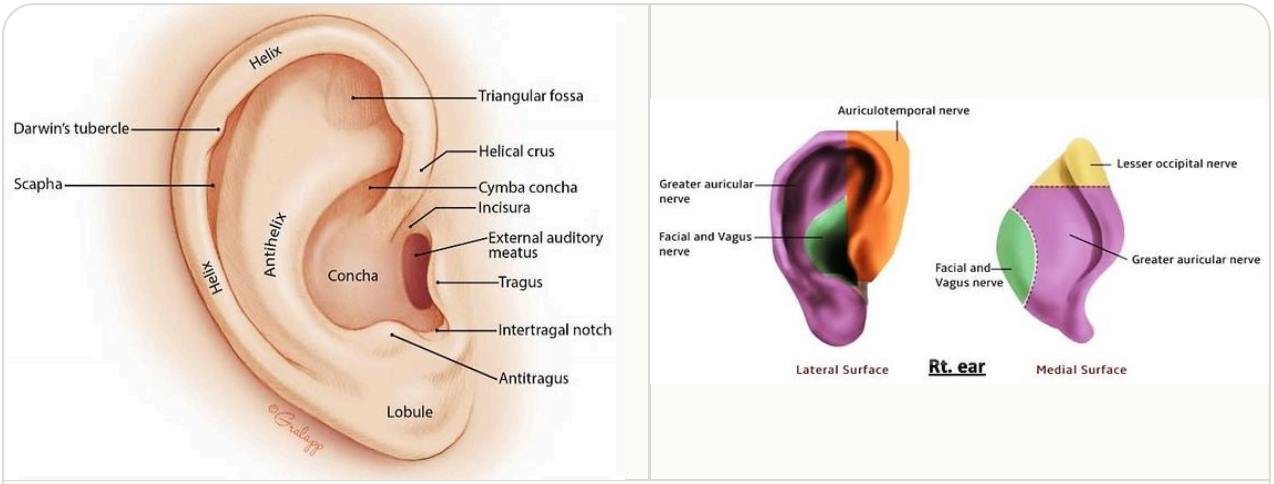



**FIGURE 2** Surface landmarks of the auricle, and its nerve supply.

Anatomy atlas · ENT Steps p. 29, ENT Steps p. 30

## 1.2 Development

The pinna develops from six *hillocks of His* around the first branchial cleft. The tragus comes from the first arch and most of the rest from the second. It reaches its adult shape by about 20 weeks of gestation and most of its adult size by about nine years.

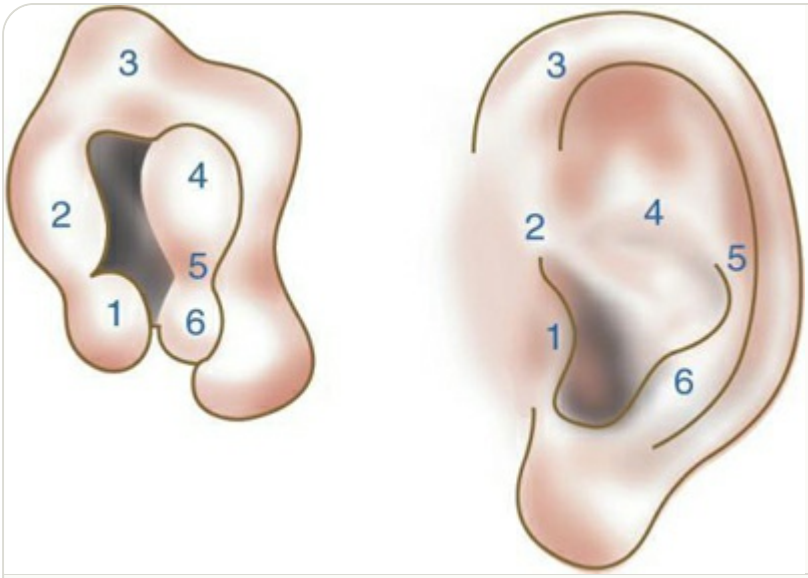



**FIGURE 3** Development of the pinna from six hillocks around the first branchial cleft.
Dhingra, fig. 1.24

WHERE THE BOOKS DIFFER · HILLOCKS

**Dhingra**
The tragus from the first arch; the rest from the second arch.

**Lee**
The first three hillocks from the first arch, the last three from the second.

# 2 CHAPTER TWO Congenital anomalies

From a tag to an absent ear.




**FIGURE 4** A normal pinna.
Dhingra, fig. 8.1

## 2.1 Absent or small

### 2.1.1 Anotia

Complete absence of the pinna, often with an absent canal.

### 2.1.2 Microtia

A small, malformed pinna, graded from mild deformity to a peanut-shaped vestige. It is often associated with canal atresia and middle ear anomalies, so hearing must be tested (see the microtia and atresia topic).



**FIGURE 5** First, anotia: no pinna or canal on the left. Second, microtia of the right ear (a peanut ear).

Dhingra, figs 8.2 and 8.3

## 2.2 Abnormal shape

- **Bat ear (prominent ear):** the commonest anomaly, from an underdeveloped antihelical fold or deep concha. The normal angle between the pinna and the head is about 15 to 30 degrees.
- **Cup and lop ear:** folding or deficiency of the upper pinna.
- **Cryptotia:** the upper pinna is buried under the scalp skin.
- **Stahl's ear:** an extra crus giving a pointed helix.
- **Satyr ear:** a pointed upper pole.
- **Darwin's tubercle:** a small, harmless nodule on the helix.
- **Coloboma:** a cleft in the pinna.

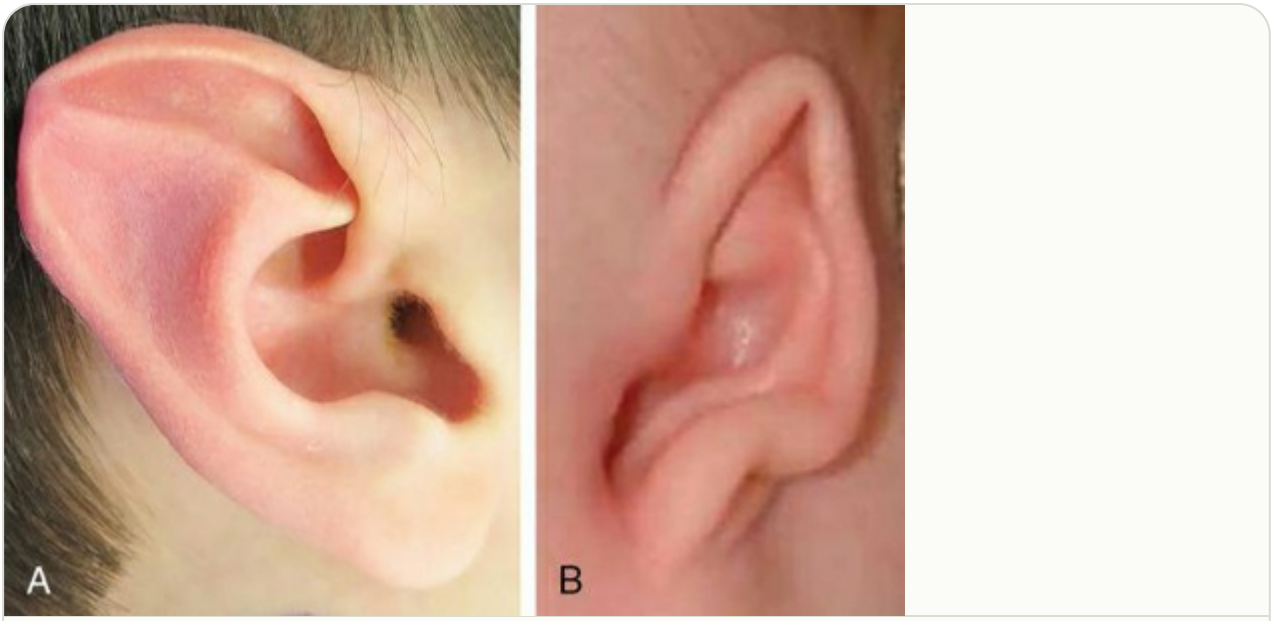



**FIGURE 6** Stahl's ear, with an extra crus.
Dhingra, fig. 8.4

## 2.3 Accessory tissue

- **Preauricular tags:** may contain cartilage.
- **Preauricular pit or sinus:** a small opening in front of the crus of the helix, from faulty fusion of the hillocks. It may become repeatedly infected.

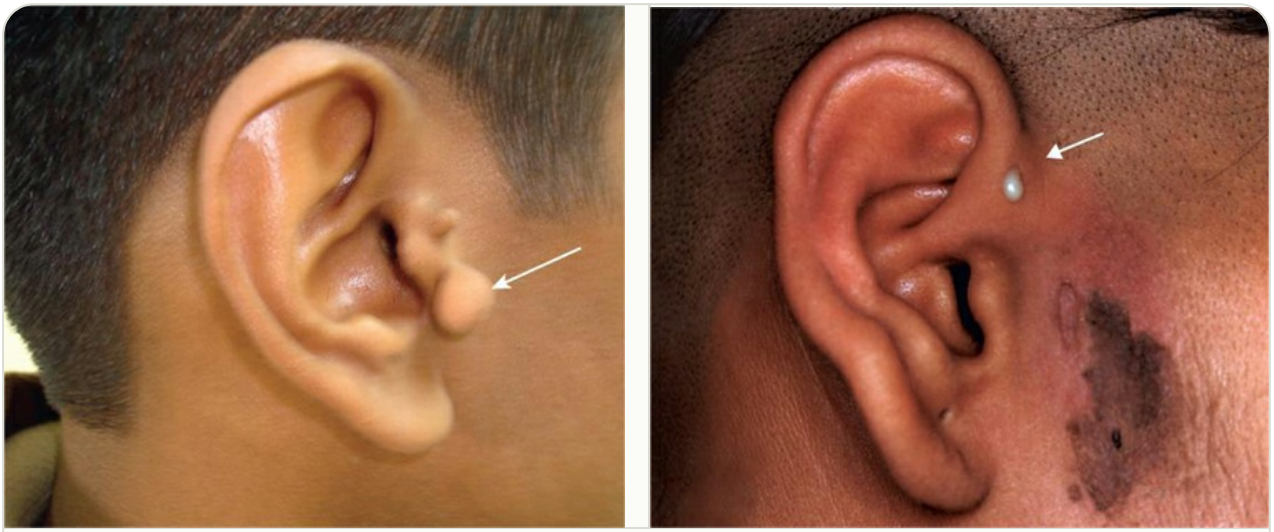


**FIGURE 7** First, preauricular appendages. Second, an infected preauricular sinus discharging pus.
Dhingra, figs 8.5 and 8.6

### 2.4 Treatment

- **Bat ear:** otoplasty, usually after age six, before teasing at school.
- **Stahl's and other shape anomalies:** a mould in the first six weeks of life, while the cartilage is soft, can correct them without surgery.
- **Preauricular sinus:** antibiotics for infection; excision of the whole tract when it has been repeatedly infected.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Otoplasty**
Otoplasty
**Microtia**
Microtia
**Congenital and Acquired Atresia and Stenosis of the External Auditory Canal**
Canal atresia and stenosis

## 3 CHAPTER THREE Haematoma

The boxer's ear.

### 3.1 Cause

Blunt trauma from boxing, wrestling or rugby shears the perichondrium from the cartilage, and blood collects between them.

### 3.2 Picture

A tense, fluctuant, smooth swelling on the lateral surface that obliterates the landmarks.

### 3.3 Treatment

Drain early: aspiration for small, fresh collections, or incision for larger or recurrent ones. Then apply a pressure bolster with the concavities packed, through-and-

through sutures or a moulded dressing, give antibiotics, and recheck for reaccumulation.

### 3.4 Complication

Untreated or infected, the cartilage necroses and new cartilage forms irregularly, giving a *cauliflower ear*.



**FIGURE 8** A cauliflower (boxer's) ear.
Dhingra, fig. 8.7

## 4 CHAPTER FOUR Other injuries

Wounds, bites and cold.

### 4.1 Lacerations

Repair early, covering the cartilage with skin and keeping the perichondrium attached.



**FIGURE 9** A laceration of the left pinna.
Dhingra, fig. 8.8

## 4.2 Avulsion

A partly avulsed segment on a pedicle is reattached. A completely avulsed part may be reattached as a composite graft if small, banked under postauricular skin, or replanted microvascularly.

## 4.3 Bites

The lobe is the commonest site; human bites are the most likely to become infected. Clean thoroughly, give antibiotics and repair.

## 4.4 Frostbite

The pinna is exposed and thin. Rewarm with moist pledgets at 38 to 42°C, give analgesia and antibiotics, avoid pressure dressings, and delay debridement for weeks to months until the dead tissue demarcates.

WHERE THE BOOKS DIFFER · REWARMING

| Dhingra | Lee |
| --- | --- |

| | |
|---|---|
| Moist pledgets at 38 to 42°C, noting that rapid rewarming is painful. | Favours slow rewarming. |

## 4.5 Keloid

After piercing, often in darker skin. Excision alone recurs; combine with steroid injection, pressure or radiotherapy.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 On call

**A SWOLLEN PINNA AFTER A BLOW**

1. Aspirate or incise the haematoma.
2. Pressure bolster with the concavities packed.
3. Antibiotics, and review within a day or two for reaccumulation.
4. Lacerations: repair early, covering the cartilage.

## 5.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Development** from the hillocks of His.
- **Congenital anomalies,** and the timing of otoplasty.
- **Haematoma:** why drain, and how to prevent cauliflower ear.
- **Microtia:** test the hearing and examine the canal and middle ear.
- **Grafts from the pinna:** tragal and conchal cartilage, perichondrium and lobule fat.

## 5.3 Common pitfalls

**COMMON PITFALLS**

- Leaving a haematoma to resolve.
- Draining without a pressure dressing.
- Excising a keloid without adjuvant treatment.

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CLINICAL · EAR

# Ototoxicity, noise, presbycusis and immune inner ear disease

Most slowly progressive sensorineural loss in both ears comes from age, noise or drugs. Ask about each specifically, stop whatever is still doing harm, and do not miss the rarer autoimmune inner ear disease, because it can respond to steroids.

**AT A GLANCE**

- Think age, noise and drugs.
- Stop harmful drugs and noise where you can.
- Noise gives a 4 kHz notch.
- Rapid bilateral loss suggests immune disease: try steroids.
- UK action values 80 and 85 dB(A) with a 3 dB exchange rate; CLB criteria for noise loss; synaptopathy still debated.

## 1 CHAPTER ONE Overview

Four causes of acquired sensorineural loss.

### 1.1 Scope

- **Ototoxicity:** hearing or balance damaged by drugs.
- **Noise-induced hearing loss:** from sustained or sudden loud sound.
- **Presbycusis:** age-related hearing loss.
- **Immune-mediated inner ear disease:** primary or as part of systemic disease.

### 1.2 Why it matters

Much of this hearing loss is preventable, by monitoring drugs and protecting ears from noise, and some is reversible.

# 2 CHAPTER TWO Ototoxicity

Drugs that damage the ear.

## 2.1 The drugs

OTOTOXIC DRUGS

| DRUG | MAIN TARGET | REVERSIBLE? |
|---|---|---|
| Streptomycin, gentamicin | Vestibular | Often permanent |
| Neomycin, kanamycin, amikacin | Cochlea | Often permanent |
| Loop diuretics | Stria vascularis | Usually |
| Salicylates | Cochlea | Yes |
| Quinine, chloroquine | Cochlear vessels | Usually; high doses may not be |
| Cisplatin, carboplatin | Outer hair cells | Often permanent |

Others include deferoxamine, some ear drops (aminoglycosides and polymyxin) through a perforation, and some topical antiseptics.

WHERE THE BOOKS DIFFER · TOBRAMYCIN

**Dhingra**
Mainly vestibulotoxic.

**Lee**
Mainly cochleotoxic.

## 2.2 Aminoglycosides

Hair cells are damaged from the basal turn upwards, so high frequencies go first. Hearing loss may appear during or after treatment. Risk is higher with renal impairment, age over 65, high blood levels, long courses, previous exposure, loop diuretics, and genetic susceptibility: mitochondrial 12S rRNA mutations make even a single dose dangerous, so ask about family members who went deaf after these drugs.

## 2.3 Cisplatin

Bilateral, permanent high-frequency loss, worse in young children under five, the elderly, and with noise or radiotherapy.

## 2.4 Prevention

Avoid ototoxic drugs where alternatives exist, monitor levels and renal function, and obtain baseline and serial audiograms, including high-frequency audiometry.

**ON THE WARD**

New tinnitus, hearing loss or imbalance on an aminoglycoside or cisplatin: alert the treating team, check levels and renal function, and test hearing.

## 2.5 Deliberate use

Gentamicin's vestibular toxicity is used therapeutically in Ménière's disease.

# 3 CHAPTER THREE Noise-induced hearing loss

Loud sound over time.

## 3.1 Mechanism

Outer hair cells in the basal turn fail first, around the region that responds to 4 kHz.

### 3.1.1 Temporary threshold shift

Hearing dulls after loud exposure and recovers.

**WHERE THE BOOKS DIFFER · RECOVERY FROM A TEMPORARY SHIFT**

| Dhingra | Lee |
| --- | --- |
| Up to two weeks. | Within 24 hours. |

### 3.1.2 Permanent threshold shift

Repeated exposure leads to irreversible loss.

### 3.1.3 Acoustic trauma

A single blast or explosion (around 140 dB or more) can rupture the drum, disrupt the ossicles and damage the cochlea at once.

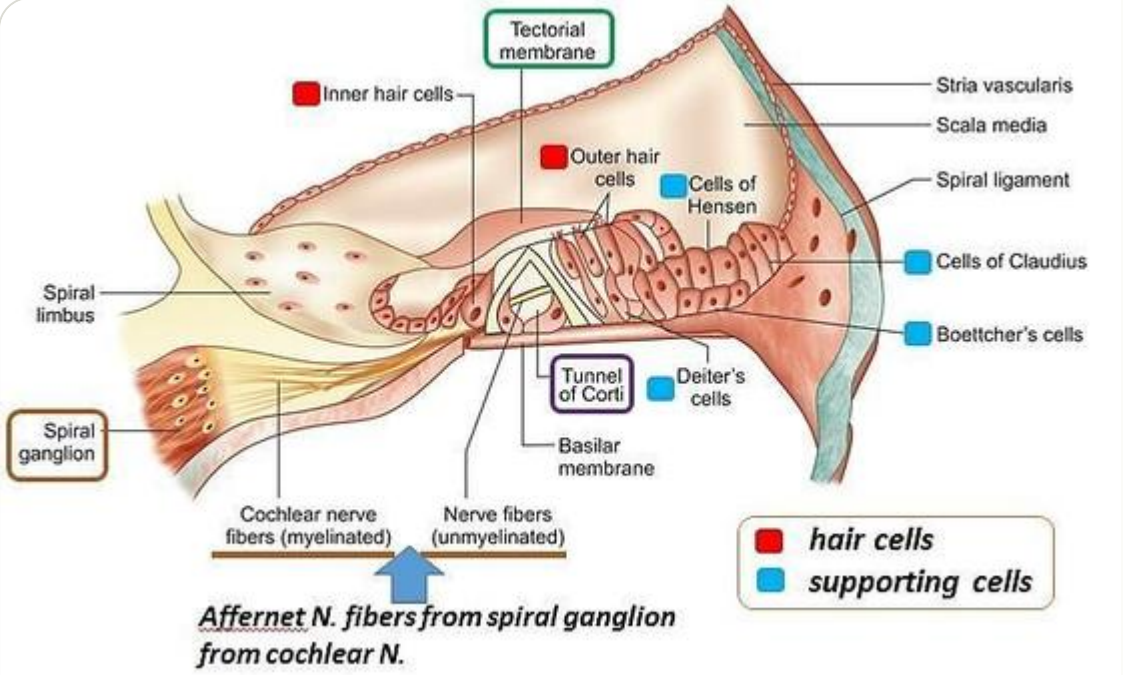



**FIGURE 1** The organ of Corti: outer hair cells are the first casualties of noise and ototoxic drugs.

Anatomy atlas · ENT Steps p. 77

## 3.2 Clinical picture

High-pitched tinnitus and difficulty hearing in noise. The audiogram shows a symmetrical *notch at 4 kHz*, which deepens and widens into the speech frequencies with continued exposure.

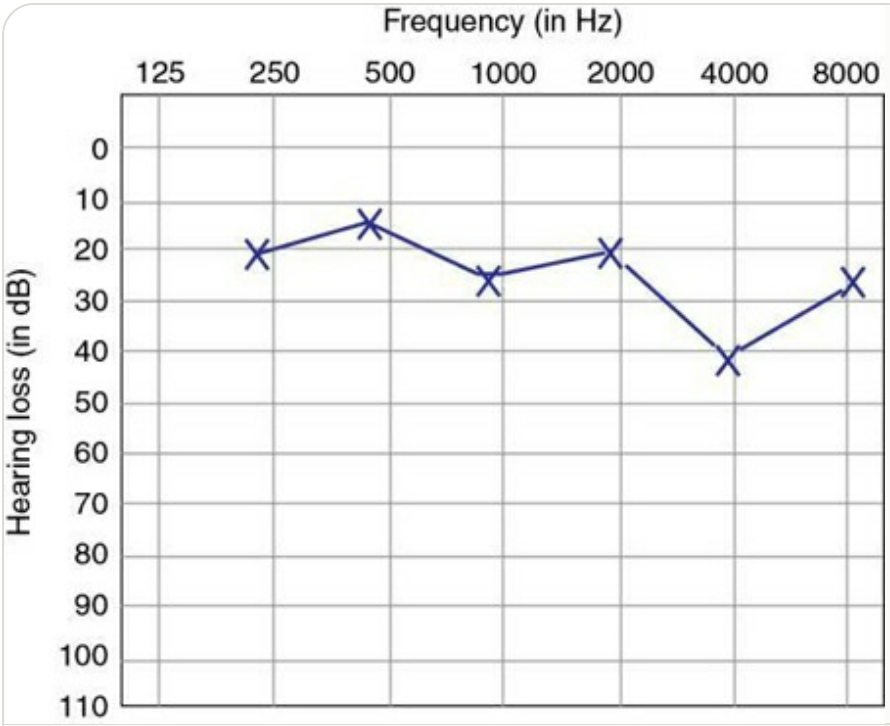



**FIGURE 2** Early noise-induced hearing loss, with a dip at 4000 Hz.
Dhingra, fig. 5.7

## 3.3 Safe limits

**NOISE EXPOSURE RULES**

- Dhingra's occupational limit: 90 dBA for 8 hours; no continuous exposure over 115 dBA; no impulse over 140 dBA.
- The *5 dB rule*: each 5 dB rise halves the safe duration.
- Hearing conservation (protection and audiometry) starts at 85 dBA.
- Limits differ by country; check local regulations.

## 3.4 Prevention and management

Remove from exposure, reduce noise at source, and use hearing protection: earplugs give about 15 to 30 dB, earmuffs 30 to 40 dB, and both together more. Protectors work better at high than low frequencies. Pre-employment and annual audiograms. Hearing aids for established loss.

**WHERE THE BOOKS DIFFER · EXPOSURE LIMITS AND THE EXCHANGE RATE**

**Dhingra**

Occupational limit of 90 dBA for 8 hours, with each 5 dB rise halving the safe duration.

**Oxford**

The UK Control of Noise at Work Regulations 2005 set a lower action value of 80 dB(A), when protection must be offered and information given, and an upper action value of 85 dB(A), when protection is compulsory and hearing surveillance required, with an absolute limit of 87 dB(A) at the ear allowing for protection. European and UK practice uses a 3 dB exchange rate (each 3 dB doubles the energy and halves the safe time); the 5 dB rule is the older American occupational convention.

### DIAGNOSIS, ACOUSTIC TRAUMA AND HIDDEN HEARING LOSS

Noise first stiffens and then kills the **outer hair cells**, producing a temporary threshold shift that becomes permanent with continued exposure; 3, 4 and 6 kHz are affected first. In medicolegal practice the **Coles, Lutman and Buffin (CLB) guidelines** require high-frequency loss beyond that expected for age, a cumulative noise immission level of at least 100 dB (roughly 85 dB(A) daily for many years), and a notch or bulge at 3 to 6 kHz at least 10 dB deeper than the thresholds at 1 or 2 kHz and at 6 or 8 kHz. **Acute acoustic trauma** from a single blast or impulse causes immediate loss without a preceding temporary shift and is treated like sudden sensorineural loss, with steroids. Animal work by Kujawa and Liberman showed that noise can destroy the synapses between inner hair cells and the cochlear nerve while thresholds recover (**cochlear synaptopathy**, or "hidden hearing loss"), which may explain difficulty in noise with a normal audiogram; in humans the effect appears variable and is still debated. Recreational noise, from music players, concerts and shooting, is an increasingly important cause in young people.

# 4 CHAPTER FOUR Presbycusis

Hearing and age.

## 4.1 Picture

Gradual, bilateral, symmetrical high-frequency loss. The patient hears speech but cannot understand it, especially in noise, and is intolerant of loud sound because of recruitment.

## 4.2 Types (Schuknecht)

TYPES OF PRESBYCUSIS

| TYPE | DAMAGE | AUDIOGRAM | DISCRIMINATION |
|---|---|---|---|
| Sensory | Hair cells, basal turn | Steep high-tone loss | Good |
| Neural | Cochlear neurones | Variable | Poor, out of proportion |
| Strial (metabolic) | Stria vascularis | Flat | Good |
| Cochlear conductive | Stiff basilar membrane | Gently sloping | Moderate |

## 4.3 Management

Hearing aids, assistive listening devices, speech reading and communication tactics; cochlear implants for severe loss. Untreated hearing loss is linked with isolation and cognitive decline.

OPERATIVE GUIDES IN THEATRE 2 operations

**Cochlear Implantation in Adults**
Cochlear implant for severe or profound loss

**Implantable Middle Ear Devices**
Implantable devices

CHAPTER FIVE

# 5 Immune-mediated inner ear disease

The treatable one.

## 5.1 Primary autoimmune inner ear disease

Rapidly progressive, bilateral, often asymmetric sensorineural loss over weeks to months, usually in middle age (40 to 50), with vestibular symptoms in about half. About 15% have another autoimmune disorder.

## 5.2 Secondary

Part of systemic disease: *Cogan syndrome* (interstitial keratitis with audiovestibular loss), granulomatosis with polyangiitis, systemic lupus, antiphospholipid syndrome, and others.

## 5.3 Diagnosis

Bilateral loss of 30 dB or more with progression on serial audiograms within three months. Blood tests: ESR, rheumatoid factor, ANA, complement; anti-Hsp70 antibodies correlate with activity but are not specific. Contrast MRI to exclude other causes. A response to steroids supports the diagnosis.

## 5.4 Treatment

Prednisolone 1 mg/kg/day, up to 60 mg, for four weeks. Responders (an improvement of 15 dB at one frequency or 10 dB at consecutive frequencies, or better discrimination) continue to a plateau and taper to the lowest maintenance dose over months. For steroid intolerance or dependence: methotrexate, cyclophosphamide (more toxic), biological agents or intratympanic steroids. Cochlear implantation if hearing is lost.

# 6 CHAPTER SIX Differential diagnosis

Others to exclude.

- Vestibular schwannoma.
- Ménière's disease.
- Syphilis.
- Multiple sclerosis.
- Familial progressive sensorineural loss.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Ototoxic drugs:** vestibulotoxic versus cochleotoxic, risk factors, monitoring.
- **Noise:** 4 kHz notch, temporary and permanent shifts, safe limits and the 5 dB rule.
- **Presbycusis:** Schuknecht's four types.
- **Autoimmune inner ear disease:** picture and steroid trial.

## 7.2 Common pitfalls

**COMMON PITFALLS**

- Not asking about drugs and noise.
- Missing a mitochondrial susceptibility to aminoglycosides in the family history.
- Calling rapidly progressive bilateral loss presbycusis.

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CLINICAL · EAR

# Eustachian tube disorders

The Eustachian tube is a valve that should open briefly with every swallow and stay closed the rest of the time. A tube that will not open leads to negative pressure, fluid and retraction of the drum; a tube that will not close lets the patient hear their own voice and breathing. In an adult with one-sided symptoms, always look at the nasopharynx.

**AT A GLANCE**

- Blocked tubes lead to fluid and retraction.
- Patulous tubes cause autophony.
- Look at the nasopharynx in adults.
- Treat the nose, adenoids, and use tubes when needed.

## 1 CHAPTER ONE Overview

A small tube with three jobs.

### 1.1 Functions

- **Ventilation:** equalising middle ear pressure with the atmosphere, for the drum to vibrate freely.
- **Protection:** from nasopharyngeal sound pressure and secretions.
- **Clearance:** drainage of middle ear secretions into the nasopharynx.

### 1.2 Disorders

**EUSTACHIAN TUBE DISORDERS**

- FAILS TO OPEN **Obstructive dysfunction** mechanical or functional
- FAILS TO CLOSE **Patulous tube** autophony
- PRESSURE CHANGE **Barotrauma** the tube locks on descent

# 2 CHAPTER TWO Applied anatomy

Adult and child.

## 2.1 The adult tube

About 36 mm long, running downwards, forwards and medially from the middle ear to the nasopharynx at about 45 degrees. The posterolateral third is bony; the anteromedial two-thirds is cartilaginous. The narrowest point, the *isthmus*, lies at the junction of the two.

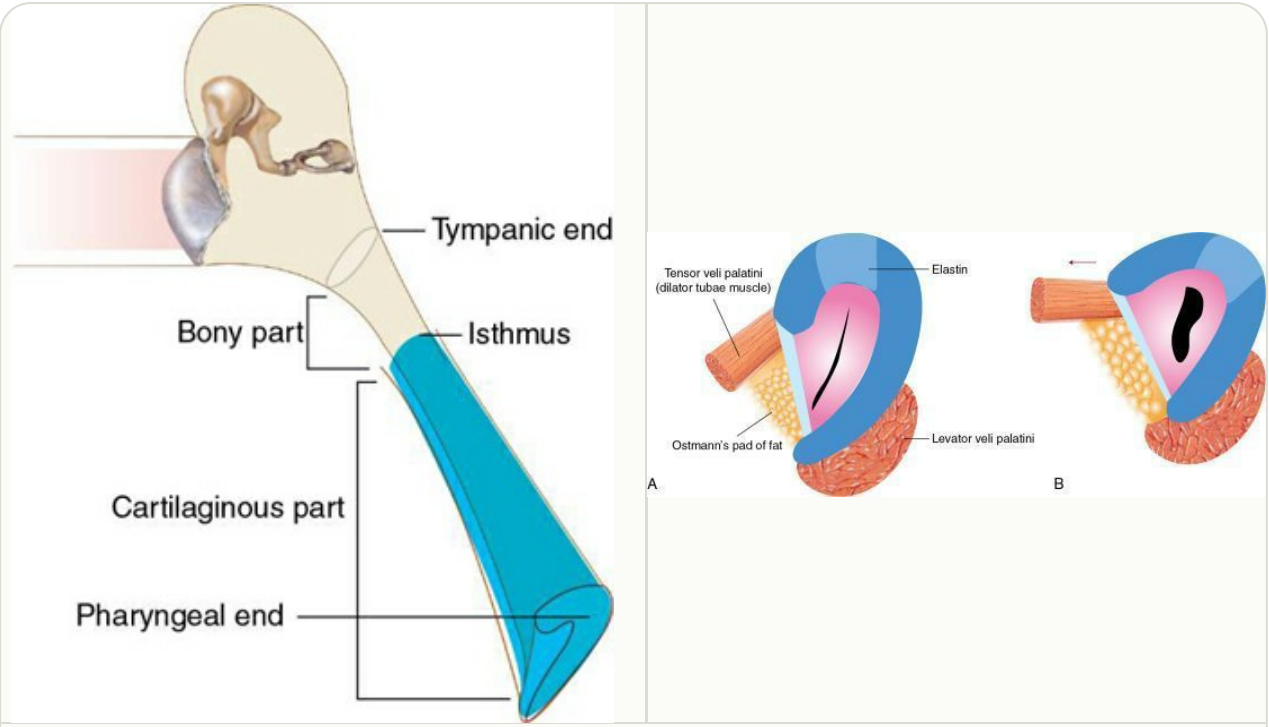



**FIGURE 1** First, a horizontal section of the tube with its bony and cartilaginous parts and the isthmus. Second, a vertical section: the cartilage forms the medial wall, roof and part of the lateral wall.

Dhingra, figs 9.1 and 9.2

## 2.2 Opening the tube

The tube is closed at rest. *Tensor veli palatini* opens it during swallowing and yawning, helped by levator veli palatini. The pharyngeal opening lies behind the inferior turbinate, bounded by the *torus tubarius*, with the fossa of Rosenmüller behind it.

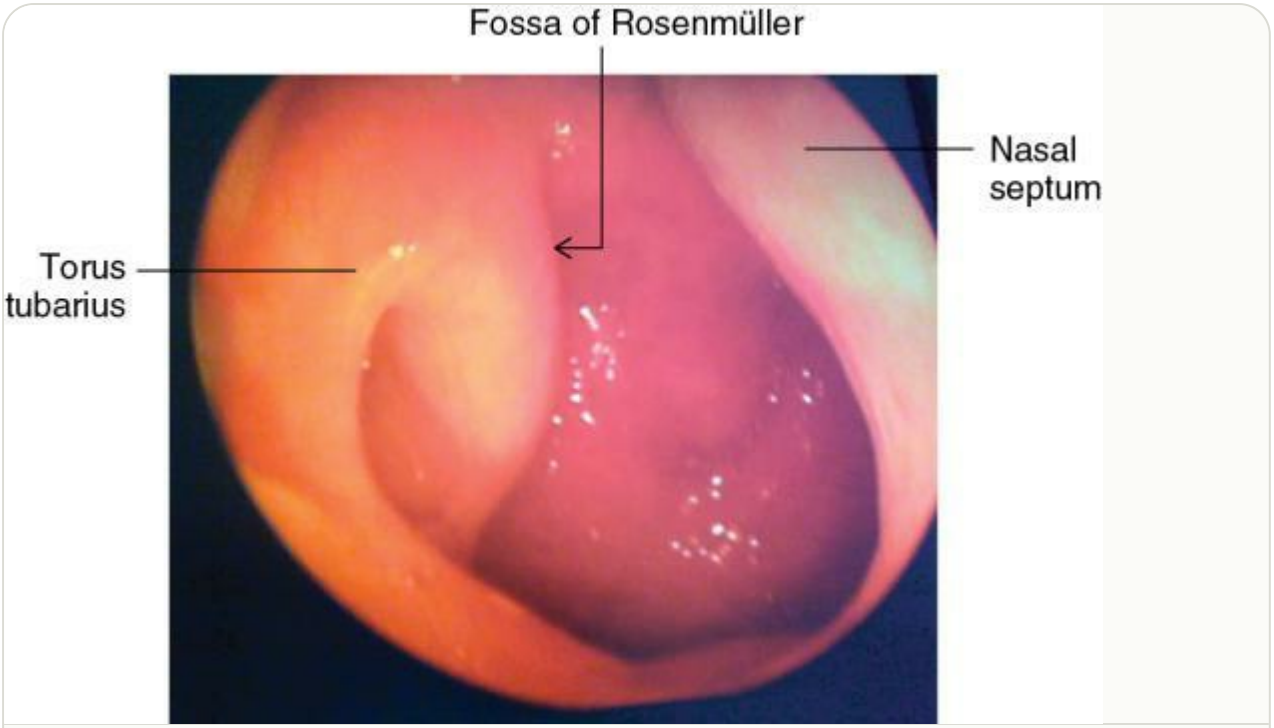



**FIGURE 2** Endoscopic view of the nasopharynx showing the right torus tubarius and the fossa of Rosenmüller.

Dhingra, fig. 9.4

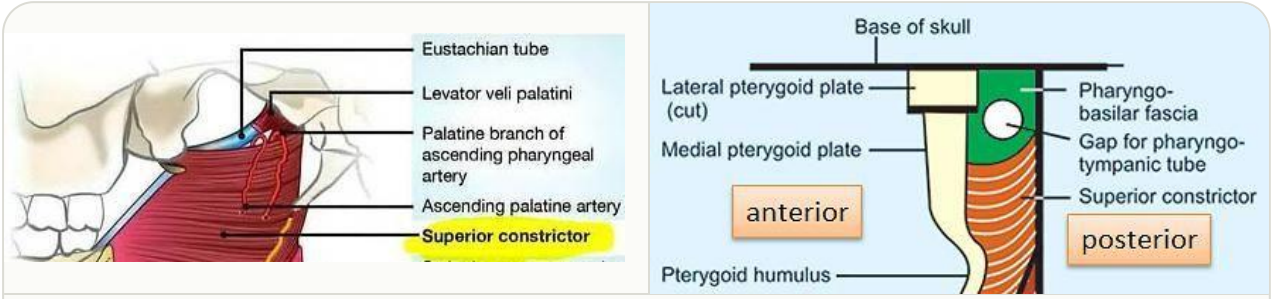



**FIGURE 3** The tube with levator and tensor veli palatini, and its course through the sinus of Morgagni.

Anatomy atlas · ENT Steps p. 204

## 2.3 The infant tube

About half the adult length, nearly horizontal (about 10 degrees), wider, with floppy cartilage and little elastin. It refluxes nasopharyngeal secretions and protects the middle ear poorly, which explains the frequency of otitis media in children. Function approaches the adult pattern by about age seven to ten.

# 3 CHAPTER THREE Obstructive dysfunction

The tube that will not open.

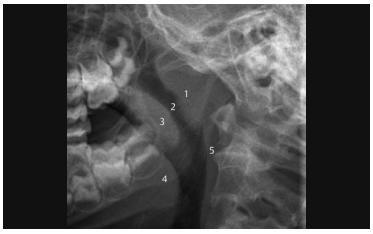
## 3.1 Causes

### 3.1.1 Mechanical

Inflammation and oedema from colds, sinusitis and allergy; adenoid hypertrophy, which also harbours bacteria; nasopharyngeal tumours, especially nasopharyngeal carcinoma in adults; and scarring after radiotherapy.

**IMAGING IN THE APP** 1 case

Adenoid enlargement obstructing the tube.




**Adenoid Hypertrophy**

Infection & inflammation · 1 image

### 3.1.2 Functional

A floppy cartilage or a weak tensor: infants, *cleft palate* (the tensor is poorly anchored) and *Down syndrome* (poor muscle tone).

## 3.2 Clinical picture

Fullness, popping, mild pain, muffled hearing, tinnitus and sometimes unsteadiness. The drum is retracted, sometimes with amber fluid or a fluid level.

## 3.3 Consequences

Negative pressure, effusion, atelectasis, retraction pockets and cholesteatoma, and ossicular erosion.

**WHERE THE BLOCK IS PREDICTS THE DAMAGE**

- Tube itself: total atelectasis of the pars tensa.
- Middle ear: posterior retraction pocket.
- Tympanic isthmi: attic retraction pocket.
- Aditus: cholesterol granuloma in the mastoid.

## 3.4 Investigations

### 3.4.1 Valsalva and Toynbee tests

Valsalva (forced expiration against a closed nose and mouth) inflates the middle ear, but only about 65% of normal people can do it. Toynbee's test (swallowing with the nose pinched) creates negative pressure.

### 3.4.2 Politzer test

Air from a Politzer bag into the nose while the child swallows or says "hick"; useful in children.

### 3.4.3 Catheterisation

A catheter passed along the nasal floor into the tubal opening; air is blown through while listening with an auscultation tube. Done gently, as it can injure the opening.

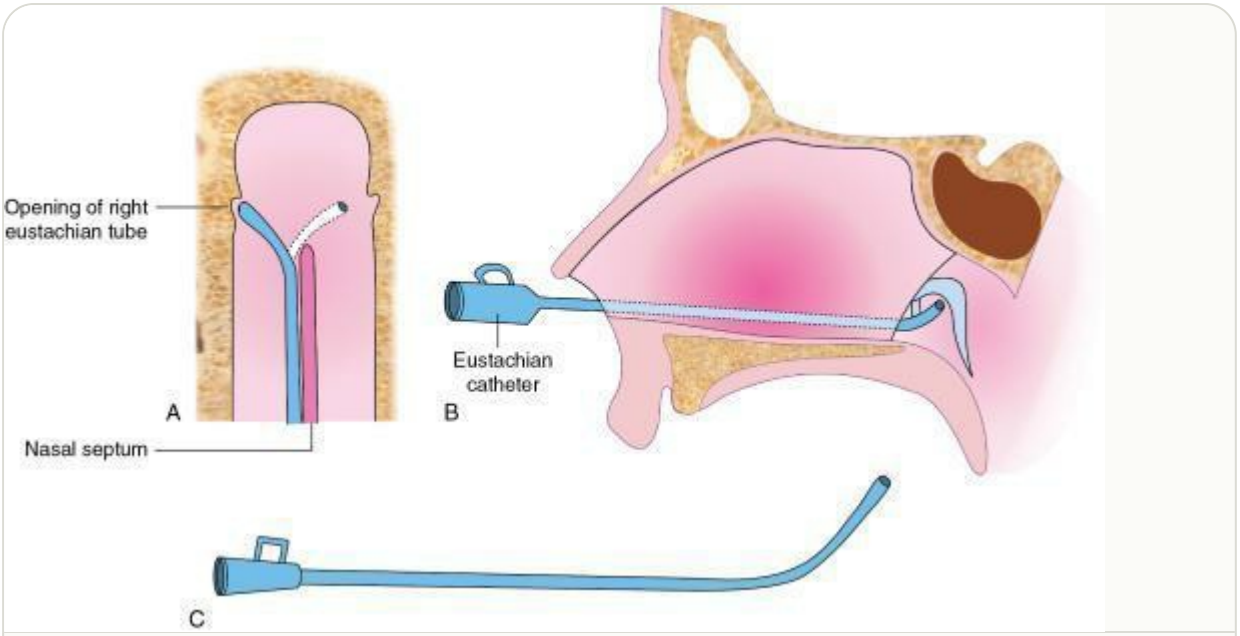



**FIGURE 4** Catheterisation of the Eustachian tube.
Dhingra, fig. 9.3

### 3.4.4 Tympanometry

Negative pressure or a flat curve; the inflation-deflation test assesses function through an intact or perforated drum.

### 3.4.5 Other tests

Dye or saccharine clearance, sonotubometry, and endoscopy of the nasopharynx.

## 3.5 Treatment

Treat the nose, sinuses, allergy and adenoids; autoinflation; a grommet for persistent effusion or retraction. *Balloon dilation* of the cartilaginous tube helps selected adults with persistent obstructive dysfunction.

**IN ADULTS**

Persistent one-sided tube symptoms or effusion in an adult: examine the nasopharynx to exclude carcinoma.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation**
Tube and balloon dilation

**Adenoidectomy**
Adenoidectomy

# 4 CHAPTER FOUR Patulous tube

The tube that will not close.

## 4.1 Causes

Idiopathic; rapid weight loss, which removes the fat pad around the tube; the last trimester of pregnancy; oral contraceptives; and neurological disease such as multiple sclerosis or stroke.

## 4.2 Clinical picture

*Autophony*: the patient hears their own voice and breathing loudly in the ear, relieved by lying down or putting the head between the knees (venous congestion closes the tube). The drum moves in and out with respiration, best seen while the patient breathes through one nostril.

## 4.3 Differential diagnosis

Superior canal dehiscence also causes autophony, but with vertigo or a conductive gap and an intact stapedial reflex.

## 4.4 Treatment

Reassurance; most acute cases settle on their own. Weight gain, nasal drops (saline, or in the past potassium iodide), cautery or injection around the tubal opening, a grommet, or plugging the tube in severe cases.

PEARL

Don't prescribe decongestants for fullness and autophony after weight loss: they make a patulous tube worse.

# 5 CHAPTER FIVE Barotrauma

When the tube locks.

On descent in an aircraft or when diving, a tube that fails to open can lock shut once the pressure difference reaches about 90 mm Hg, causing pain, effusion, haemorrhage or perforation (see the tympanic membrane topic).

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Anatomy:** length, angle, bony and cartilaginous parts, isthmus, tensor veli palatini.
- **Infant versus adult tube.**
- **Tests of tubal function:** Valsalva, Toynbee, Politzer, catheterisation, tympanometry.
- **Patulous tube:** causes, symptoms, signs.
- **Unilateral effusion in an adult:** look at the nasopharynx.

## 6.2 Common pitfalls

COMMON PITFALLS

- Missing nasopharyngeal carcinoma in an adult with one-sided fluid.
- Treating a patulous tube with decongestants.

- Performing Valsalva with an infected nose or an atrophic drum.

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CLINICAL · EAR

# Vertigo: BPPV, neuritis and central causes

Vertigo is the illusion of movement, usually spinning, and in most patients it comes from the inner ear. Peripheral causes account for about 85%, but the duty of the doctor is to catch the dangerous central minority. Four questions sort most cases: how long the attacks last, what triggers them, whether hearing is affected, and whether there are neurological signs.

**AT A GLANCE**

- Timing, triggers, hearing and neurology sort most cases.
- BPPV is seconds; Ménière's is hours; neuritis is days.
- HINTS and focal signs catch stroke.
- Epley for BPPV; early rehabilitation for neuritis.
- Horizontal canal BPPV: supine roll test and barbecue roll or Gufoni; anterior canal: Yacovino manoeuvre.

## 1 CHAPTER ONE Overview

A symptom, not a diagnosis.

### 1.1 Definitions

#### 1.1.1 Vertigo

A false sense of rotation or movement of oneself or the surroundings, from asymmetry in the vestibular system.

#### 1.1.2 Other kinds of dizziness

- **Disequilibrium:** unsteadiness without a sense of rotation.
- **Oscillopsia:** the visual world jumps with head movement, typical of bilateral vestibular loss.
- **Presyncope:** light-headedness from reduced cerebral perfusion (cardiac, orthostatic, hypoglycaemic).
- **Physiological:** motion sickness and height vertigo.

## 1.2 Peripheral or central

**CAUSES OF VERTIGO**

- PERIPHERAL **BPPV, vestibular neuritis** labyrinthitis, Ménière's disease, perilymph fistula, superior canal dehiscence, ototoxicity, trauma, syphilis
- CENTRAL **Posterior circulation stroke** vertebrobasilar insufficiency, Wallenberg syndrome, vestibular migraine, multiple sclerosis, cerebellar and brainstem tumours, epilepsy
- OTHER **Cervical** ocular, psychogenic, cardiovascular

# 2 CHAPTER TWO The history

Timing is everything.

## 2.1 Duration

**DURATION OF ATTACKS**

| DURATION | TYPICAL CAUSES |
|---|---|
| Seconds | Benign paroxysmal positional vertigo |
| Minutes | Vertebrobasilar ischaemia, vestibular migraine |
| Minutes to hours | Ménière's disease, recurrent vestibulopathy, vestibular migraine |
| Days to weeks | Vestibular neuritis, labyrinthitis, stroke |
| Variable | Perilymph fistula, superior canal dehiscence |

## 2.2 Triggers

Head position (BPPV), loud sound or pressure (superior canal dehiscence, fistula), straining, neck movement, standing up (orthostatic hypotension), and migraine triggers.

### 2.3 Hearing

Hearing loss or tinnitus points to the labyrinth (labyrinthitis, Ménière's disease, fistula) or the cerebellopontine angle, or to an AICA stroke.

### 2.4 Neurological symptoms

Double vision, slurred speech, numbness, weakness, difficulty swallowing, new severe headache: central until proven otherwise.

## 3 CHAPTER THREE Examination

Eyes, head and gait.

### 3.1 Nystagmus

Look in the primary position and on gaze, with and without fixation (Frenzel glasses). Peripheral nystagmus is horizontal-torsional, beats away from the affected side and is suppressed by fixation; it is stronger looking towards the fast phase (*Alexander's law*).

### 3.2 Head impulse test

A rapid head turn: a catch-up saccade means the vestibulo-ocular reflex on that side has failed, a peripheral lesion.

### 3.3 HINTS

In acute continuous vertigo with nystagmus: a *normal* head impulse, direction-changing nystagmus, or a vertical skew deviation points to a stroke.

### 3.4 Positional tests

The Dix-Hallpike manoeuvre for posterior canal BPPV, and the supine roll test for the

lateral canal.

## 3.5 Other bedside tests

Romberg and tandem gait, head shake, dynamic visual acuity (losing more than five lines suggests bilateral loss), fistula test, and a full neurological and cranial nerve examination.

PERIPHERAL OR CENTRAL FINDINGS

| FINDING | ACUTE PERIPHERAL | CENTRAL |
|---|---|---|
| Nystagmus | Mixed horizontal-torsional | Pure vertical or torsional, direction-changing |
| Fixation | Suppresses | Does not suppress |
| Head impulse | Abnormal | Normal |
| Pursuit | Normal | Saccadic |
| Saccades | Normal | Dysmetric |
| Gait | Can walk | Often cannot stand |
| Other CNS signs | Absent | Often present |

# 4 CHAPTER FOUR Peripheral causes

The common ones.

## 4.1 Benign paroxysmal positional vertigo

### 4.1.1 Mechanism

Otoconia dislodged from the utricle settle in a semicircular canal, most often the posterior canal, and move with gravity when the head changes position (*canalolithiasis*). It often follows head injury or ear infection, or occurs in older people without cause.

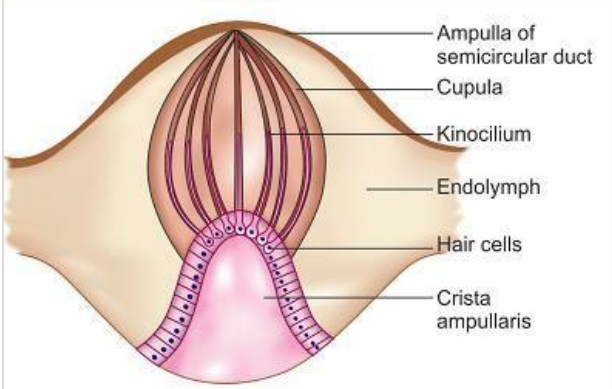



**Fig. 41:** Crista, hair cells and cupula. Cut section of ampulla of semicircular duct

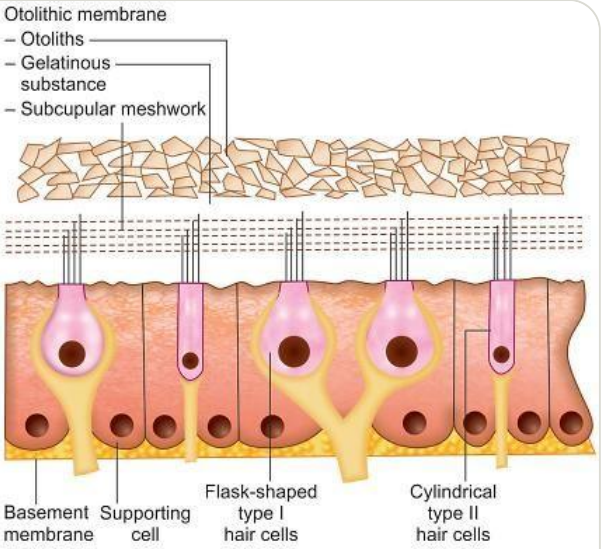



**Fig. 43:** Macula of otolith organs: utricle and saccule

**FIGURE 1** The crista ampullaris with its cupula, and the macula of the utricle and saccule whose otoconia cause BPPV.

Anatomy atlas · ENT Steps p. 78, ENT Steps p. 80

### 4.1.2 Picture

Brief vertigo, under a minute, on lying down, rolling over or looking up. No hearing loss.

### 4.1.3 Dix-Hallpike nystagmus

Rotatory and geotropic, after a latency of a few seconds, rising then fading within a minute, reversing on sitting up, and fatiguing on repetition.

### 4.1.4 Treatment

The *Epley manoeuvre* repositions the debris into the utricle; about 80% are cured by one manoeuvre. Semont manoeuvre and Brandt-Daroff exercises are alternatives. Most cases resolve in months anyway. Surgery (posterior canal occlusion or singular neurectomy) is rarely needed.

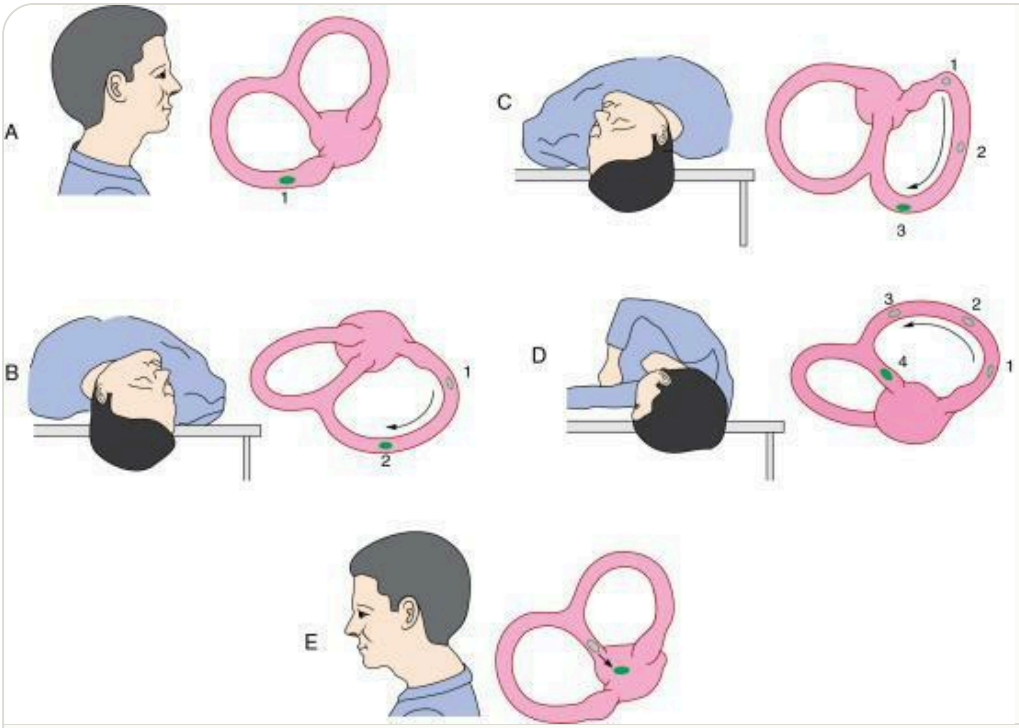



**FIGURE 2** The Epley manoeuvre for posterior canal BPPV, with the position of the patient and of the otolith debris in the canal at each step.

Dhingra, fig. 7.1

### BEYOND THE POSTERIOR CANAL

The Oxford Textbook gives the current picture. The **posterior canal** causes about 90 to 95% of BPPV, because it is the lowest canal when lying down; the **horizontal (lateral) canal** causes 5 to 15%, and the anterior canal very few. Repositioning cures about 80% after one session and over 90% after up to four, and patients can be taught to do it themselves. No postural restrictions are needed afterwards, because they do not prevent recurrence. Recurrence is common, in up to a third of patients within four years, and the occasional Epley manoeuvre converts posterior into horizontal canal BPPV.

**Horizontal canal BPPV** is diagnosed with the supine roll test, which produces horizontal nystagmus on turning to either side. In the commoner *geotropic* form (beating towards the ground, from free particles in the canal), the side with the stronger nystagmus is the affected ear. In the *apogeotropic* form (beating away from the ground, thought to be cupulolithiasis or particles near the cupula), the weaker side is the affected ear. It is treated with the **Lempert 360° (barbecue or log) roll** towards the healthy side, the **Gufoni manoeuvre**, or prolonged lying on one side; these work moderately well for the geotropic form, and the evidence for the apogeotropic form is weak. **Anterior canal BPPV** produces down-beating nystagmus with a torsional component and can be treated by the **Yacovino (deep head-hanging) manoeuvre**, which treats either side because the head is not

turned; down-beating positional nystagmus that does not settle must be imaged to exclude a cerebellar lesion.

Vestibular sedatives and benzodiazepines slow central compensation and increase falls in the elderly, who carry a much higher fall risk when dizzy, so they have no role in BPPV. Atypical nystagmus, failure after several manoeuvres and frequent recurrence call for further investigation.

POSITIONAL VERTIGO: PERIPHERAL OR CENTRAL

| FEATURE | PERIPHERAL | CENTRAL |
|---|---|---|
| Latency | Present | Often none |
| Duration | Under a minute | Persists |
| Fatigue | Yes | No |
| Direction | Fixed, torsional | Changing or vertical |
| Vertigo | Marked | Often mild |

## 4.2 Vestibular neuritis

### 4.2.1 Mechanism

Probably viral inflammation of the vestibular nerve or ganglion.

### 4.2.2 Picture

Sudden, severe, continuous vertigo with nausea and vomiting, lasting days and settling over two to three weeks, without hearing loss. Nystagmus beats away from the affected side; the head impulse is abnormal towards it; calorics are reduced. BPPV follows in up to 15%.

### 4.2.3 Treatment

Short-course vestibular sedatives, possibly steroids early, then early vestibular rehabilitation. Failure to improve at two to three weeks means a central lesion, such as a cerebellar infarct, must be excluded.

### 4.3 Labyrinthitis

As neuritis, but with hearing loss, from viral or bacterial infection of the labyrinth itself.

### 4.4 Ménière's disease

Attacks of vertigo lasting hours with fluctuating hearing loss, tinnitus and fullness.

### 4.5 Recurrent vestibulopathy

Ménière-like spells without hearing symptoms. About 60% remit; a minority become Ménière's disease or BPPV.

### 4.6 Perilymph fistula

Episodic vertigo and fluctuating hearing after stapes surgery, barotrauma, heavy lifting or coughing. Bed rest with the head up and laxatives; explore if worsening.

### 4.7 Superior canal dehiscence

Vertigo provoked by sound and pressure, autophony and a low-frequency conductive loss with intact reflexes (see the separate topic).

### 4.8 Bilateral vestibular loss

Oscillopsia and unsteadiness in the dark, without true vertigo, most often from aminoglycosides. Cerebellar ataxia with bilateral vestibular loss is called *CABV*, or *CANVAS* when sensory neuropathy is added.

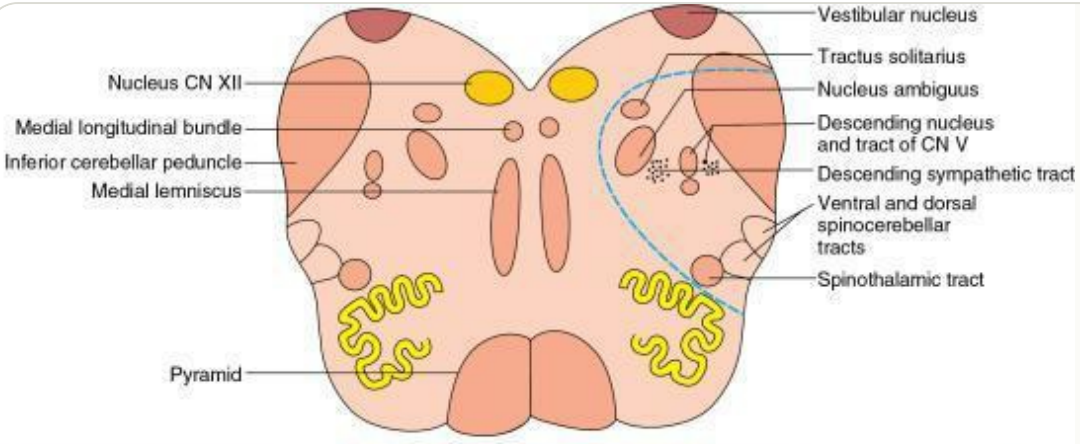
## 5 CHAPTER FIVE Central causes

The ones not to miss.

# 5.1 Posterior circulation stroke

## POSTERIOR CIRCULATION STROKE SYNDROMES

| FEATURE | PICA | AICA | SCA |
|---|---|---|---|
| Vertigo, nystagmus, ataxia | Yes | Yes | Yes |
| Hearing loss, tinnitus | No | Yes | No |
| Facial palsy | Sometimes | Yes | No |
| Dysphagia, hoarseness | Yes | No | No |
| Contralateral loss of vibration and position | No | No | Yes |




| | |
|---|---|
| • Inferior cerebellar peduncle | Vertigo, nausea, vomiting and nystagmus |
| • Spinocerebellar tracts | Ataxia |
| • Nucleus ambiguus (CN X, IX) | Hoarseness and dysphagia |
| • Descending sympathetic tract | Horner syndrome |
| • Uncrossed fibres of spinothalamic tract | Loss of pain and temperature on ipsilateral face |
| • Descending nucleus and tract of CN V | Pain and numbness over ipsilateral face |
| • Crossed fibres of spinothalamic tract | Contralateral loss of pain and temperature of arm, trunk and leg |

**FIGURE 3** Lateral medullary (Wallenberg) syndrome from occlusion of the posterior inferior cerebellar artery.

Dhingra, fig. 7.2

### 5.1.1 Wallenberg syndrome

Violent vertigo with hoarseness, dysphagia, ipsilateral Horner syndrome, ipsilateral facial and contralateral body loss of pain and temperature, and ataxia.

### 5.2 Vertebrobasilar insufficiency

Attacks lasting minutes in patients over 50, with visual disturbance, diplopia or drop attacks, sometimes on neck movement.

### 5.3 Vestibular migraine

Attacks of vertigo, minutes to hours, with or near migraine headache, often in young women with a family history; response to migraine treatment supports the diagnosis.

### 5.4 Multiple sclerosis

Young adults with vertigo and pendular or dissociated nystagmus, and other neurological episodes.

### 5.5 Other central causes

Cerebellar and brainstem tumours, Chiari malformation, idiopathic intracranial hypertension, encephalitis and epilepsy.

## 6 CHAPTER SIX Other causes

When it isn't the ear or the brain.

### 6.1 Cervical vertigo

Vertigo with neck movement after a neck injury; controversial and probably over-diagnosed.

### 6.2 Cardiovascular

Presyncope from arrhythmia, orthostatic hypotension or hypoglycaemia; not true

rotatory vertigo.

### 6.3 Psychogenic

Vague floating or swaying, with anxiety, no nystagmus and normal tests. Persistent postural-perceptual dizziness often follows an acute vestibular event.

## 7 CHAPTER SEVEN Management

Principles.

**ACUTE VERTIGO ON CALL**

1. Ask about timing, triggers, hearing and neurological symptoms.
2. Examine nystagmus, head impulse, skew and gait (HINTS).
3. Any focal sign, new severe headache, inability to stand, or a normal head impulse with nystagmus: treat as stroke until proven otherwise.
4. Dix-Hallpike only once the neck and the patient's condition allow it.
5. Give antiemetics or vestibular sedatives for a few days only.

### 7.1 Vestibular rehabilitation

Head and eye exercises that drive central compensation, the mainstay after any peripheral loss.

### 7.2 Avoid long-term sedatives

They slow compensation.

**PEARLS**

- An acoustic neuroma usually causes unsteadiness, not severe vertigo.
- Fatigable, delayed, torsional positional nystagmus is peripheral; immediate, persistent or vertical nystagmus is central.

- A positive fistula test: perilymph fistula, cholesteatoma with a fistula, Ménière's disease, superior canal dehiscence, or syphilis.

**OPERATIVE GUIDES IN THEATRE** 2 operations

**Surgery for Vertigo**
Surgery for vertigo

**Superior Semicircular Canal Dehiscence**
Superior canal dehiscence

# 8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

## 8.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Peripheral versus central** vertigo and nystagmus.
- **BPPV:** mechanism, Dix-Hallpike findings, Epley manoeuvre.
- **Vestibular neuritis** versus labyrinthitis.
- **HINTS** in acute vertigo.
- **Wallenberg syndrome.**

## 8.2 Common pitfalls

**COMMON PITFALLS**

- Labelling a stroke as neuritis because the patient is vomiting.
- Missing a normal head impulse in continuous vertigo.
- Keeping patients on vestibular sedatives for weeks.
- Diagnosing BPPV without seeing the nystagmus.

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CLINICAL · EAR

# The deaf child and congenital hearing loss

Speech and language are built between birth and five years, and they are built on hearing. A child who does not speak has usually not heard. Children identified and helped before six months of age do far better, which is why every newborn is screened, and why a failed screen is never brushed aside.

**AT A GLANCE**

- Identify and treat before speech develops.
- Screen all newborns: half have no risk factor.
- OAE and ABR, then age-appropriate testing.
- Aids early; implant for severe-profound loss.
- Check ECG, thyroid, kidney and eyes by syndrome.

## 1 CHAPTER ONE Overview

The critical years.

### 1.1 Why early matters

The auditory pathways develop in response to sound during the first years of life. Without input, the auditory cortex is taken over by other functions and language develops late or not at all. Early fitting of hearing aids or a cochlear implant allows near-normal language in many children.

### 1.2 Prelingual and postlingual

*Prelingual* deafness occurs before speech has developed and has the greatest effect on language; *postlingual* deafness occurs after.

### 1.3 The 1-3-6 rule

Screen by one month, diagnose by three months, and start intervention by six

months.

# 2 CHAPTER TWO Causes

Genes, infection and the newborn period.

## 2.1 Proportions

About half of congenital sensorineural loss is genetic, about a quarter environmental and about a quarter unexplained. Of the genetic cases, about 70% are non-syndromic (mostly autosomal recessive) and 30% syndromic.

## 2.2 Genetic

### 2.2.1 Non-syndromic

*GJB2* mutations affecting connexin 26 (DFNB1) account for up to half of recessive non-syndromic deafness. Mitochondrial mutations (12S rRNA, 1555A>G) cause deafness after even a single dose of an aminoglycoside.

### 2.2.2 Syndromic

SYNDROMES TO KNOW

| SYNDROME | CLUE | HEARING LOSS |
|---|---|---|
| Waardenburg | White forelock, odd-coloured eyes, dystopia canthorum (type 1) | Sensorineural |
| Usher | Retinitis pigmentosa | Sensorineural |
| Pendred | Goitre, enlarged vestibular aqueduct | Sensorineural |
| Jervell and Lange-Nielsen | Fainting, long QT | Sensorineural |
| Alport | Nephritis, X-linked | Progressive sensorineural |
| Treacher Collins | Lower lid coloboma, small jaw and malar, microtia | Conductive |

| SYNDROME | CLUE | HEARING LOSS |
|---|---|---|
| Crouzon and Apert | Craniosynostosis, proptosis (syndactyly in Apert) | Conductive or mixed |
| Branchio-oto-renal | Neck fistulae, pits, renal anomalies | Conductive or mixed |
| Stickler | Small jaw, cleft, myopia | Conductive or sensorineural |
| Goldenhar | Facial asymmetry, epibulbar dermoid | Conductive or mixed |
| Klippel-Feil | Short neck, fused cervical spine | Sensorineural or mixed |
| Down | Effusion, narrow canals | Conductive |



**FIGURE 1** Waardenburg syndrome: white forelock, heterochromia of the irides and depigmented skin.
Dhingra, fig. 19.1

**SYNDROME PEARLS**

- Usher type 1: profound loss and absent vestibular function; type 2: moderate loss, normal or reduced vestibular function.
- Pendred: the goitre appears around age eight; the perchlorate discharge test is positive.
- Waardenburg type 1 has dystopia canthorum; type 2 does not.

- Treacher Collins: lower lid coloboma; Goldenhar: upper lid.

## 2.3 Inner ear malformations

- **Michel:** complete absence of the inner ear; no aid or implant can help.
- **Mondini:** a cochlea of only one and a half turns, often with an enlarged aqueduct.
- **Scheibe:** cochleosaccular dysplasia, the commonest.
- **Alexander:** the basal turn alone, losing high tones.
- **Bing-Siebenmann:** the membranous labyrinth fails to develop.
- **Enlarged vestibular aqueduct:** fluctuating, progressive loss after minor head injury.

WHERE THE BOOKS DIFFER · ENLARGED VESTIBULAR AQUEDUCT

| Dhingra | Lee |
|---|---|
| Wider than 2 mm. | 1.5 mm or more. |

Avoid contact sports and head injury; surgical closure of the aqueduct causes hearing loss.

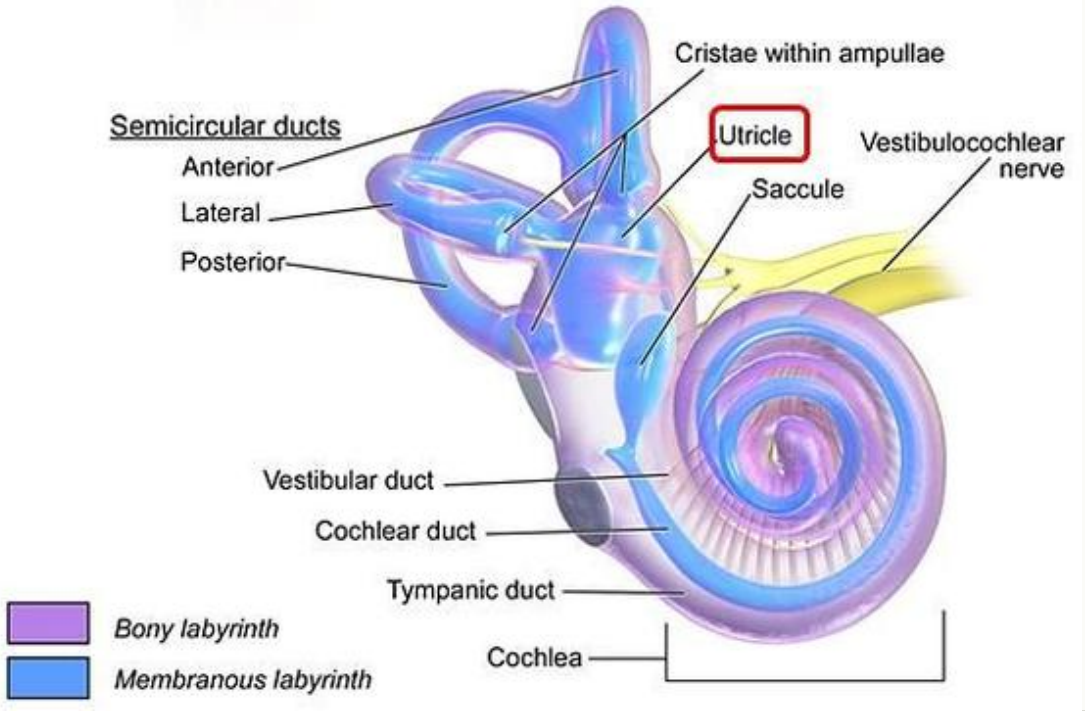



FIGURE 2 The membranous labyrinth within the bony labyrinth, the structure that fails to form in these malformations.

Anatomy atlas · ENT Steps p. 72

## 2.4 Environmental

**ENVIRONMENTAL CAUSES**

- PRENATAL **TORCH infections** cytomegalovirus, rubella, toxoplasmosis, syphilis, Zika; aminoglycosides, quinine, thalidomide, alcohol, radiation
- PERINATAL **Hypoxia, prematurity** birth weight under 1500 g, jaundice (bilirubin over 20 mg%), sepsis, NICU stay, ventilation
- POSTNATAL **Meningitis** viral infections, otitis media with effusion, ototoxic drugs, trauma, noise

### 2.4.1 Cytomegalovirus

The commonest non-genetic cause. The loss can be absent at birth and appear later, so a normal newborn screen does not exclude it.

### 2.4.2 Meningitis

Labyrinthitis ossificans can close the cochlea within weeks; test hearing and image quickly.

## 2.5 Risk factors

Family history, craniofacial anomaly or syndrome stigmata, NICU stay of five days or more, ventilation, low Apgar scores, severe jaundice, congenital infection, and meningitis. Half of children with sensorineural loss have no risk factor at all, which is why screening must be universal.

# 3 CHAPTER THREE Evaluation

Screen, confirm, find the cause.

## 3.1 Newborn screening

### 3.1.1 Otoacoustic emissions

Test the outer hair cells; quick and cheap. Absent with fluid or debris, or a cochlear loss. Normal in auditory neuropathy, which they therefore miss.

#### 3.1.2 Automated ABR

Tests the pathway to the brainstem; used for NICU babies, who are at risk of auditory neuropathy.

## 3.2 Diagnostic audiology

#### 3.2.1 ABR

Ear-specific thresholds, under natural sleep or sedation.

#### 3.2.2 Behavioural tests by age

- Under six months: behavioural observation.
- Six to eighteen months: distraction testing.
- Six months to two years: visual reinforcement audiometry.
- Two to five years: play audiometry.
- Older: conventional and speech audiometry.

## 3.3 Finding the cause

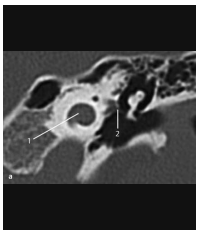
Imaging (CT and MRI) before implantation; ECG for long QT; urinalysis; thyroid function; eye examination; CMV PCR on newborn saliva, urine or dried blood spot (within 21 days); and genetic testing (GJB2, GJB6 and gene panels).

**LONG QT**

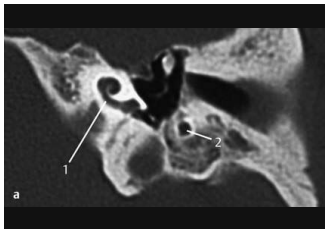
A deaf child who faints needs an urgent ECG: Jervell and Lange-Nielsen syndrome can cause sudden death. Treat with beta-blockers.

**IMAGING IN THE APP** 5 cases

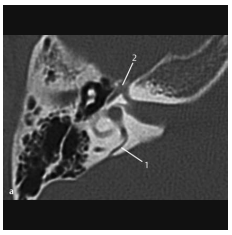
Inner ear and ossicular malformations, an enlarged vestibular aqueduct, Treacher Collins syndrome, and a gusher ear.



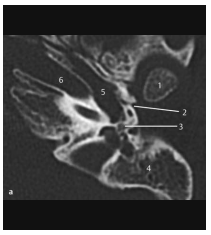
**Malformations of the Inner Ear**
Congenital · 3 images



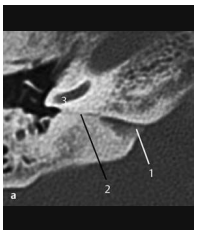
**Malformations of the Inner Ear and Ossicular Chain**
Congenital · 3 images



**Vestibular Aqueduct, Enlargements**
Congenital · 11 images



**Malformations, Treacher Collins**
Congenital · 2 images



**Gusher**
Infection & inflammation · 3 images

### 3.4 Signs at home

No startle to loud sound, sleeping through noise, no babbling or no words by one to two years, and poor school performance.

## 4 CHAPTER FOUR Management

Hearing first, then language.

### 4.1 Counselling

Parents need a clear explanation, support and early contact with educators.

### 4.2 Hearing aids

As early as possible, in both ears if there is usable hearing.

### 4.3 Cochlear implant

For severe to profound loss with little benefit from aids, ideally around the first birthday, or urgently after meningitis.

**OPERATIVE GUIDES IN THEATRE** 5 operations

**Pediatric Cochlear Implantation**
Paediatric implant
**Congenital Malformations of the Middle Ear**
Congenital middle ear malformations
**Congenital and Acquired Atresia and Stenosis of the External Auditory Canal**
Atresia
**Microtia**
Microtia
**Bone-Anchored Hearing Devices**
Bone-anchored devices

## 4.4 Communication and education

Auditory-oral approaches, sign language, or total communication; radio aids at school; speech and language therapy; vocational guidance later.

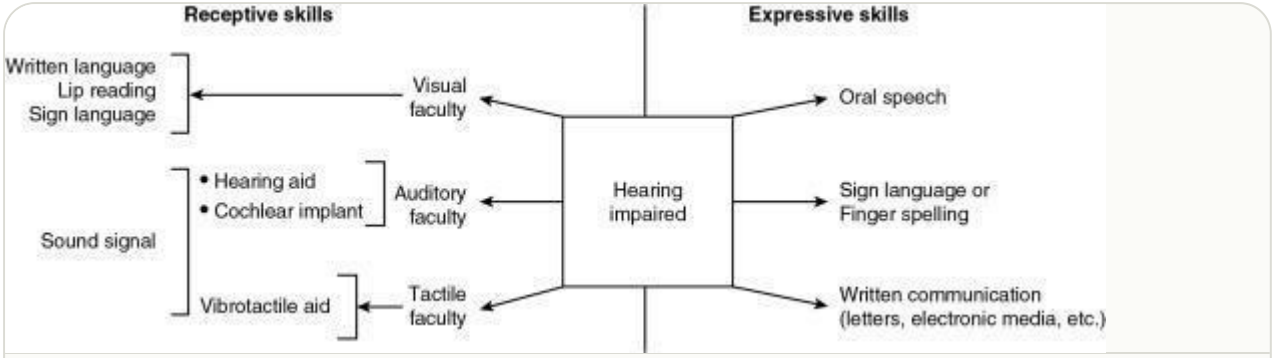



**FIGURE 3** The faculties a hearing-impaired child can use for receptive and expressive communication.
Dhingra, fig. 19.3

## 4.5 Specific treatments

- **Symptomatic congenital CMV:** valganciclovir for six months.
- **Toxoplasmosis:** pyrimethamine and sulfadiazine can prevent hearing loss.
- **Long QT:** beta-blockers.
- **Pendred goitre:** thyroid hormone.
- **Enlarged aqueduct:** avoid head injury and contact sport.

### 4.6 Follow-up

Children with risk factors need continued hearing checks even after passing the newborn screen; treated CMV infants continue audiology until about age six.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

### 5.1 On call

**A FAILED NEWBORN HEARING SCREEN**

1. Repeat the screen; if it fails again, arrange a diagnostic ABR. Don't reassure.
2. A deaf child with fainting: urgent ECG for long QT.
3. After meningitis: audiology and imaging quickly, before ossification closes the implant window.

### 5.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Causes:** genetic, prenatal, perinatal, postnatal.
- **High-risk registry.**
- **OAE versus ABR,** and auditory neuropathy.
- **Behavioural tests by age.**
- **The syndromes table.**
- **Inner ear malformations,** especially Michel and Mondini.

### 5.3 Common pitfalls

**COMMON PITFALLS**

- Reassuring parents after a failed screen.
- Relying on OAE in a NICU baby.
- Missing late-onset CMV loss after a normal newborn screen.
- Forgetting the ECG.

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CLINICAL · EAR

# External ear tumours and bony growths

The pinna faces the sun every day of a person's life, and most cancers of the ear start there. Any lesion of the pinna that does not heal deserves a biopsy. In the canal, a hard lump is often just bone, an osteoma or exostoses, while a painful, bleeding canal may be carcinoma.

**AT A GLANCE**

- Most ear cancers start on the sun-exposed pinna.
- Biopsy any non-healing lesion.
- Canal SCC is staged by the Pittsburgh system.
- Bony growths need treatment only if they obstruct.

## 1 CHAPTER ONE Overview

Tumours of the pinna and canal.

### 1.1 Classification

**EXTERNAL EAR TUMOURS**

- PINNA, BENIGN **Cysts, keloid** haemangioma, papilloma, cutaneous horn, keratoacanthoma, neurofibroma
- PINNA, MALIGNANT **Basal cell carcinoma, squamous cell carcinoma** melanoma
- CANAL, BENIGN **Osteoma, exostoses** ceruminous adenoma, papilloma
- CANAL, MALIGNANT **Squamous cell carcinoma** basal cell carcinoma, adenoid cystic and ceruminous carcinoma, melanoma

### 1.2 Where ear cancers arise

About 85% on the pinna, 10% in the canal and 5% in the middle ear.

# 2 CHAPTER TWO
# Benign lesions of the pinna

Lumps that are not cancer.

## 2.1 Keloid

Overgrowth of scar after piercing, more common in darker skin. Excision alone recurs; combine with steroid injection, pressure clips or, rarely, radiotherapy.




**FIGURE 1** Keloid after ear piercing.
Dhingra, fig. 16.1

**WHERE THE BOOKS DIFFER · RADIOTHERAPY FOR KELOID**

| Dhingra | Lee |
|---|---|
| Postoperative radiation is an option. | Radiation is rarely used. |

## 2.2 Haemangioma and vascular malformations

A *strawberry haemangioma* grows in the first year and regresses by about five years; a *port-wine stain* does not. Venolymphatic malformations may need excision.

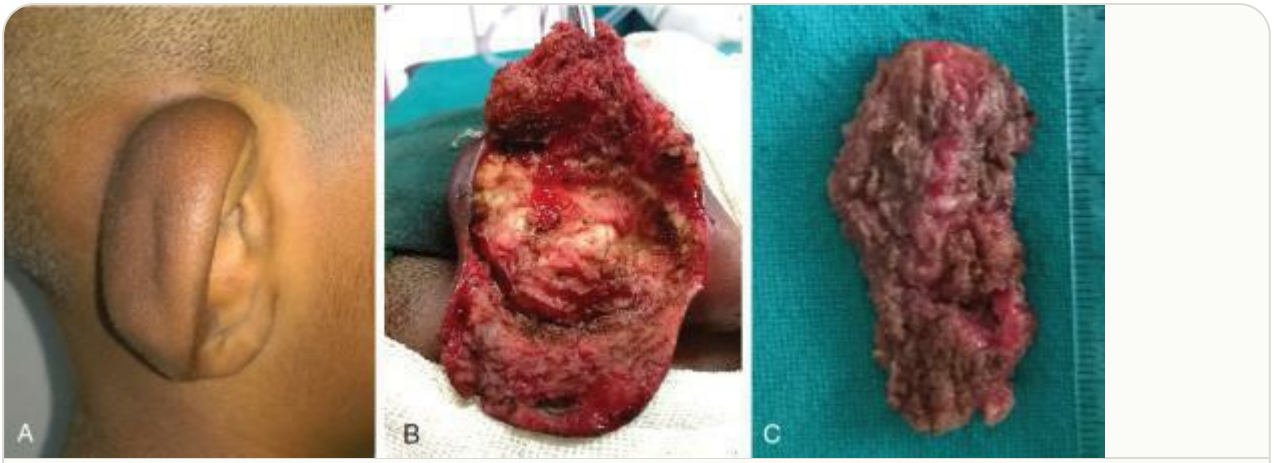



**FIGURE 2** A venolymphatic malformation of the pinna in a child: before, during operation, and the excised specimen.

Dhingra, fig. 16.2

### 2.3 Keratoacanthoma

A rapidly growing crater with a keratin plug that may regress on its own; it looks malignant and is usually excised.

### 2.4 Others

Sebaceous and dermoid cysts, papillomas, cutaneous horns and neurofibromas.

## 3 CHAPTER THREE
# Malignant tumours of the pinna

Sun-exposed skin.

### 3.1 Risk factors

Sun exposure, fair skin, male sex, older age and immunosuppression.

### 3.2 Squamous cell carcinoma

An ulcer, usually on the helix, with raised, everted edges. It spreads to the preauricular, postauricular, parotid and upper cervical nodes.



**FIGURE 3** Squamous cell carcinoma of the pinna.
Dhingra, fig. 16.3

## 3.3 Basal cell carcinoma

A nodule with a pearly, rolled, beaded edge and central crusting, usually behind the ear or on the helix; it rarely spreads to nodes but invades locally.

**WHERE THE BOOKS DIFFER · THE COMMONEST EAR MALIGNANCY**

| Dhingra | Lee |
|---|---|
| Emphasises squamous carcinoma on the helix. | Basal cell carcinoma is the commonest, at about 45%. |

## 3.4 Melanoma

A pigmented, irregular lesion; spreads early to the parotid and neck nodes.

## 3.5 Treatment

- **Squamous carcinoma:** excision with about 1 cm margins for small lesions; larger lesions, those within 1 cm of the canal, or with nodes need partial or total amputation

of the pinna with parotidectomy and neck dissection.

- **Basal cell carcinoma:** radiotherapy for superficial lesions not involving cartilage; excision (Mohs surgery where available) for deeper lesions.
- **Melanoma:** wedge excision if small on the helix; wider resection with parotid and neck surgery as staged.

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Osteomas and Exostoses of the External Auditory Canal**
Osteoma and exostoses

**Resection of the Temporal Bone**
Temporal bone resection

**Management of Skin Cancer of the Head and Neck**
Skin cancer of the head and neck

CHAPTER FOUR

# 4 Bony growths of the canal

Often just bone.

### OSTEOMA OR EXOSTOSES?

| | OSTEOMA | EXOSTOSES |
|---|---|---|
| **Number** | Single | Multiple |
| **Sides** | One | Usually both |
| **Shape** | Smooth, pedunculated | Sessile, broad-based |
| **Site** | Outer canal, at the bony-cartilaginous junction | Deep canal, near the drum |
| **Cause** | True benign tumour | Reaction to cold water (swimmers, surfers) |



**FIGURE 4** An osteoma arising from the anterior wall of the right ear canal (arrow).
Dhingra, fig. 16.4

WHERE THE BOOKS DIFFER · SITE OF EXOSTOSES

| Dhingra | Lee |
| --- | --- |
| Deep in the canal, near the drum. | On the anterior and posterior canal walls. |

### 4.1 Treatment

None if small. When they block the canal, trap wax or cause recurrent otitis externa: an osteoma can be removed by fracturing its pedicle or drilling; exostoses are drilled away through the canal or a postauricular approach.

**PEARL**

Use a drill, not a gouge and hammer, for exostoses: the facial nerve lies close to the posterior canal wall.

## 5 CHAPTER FIVE Tumours of the canal

Glands and skin.

## 5.1 Ceruminous gland tumours

A firm, skin-covered mass in the outer canal. Malignant ones (adenoid cystic carcinoma, ceruminous adenocarcinoma) outnumber benign ones about two to one. Treated by wide excision, with radiotherapy for malignant tumours.

## 5.2 Squamous cell carcinoma of the canal

Often follows long-standing chronic discharge. It presents with bloody discharge, pain and granulation that will not heal; facial palsy means advanced disease.

### 5.2.1 Staging (Pittsburgh)

- **T1:** confined to the canal, without bone erosion.
- **T2:** limited bone erosion, or soft tissue extension under 0.5 cm.
- **T3:** full-thickness bone erosion with soft tissue under 0.5 cm, or middle ear or mastoid involvement.
- **T4:** cochlea, petrous apex, medial wall, carotid canal, jugular foramen, dura, soft tissue over 0.5 cm, or facial palsy.

### 5.2.2 Treatment

Sleeve or lateral temporal bone resection, with parotidectomy and neck dissection as needed, and postoperative radiotherapy.

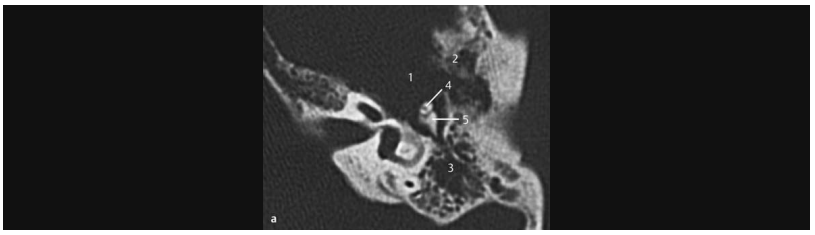
# 6 CHAPTER SIX Assessment

Biopsy and image.

- Biopsy any non-healing lesion of the pinna or canal.
- CT and MRI for deep or canal tumours.

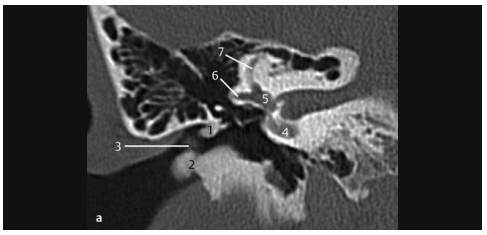
**IMAGING IN THE APP** 2 cases

Canal carcinoma and exostoses.



Malignancy of the External Auditory Canal Squamous Cell Carcinoma

Tumours · 2 images



Exostoses of the External Auditory Canal

Bone disease · 2 images

- Examine the anterior and posterior auricular, parotid and upper cervical nodes.

## 6.1 Differential diagnosis

Chondrodermatitis nodularis helicis, perichondritis, necrotising otitis externa and granulation from chronic otitis.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Squamous versus basal cell carcinoma** of the pinna, and their treatment.
- **Osteoma versus exostoses.**
- **Pittsburgh staging** of canal carcinoma.
- **Biopsy any non-healing lesion.**

## 7.2 Common pitfalls

**COMMON PITFALLS**

- Treating a non-healing helical ulcer with creams for months.

- Using a gouge on exostoses.
- Missing canal carcinoma in a patient with chronic discharge and new pain.

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CLINICAL · EAR

# Hearing tests: how to do and read them

Each hearing test answers a different question. Tuning forks give the type of loss at the bedside in seconds; the audiogram measures how much; the tympanogram shows what the middle ear is doing; and objective tests such as otoacoustic emissions and the brainstem response work when the patient cannot or will not respond.

**AT A GLANCE**

- Forks give type; audiogram gives amount; tympanogram shows the middle ear.
- Mask when ears differ.
- OAE and ABR suit those who can't respond.
- Reflexes help locate cochlear, VIII, VII and brainstem lesions.
- Read every audiogram in order: air, bone, compare ears, masking, fit the clinical picture.

## 1 CHAPTER ONE Overview

Which test answers which question.

### 1.1 The toolkit

**HEARING TESTS**

- BEDSIDE **Voice and tuning forks** conductive or sensorineural, roughly how much
- BEHAVIOURAL **Pure-tone and speech audiometry** thresholds and understanding
- MIDDLE EAR **Tympanometry and stapedial reflexes** drum, ossicles, tube, reflex arc
- OBJECTIVE **OAE, ABR, ASSR, ECochG** cochlea and neural pathway without a response

### 1.2 Choosing a test

Forks at the bedside; audiogram and tympanogram in clinic; OAE and ABR in newborns and uncooperative patients; ASSR for profound loss, beyond the range of

ABR.

CHAPTER TWO

# 2 Sound and the decibel

The language of audiology.

## 2.1 Decibel scales

- **dB SPL:** sound pressure relative to 20 µPa.
- **dB HL:** relative to average normal hearing at each frequency, so 0 dB HL is normal on the audiogram.
- **dB SL:** relative to the patient's own threshold.

## 2.2 Everyday levels

Whisper about 30 dB, conversation about 60 dB, a shout about 90 dB, discomfort about 120 dB and pain about 130 dB.

## 2.3 How the middle ear amplifies

The area ratio of the drum to the footplate (about 17 to 1) and the lever action of the ossicles (about 1.3 to 1) give a gain of about 22 times, around 25 dB. Without it, most sound energy would bounce off the fluid of the inner ear.

## 2.4 Where frequencies are heard

The base of the cochlea hears high frequencies, the apex low frequencies. The canal resonates at about 3000 Hz.

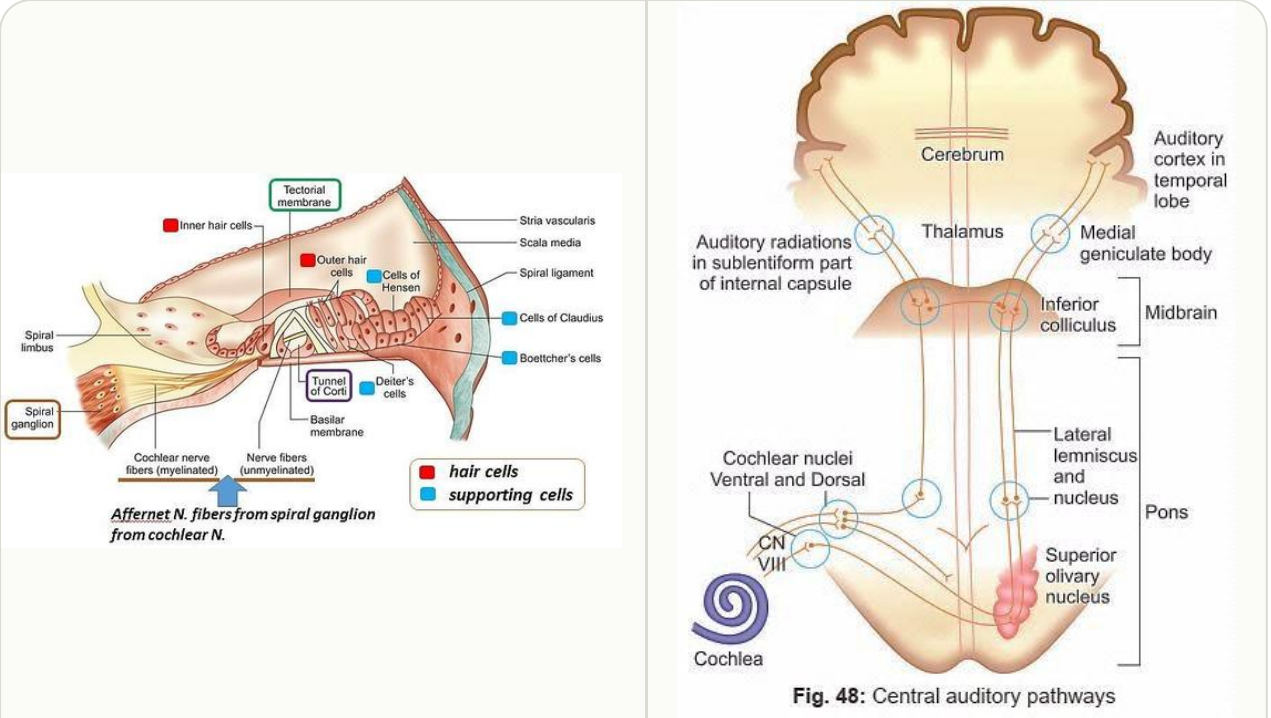



**FIGURE 1** The organ of Corti, and the central auditory pathway that the ABR waves trace.
Anatomy atlas · ENT Steps p. 77, ENT Steps p. 84

# 3 CHAPTER THREE Bedside tests

Voice and tuning forks.

## 3.1 Voice tests

A whisper heard at 6 m is taken as normal; the other ear is masked by rubbing the tragus.

## 3.2 Tuning fork tests

The 512 Hz fork is the routine choice: lower forks are felt as vibration, higher ones fade too quickly.

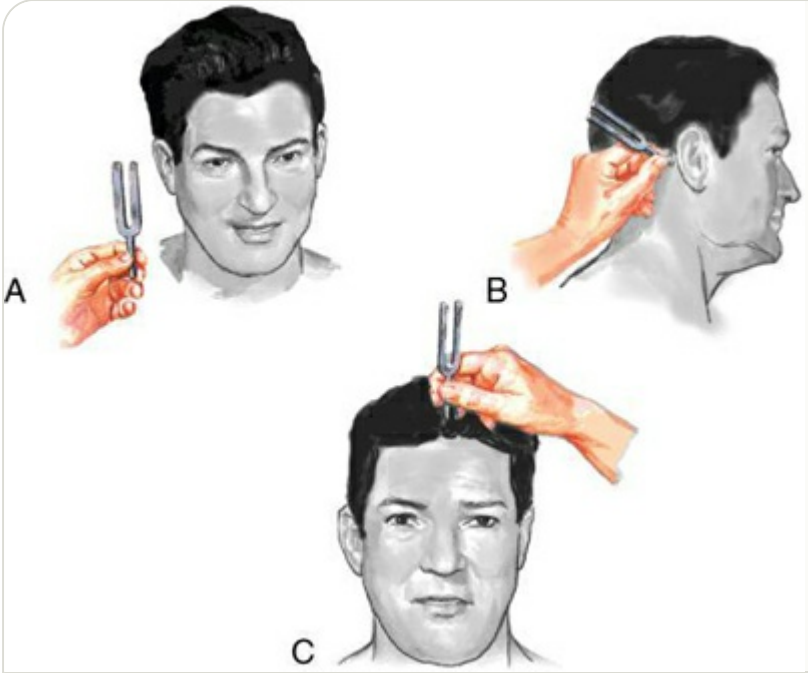



**FIGURE 2** Tuning fork tests: (A) air conduction, (B) bone conduction, (C) the Weber test. Dhingra, fig. 4.1

### 3.2.1 Rinne test

Compares air conduction with bone conduction in the same ear. *Positive* (air louder) is normal, and also seen in sensorineural loss; *negative* (bone louder) means a conductive loss of at least about 20 dB.

### 3.2.2 Weber test

The fork on the forehead or vertex lateralises to the poorer ear in a conductive loss and to the better ear in a sensorineural loss. It is sensitive, lateralising with quite small differences between the ears.

### 3.2.3 Absolute bone conduction and Schwabach tests

Compare the patient's bone conduction with the examiner's (with the canal occluded in the ABC test). Reduced in sensorineural loss.

### 3.2.4 Bing and Gellé tests

Bing: occluding the canal makes the tone louder in normal and sensorineural ears but not in a conductive loss. Gellé: raising canal pressure normally reduces hearing, but not with a fixed or broken chain; tympanometry has replaced it.

#### READING TUNING FORKS

| TEST | NORMAL | CONDUCTIVE | SENSORINEURAL |
|---|---|---|---|
| Rinne | Air louder | Bone louder | Air louder |

| TEST | NORMAL | CONDUCTIVE | SENSORINEURAL |
|---|---|---|---|
| Weber | Central | To poorer ear | To better ear |
| ABC | Same as examiner | Same | Reduced |
| Schwabach | Equal | Lengthened | Shortened |

### 3.2.5 Estimating the gap from Rinne

A negative Rinne at 256 Hz only suggests a gap of about 20 to 30 dB; at 256 and 512 Hz, 30 to 45 dB; at all three forks, 45 to 60 dB.

WHERE THE BOOKS DIFFER · GAP THAT TURNS EACH FORK NEGATIVE

**Dhingra**
15 dB at 256, 30 dB at 512 and 45 dB at 1024 Hz.

**Lee**
15, 25 and 35 dB.

**FALSE-NEGATIVE RINNE**

In a severe one-sided sensorineural loss, the fork on the mastoid is heard by the other cochlea, so bone seems louder. Mask the good ear with a Barany noise box, and use the Weber.

# 4 CHAPTER FOUR Pure-tone audiometry

Thresholds, frequency by frequency.

## 4.1 How it is done

In a sound-treated room, the softest tone the patient hears at each frequency from 250 to 8000 Hz is recorded, first by air conduction through headphones, then by bone conduction with a vibrator on the mastoid.



**FIGURE 3** A two-room audiometry set-up: the audiologist watches the patient through a glass partition.

Dhingra, fig. 4.2

## 4.2 The symbols

Frequency runs across the top from low (250 Hz) to high (8000 Hz), and loudness runs down the side, so a worse hearing threshold sits *lower* on the chart. By convention the right ear is red and the left ear blue. Air conduction is a circle (right) or a cross (left), joined by a line; bone conduction is an angle bracket pointing away from the ear tested, < for right and > for left, and a square bracket, [ or ], when the other ear was masked. An arrow below a symbol means no response at the machine's maximum output, which is about 120 dB for air but only about 60 to 70 dB for bone.

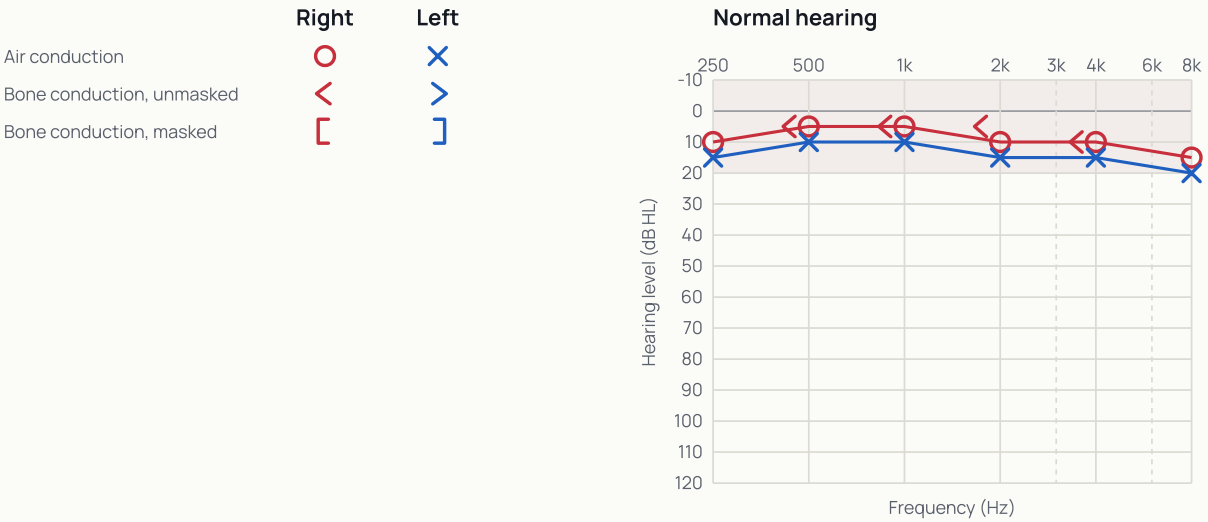



**FIGURE 4** The symbols, and a normal audiogram: every threshold sits in the shaded band of 20 dB HL or better, and air and bone conduction coincide.

Drawn for ENT Atlas

## 4.3 Reading it, step by step

Read every audiogram in the same order, so nothing is missed.

### READING A PURE-TONE AUDIOGRAM

**1 Check the chart**

→ right and left, masked or unmasked, reliability noted by the tester

**2 Air conduction**

→ normal (20 dB or better)

→ **how much loss**
mild, moderate, severe, profound

→ **which frequencies**
flat, sloping, rising, notched, cookie-bite

**3 Bone conduction**

→ **equal to air**
sensorineural

→ **better than air by more than 10 dB**

conductive component

→ **both raised with a gap**
mixed

4 Compare the ears

→ symmetrical

→ asymmetric by 15 dB or more at two adjacent frequencies, or 20 dB at one: MRI of the internal auditory meatus

5 Check masking

→ **ears differ by 40 dB or more on air, or any air–bone gap**
masked thresholds needed — was it done?

6 Fit the clinical picture

→ otoscopy, tuning forks and tympanometry must agree

→ if they do not, repeat or think non-organic

**Bone conduction is cochlear function.** The air–bone gap is the size of the conductive loss: a gap of up to about 60 dB is possible from the middle ear alone (complete ossicular discontinuity behind an intact drum), so a larger gap means the bone thresholds are wrong or the loss is mixed. Bone conduction can be *better than normal* in a third-window lesion such as superior canal dehiscence, giving an apparent conductive loss with normal tympanometry and present reflexes.

## 4.4 Masking

Sound presented to one ear can cross the skull and be heard by the other, so the non-test ear is masked with narrow-band noise. Interaural attenuation is 40 to 65 dB for air conduction with headphones (more with insert earphones) and close to 0 dB for bone, because the vibrator shakes the whole skull. Mask whenever the ears differ by 40 dB or more on air conduction, and for bone conduction whenever there is an air–bone gap in the test ear.

The usual method is the **plateau (Hood) technique**: noise in the non-test ear is raised in 10 dB steps while the threshold in the test ear is re-measured; while the threshold keeps rising with the noise, the other ear was answering, and once it stays the same across several steps (the plateau), the true threshold has been found. Without masking, a dead ear produces a **shadow curve**, a threshold that mirrors the good ear about 40 to 60 dB lower for air and almost level with it for bone. The **masking dilemma** arises in large bilateral conductive losses, as in bilateral atresia, when the noise needed to mask one ear crosses over and masks the test ear too; insert earphones and ABR by bone conduction help.

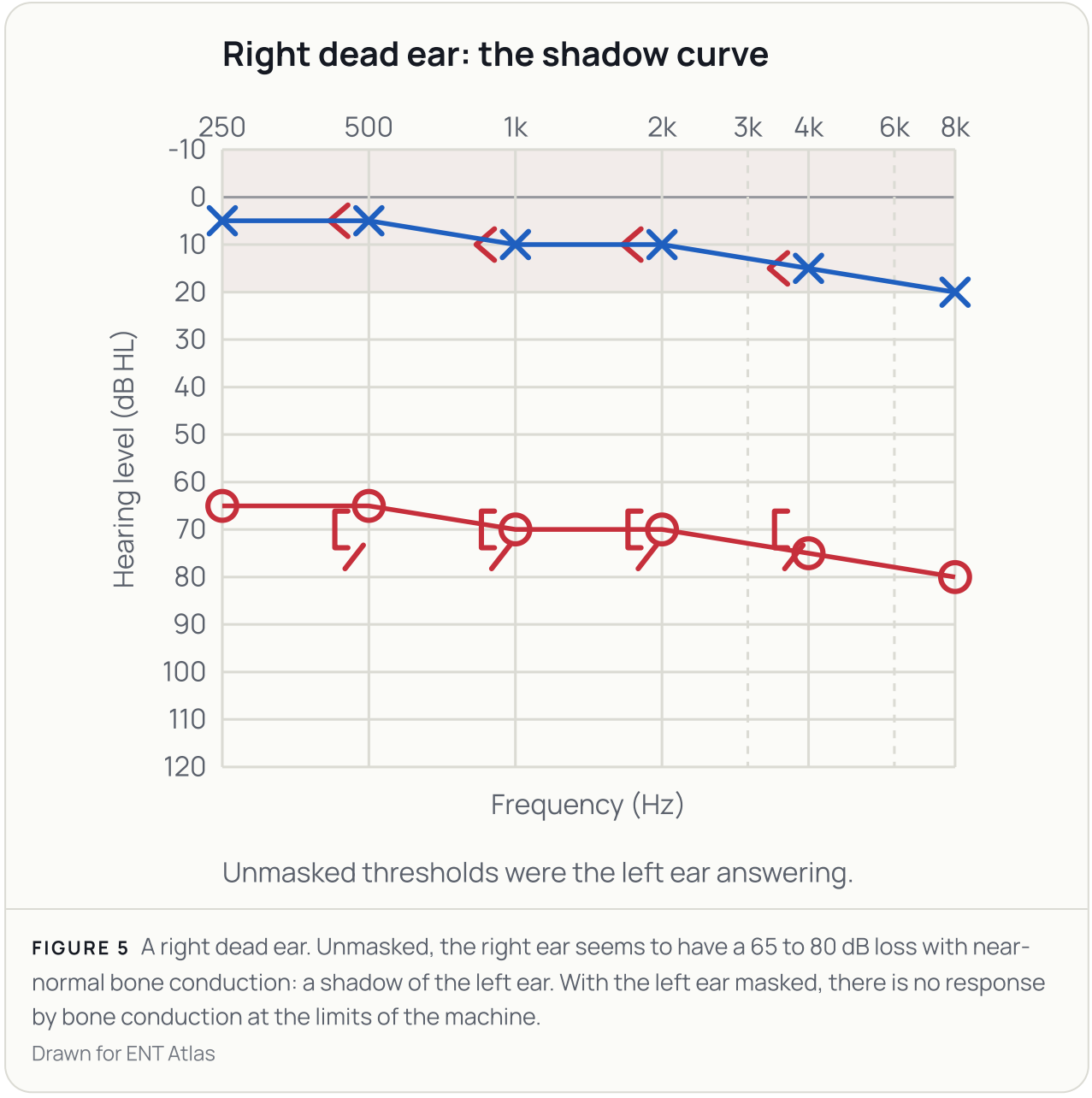



**FIGURE 5** A right dead ear. Unmasked, the right ear seems to have a 65 to 80 dB loss with near-normal bone conduction: a shadow of the left ear. With the left ear masked, there is no response by bone conduction at the limits of the machine.

Drawn for ENT Atlas

### 4.5 Repeatability

Normal repeat audiograms agree within about 5 dB, so a change of 10 dB or more at two frequencies is significant. Grades of loss use the average of 500, 1000, 2000 and 4000 Hz in the better ear (see the hearing loss article).

---

CHAPTER FIVE

# 5 Patterns on the audiogram

Worked examples.

### 5.1 Conductive and mixed losses

**Otosclerosis** produces a conductive loss, often worse in the low frequencies, with a characteristic dip of bone conduction of about 10 to 15 dB at 2 kHz, the **Carhart notch**. The notch is mechanical, not cochlear: it disappears after successful stapes surgery. Tympanometry is type A or As, and the stapedial reflexes are absent.

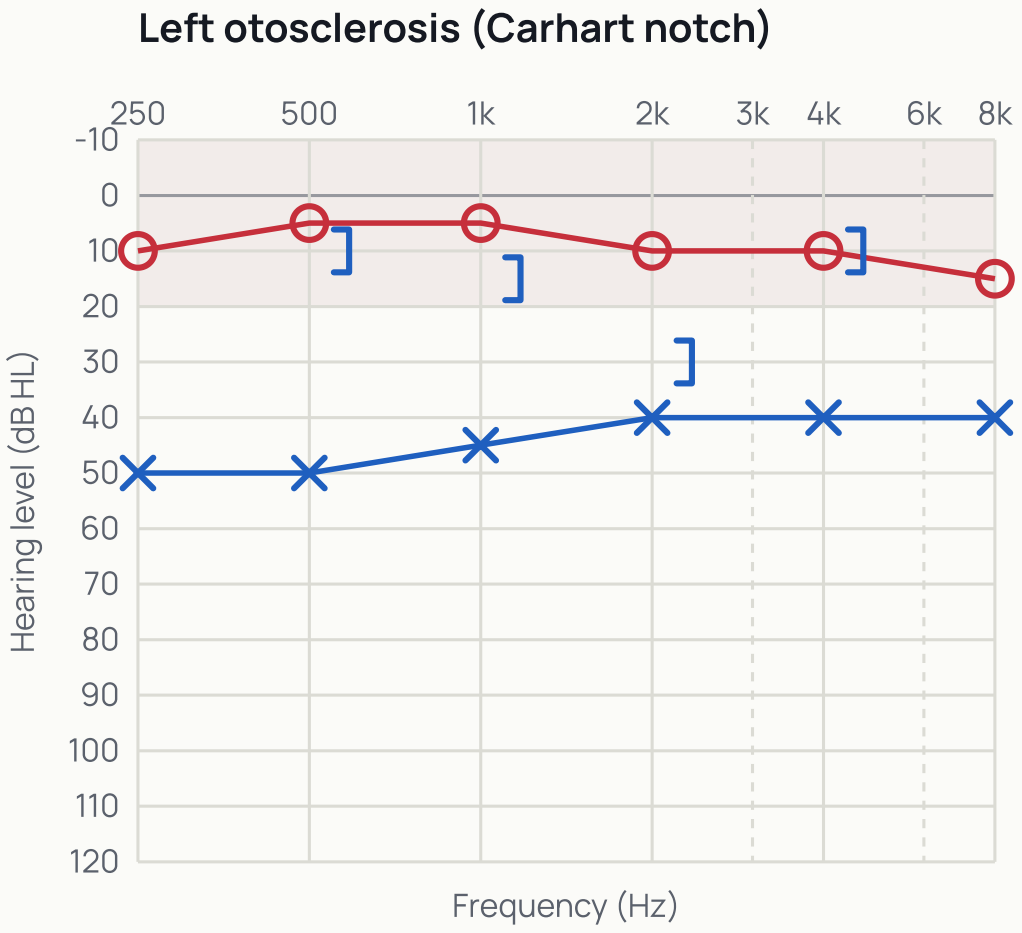



**FIGURE 6** Left otosclerosis. Air conduction is about 40 to 50 dB; masked bone conduction is near normal except for the dip at 2 kHz (Carhart notch). Air–bone gap about 35 dB.

Drawn for ENT Atlas

A **mixed loss** combines a raised bone conduction with an air–bone gap, as in chronic otitis media with cochlear damage from long-standing infection, otosclerosis with cochlear involvement, or a temporal bone fracture. Both components matter: the gap may be closed by surgery, but the sensorineural part sets the best possible result, so a patient with bone conduction of 40 dB will still need a hearing aid after a perfect tympanoplasty.

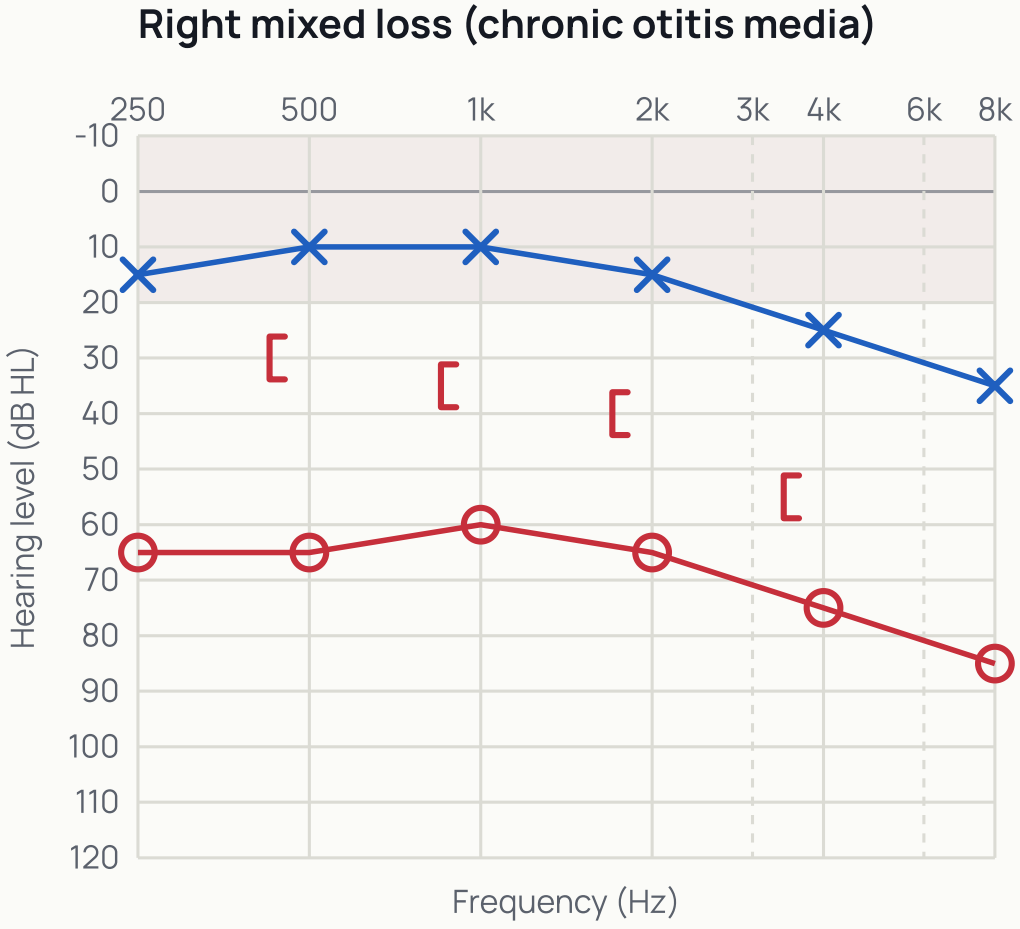



**FIGURE 7** Right mixed loss. Masked bone conduction is 30 to 55 dB (the sensorineural part), and air conduction 60 to 85 dB (the gap of about 30 dB is the conductive part).

Drawn for ENT Atlas

## 5.2 Sensorineural patterns

The *shape* of a sensorineural loss points to its cause.

### SENSORINEURAL SHAPES AND WHAT THEY SUGGEST

| SHAPE | TYPICAL CAUSE | CLUES |
|---|---|---|
| Bilateral, symmetrical, sloping to the high frequencies | Presbycusis, ototoxicity | Age; aminoglycosides, cisplatin |
| Notch at 3, 4 or 6 kHz with recovery at 8 kHz | Noise | Occupational or recreational noise; usually bilateral |
| Low-frequency (rising) loss, fluctuating | Ménière's disease, early hydrops | Episodic vertigo, fullness, roaring tinnitus |

| SHAPE | TYPICAL CAUSE | CLUES |
|---|---|---|
| Mid-frequency "cookie-bite" (U-shaped) loss | Genetic, congenital | Family history; stable |
| Flat, severe, sudden | Sudden sensorineural loss, viral, vascular | Onset within 72 hours |
| Asymmetric | Vestibular schwannoma until proven otherwise; also Ménière's, sudden loss, noise from one side (shooting) | MRI of the internal auditory meatus |
| Dead ear | Labyrinthitis, meningitis, fracture, surgery | Shadow curve without masking |

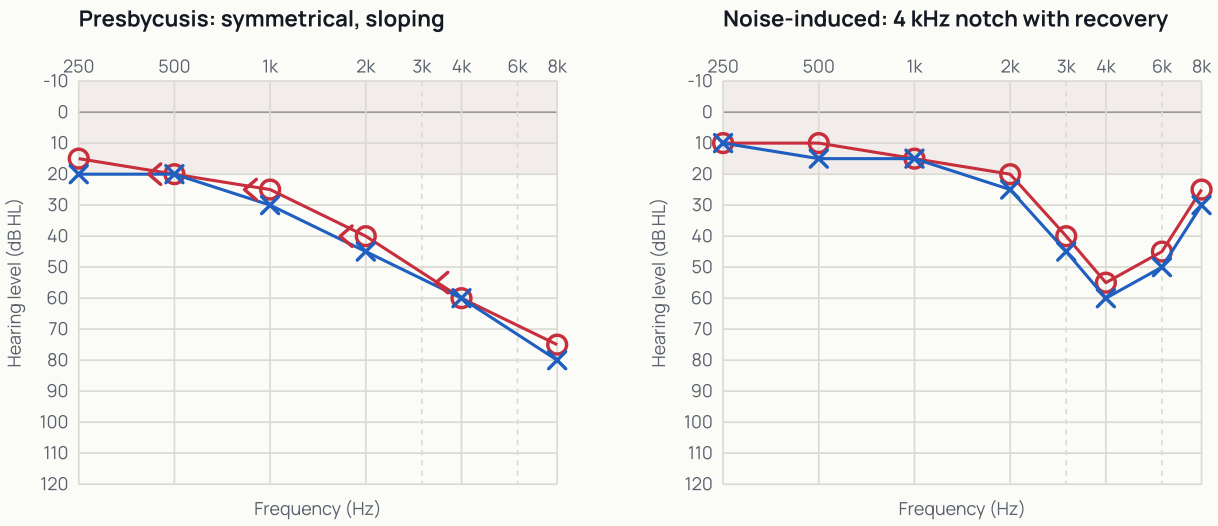
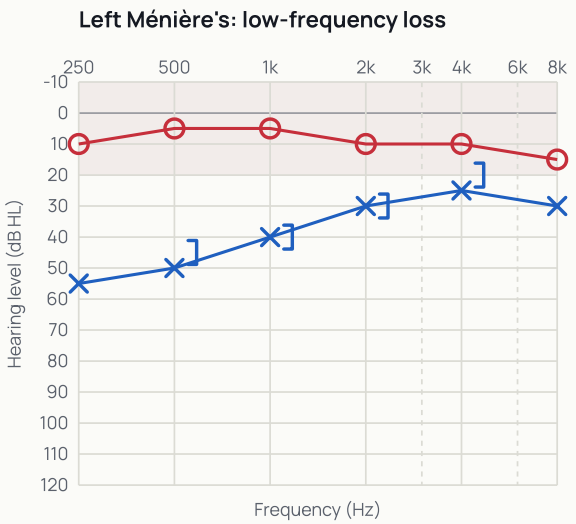



**FIGURE 8** Left: presbycusis, a symmetrical high-frequency sloping loss. Right: noise damage, a notch at 4 kHz that recovers at 8 kHz; the recovery separates noise from ageing, although both often coexist.

Drawn for ENT Atlas



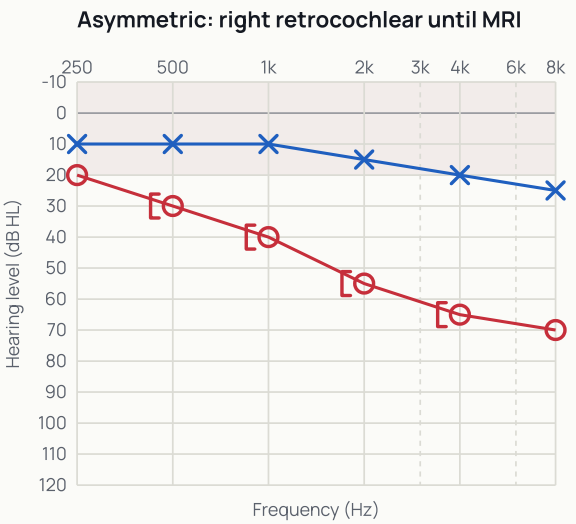




**FIGURE 9** Left: early Ménière's disease in the left ear, a low-frequency (rising) sensorineural loss that fluctuates between tests. Right: an asymmetric right sensorineural loss; with no explanation, this needs an MRI.

Drawn for ENT Atlas

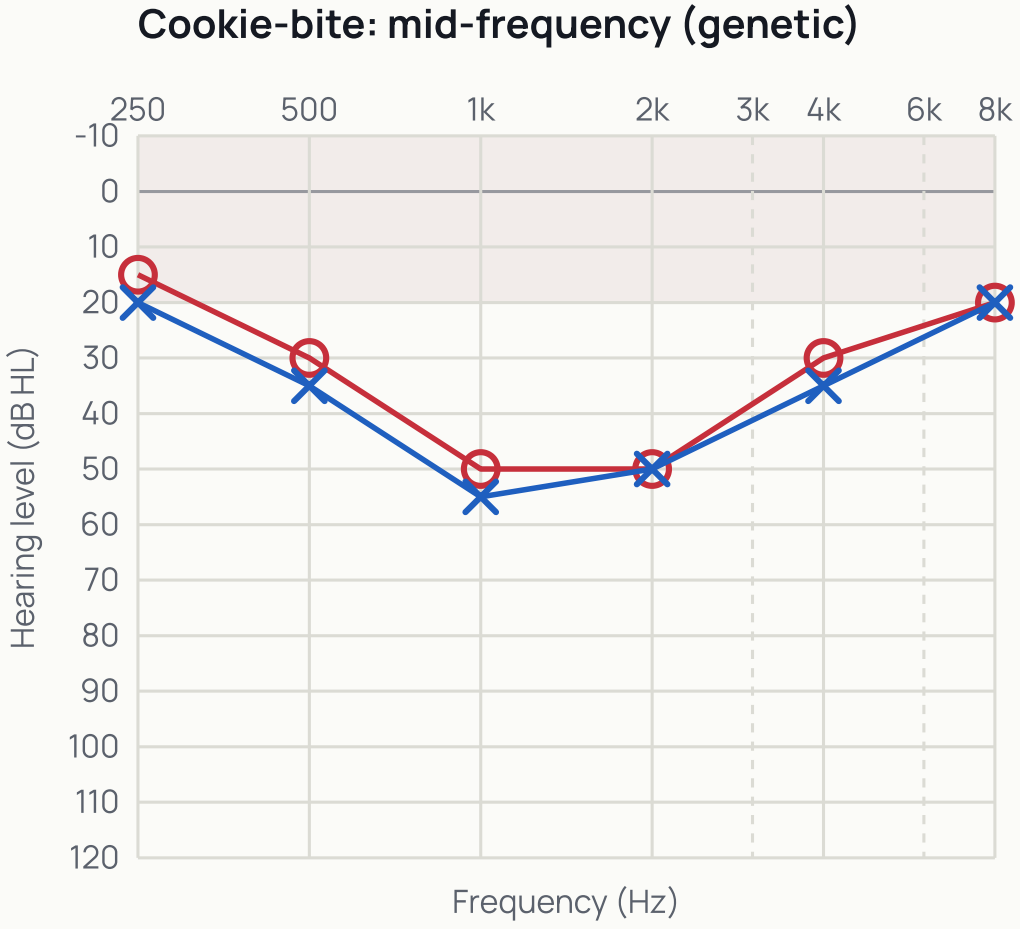



**FIGURE 10** A mid-frequency "cookie-bite" loss, typical of some genetic hearing losses. Drawn for ENT Atlas

## 5.3 When the audiogram does not fit

**Non-organic (functional) hearing loss** is suggested when thresholds vary between tests, the patient converses easily despite a "severe" loss, the speech reception threshold is much better than the pure-tone average, a unilateral "dead ear" shows no shadow curve, or acoustic reflexes are present at levels near or below the claimed threshold. Objective tests (otoacoustic emissions, ABR or cortical evoked response audiometry) establish the true thresholds; the Stenger test detects a feigned unilateral loss.

# 6 CHAPTER SIX Speech audiometry

Understanding, not just hearing.

## 6.1 Speech reception threshold

The level at which half of two-syllable spondee words are repeated correctly; it lies within 10 dB of the pure-tone average.

## 6.2 Speech discrimination

Single-syllable words presented 30 to 40 dB above the SRT. Good in conductive loss, reduced in cochlear loss, very poor in retrocochlear loss.

## 6.3 Roll-over

Discrimination falling as loudness increases, a sign of a retrocochlear lesion.

## 6.4 Reading the speech audiogram

The speech audiogram plots the percentage of words repeated correctly against loudness. In a **conductive loss** the curve has a normal shape but is shifted to the right: once the sound is loud enough, the patient understands everything. In a **cochlear loss** the curve is shifted and may never reach 100%, because the damaged cochlea distorts the sound. In a **retrocochlear loss** the maximum is poor and falls again at high levels (roll-over), and speech discrimination is much worse than the pure-tone thresholds would suggest. Speech scores matter clinically: they predict hearing aid benefit and help select candidates for cochlear implants, which are assessed with sentence tests in quiet and in noise.

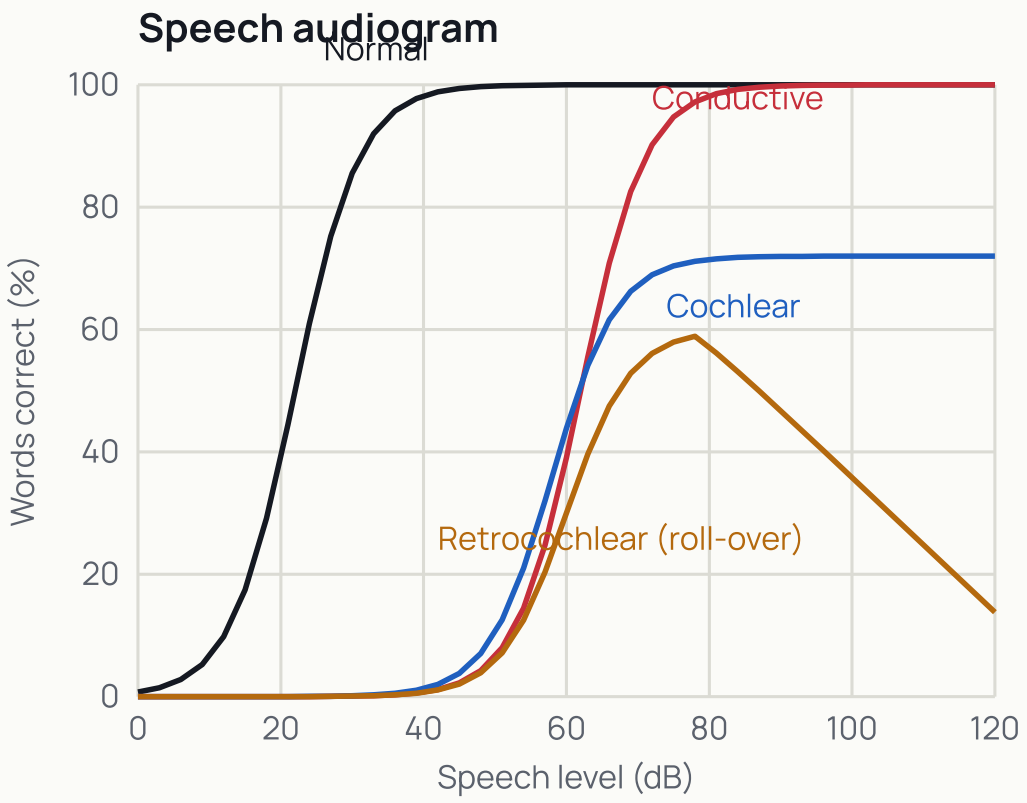



**FIGURE 11** Speech audiogram patterns: normal, conductive (shifted but reaching 100%), cochlear (shifted and reduced maximum) and retrocochlear (poor maximum with roll-over).
Drawn for ENT Atlas

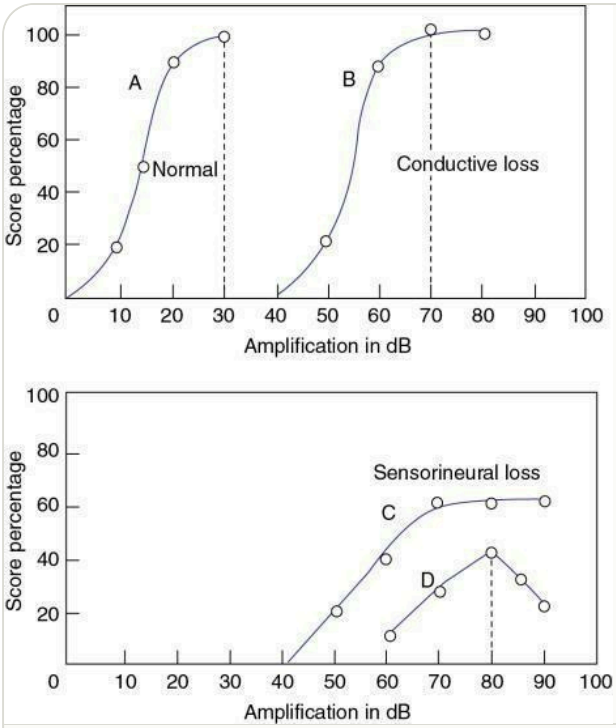



**FIGURE 12** Speech audiogram.
Dhingra, fig. 4.3

# 7 CHAPTER SEVEN Impedance audiometry

The middle ear and reflexes.

## 7.1 Tympanometry

A probe in the canal changes the pressure while measuring how much sound is reflected from the drum. The peak of compliance shows the middle ear pressure and the mobility of the system.

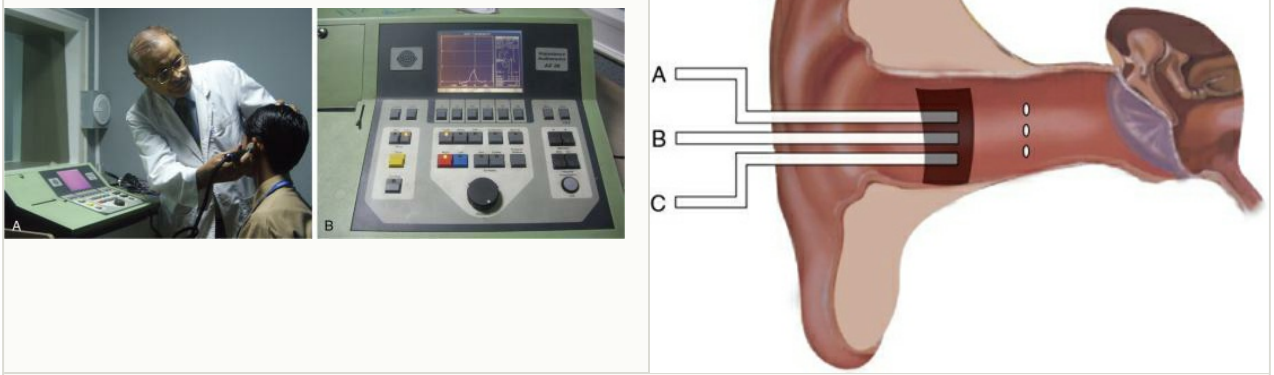



**FIGURE 13** First, impedance audiometry in progress. Second, the principle of the impedance audiometer.

Dhingra, figs 4.4 and 4.5

## TYMPANOGRAM TYPES

| TYPE | FINDING | THINK OF |
|---|---|---|
| A | Normal peak | Normal ear |
| As | Low peak | Otosclerosis, malleus fixation |
| Ad | High peak | Ossicular discontinuity, lax drum |
| B | Flat | Fluid, thick drum; perforation or patent grommet if the volume is high |
| C | Peak below −100 mm H2O | Retraction, tube dysfunction |

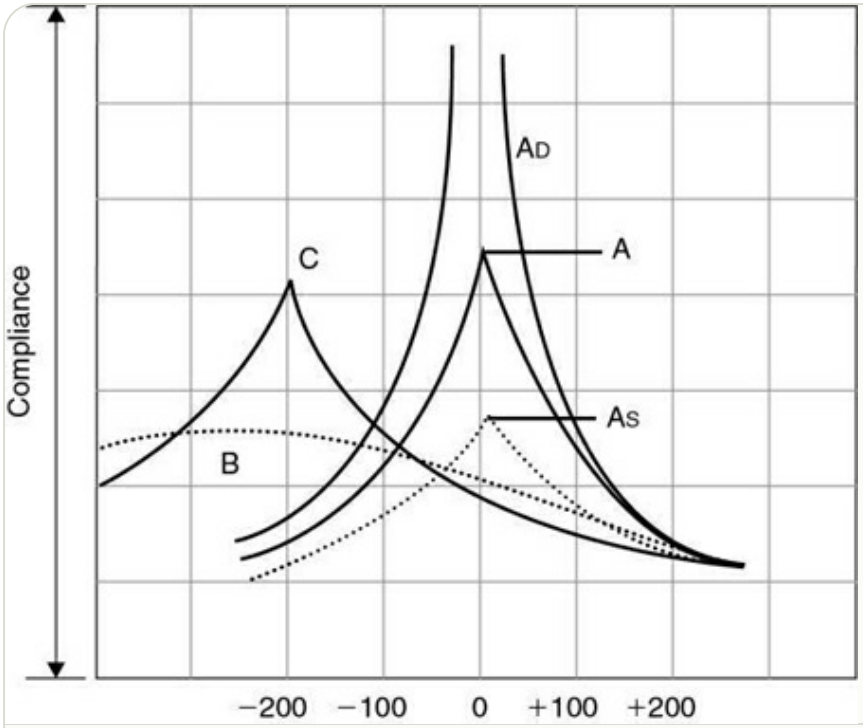



**FIGURE 14** Types of tympanogram.
Dhingra, fig. 4.6

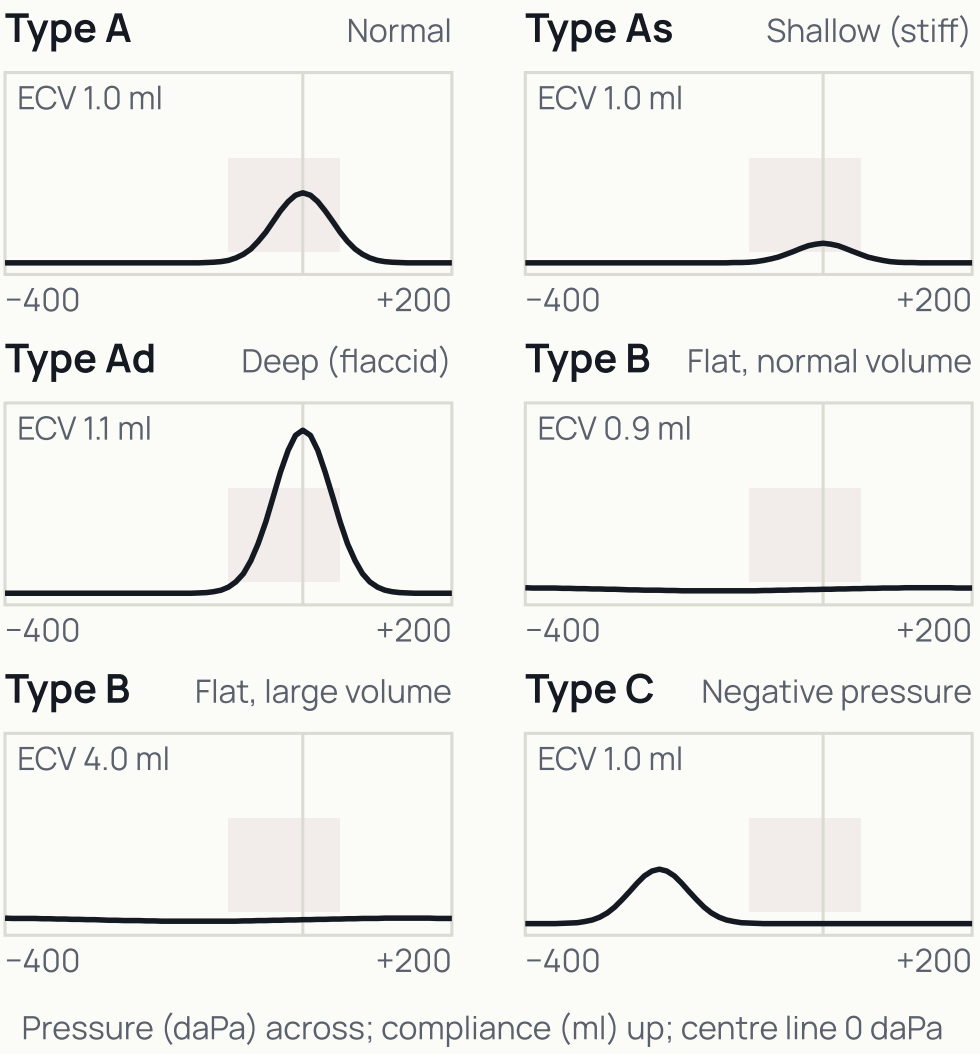



**FIGURE 15** Reading a tympanogram: the peak height (compliance), the peak position (middle ear pressure) and the ear canal volume (ECV). The shaded box is the normal range, roughly −100 to +50 daPa and 0.3 to 1.5 ml.

Drawn for ENT Atlas

## 7.2 Reading a tympanogram

Read three numbers, not just the shape. The **peak pressure** shows the middle ear pressure; normal is about −100 to +50 daPa (1 daPa is about 1 mm $H_2O$). The **peak compliance** shows how mobile the drum and chain are; normal in adults is about 0.3 to 1.5 ml. The **ear canal volume** is the air between the probe and the drum; normal is about 0.6 to 1.5 ml in adults and 0.3 to 0.9 ml in children.

### A FLAT TYMPANOGRAM: WHAT THE VOLUME TELLS YOU

| EAR CANAL VOLUME | MEANING |
|---|---|
| Normal | Middle ear fluid or a thickened, immobile drum |

| EAR CANAL VOLUME | MEANING |
|---|---|
| High (over about 2 ml adults, 1 ml children) | Perforation or a patent ventilation tube |
| Low | Wax or debris against the drum, or the probe against the canal wall |

An ear canal volume over about 2.5 mL in adults (2 mL in children) is a strong sign of a perforation or patent grommet. Infants under six months need a 1000 Hz probe tone, because the soft canal wall makes the standard 226 Hz test unreliable. A patulous Eustachian tube produces a trace that moves with breathing; a glomus tumour can produce pulsations in time with the heartbeat.

## 7.3 Stapedial (acoustic) reflex

A loud sound, 70 to 100 dB above threshold, contracts the stapedius in both ears.

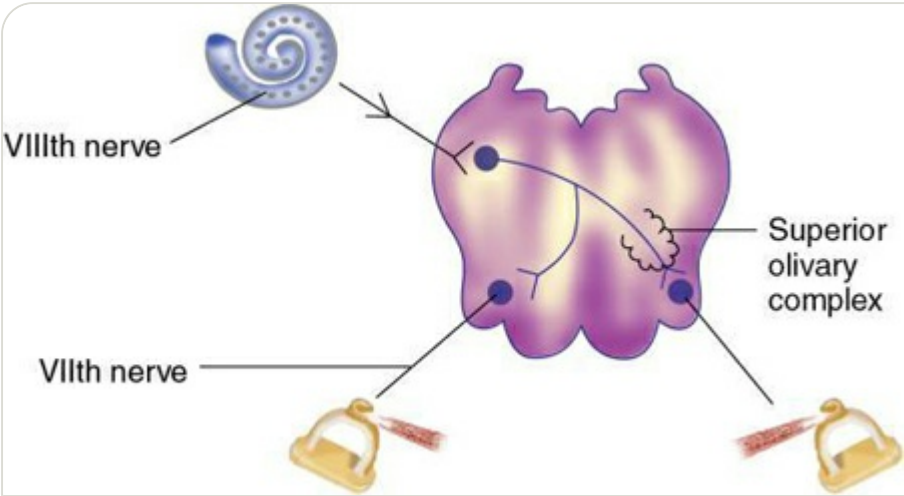



**FIGURE 16** The acoustic reflex pathway.
Dhingra, fig. 4.7

### 7.3.1 What reflexes tell you

Reflexes are measured in four conditions: ipsilateral and contralateral for each ear. Record the **stimulus ear** and the **probe ear** carefully: a contralateral right reflex means sound in the right ear, response measured in the left.

- A reflex at only 40 to 60 dB above threshold: recruitment, a cochlear lesion.
- Reflex decay to half within 10 seconds: an eighth nerve lesion.
- Absent with normal hearing: a facial nerve lesion above the nerve to stapedius.

- Present on the same side but absent on the other: a brainstem lesion.
- Absent: stapes fixation or any middle ear disease.

**REFLEX PATTERNS AND THE SITE OF THE LESION**

| FINDING | SITE |
|---|---|
| Absent whenever the probe is in the affected ear | Middle ear on that side (the probe cannot detect a reflex) |
| Absent whenever the affected ear is stimulated | Severe cochlear loss or eighth nerve lesion on that side |
| Absent whenever the probe is in the affected ear, with normal hearing | Facial nerve on that side, above the nerve to stapedius |
| Ipsilateral reflexes present, both contralateral reflexes absent | Brainstem lesion crossing the midline |
| Present with an apparent severe loss | Non-organic loss |

# 8 CHAPTER EIGHT Site-of-lesion tests

## Cochlear or retrocochlear?

Older behavioural tests, now largely replaced by MRI:

- **Alternate binaural loudness balance:** recruitment in cochlear loss.
- **Short increment sensitivity index (SISI):** high scores in cochlear loss.
- **Tone decay:** over 25 dB in retrocochlear loss.
- **Bekesy audiometry.**

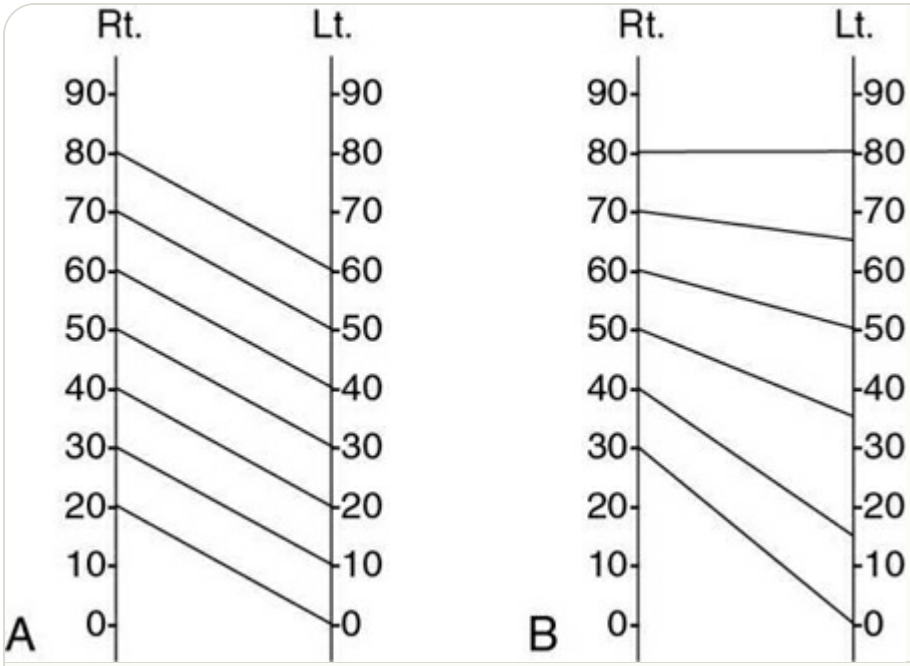



**FIGURE 17** Alternate binaural loudness balance test for recruitment.
Dhingra, fig. 4.8

# 9 CHAPTER NINE Objective tests

When the patient cannot respond.

## 9.1 Otoacoustic emissions

Sounds produced by the outer hair cells and recorded in the canal. Transient emissions disappear with losses over about 30 dB, distortion-product emissions over about 50 dB. Absent with middle ear fluid or debris; present in auditory neuropathy. Used for newborn screening.

Results are reported as **pass** (clear response) or **refer**. A refer is not a diagnosis of deafness: it may mean vernix or fluid in the canal, a noisy baby or environment, or a true loss, which is why screening repeats the test and then uses automated ABR. Because emissions reflect outer hair cell function, they can show early damage from noise or ototoxic drugs before the audiogram changes, and they monitor patients on cisplatin.

## 9.2 Auditory brainstem response

Five to seven waves recorded from the scalp within 10 ms of a click: I and II from the eighth nerve, III the cochlear nucleus, IV the superior olive, V the lateral lemniscus and inferior colliculus (*E. COLI*). Used for thresholds in infants (pass at about 30 to 40 dB nHL) and to look for retrocochlear lesions, though it misses small tumours. Unaffected by sleep and sedation.

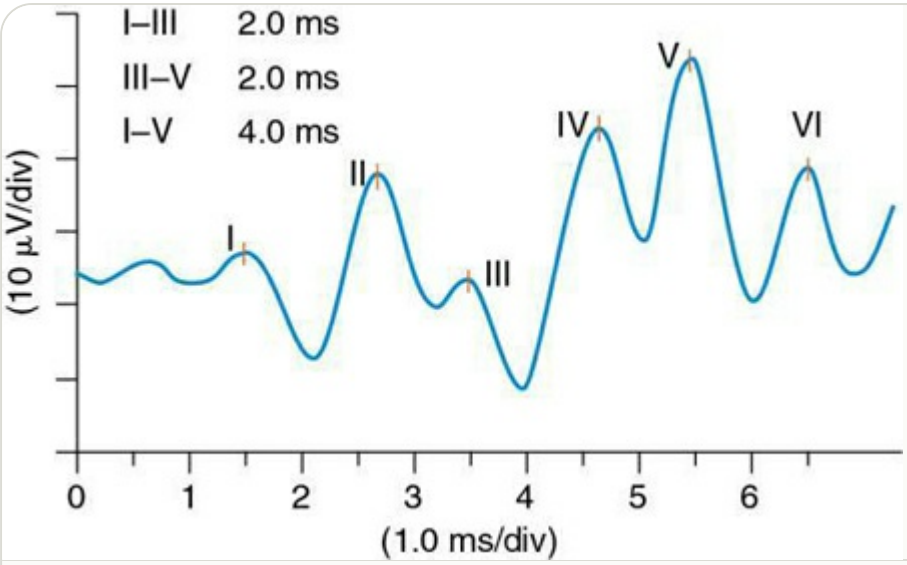



**FIGURE 18** Brainstem auditory evoked potentials.
Dhingra, fig. 4.10

## 9.3 Reading an ABR

Look at three things. **Wave latencies:** wave V normally appears at about 5.5 to 6 ms with a loud click. **Interpeak intervals:** I–III about 2 ms, III–V about 2 ms, and I–V about 4 ms; the I–V interval measures conduction along the nerve and brainstem and is *independent of the middle ear*. **Threshold:** the softest level at which wave V can still be identified, used to estimate hearing in infants. In a conductive loss all the waves are delayed but the I–V interval is normal; in a cochlear loss wave I may be small or absent and the latency of V falls steeply as loudness rises (recruitment); in a retrocochlear lesion the I–V interval is prolonged or the waves after I are lost, and wave V on the affected side is delayed compared with the other ear.

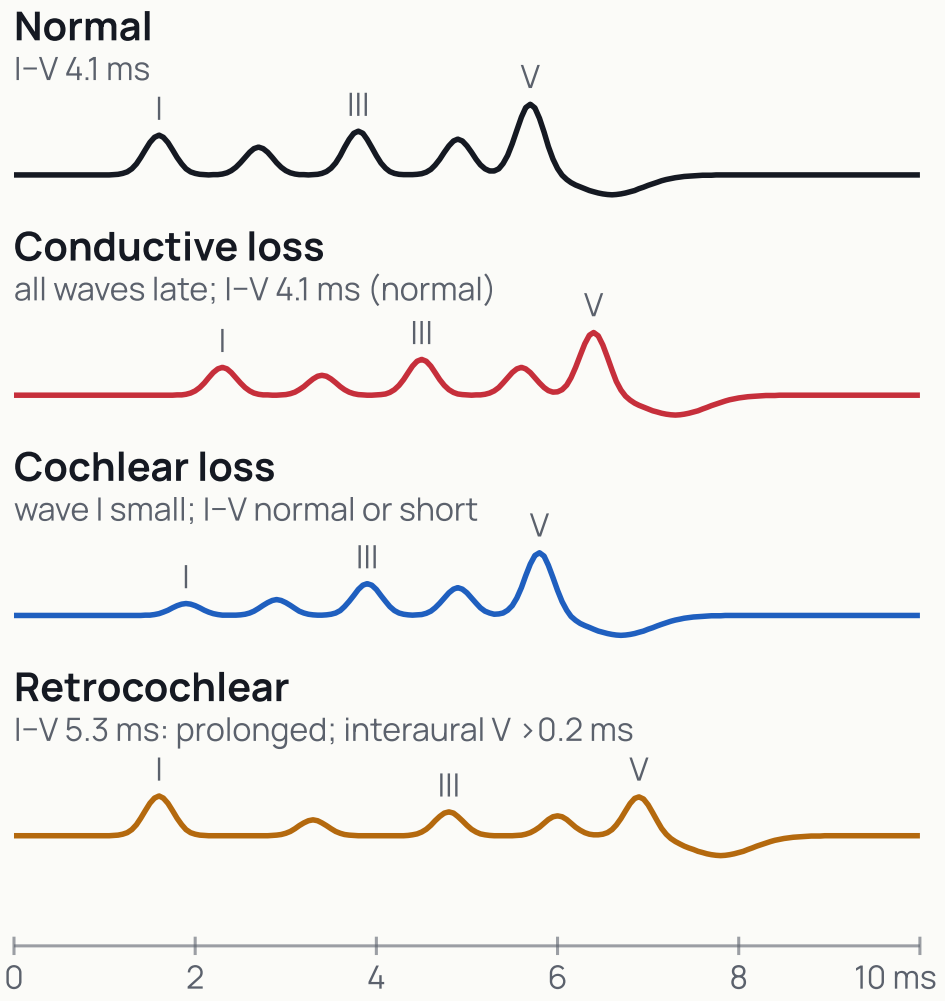



**FIGURE 19** ABR traces. Conductive loss delays every wave; a retrocochlear lesion prolongs the I–V interval. Values are typical, not normative.

Drawn for ENT Atlas

**WHERE THE BOOKS DIFFER · ABNORMAL INTERAURAL WAVE V DIFFERENCE**

| Dhingra | Lee |
|---|---|
| Over 0.2 ms. | 0.4 to 0.5 ms. |

## 9.4 Auditory steady-state response

Frequency-specific thresholds into the profound range, beyond the reach of ABR (about 80 dB).

## 9.5 Electrocochleography

Records the summating potential, action potential and cochlear microphonic. An enlarged SP to AP ratio suggests Ménière's disease (see that topic).

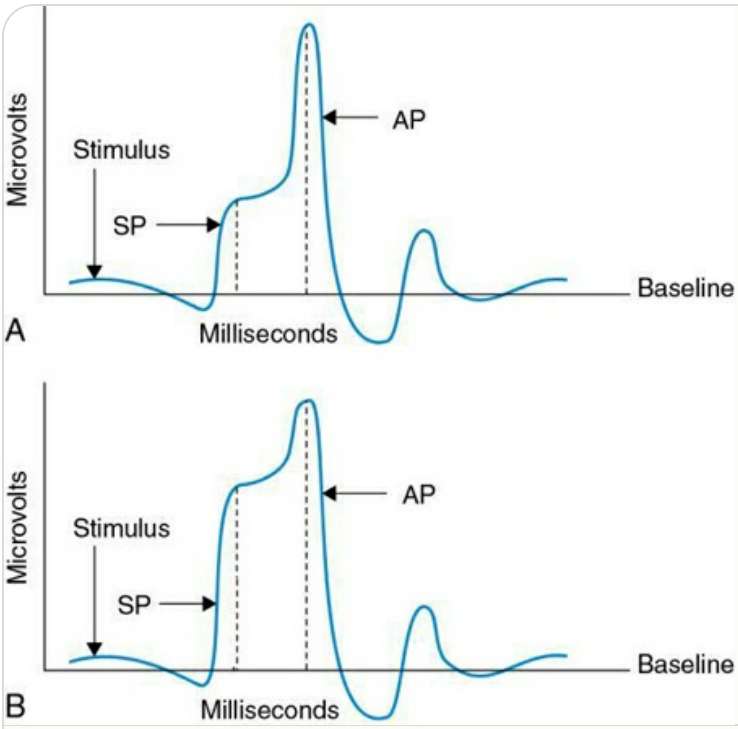



**FIGURE 20** Electrocochleography: normal ear and an ear with Ménière's disease.
Dhingra, fig. 4.9

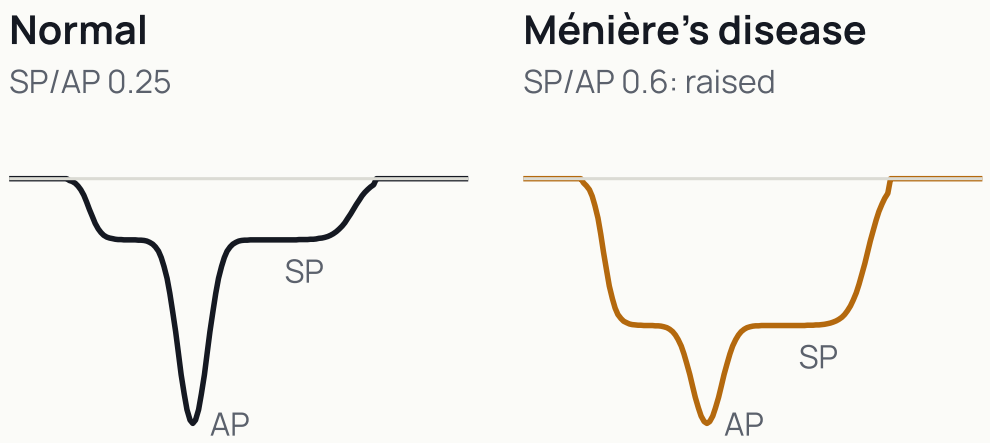



**FIGURE 21** Reading electrocochleography: the summating potential (SP) is the shoulder before the sharp action potential (AP). An SP/AP ratio above about 0.4 to 0.5 suggests endolymphatic hydrops; a normal result does not exclude it.

Drawn for ENT Atlas

## 9.6 Auditory neuropathy

Present otoacoustic emissions and cochlear microphonic with an absent or abnormal ABR: the hair cells work but the nerve does not carry the signal synchronously.

# 10 CHAPTER TEN Putting the tests together

Patterns in common conditions.

### THE TEST BATTERY IN COMMON CONDITIONS

| CONDITION | TUNING FORKS | AUDIOGRAM | TYMPANOGRAM | REFLEXES | OTHER |
|---|---|---|---|---|---|
| Wax or otitis media with effusion | Rinne negative, Weber to the ear | Conductive, flat or low-frequency | B, normal volume (effusion) | Absent | OAE absent |

| CONDITION | TUNING FORKS | AUDIOGRAM | TYMPANOGRAM | REFLEXES | OTHER |
|---|---|---|---|---|---|
| Perforation | Rinne negative if gap over about 20 dB | Conductive | B, high volume | Absent | — |
| Otosclerosis | Rinne negative | Conductive, Carhart notch | A or As | Absent | Bone better than normal at 2 kHz after surgery |
| Ossicular discontinuity | Rinne negative | Conductive, up to 60 dB | Ad | Absent | — |
| Superior canal dehiscence | Weber to the ear | Low-frequency gap, bone better than normal | A | Present | VEMP thresholds low; CT |
| Presbycusis | Rinne positive | Bilateral high-frequency sensorineural | A | Present, may be at reduced levels | Speech scores fair |
| Ménière's disease | Weber away | Low-frequency sensorineural, fluctuating | A | Present, recruitment | ECochG SP/AP raised |
| Vestibular schwannoma | Weber away | Asymmetric sensorineural | A | Absent or decay | Poor speech, roll-over; ABR I–V prolonged; MRI |
| Auditory neuropathy | Variable | Any, often fluctuating | A | Absent | OAE present, ABR absent or abnormal |
| Non-organic loss | Inconsistent | Variable, no shadow curve | A | Present near claimed threshold | SRT better than PTA; objective tests normal |

# 11 CHAPTER ELEVEN Testing children

By age.

- Birth to six months: behavioural observation, OAE and ABR.
- Six to thirty months: visual reinforcement audiometry.
- From about thirty months: play audiometry.
- Older children: conventional audiometry. Normal thresholds in children are 15 dB or better.

**Visual reinforcement audiometry** conditions the child to turn towards a sound, rewarded by a light or puppet, and uses insert earphones or free field; free-field results reflect the better ear only. **Play audiometry** turns the test into a game, such as putting a peg in a board each time a sound is heard. Tympanometry and otoacoustic emissions support behavioural tests at every age, and ABR under sleep or sedation is used when behavioural results are unreliable.

# 12 CHAPTER TWELVE For the examination

What the examiner wants to hear.

## 12.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Rinne and Weber:** technique and interpretation, and the false-negative Rinne.
- **Masking:** when and why; the plateau method, shadow curve and masking dilemma.
- **Reading an audiogram aloud:** describe each ear, the type, degree and shape of loss, and what you would do next.
- **Audiogram patterns:** Carhart notch, noise notch, low-frequency loss, asymmetry, dead ear.
- **Tympanogram types** and the ear canal volume.
- **Reflex patterns** and the site of the lesion.
- **Reading an ABR:** latencies, the I–V interval and wave V asymmetry.
- **Uses of the stapedial reflex.**

- **OAE versus ABR,** and the waves of the ABR.
- **Auditory neuropathy.**

## 12.2 Common pitfalls

COMMON PITFALLS

- Reading a false-negative Rinne as a conductive loss.
- Forgetting to mask.
- Calling a flat tympanogram fluid without checking the canal volume.
- Taking a shadow curve for a real moderate loss in a dead ear.
- Planning stapes surgery for an air–bone gap with present reflexes before excluding a third window.
- Accepting a severe loss in a patient who follows quiet conversation.

Written for this app from Oxford Textbook of Otolaryngology ch. 5, Oxford Specialist Handbook ch. 14, Oxford Handbook of ENT ch. 4, Dhingra ch. 3, 4 and K. J. Lee ch. 14, 15. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Vestibular testing: bedside and laboratory

Vestibular tests tell you which side is weak and whether the problem is peripheral or central; they never tell the whole story. Normal results do not exclude vestibular disease, so every test should be chosen and read with the history in mind. The bedside examination comes first, and often it is enough.

**AT A GLANCE**

- Bedside tests come first: nystagmus, head impulse, Dix-Hallpike, gait.
- Calorics test the lateral canal; VEMP tests the otoliths.
- Fixation failure and direction change suggest central.
- Normal tests don't exclude vestibular disease.
- vHIT gain ≥0.8 is normal; catch-up saccades mean a weak canal; oVEMP is the best screen for dehiscence.

## 1 CHAPTER ONE Overview

What the balance system is made of.

### 1.1 The end organs

The three semicircular canals sense angular (rotational) acceleration; the two otolith organs, utricle and saccule, sense linear acceleration and head tilt.

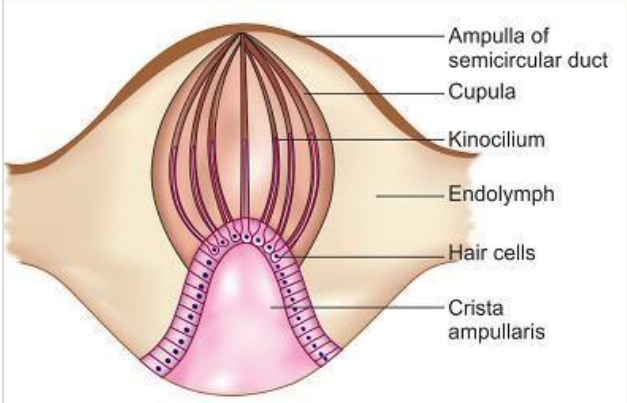



**Fig. 41:** Crista, hair cells and cupula. Cut section of ampulla of semicircular duct

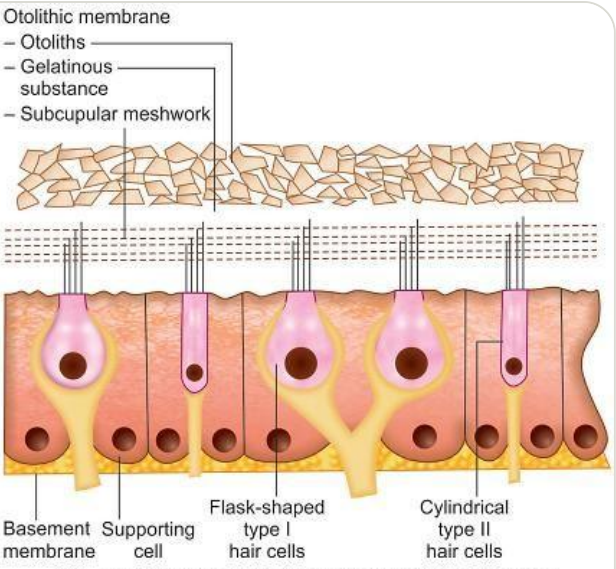



**Fig. 43:** Macula of otolith organs: utricle and saccule

**FIGURE 1** The crista ampullaris of a canal, and the macula of an otolith organ.

Anatomy atlas · ENT Steps p. 78, ENT Steps p. 80

## 1.2 The nerves

The *superior vestibular nerve* supplies the lateral and superior canals and the utricle; the *inferior vestibular nerve* supplies the posterior canal and the saccule.

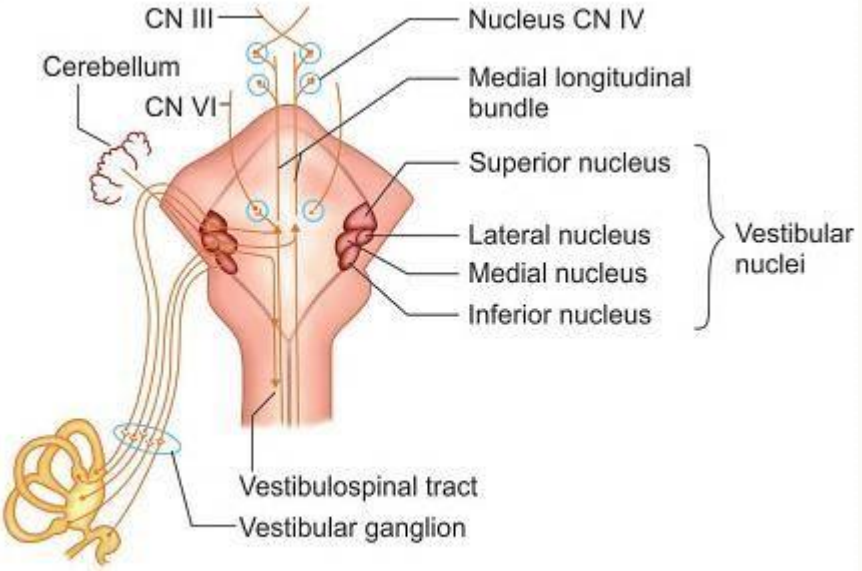



**Fig. 49:** Vestibular pathways

**FIGURE 2** Central vestibular pathways.

Anatomy atlas · ENT Steps p. 86

## 1.3 Physiology to remember

At rest both labyrinths fire equally. If the right labyrinth fails, the brain reads the imbalance as turning to the left: the eyes drift slowly to the right and flick back to the left. So nystagmus beats *away* from a dead labyrinth (a paretic lesion) and *towards* an irritated one.

## 1.4 Ewald's laws

The eyes move in the plane of the stimulated canal. In the lateral canal, flow towards the ampulla (ampullopetal) stimulates more than flow away from it; in the vertical canals the reverse is true.

## 1.5 What each test measures

WHAT EACH TEST MEASURES

| TEST | MEASURES |
|---|---|
| Caloric test | Each lateral canal, at low frequency |
| Rotary chair | Both lateral canals together |
| Video head impulse test | Each of the six canals, at high frequency |
| Cervical VEMP | Saccule and inferior vestibular nerve |
| Ocular VEMP | Utricle and superior vestibular nerve |
| Posturography | How the patient uses vestibular, visual and proprioceptive input |

# 2 CHAPTER TWO
# Bedside examination

First, and often enough.

## 2.1 Spontaneous nystagmus

Look in the primary position and in each direction of gaze, with and without fixation (Frenzel glasses remove fixation). Peripheral nystagmus is horizontal-torsional, suppressed by fixation, and stronger looking towards the fast phase.

### 2.1.1 Degrees (Alexander's law)

- **First degree:** only when looking towards the fast phase.
- **Second degree:** also looking straight ahead.
- **Third degree:** in all directions of gaze.

### 2.1.2 Central patterns

- **Downbeat:** craniocervical junction (Chiari malformation, cerebellar degeneration).
- **Upbeat:** brainstem.
- **Pure torsional:** brainstem.
- **Pendular:** multiple sclerosis and congenital nystagmus.

## 2.2 Head impulse test

A rapid head turn; a catch-up saccade shows a failed vestibulo-ocular reflex on that side.

## 2.3 Dynamic visual acuity

Losing more than five lines on a chart while the head is shaken suggests bilateral vestibular loss.

## 2.4 Positional tests

### 2.4.1 Dix-Hallpike manoeuvre

With the patient sitting, turn the head 45 degrees to one side, then lie them back quickly with the head 30 degrees below horizontal. Watch for latency, duration, direction and fatigue.

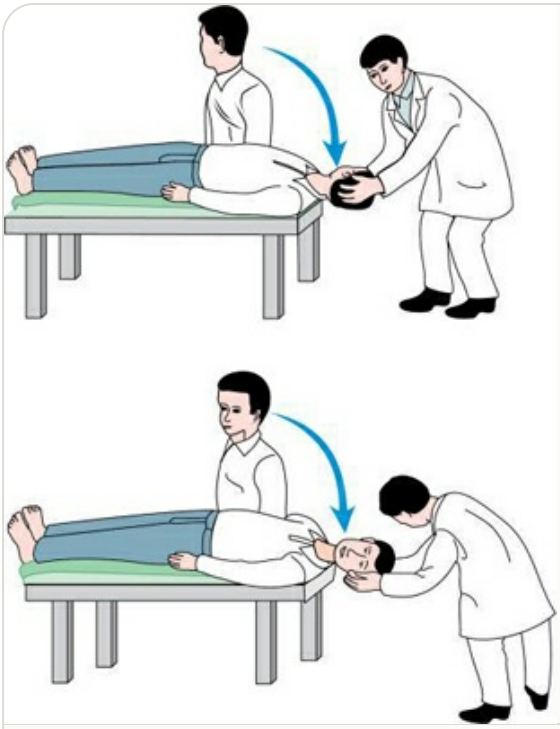


**FIGURE 3** The Dix-Hallpike manoeuvre.
Dhingra, fig. 6.1

POSITIONAL NYSTAGMUS

| FEATURE | PERIPHERAL | CENTRAL |
|---|---|---|
| Latency | 2–20 seconds | None |
| Duration | Under a minute | Over a minute |
| Direction | Fixed, towards the undermost ear | Changes |
| Fatigue | Yes | No |
| Vertigo | Severe | None or slight |

## 2.5 Fistula test

Pressure on the tragus or with a Siegle speculum causes nystagmus when the labyrinth is open: lateral canal erosion, a fenestration cavity, superior canal dehiscence, or a window leak. It needs a working labyrinth: a dead ear, or

cholesteatoma sealing the fistula, gives a false negative. *Hennebert's sign* is a positive test without a fistula, in congenital syphilis and some Ménière's disease.

### 2.6 Balance and cerebellar tests

Romberg, tandem walking, Unterberger's stepping test, and past-pointing: falling and past-pointing go towards the side of the slow phase. Cerebellar hemisphere disease causes dysmetria, dysdiadochokinesia and rebound; midline disease causes a wide gait and truncal ataxia.

## 3 CHAPTER THREE Laboratory tests

Measuring each part.

### 3.1 Caloric test

Warm or cold water (or air) in the canal sets up convection currents in the lateral canal. With the patient lying and the head raised 30 degrees, the lateral canal is vertical and responds best.

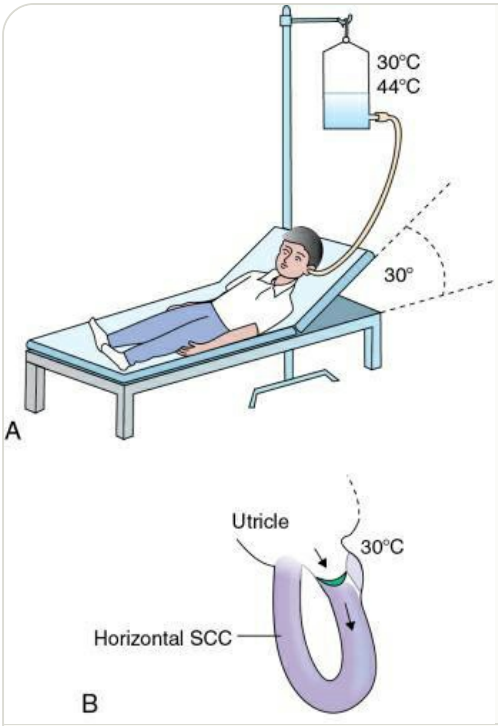



**FIGURE 4** The Fitzgerald-Hallpike caloric test: the head is raised 30 degrees to make the lateral canal vertical.

Dhingra, fig. 6.2

### 3.1.1 COWS

Cold water makes nystagmus beat to the opposite side; warm to the same side.

### 3.1.2 Reading it

- **Canal paresis:** one side responds less (a difference of more than about 15 to 30%).
- **Directional preponderance:** nystagmus stronger in one direction (more than about 25 to 30%).
- **Bilateral weakness:** bilateral vestibular loss.

### 3.1.3 A worked example

Each ear is irrigated with warm (44 °C) and cool (30 °C) water or air, and the peak slow-phase velocity of the nystagmus is measured in degrees per second. **Jongkees' formula** compares the two sides: canal paresis = [(RW + RC) − (LW + LC)] ÷ (RW + RC + LW + LC) × 100. For example, with right warm 30, right cool 28, left warm 12 and left cool 10°/s, the result is (58 − 22) ÷ 80 × 100 = 45%, a left canal paresis (most labs use a cut-off of about 20 to 25%). Directional preponderance compares right-beating (RW + LC) with left-beating (LW + RC) responses in the same way. If the sum of all four responses is very small (under about 6 to 8°/s per irrigation), the result is bilateral weakness, confirmed by rotary chair or head

impulse testing. Failure of fixation to suppress the nystagmus (fixation index) points to a central lesion.

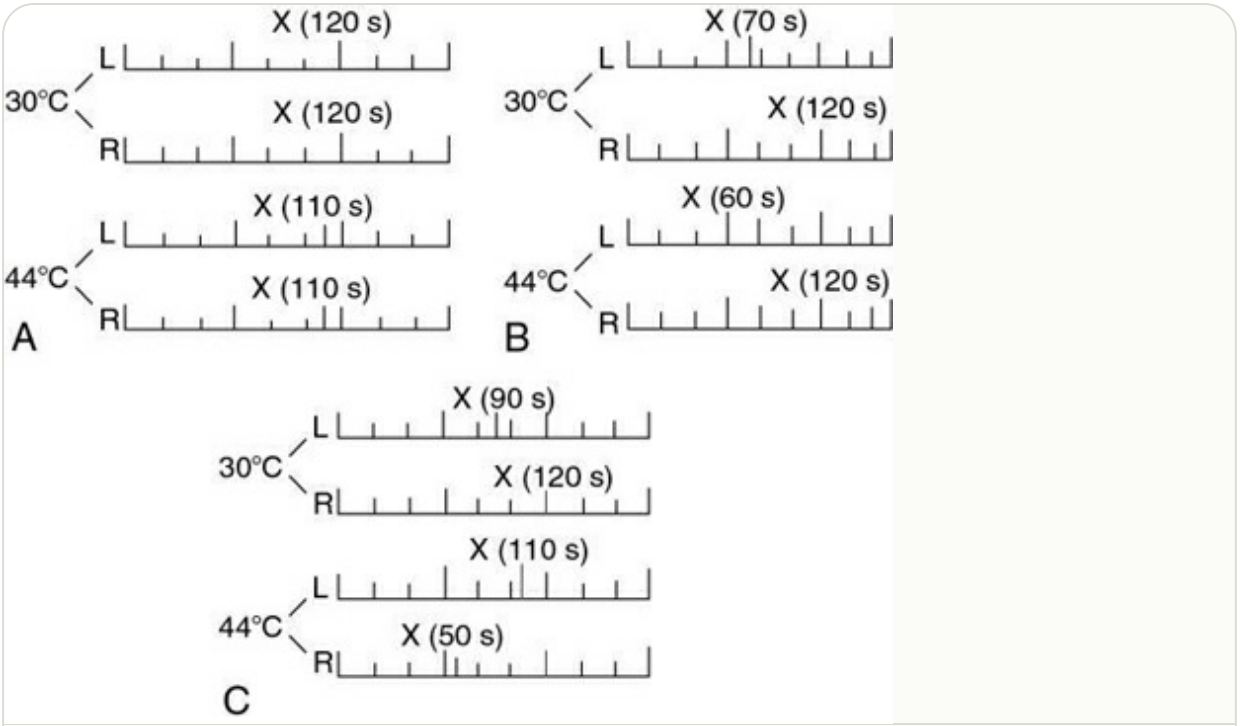



**FIGURE 5** Calorigrams: (A) normal, (B) left canal paresis, (C) left directional preponderance. Dhingra, fig. 6.3

WHERE THE BOOKS DIFFER · THE CALORIC TEST

| Dhingra | Lee |
|---|---|
| Water at 30°C and 44°C for 40 seconds. | For 30 seconds; considers directional preponderance of questionable value. |

### 3.1.4 Kobrak test

Small volumes of iced water, increasing from 5 mL; no response suggests a dead labyrinth.

**CALORICS AND PERFORATIONS**

Don't use water with a perforation; use air.

## 3.2 Electronystagmography and videonystagmography

Record eye movements during calorics, positional tests, saccades, pursuit and optokinetic testing. Central signs: direction-changing nystagmus, failure of fixation

suppression, abnormal saccades and pursuit, and hyperactive calorics.

## 3.3 Video head impulse test

Measures the gain of the reflex for each canal separately; low gain with overt or covert saccades means a weak canal.

Small, fast, unpredictable head turns are made while a lightweight camera records the eye. The **gain** is eye velocity divided by head velocity: normal for the lateral canals is about 0.8 or more (vertical canals about 0.7). A weak canal has low gain and **catch-up saccades**: *overt* saccades after the head stops, which the examiner can see at the bedside, and *covert* saccades during the movement, which only video can detect. The test is quick, tests all six canals, and measures the high frequencies of head movement used in daily life, whereas calorics test only the lateral canal at a very low frequency; the two are complementary, and Ménière's disease often shows abnormal calorics with a normal vHIT. A normal vHIT in acute continuous vertigo is a warning sign of a central cause.

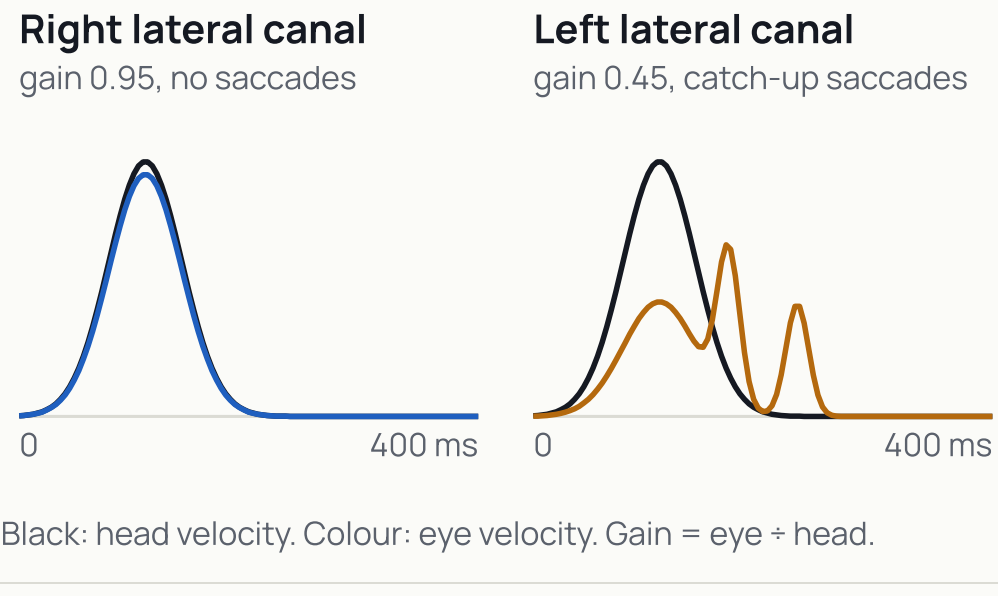



**FIGURE 6** Reading a video head impulse test. On the right the eye matches the head (gain near 1). On the left the eye lags (gain 0.45) and catch-up saccades follow: a left lateral canal weakness, as in vestibular neuritis.

Drawn for ENT Atlas

## 3.4 Vestibular evoked myogenic potentials

Loud sounds stimulate the saccule (cervical VEMP from sternocleidomastoid) and utricle (ocular VEMP from the inferior oblique). They work even in deaf ears. In superior canal dehiscence thresholds are abnormally low and ocular amplitudes high.

### READING VEMPS

| FINDING | CERVICAL VEMP (SACCULE, INFERIOR VESTIBULAR NERVE) | OCULAR VEMP (UTRICLE, SUPERIOR VESTIBULAR NERVE) |
|---|---|---|
| Superior canal dehiscence | Low threshold (often under 70 dB nHL) | Large amplitude, low threshold: the best screening test |
| Vestibular neuritis, superior nerve | Usually preserved | Absent on the affected side |
| Vestibular neuritis, inferior nerve | Absent on the affected side | Preserved |
| Ménière's disease | Reduced or absent, or tuning shifted to higher frequencies | Variable |
| Vestibular schwannoma | Often reduced or absent | Often reduced or absent |
| Conductive loss | Air-conducted VEMP lost; bone-conducted VEMP preserved | Same |

## 3.5 Rotary chair

Tests both labyrinths together over a range of frequencies; useful in bilateral loss, in children, and when a canal cannot be irrigated.

## 3.6 Posturography

Shows how the patient uses each sense to balance; useful in rehabilitation and in non-organic imbalance.

# 4 CHAPTER FOUR Using the tests

When to order them.

- When the side or site of the lesion is unclear.
- Before a destructive procedure, to check the other ear.
- Planning rehabilitation and tracking recovery.
- Monitoring gentamicin effect; daily head impulse and visual acuity help detect toxicity.

**PEARLS**

- A canal paresis with directional preponderance away from it suggests Ménière's disease; preponderance towards the same side, a schwannoma.
- Fixation suppressing the nystagmus is peripheral; failure to suppress suggests a central cause.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 On call

**THE DIZZY PATIENT AT THE BEDSIDE**

1. Spontaneous nystagmus, with and without fixation.
2. Head impulse test and test of skew.
3. Gait and cerebellar signs.
4. Dix-Hallpike when the neck and condition allow.

## 5.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Ewald's laws** and why nystagmus beats away from a dead ear.
- **Dix-Hallpike** technique and peripheral versus central positional nystagmus.
- **Caloric test:** technique, COWS, canal paresis and directional preponderance.
- **Fistula test** and Hennebert's sign.
- **VEMP** and what it tests.

## 5.3 Common pitfalls

COMMON PITFALLS

- Using water calorics with a perforation.
- Taking normal tests as proof of no vestibular disease.
- Missing central signs because the eyes were tested only with fixation.

Written for this app from Oxford Textbook of Otolaryngology ch. 7, Oxford Specialist Handbook ch. 14, Oxford Handbook of ENT ch. 4, Dhingra ch. 6 and K. J. Lee ch. 16. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Hearing rehabilitation: aids, bone devices and implants

Hearing rehabilitation means matching the device to the loss. A hearing aid helps whatever amplification can help; a bone-conduction device helps when the canal or middle ear is in the way; a cochlear implant takes over when the cochlea can no longer do the job; and a brainstem implant is kept for the patient who has no cochlear nerve.

**AT A GLANCE**

- Match device to the loss.
- Aids need speech recognition and dynamic range.
- Bone devices bypass the canal and middle ear.
- Implant for severe-profound with poor speech.
- Brainstem implant only if there's no nerve.
- Bone-conduction devices: skin-drive, percutaneous or active transcutaneous; ABI when the cochlear nerve cannot be used.

## 1 CHAPTER ONE Overview

The right device for the loss.

**WHICH DEVICE?**

| SITUATION | CONSIDER |
|---|---|
| Sensorineural loss with useful speech understanding | Air-conduction hearing aid |
| Chronic discharge, canal atresia | Bone-anchored device |
| Single-sided deafness | CROS aid, bone-anchored device, cochlear implant |
| Moderate to severe loss, cannot tolerate an aid | Middle ear implant |
| Severe to profound loss with poor speech understanding | Cochlear implant |
| No cochlear nerve, NF2 | Auditory brainstem implant |

### 1.1 Beyond devices

Assistive listening devices (loop systems, FM and remote microphones), alerting devices, speech reading, auditory training and speech conservation.

### 1.2 Two ears are better than one

Bilateral fitting gives the *head shadow* effect, *binaural summation* (sound seems about 3 dB louder) and *squelch* (better hearing in noise), and restores sound localisation.

## 2 CHAPTER TWO Hearing aids

Amplification.

### 2.1 Components

A microphone, amplifier and receiver (speaker), powered by a battery, with digital processing that shapes gain by frequency and compresses loud sounds.

## 2.2 Styles

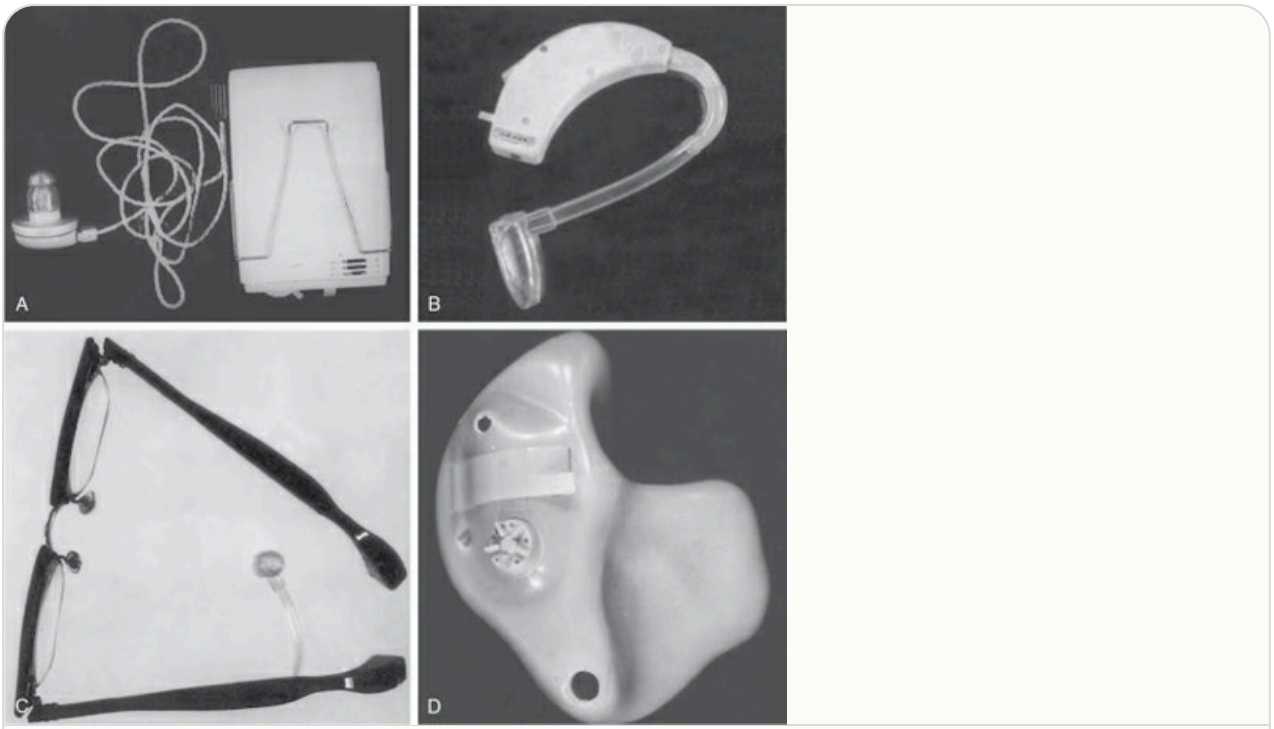



**FIGURE 1** Types of hearing aid: (A) body-worn, (B) behind-the-ear, (C) spectacle and (D) in-the-ear.

Dhingra, fig. 20.1

- **Body-worn:** high power; now rare.
- **Behind-the-ear:** the most versatile, from mild to profound loss, and the choice for children because the mould can be changed as the ear grows.
- **Receiver-in-canal and open fit:** for high-frequency loss with good low tones; avoid the occlusion effect.
- **In-the-ear, in-the-canal, completely-in-canal:** cosmetic, for mild to moderate loss in adults with good dexterity.

## 2.3 Who does well

Speech discrimination of at least about 50%, and a wide *dynamic range* between the threshold and the level of discomfort.

## 2.4 Problems

### 2.4.1 Recruitment

Abnormal growth of loudness in a cochlear loss narrows the dynamic range: soft sounds are missed but loud sounds are intolerable. Compression circuits help.

#### 2.4.2 Occlusion effect

A blocked canal makes the patient's own voice boom. Venting reduces it, but venting also loses low-frequency gain.

#### 2.4.3 Feedback

Whistling from sound leaking back to the microphone; better moulds and feedback cancellation help.

---

CHAPTER THREE

# 3 Bone-conduction devices

Around the canal and middle ear.

## 3.1 How they work

Sound vibrates the skull directly and reaches both cochleas, bypassing the canal and middle ear.

---

## 3.2 Types

Conventional bone-conduction aids on a headband or spectacles; *percutaneous* bone-anchored devices fixed to an osseointegrated titanium implant through the skin; and *transcutaneous* (magnetic) devices under intact skin.




**FIGURE 2** First, a bone-anchored device: titanium fixture, abutment and sound processor. Second, the device worn.
Dhingra, figs 20.2 and 20.3

## 3.3 Indications

- Chronic discharging ears that cannot tolerate an aid.
- Canal atresia and microtia.
- Conductive or mixed loss where surgery is unsuitable.
- Single-sided deafness with a good other ear (about 20 dB or better), sending sound across the skull.

## 3.4 Practical points

Osseointegration takes about three months, longer in children. Complications include failed integration and skin reactions around the abutment. A bone-anchored device does not restore localisation.

**WHERE THE BOOKS DIFFER · SINGLE-SIDED DEAFNESS**

| Dhingra | Lee |
|---|---|
| Bone-anchored devices have replaced CROS aids. | CROS aids remain an option. |

### 3.5 CROS and BiCROS aids

A microphone on the deaf side sends sound wirelessly to the better ear; BiCROS adds amplification for a better ear that also has some loss.

---

# 4 CHAPTER FOUR Middle ear implants

Driving the ossicles directly.

A transducer fixed to an ossicle or the round window, driven electromagnetically or piezoelectrically. For adults with moderate to severe sensorineural or mixed loss who are unhappy with conventional aids, and some conductive losses. The canal is left open.

# 5 CHAPTER FIVE Cochlear implants

Bypassing the cochlea.

### 5.1 Principle

An external microphone and processor convert sound into electrical patterns, sent across the skin to an internal receiver, which stimulates the spiral ganglion cells through an electrode array in the scala tympani. The cochlea's place coding is reproduced: basal electrodes for high pitches, apical for low.

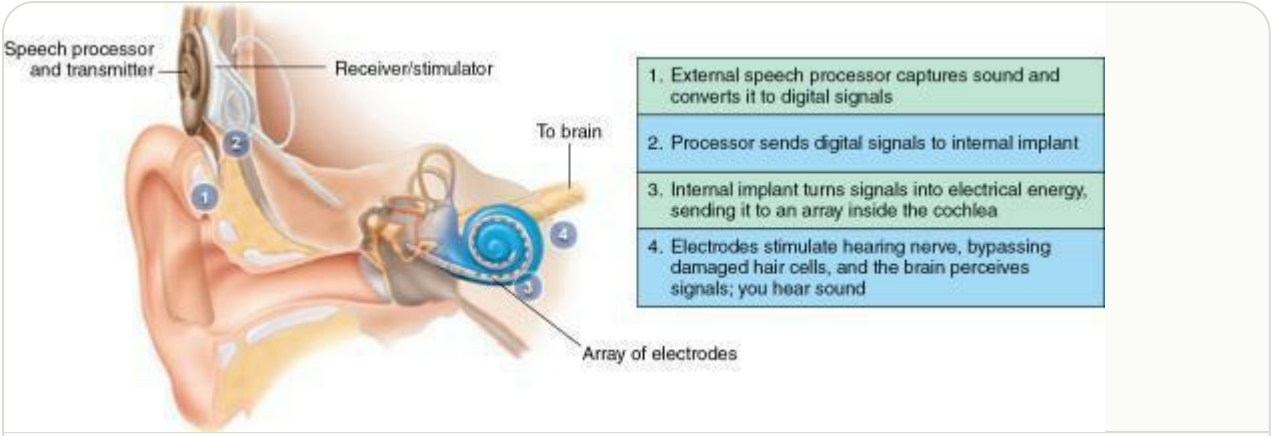



**FIGURE 3** Principle of the cochlear implant.
Dhingra, fig. 20.9

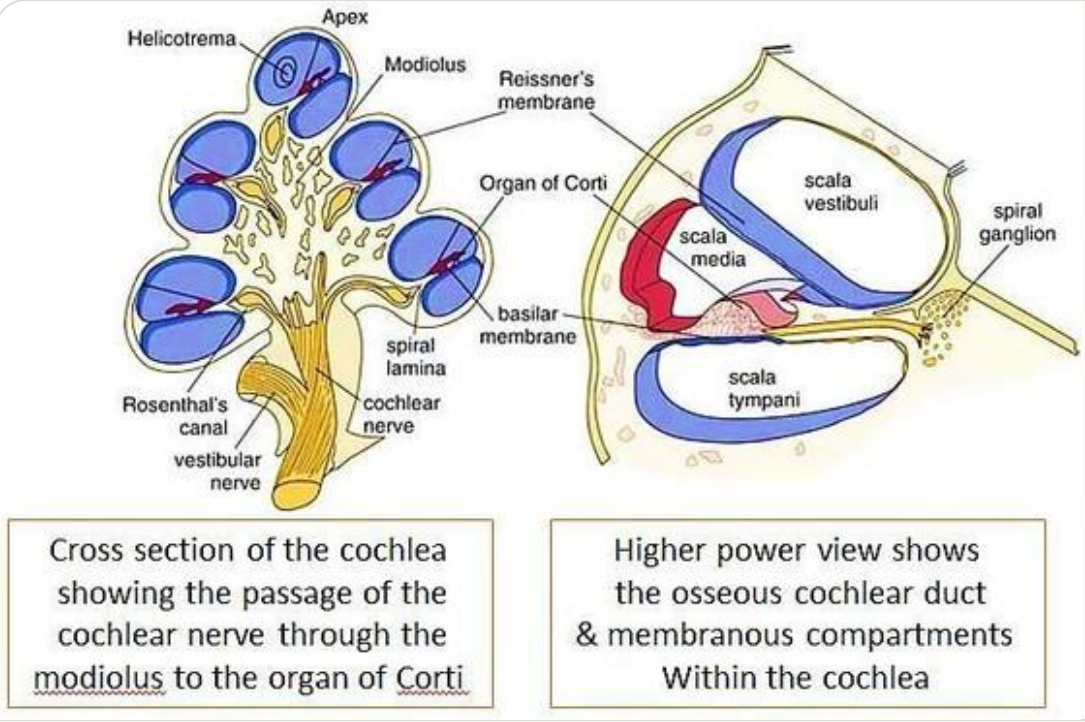



**FIGURE 4** Cross-section of the cochlea: the electrode lies in the scala tympani, next to the spiral ganglion.

Anatomy atlas · ENT Steps p. 69

## 5.2 Candidates

### 5.2.1 Adults

Severe to profound bilateral sensorineural loss with limited benefit from well-fitted aids. Lee gives moderate to profound loss with an aided sentence score of 50% or less in the ear to be implanted and 60% or less best-aided.

### 5.2.2 Children

From about twelve months, with severe to profound loss and little benefit after a three to six month trial of aids. The trial is waived after meningitis, when ossification threatens.

WHERE THE BOOKS DIFFER · AGE OF IMPLANTATION

| Dhingra | Lee |
| --- | --- |
| From twelve months. | Earlier implantation is safe and may help language. |

### 5.2.3 Outcomes

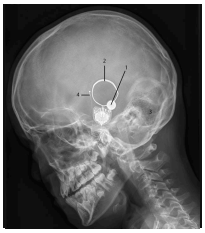
> Duration of deafness and pre-operative speech scores explain most of the variation. Postlingually deafened adults often achieve high sentence scores; children implanted early commonly join mainstream schools; prelingually deaf adults gain mostly sound awareness, because the window for language (birth to about two years) has passed.

## 5.3 Assessment

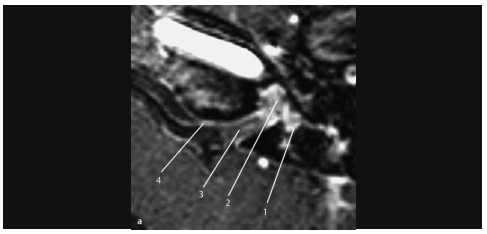
Medical review; audiology with an aid trial; MRI to show the cochlear nerve and a fluid-filled cochlea; HRCT for bony anatomy, malformations, a narrow internal auditory canal (under 3 mm) and surgical hazards; psychological and speech assessment.

**IMAGING IN THE APP** 4 cases

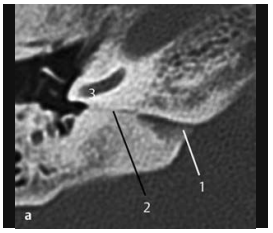
A cochlear implant in place, labyrinthitis ossificans, a gusher ear and iatrogenic fenestration.



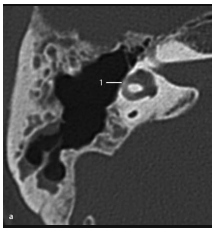
**Cochlear Implantation**
After surgery · 8 images



**Labyrinthitis Ossificans (Ossification after Meningitis)**
Bone disease · 14 images



**Gusher**
Infection & inflammation · 3 images



**Iatrogenic Fenestration**
After surgery · 2 images

## 5.4 Contraindications

Complete cochlear or labyrinthine aplasia (Michel), complete ossification, and an absent cochlear nerve. Retrocochlear disease is a relative contraindication.

## 5.5 The operation

Cortical mastoidectomy, posterior tympanotomy through the facial recess, entry into the scala tympani through the round window or a cochleostomy, slow insertion

of the array, sealing, and testing of the device. *Soft surgery* (a flexible array, the round window route, slow insertion, no suction of perilymph, and steroids) preserves residual hearing. In an ossified cochlea, the basal turn is drilled out or a split array used.

**WHERE THE BOOKS DIFFER · ROUTE INTO THE COCHLEA**

| Dhingra | Lee |
| --- | --- |
| Describes a cochleostomy. | Prefers the round window. |

**OPERATIVE GUIDES IN THEATRE** 5 operations

**Bone-Anchored Hearing Devices**
Bone-anchored devices

**Implantable Middle Ear Devices**
Implantable middle ear devices

**Cochlear Implantation in Adults**
Adult implant

**Pediatric Cochlear Implantation**
Paediatric implant

**Ossicular Chain Reconstruction**
Ossiculoplasty

## 5.6 After surgery

Activation at about two to four weeks, then repeated *mapping* of each electrode and intensive auditory therapy; sound quality improves over three to six months.

## 5.7 Vaccination

Implant recipients are vaccinated against pneumococcus (PCV13, then PPSV23 at least eight weeks later), because meningitis is commoner in implanted children, especially with malformations or a CSF leak.

## 5.8 Complications

- Flap breakdown and device extrusion.
- Device failure, about 4% over a lifetime; revision through the same track.
- Facial nerve injury (under 1%) and facial stimulation, managed by switching off

electrodes.

- Taste change, vertigo, tinnitus.
- CSF gusher, especially with malformations.
- Meningitis.

**PEARLS**

- Bilateral implants restore head shadow, summation and squelch, and localisation.
- Inner ear malformations raise the risk of an anomalous facial nerve, gusher and meningitis.
- Auditory neuropathy can do well with an implant.
- Acute otitis media in an implanted child is treated promptly, intravenously if severe.

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CHAPTER SIX

# 6 Auditory brainstem implant

When there is no nerve.

An electrode placed on the cochlear nucleus, through the lateral recess of the fourth ventricle, for patients without a usable cochlear nerve: neurofibromatosis type 2, nerve aplasia, or complete cochlear ossification. Results are more modest than with cochlear implants.

**THE IMPLANT FAMILY TODAY (OXFORD TEXTBOOK)**

Implantable devices act at different levels of the pathway. **Bone-conduction hearing devices** bypass the outer and middle ear. *Skin-drive* (passive transcutaneous) devices, such as Baha Attract and the adhesive Adhear, lose energy through the skin and are being superseded. *Direct-drive* devices are either *percutaneous*, with an abutment through the skin (Baha Connect, Ponto), or *active transcutaneous*, with the vibrator implanted under intact skin (Bonebridge, Osia). UK criteria require a conductive or mixed loss, or single-sided deafness, with bone conduction in the fitting range and stable (no more than 15 dB change at two or more frequencies over two years), failure or unsuitability of a conventional or CROS aid after at least four weeks, and benefit from a trial on a headband for at least 14 days. **Active middle ear implants**, most often the Vibrant Soundbridge, drive the incus, stapes or round window with a floating mass transducer, and suit patients who cannot wear an aid but have a cochlea worth amplifying.

**Cochlear implantation** now aims to preserve the cochlea and any residual hearing: a round window or cochleostomy approach into the scala tympani, slow insertion over minutes to avoid basilar membrane rupture, and steroids. Children with profound bilateral loss are implanted on both sides, simultaneously, ideally between 9 and 12 months. In aural atresia, a headband bone-conduction aid is started by about six months, with a definitive implant at about four to five years. **Auditory brainstem implants** stimulate the cochlear nucleus when a cochlear implant cannot work: neurofibromatosis type 2, cochlear or nerve aplasia, and severe cochlear ossification after meningitis or fracture. In adults, treating hearing loss may slow cognitive decline.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Hearing aid types** and who benefits.
- **Recruitment** and dynamic range.
- **Bone-anchored devices:** indications and osseointegration.
- **Cochlear implant:** principle, candidacy, assessment, contraindications, operation, complications.
- **Auditory brainstem implant:** indications.

## 7.2 Common pitfalls

COMMON PITFALLS

- Fitting an aid to an asymmetric loss without excluding a retrocochlear cause.
- Delaying implantation after meningitis.
- Forgetting pneumococcal vaccination.

Written for this app from Oxford Textbook of Otolaryngology ch. 16, Oxford Specialist Handbook ch. 14, Dhingra ch. 20 and K. J. Lee ch. 19, 20. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Superior canal dehiscence

In superior canal dehiscence, loud sounds or pressure make the world move, and the patient may hear their own eyes moving, their footsteps and their pulse. The audiogram can look exactly like otosclerosis, but the stapedial reflexes are normal: check them before any stapes operation.

**AT A GLANCE**

- Sound and pressure cause vertigo; body sounds are too loud.
- Normal reflexes and a low cVEMP threshold are the clue.
- Oblique CT confirms.
- Plug or resurface if disabling.

## 1 CHAPTER ONE Overview

A third window into the inner ear.

### 1.1 Definition

A defect in the bone over the superior semicircular canal, where it rises towards the floor of the middle cranial fossa, leaving the membranous canal covered only by dura.

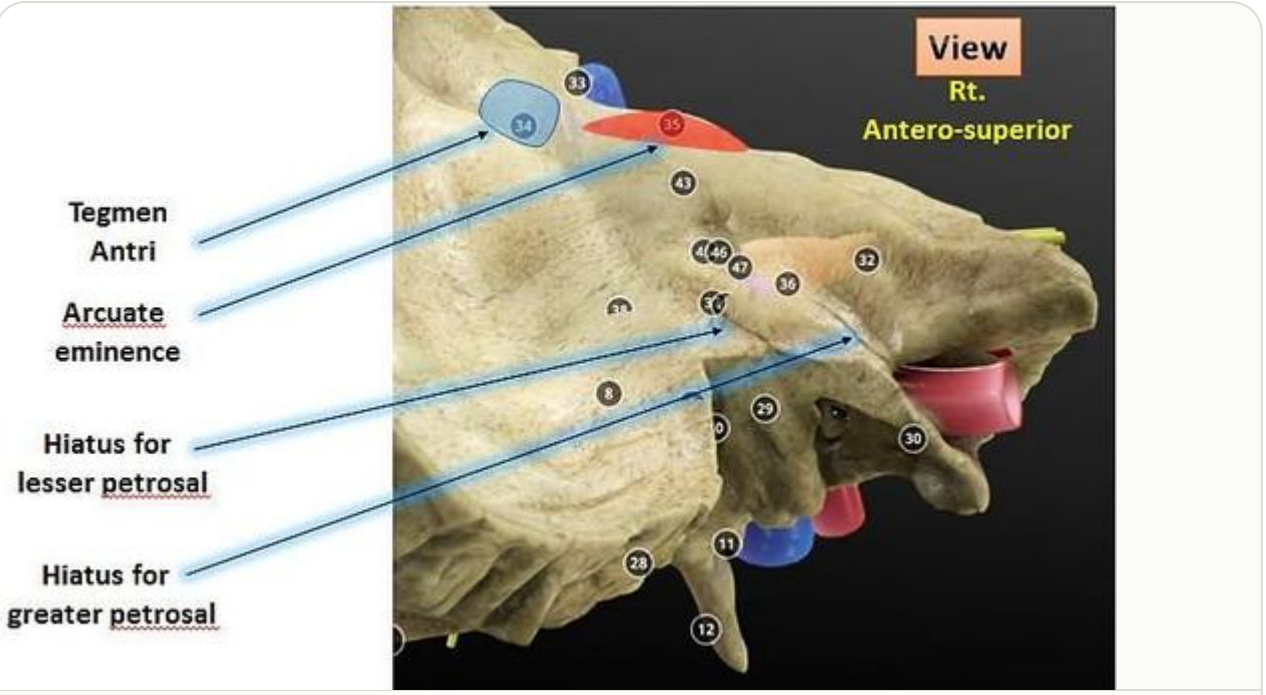



**FIGURE 1** The anterior surface of the petrous bone with the arcuate eminence, over the superior canal.

Anatomy atlas · ENT Steps p. 21

## 1.2 Cause

Probably a developmental thinness of the bone, with the final layer lost later in life through minor trauma, straining or the pulsation of the brain. It is often bilateral on CT, though usually symptomatic on one side.

## 1.3 The third window

Normally the inner ear has two mobile windows, the oval and the round. The dehiscence adds a third, so sound and pressure energy escape through it. Pressure in the middle ear or the cranial cavity now moves endolymph in the superior canal, causing vertigo; sound energy is diverted away from the cochlea, causing an apparent conductive loss; and bone-conducted sound is heard more loudly.

# 2 CHAPTER TWO Clinical features

Sound, pressure and the patient's own body.

## 2.1 Vestibular

- **Tullio phenomenon:** vertigo or oscillopsia triggered by loud sound.
- **Hennebert's sign:** vertigo with pressure in the canal, on Valsalva, coughing or straining.
- Chronic imbalance.

The eye movements induced lie in the plane of the superior canal.

## 2.2 Auditory

- **Autophony:** the patient's own voice sounds loud.
- **Hyperacusis for bone-conducted sounds:** hearing their footsteps, chewing, heartbeat, or even their eye movements.
- Pulsatile and gaze-evoked tinnitus.
- Aural fullness.

# 3 CHAPTER THREE Investigations

The clues.

## 3.1 Audiogram

A low-frequency air-bone gap, with bone conduction thresholds that may be better than 0 dB (supranormal). Stapedial reflexes are *present*, which an otosclerotic or other middle ear conductive loss would abolish.

### 3.2 Vestibular evoked myogenic potentials

The cervical VEMP threshold is abnormally low and the ocular VEMP amplitude abnormally high. A true middle ear conductive loss normally abolishes VEMPs, so their presence with a gap is a strong clue.

### 3.3 Electrocochleography

May show a raised SP to AP ratio.

### 3.4 CT

High-resolution CT reformatted in the plane of the superior canal (Pöschl view) confirms the dehiscence. Thin bone can look dehiscent on CT, so imaging is read with the symptoms and VEMPs.

## 4 CHAPTER FOUR Differential diagnosis

Mimics.

HOW TO TELL THEM APART

| CONDITION | DISTINGUISHING FEATURES |
|---|---|
| Otosclerosis | Stapedial reflexes absent, VEMPs absent, no vestibular symptoms with sound |
| Patulous Eustachian tube | Autophony with breathing, drum moves with respiration, no air-bone gap |
| Perilymph fistula | Follows surgery, trauma or straining; fluctuating hearing |
| Ménière's disease | Attacks of vertigo for hours, fluctuating low-frequency sensorineural loss |
| Otosyphilis | Positive serology, Hennebert's sign |

# 5 CHAPTER FIVE Management

Only if it is disabling.

## 5.1 Conservative

Reassurance and explanation; avoiding triggers such as loud noise and straining. Many patients need nothing more.

## 5.2 Surgery

For disabling vertigo or autophony: *plugging* the canal or *resurfacing* the defect, through the middle cranial fossa or a transmastoid approach. Plugging is more reliable for vestibular symptoms but sacrifices the canal. Round window reinforcement is an alternative in some centres.

**OPERATIVE GUIDES IN THEATRE** 1 operation

**Superior Semicircular Canal Dehiscence**
Superior canal dehiscence

## 5.3 Risks

Sensorineural hearing loss and vestibular loss after repair, and the usual risks of a middle fossa craniotomy.

**BEFORE STAPES SURGERY**

A conductive loss with a normal drum and present stapedial reflexes is not otosclerosis. Operating on the stapes in superior canal dehiscence does not help and may harm.

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **The third window** and how it explains each symptom.
- **Tullio phenomenon** and Hennebert's sign.
- **Conductive gap with present reflexes,** and low VEMP thresholds.
- **CT in the plane of the canal.**
- **Plugging or resurfacing.**

## 6.2 Common pitfalls

**COMMON PITFALLS**

- Offering stapes surgery for a conductive loss with present reflexes.
- Diagnosing from CT alone, without symptoms and VEMPs.

Written for this app from Oxford Textbook of Otolaryngology ch. 7, 12, Oxford Specialist Handbook ch. 18, K. J. Lee ch. 1, 16, 19. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Ear and mastoid operations: terms and principles

Ear operations are easiest to learn by what each one removes and what it keeps. The order of priorities never changes: a safe ear first, a dry ear second, hearing third. This page explains the terms and principles; the step-by-step guides are in Theatre.

**AT A GLANCE**

- Know each operation by what it removes and keeps.
- Cortical keeps the wall; modified radical keeps the hearing parts; radical removes them.
- Myringoplasty repairs the drum; tympanoplasty rebuilds hearing.
- Safety before hearing.
- Austin–Kartush classification (0, A–F) and MERI predict ossiculoplasty results.

## 1 CHAPTER ONE Overview

The language of ear surgery.

### 1.1 What hearing needs

For the middle ear to work, it needs an intact drum, a mobile ossicular chain, two mobile windows with *acoustic separation* (sound must reach the oval window but not the round window at the same time), an aerated middle ear with a working Eustachian tube, and a working cochlea. Every reconstruction tries to restore as many of these as possible.

### 1.2 Approaches

- **Transcanal (Rosen):** through the canal with a speculum, for stapes surgery, small perforations and exploratory tympanotomy.
- **Endaural (Lempert):** incisions in the canal extended up through the *incisura terminalis*, where there is no cartilage.
- **Postaural (Wilde):** behind the ear, for mastoid surgery and anterior perforations. In children under two, the mastoid tip is absent and the facial nerve is superficial, so the

incision is shorter and more horizontal.

# 2 CHAPTER TWO Myringotomy and grommets

Draining and ventilating.

## 2.1 Indications

Acute otitis media with a bulging, painful drum or complications; incomplete resolution; recurrent acute otitis media; otitis media with effusion; barotrauma; atelectasis; and access for intratympanic treatment.

## 2.2 Site

Posteroinferior for draining acute pus; anteroinferior or inferior for effusion and tubes. An anterosuperior tube stays in longest.

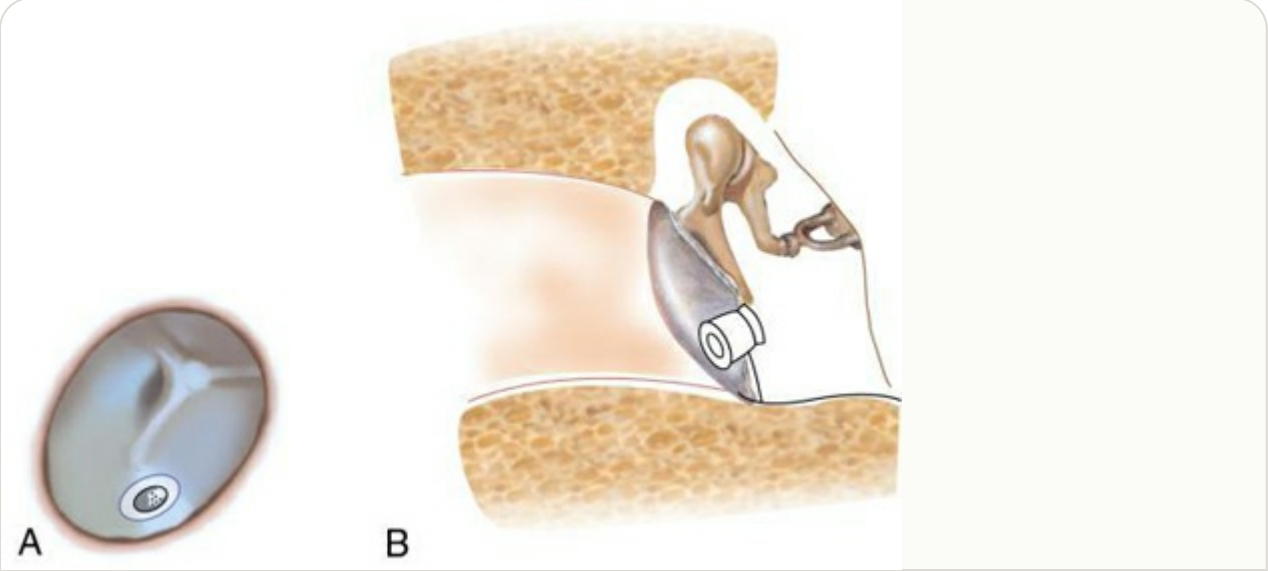



**FIGURE 1** Grommets in the tympanic membrane.
Dhingra, fig. 10.4

## 2.3 Complications

Injury to the ossicles or a high jugular bulb, persistent perforation, tube blockage or extrusion, otorrhoea, granuloma and tympanosclerosis.

**PEARL**

Look for a vascular anomaly (a high jugular bulb or aberrant carotid) before cutting the drum.

# 3 CHAPTER THREE Repairing the drum and chain

Myringoplasty and tympanoplasty.

## 3.1 Myringoplasty

Repair of a perforation alone, without inspecting or reconstructing the middle ear. Dhingra distinguishes it from a type I tympanoplasty, in which the middle ear is examined.

### 3.1.1 Grafts

Temporalis fascia is the commonest; tragal perichondrium, cartilage (resists retraction) and vein are alternatives.

### 3.1.2 Techniques

- **Underlay:** the graft goes medial to the drum remnant and handle of malleus. Risks: a narrow middle ear space and anterior reperforation.
- **Overlay:** the graft goes lateral to the fibrous layer after removing the epithelium. Risks: blunting of the anterior angle, epithelial pearls and lateralisation of the graft.

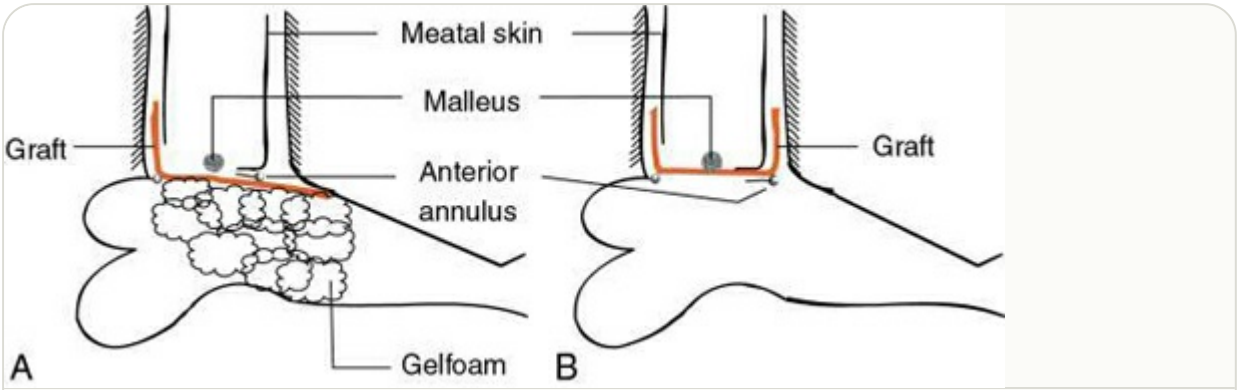



**FIGURE 2** Myringoplasty: (A) the underlay technique, with the graft medial to the annulus and supported by gelfoam; (B) the overlay technique.

Dhingra, fig. 5.3

### 3.1.3 Small holes

A paper patch after cauterising the rim, or a fat plug.

### 3.1.4 Contraindications

Active discharge, uncontrolled nasal allergy, otitis externa, squamous epithelium growing into the middle ear, an only hearing ear, and young children under about three.

## 3.2 Tympanoplasty

Eradication of middle ear disease and reconstruction of the hearing mechanism, with or without mastoidectomy.

### 3.2.1 Wullstein's types

- **Type I:** graft on the handle of malleus, chain intact.
- **Type II:** graft on the incus, malleus missing.
- **Type III:** graft on the stapes head (myringostapediopexy).
- **Type IV:** only a mobile footplate; the graft shields the round window, creating a small air pocket.
- **Type V:** fixed footplate; a window is made in the lateral canal (Va) or the footplate removed (Vb).

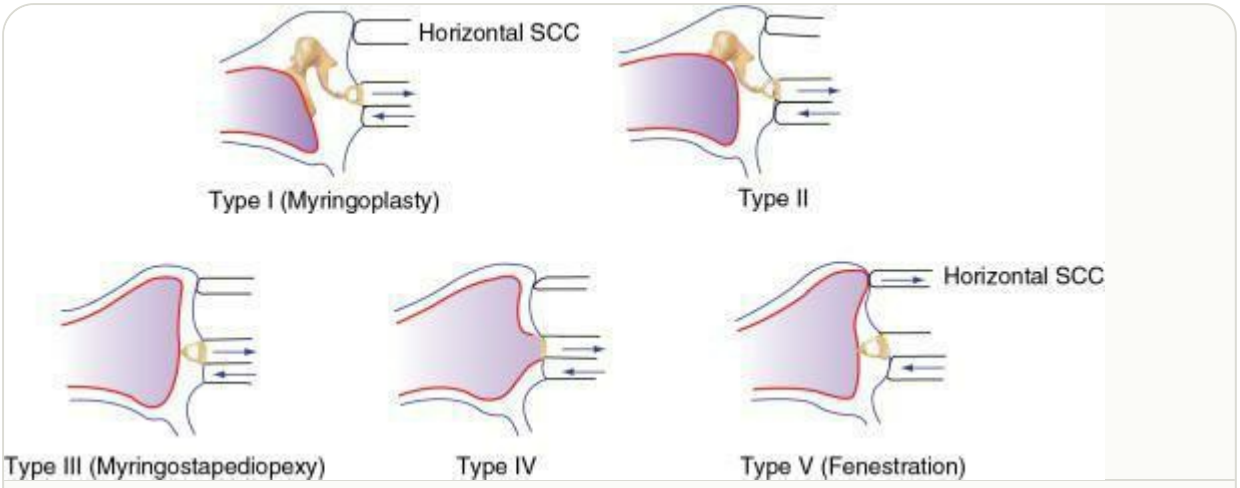



**FIGURE 3** Types of tympanoplasty: the graft rests progressively on the malleus, incus, stapes and footplate.

Dhingra, fig. 5.2

### 3.2.2 Ossicular reconstruction

Reshaped autograft incus, or prostheses: a *PORP* (partial) sits on the stapes head; a *TORP* (total) on the footplate. Hydroxyapatite and titanium are common materials; a cartilage cap prevents extrusion. A PORP gives better hearing than a TORP.

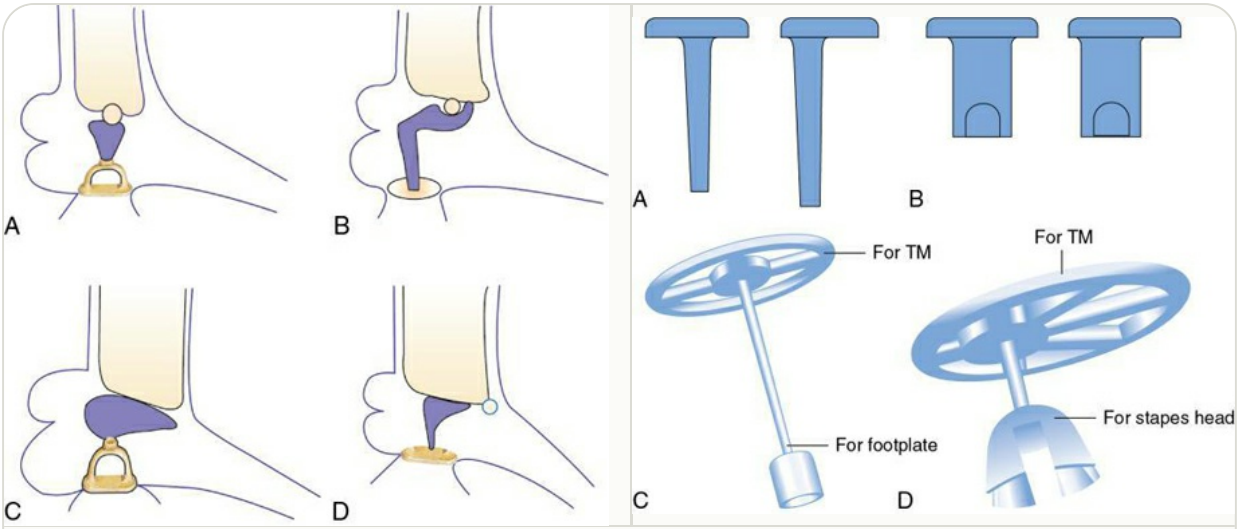



**FIGURE 4** First, sculptured autograft ossicles. Second, hydroxyapatite and titanium partial and total prostheses.

Dhingra, figs 5.4 and 5.5

### 3.2.3 Timing

Reconstruction is often delayed about six months after canal wall up surgery for cholesteatoma, to coincide with the second look; it is done primarily after trauma or for fixation.

# 4 CHAPTER FOUR Mastoid operations

What they remove.

## 4.1 Landmarks

*MacEwen's (suprameatal) triangle* marks the antrum, about 12 to 15 mm deep in adults. *Körner's septum*, a persistent petrosquamous plate, can be mistaken for the floor of the antrum. Other landmarks: the lateral semicircular canal, sinus plate, tegmen, digastric ridge and facial nerve.

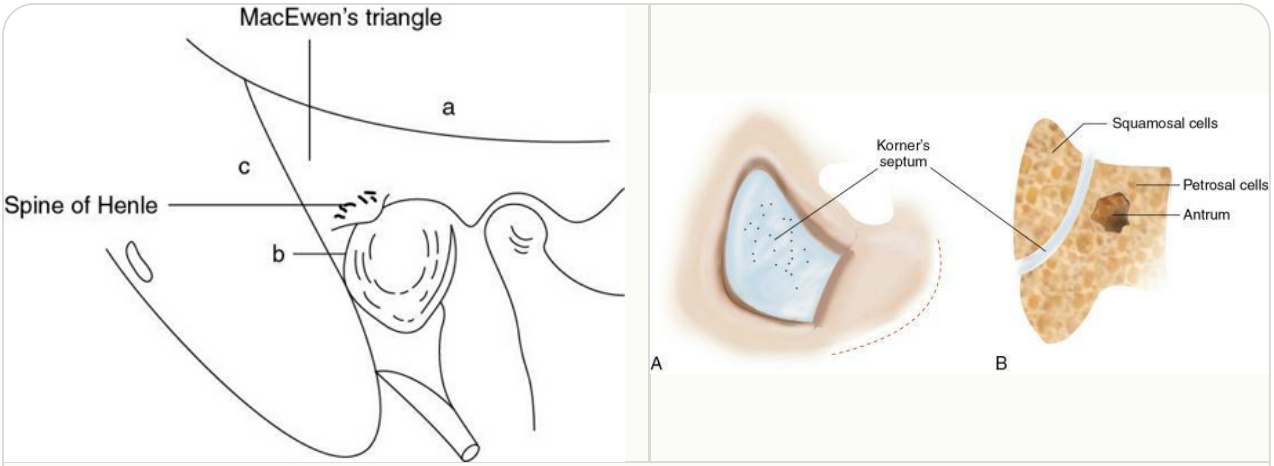



**FIGURE 5** First, MacEwen's triangle. Second, Körner's septum, which may hide the antrum.
Dhingra, figs 1.12 and 1.14

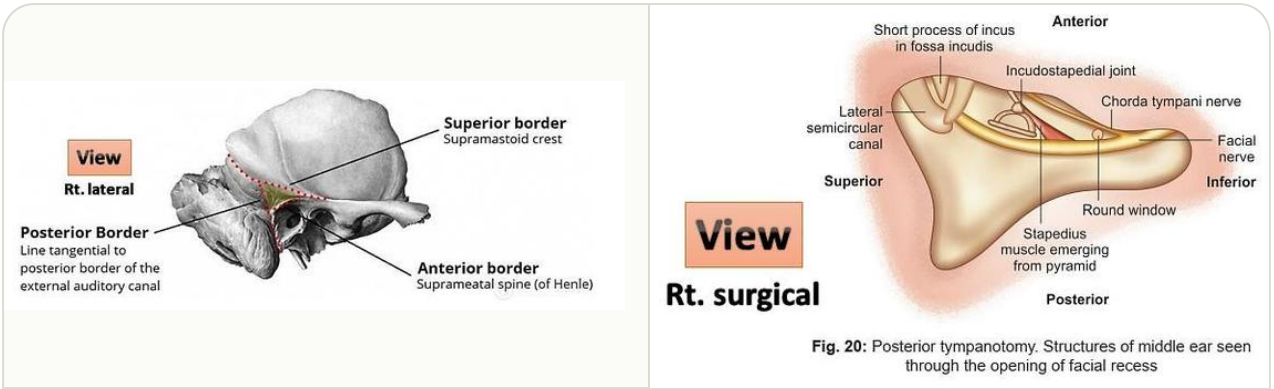



**FIGURE 6** MacEwen's triangle, and the middle ear seen through a posterior tympanotomy.
Anatomy atlas · ENT Steps p. 56, ENT Steps p. 46

## 4.2 Cortical (Schwartze) mastoidectomy

Removes the mastoid air cells and opens the antrum; the posterior canal wall and middle ear are left alone. For coalescent or masked mastoiditis, and as access for endolymphatic sac surgery, facial nerve decompression, cochlear implants and translabyrinthine surgery.

## 4.3 Modified radical mastoidectomy

The posterior canal wall is lowered and the attic, antrum and mastoid are exteriorised into one cavity, *keeping* the drum remnant, the working ossicles and the Eustachian tube. For cholesteatoma of the attic and antrum.

## 4.4 Radical mastoidectomy

Everything is exteriorised: the drum, malleus and incus and the middle ear lining are removed, and the Eustachian tube is plugged. For disease that cannot be cleared safely otherwise, failed surgery, and some tumours. Hearing loss is 50 dB or more.

MASTOID OPERATIONS COMPARED

| OPERATION | POSTERIOR CANAL WALL | OSSICLES | HEARING |
|---|---|---|---|
| Cortical | Kept | Kept | Usually unchanged |
| Modified radical | Removed | Working parts kept | Can be reconstructed |
| Radical | Removed | All but the stapes removed | Loss of 50 dB or more |

## 4.5 Canal wall up and canal wall down

See the cholesteatoma topic for the choice between keeping and removing the posterior wall.

## 4.6 Meatoplasty

Every canal wall down cavity needs a wide meatus for ventilation and cleaning.

## 4.7 Obliteration

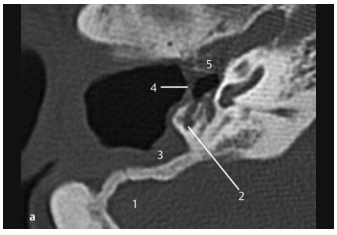
Filling a cavity with pedicled muscle, bone pâté or other material reduces discharge and care; disease must never be buried under it.

## 4.8 Complications of mastoid surgery

Facial nerve injury, lateral canal fistula with vertigo and hearing loss, dural injury and CSF leak, sigmoid sinus bleeding, and incus dislocation. About a quarter of radical cavities continue to discharge.

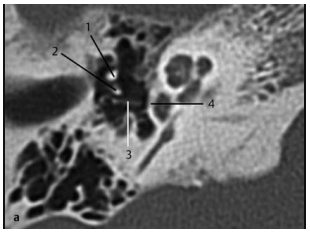
IMAGING IN THE APP 4 cases

A discharging cavity, piston position after stapedectomy, iatrogenic fenestration, and dural prolapse with CSF leak.



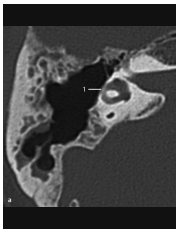
**Modified Radical Cavity, Persistent Infections**

After surgery · 2 images



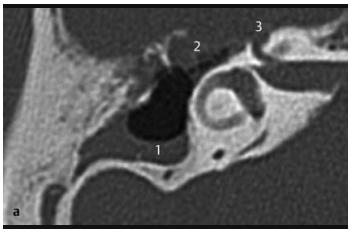
**Stapedectomy, Control of Piston Position**

After surgery · 4 images



**Iatrogenic Fenestration**

After surgery · 2 images



**Dural Prolapse and Cerebrospinal Fluid Leakage**

After surgery · 2 images

# 5 CHAPTER FIVE Before and after

Practical points.

## 5.1 Before

An audiogram and tuning forks before any ear operation; CT for cholesteatoma, revisions and suspected complications; check the other ear and Eustachian tube function.

## 5.2 After

Keep water out of the ear while a tube or graft is healing; review repaired ears at about three and six weeks; the graft is fully epithelialised in six to eight weeks. Clean mastoid cavities every four to six months in the first year, then yearly.

**OPERATIVE GUIDES IN THEATRE** 5 operations

**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation**
Tubes

**Myringoplasty and Tympanoplasty**
Tympanoplasty

**Ossicular Chain Reconstruction**
Ossicular reconstruction

**Mastoid Surgery**
Mastoid surgery
**Otosclerosis**
Stapes surgery

**OSSICULOPLASTY: CLASSIFICATION AND OUTCOMES (OXFORD TEXTBOOK)**

The commonest defect is erosion of the long process of the incus and the incudostapedial joint, from chronic otitis media or retraction; trauma most often damages the incus. An air–bone gap over about 30 dB is a good indication for reconstruction, although smaller gaps are accepted when surgery is needed anyway for infection or cholesteatoma.

The **Austin–Kartush** classification describes what remains: **0** all ossicles present; **A** malleus and stapes present (incus eroded); **B** malleus present, stapes superstructure absent; **C** malleus absent, stapes present; **D** both absent; **E** fixed ossicular head; **F** fixed stapes. The **Middle Ear Risk Index (MERI)** adds scores for otorrhoea, perforation, cholesteatoma, ossicular status, granulation or effusion, previous surgery and smoking to predict the result.

Reconstruction depends on the stapes. With the stapes present, options are incus interposition or a reshaped ossicle, a **partial ossicular replacement prosthesis (PORP)**, bone cement (hydroxyapatite or glass ionomer) bridging a small incus defect, or myringostapediopexy. With only the footplate, a **total ossicular replacement prosthesis (TORP)** or autologous columella is used, sometimes with a footplate shoe for stability. Modern prostheses are titanium or hydroxyapatite and are covered by a cartilage plate to prevent extrusion. An air–bone gap under 20 dB is reached in about 60 to 90% with a stapes and 30 to 80% without. An active middle ear implant such as the Vibrant Soundbridge is an alternative when reconstruction repeatedly fails.

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Myringoplasty versus tympanoplasty.**
- **Wullstein's five types.**

- **Underlay versus overlay.**
- **Cortical, modified radical and radical mastoidectomy:** what each removes and keeps.
- **Mastoid landmarks:** MacEwen's triangle and Körner's septum.
- **Indications for myringotomy** and the choice of site.

## 6.2 Common pitfalls

COMMON PITFALLS

- Repairing a drum while the ear is still discharging.
- Mistaking Körner's septum for the floor of the antrum.
- Burying cholesteatoma under an obliteration.

Written for this app from Oxford Textbook of Otolaryngology ch. 19, 20, 21, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 20, Dhingra ch. 5, 78, 79, 80, 81, 82, 83 and K. J. Lee ch. 19. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Microtia and congenital canal atresia

A baby born with a small or absent ear raises two questions, and hearing comes first. Test early, amplify by six months, and only later think about building an ear or opening a canal. When surgery is planned, image first: the facial nerve is often in the wrong place.

**AT A GLANCE**

- Test early and amplify by 6 months.
- Thin CT before any surgery.
- Pinna is reconstructed before the canal.
- Expect an abnormal facial nerve.

## 1 CHAPTER ONE Overview

A first-arch malformation.

### 1.1 Definitions

- **Microtia:** a small, malformed pinna; *anotia* is its complete absence.
- **Congenital aural atresia:** absence or narrowing of the external canal, usually with a malformed middle ear.

The two usually occur together, on the same side.

### 1.2 Epidemiology

Microtia occurs in roughly 1 in 6,000 to 1 in 12,000 births, more often in boys and on the right. About 10% are bilateral.

### 1.3 Embryology

The pinna develops from the hillocks around the first branchial cleft, the canal from the first branchial groove, and the malleus and incus from the first and second

arches. The inner ear develops separately from the otic placode, so it is usually normal.

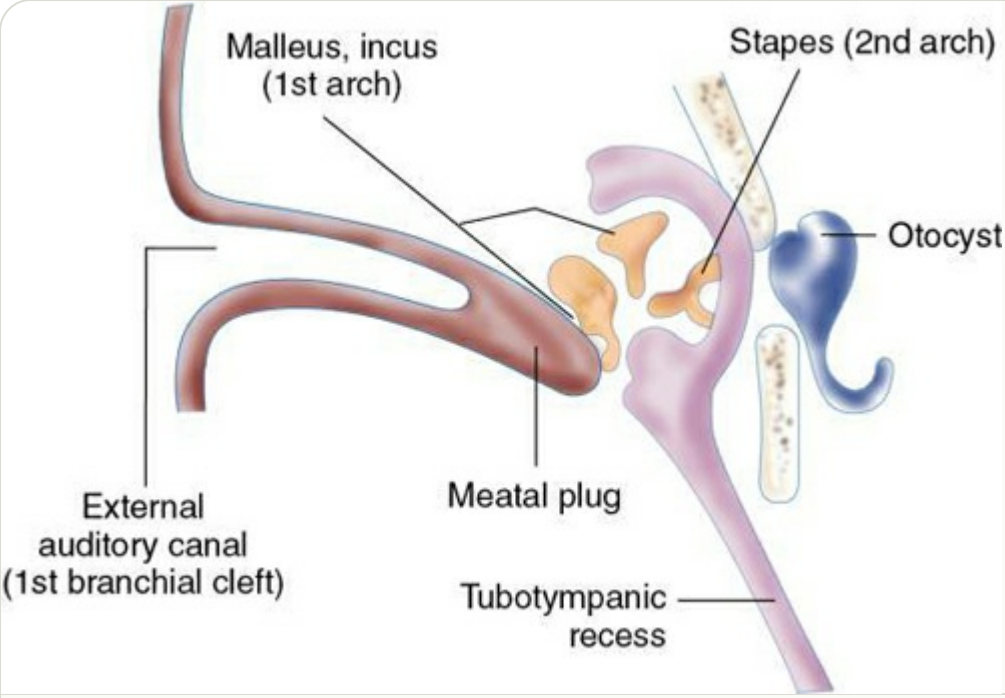



**FIGURE 1** Development of the external auditory canal and middle ear.
Dhingra, fig. 1.25

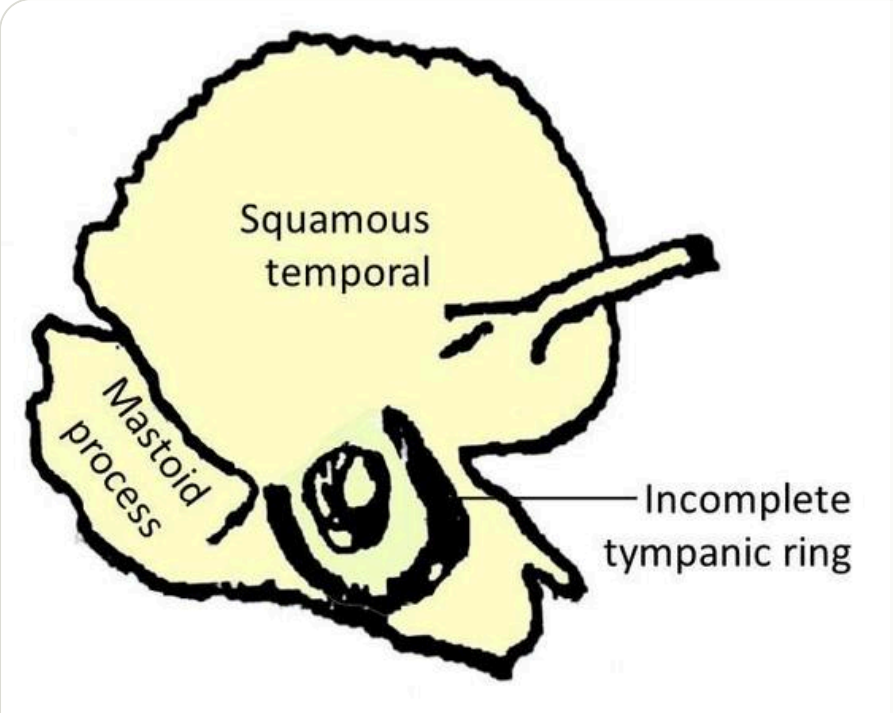



**FIGURE 2** The temporal bone at birth: an incomplete tympanic ring and no mastoid process.
Anatomy atlas · Doss p. 37

### 1.4 Associated syndromes

Treacher Collins, Goldenhar and hemifacial microsomia, Crouzon, and branchio-oto-renal syndrome.

---

# 2 CHAPTER TWO Classification

Pinna and middle ear.

### 2.1 Microtia

- **Grade I:** a recognisable but smaller or deformed pinna.
- **Grade II:** a deformed framework with some parts missing.
- **Grade III:** a soft-tissue remnant, the *peanut ear*.
- **Grade IV:** anotia.



**FIGURE 3** Grade III microtia of the right ear (peanut ear).
Dhingra, fig. 8.3

### 2.2 Middle ear (Lee)

- **Class I:** normal pinna with an ossicular defect, the commonest.
- **Class II:** microtia and atresia with an aerated middle ear and mastoid.

- **Class III:** poorly aerated, underdeveloped middle ear and mastoid.

> **PEARL**
> A normal pinna with atresia implies the insult happened later in development, after the pinna had formed.

# 3 CHAPTER THREE Clinical features

Hearing and appearance.

## 3.1 Hearing

- Atresia: a conductive loss of about 60 to 70 dB.
- Ossicular malformation alone: about 25 to 50 dB.
- Congenital stapes fixation: about 50 to 65 dB.
- Bilateral atresia threatens speech and language development.

## 3.2 Associated findings

Preauricular tags, pits and sinuses, facial asymmetry, jaw underdevelopment, and syndromic features. Airway problems in some syndromes.

# 4 CHAPTER FOUR Investigations

Hearing first, imaging before surgery.

## 4.1 Audiology

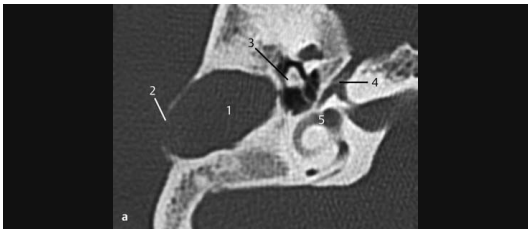
OAE and ABR from birth, including bone-conduction ABR, to measure cochlear function on each side.

## 4.2 Imaging

Thin-section CT of the temporal bones before any surgery, usually around five or six years: the size and aeration of the middle ear and mastoid, the ossicles, the oval and round windows, the inner ear, and the course of the facial nerve. MRI if there is a sensorineural loss.

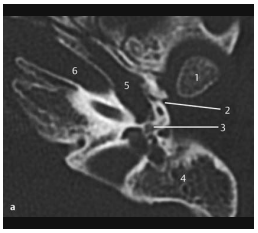
**IMAGING IN THE APP** 3 cases

Canal atresia with cholesteatoma, Treacher Collins syndrome, and inner ear and ossicular malformations.



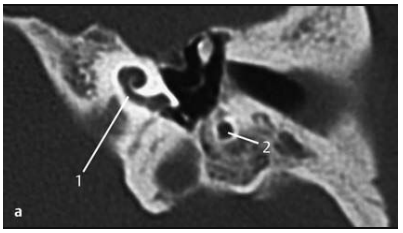

**Inclusion Cholesteatoma and Atresia of the External Auditory Canal**

Congenital · 3 images




**Malformations, Treacher Collins**

Congenital · 2 images




**Malformations of the Inner Ear and Ossicular Chain**

Congenital · 3 images

## 4.3 Other tests

Renal ultrasound when preauricular pits accompany neck fistulae (branchio-oto-renal syndrome).

# 5 CHAPTER FIVE Management

Hearing, then the ear.

## 5.1 Hearing

- **Bilateral atresia:** a bone-conduction aid on a headband within the first months, by six months at the latest.
- **Unilateral atresia with a normal other ear:** speech usually develops normally;

preferential seating at school and monitoring; amplification is optional.

- A bone-anchored implant from around five or six years.

## 5.2 The pinna

Reconstruction with autologous rib cartilage in stages, usually at about five to eight years when there is enough cartilage, or a porous polyethylene framework, or an osseointegrated prosthesis.

## 5.3 The canal

### 5.3.1 Selection

Canal surgery suits ears with a conductive loss over 30 dB, bone conduction better than 20 dB, and a well-aerated middle ear with a favourable facial nerve on CT. The pinna is reconstructed first so the canal can be placed correctly.

### 5.3.2 Risks

Facial nerve injury, restenosis of the new canal, lateralisation of the graft and sensorineural loss.

**OPERATIVE GUIDES IN THEATRE** 5 operations

**Microtia**
Microtia

**Congenital and Acquired Atresia and Stenosis of the External Auditory Canal**
Atresia and stenosis

**Congenital Malformations of the Middle Ear**
Congenital middle ear malformations

**Bone-Anchored Hearing Devices**
Bone-anchored hearing

**Otoplasty**
Otoplasty

## 5.4 Preauricular sinus

Leave it alone unless it becomes infected; then excise the whole tract.

**THE FACIAL NERVE**

In atresia the facial nerve often runs more anteriorly and laterally than usual, crossing the area where the canal would be drilled. First branchial cleft anomalies also run close

to it.

CHAPTER SIX

# 6 For the examination

What the examiner wants to hear.

## 6.1 On call

**A NEWBORN WITH BILATERAL ATRESIA**

1. Early audiology with bone-conduction testing.
2. A bone-conduction aid as soon as possible.
3. Look for syndromic features and airway problems.

## 6.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Embryology:** why the inner ear is spared.
- **Grades of microtia.**
- **Hearing before cosmetics,** and the timing of each intervention.
- **Selection for canal surgery,** and the facial nerve risk.

## 6.3 Common pitfalls

**COMMON PITFALLS**

- Delaying amplification in bilateral atresia.
- Opening a canal before reconstructing the pinna.
- Drilling without a CT map of the facial nerve.

Written for this app from Oxford Textbook of Otolaryngology ch. 13, 83, Oxford Specialist Handbook ch. 23, Dhingra ch. 8, 19 and K. J. Lee ch. 13, 17, 51. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · EAR

# Ear signs and eponyms at a glance

Otology is full of names. This page gathers the signs, syndromes and anatomical landmarks that examiners love, each explained in a line or two and pointing to the topic where it belongs. Use it for revision the night before a viva.

**AT A GLANCE**

- A one-stop revision sheet of signs, syndromes and landmarks.
- Each links to the topics that explain it.

## 1 CHAPTER ONE Signs

What you see or elicit.

### 1.1 Abscesses around the mastoid

#### 1.1.1 Bezold abscess

Pus breaking through the medial side of a pneumatised mastoid tip and collecting under the sternocleidomastoid. See Mastoiditis.

#### 1.1.2 Citelli abscess

Pus tracking from the tip along the posterior belly of digastric, behind the mastoid or in the digastric triangle.

#### 1.1.3 Luc abscess

A meatal abscess: pus from the antrum tracking under the periosteum of the bony canal.

### 1.2 Signs of intracranial spread

#### 1.2.1 Griesinger's sign

Oedema over the mastoid emissary vein in lateral sinus thrombosis.

#### 1.2.2 Tobey-Ayer test

Compressing the jugular vein on the healthy side raises CSF pressure, on the thrombosed side does not; historical.

#### 1.2.3 Crowe-Beck test

Compressing the healthy jugular vein engorges the retinal veins in lateral sinus thrombosis; historical.

#### 1.2.4 Gradenigo's triad

Ear discharge, deep pain behind the eye and a sixth nerve palsy in petrous apicitis.

## 1.3 Labyrinth

#### 1.3.1 Hennebert's sign

A positive fistula test with an intact drum: congenital syphilis, Ménière's disease, superior canal dehiscence.

#### 1.3.2 Tullio phenomenon

Vertigo triggered by loud sound: superior canal dehiscence, perilymph fistula, hydrops.

## 1.4 Otosclerosis

#### 1.4.1 Schwartze sign

A red blush on the promontory through the drum, from an active, vascular focus.

#### 1.4.2 Carhart's notch

A dip of bone conduction maximal at 2000 Hz, mechanical, disappearing after stapes surgery.

#### 1.4.3 Paracusis Willisii

Hearing better in noisy surroundings.

## 1.5 Tumours

#### 1.5.1 Brown's sign

A glomus tumour pulsates, then blanches, as pressure in the canal is raised.

#### 1.5.2 Rising sun

A red glomus tumour seen rising from the floor of the middle ear behind the drum.

#### 1.5.3 Hitselberger's sign

Reduced sensation of the posterior canal wall, from pressure on the sensory fibres of the facial nerve by a vestibular schwannoma.

## 1.6 Facial nerve

#### 1.6.1 Bell's phenomenon

The eye rolls up and out when the patient tries to close it.

## 1.7 Mastoiditis

#### 1.7.1 Reservoir sign

The canal refills with pus straight after it is mopped.

#### 1.7.2 Light-house sign

Pulsatile discharge through a pars tensa perforation.

# 2 CHAPTER TWO Syndromes

Named collections of findings.

## 2.1 Facial nerve

- **Ramsay Hunt:** herpes zoster oticus with facial palsy, often with hearing loss and vertigo.
- **Melkersson-Rosenthal:** recurrent facial palsy, swelling of the lips and a fissured tongue.
- **Heerfordt:** sarcoidosis with uveitis, parotid swelling and facial palsy.
- **Möbius:** congenital bilateral facial palsy with sixth nerve palsy.
- **Bogorad (crocodile tears):** tearing while eating after facial nerve regeneration.
- **Frey:** gustatory sweating after parotid surgery.

- **Bannwarth:** facial palsy in Lyme meningoradiculitis.

## 2.2 Inner ear

- **Cogan:** interstitial keratitis with audiovestibular loss.
- **Vogt-Koyanagi-Harada:** uveitis, alopecia, white hair and deafness.
- **Lermoyez:** hearing improves with the vertigo attack.
- **Tumarkin's otolithic crisis:** sudden drop attacks in Ménière's disease.
- **Mal de débarquement:** a rocking sensation persisting after travel by sea or air.
- **van der Hoeve:** osteogenesis imperfecta, blue sclerae and deafness.

## 2.3 Pain

- **Costen:** a faulty bite with earache, fullness, tinnitus and dizziness.
- **Eagle:** an elongated styloid process with throat and ear pain.

## 2.4 Jugular foramen

- **Vernet:** palsy of the ninth, tenth and eleventh nerves.
- **Collet-Sicard:** ninth to twelfth nerves.
- **Villaret:** ninth to twelfth nerves with Horner syndrome.

# 3 CHAPTER THREE Spaces and landmarks

Named anatomy.

## 3.1 Middle ear and mastoid

- **Prussak's space:** between the pars flaccida and the neck of the malleus; where attic cholesteatoma begins.
- **Trautmann's triangle:** bounded by the bony labyrinth, the sigmoid sinus and the superior petrosal sinus (dura); a route to the posterior fossa.

- **Citelli's angle:** the sinodural angle, between the sigmoid sinus and the middle fossa dura.
- **MacEwen's triangle:** the suprameatal triangle, marking the antrum.
- **Körner's septum:** a persistent petrosquamous plate in the mastoid, which can be mistaken for the floor of the antrum.

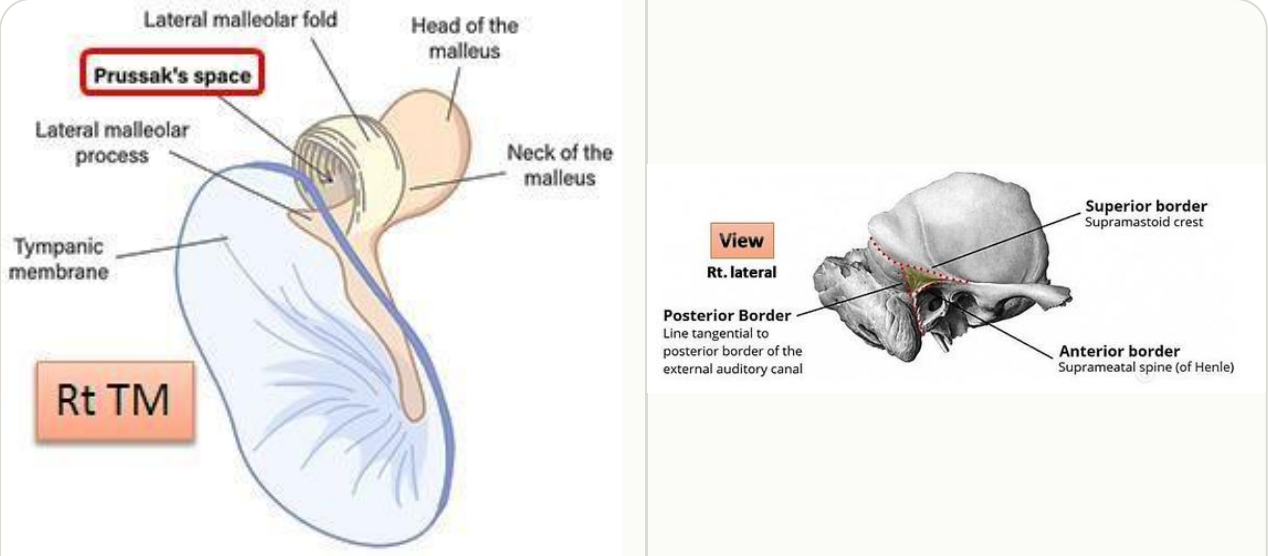



**FIGURE 1** Prussak's space, and MacEwen's triangle.
Anatomy atlas · ENT Steps p. 54, ENT Steps p. 56

## 3.2 Canal

- **Foramen of Huschke:** a defect in the anteroinferior bony canal in children, a route between the canal and the parotid.
- **Fissures of Santorini:** clefts in the cartilaginous canal through which infection passes to the parotid.

## 3.3 Skull base

- **Dorello's canal:** where the sixth nerve passes under the petroclinoid ligament at the petrous tip.
- **Meckel's cave:** the trigeminal ganglion on the petrous apex.
- **Hyrtl's fissure:** an infant fissure from the hypotympanum to the subarachnoid space.

### 3.4 Nerves

- **Arnold's nerve:** the auricular branch of the vagus; explains the ear cough reflex.
- **Jacobson's nerve:** the tympanic branch of the glossopharyngeal; origin of glomus tympanicum and route for referred pain.

---

# 4 CHAPTER FOUR
# For the examination

Using this page.

**HOW TO USE THE NAMES**

- Say what the sign is, then what it means, then which disease it points to.
- Know which tests are historical (Tobey-Ayer, Crowe-Beck) and what has replaced them (imaging).
- Link each landmark to the operation where it matters.

Written for this app from Dhingra ch. 1, 12, 14, 17 and K. J. Lee ch. 1, 13, 16. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# Imaging for this region

*The core imaging conditions of this region, with their key facts and a representative scan. For the normal anatomy and a search pattern, read Lesson 5, The temporal bone on CT; Lesson 6, The inner ear, IAC and CPA on MRI; Lesson 7, The skull base and its foramina in the imaging volume.*

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

## Malignant External Otitis or Necrotizing External Otitis

### IMAGING FINDINGS

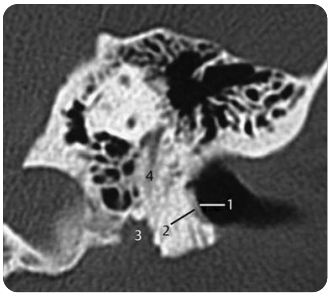
- Soft tissue in the external canal with erosion of its floor and the tympanic bone
- Spread below the skull base: loss of fat at the stylomastoid foramen, parapharyngeal and masticator spaces, temporomandibular joint
- Later: skull base osteomyelitis of the clivus, jugular foramen and petrous apex; venous thrombosis

### TOP DIFFERENTIALS

- Carcinoma of the external canal
- Canal cholesteatoma
- Keratosis obturans

### CHECKLIST

- Facial nerve and lower cranial nerves
- Sigmoid sinus and jugular vein; follow-up MRI changes lag behind clinical cure




CT, coronal. Necrotizing lesion in the floor of the external auditory canal (1). Clinically, erosive bone lesions (2) with purulent debris and denuded bone…



Axial bone CT shows EAC opacification with focal anterior wall and floor of middle ear erosion in this diabetic patient with painful otorrhea and earl…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1010.

MIDDLE EAR & MASTOID · TRAUMA

# Luxation of the Ossicular Chain

## IMAGING FINDINGS

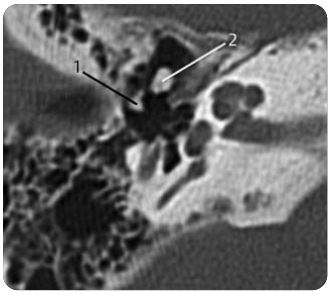
- Separation of the incudostapedial joint; incus rotated or displaced
- Loss of the "ice-cream cone" between malleus head and incus body
- Associated temporal bone fracture through the canal and middle ear

## TOP DIFFERENTIALS

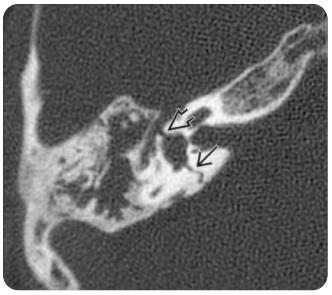
- Ossicular erosion from chronic otitis
- Congenital ossicular malformation
- Partial volume on a single slice

## CHECKLIST

- Compare with the other ear on the same slice
- Check facial canal, tegmen and otic capsule




CT, axial. Traumatic luxation with dislocation of the connection between incus (1) and malleus (2) is seen. The typical icecone configuration of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

MIDDLE EAR & MASTOID · BONE DISEASE

# Otosclerosis, Radiologic Signs around the Oval Niche

## IMAGING FINDINGS

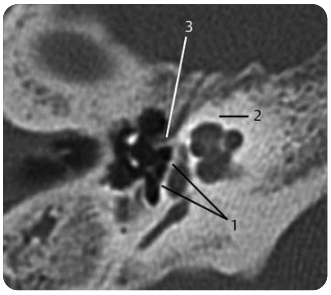
- Lucent focus at the fissula ante fenestram, anterior to the oval window
- Thickened footplate or obliterated oval window niche
- Cochlear form: pericochlear lucent halo ("double ring")

## TOP DIFFERENTIALS

- Tympanosclerosis
- Osteogenesis imperfecta
- Paget disease of the temporal bone

## CHECKLIST

- Compare both ears: usually bilateral
- Report round window involvement and cochlear disease before stapes surgery




CT, axial. At the edges of the footplate, two densities are visible (1), indicating osseous thickening, which may be considered as otosclerotic foci,…



Coronal right temporal bone CT shows a lytic focus anterior to the oval window, the typical appearance and location of an otospongiotic plaque o…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1120.

MIDDLE EAR & MASTOID · TUMOURS

# Glomus Tympanicum (Paraganglioma)

## IMAGING FINDINGS

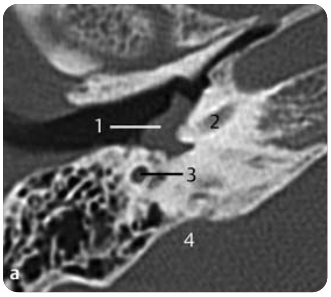
- Small enhancing mass on the cochlear promontory
- Floor of the middle ear intact
- Flow voids only in larger tumours

## TOP DIFFERENTIALS

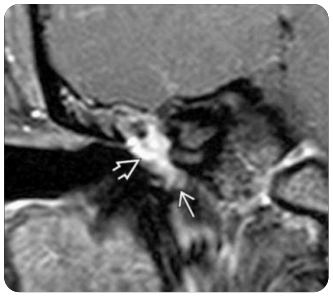
- Glomus jugulare (floor eroded, from the jugular foramen)
- Aberrant internal carotid artery
- Dehiscent jugular bulb

## CHECKLIST

- Check the hypotympanic floor and jugular bulb
- Ossicles and extension into the mastoid




CT, axial. On this axial slice at the level of the hypotympanum, a smooth-bordered lesion (1) is seen, without any destruction of the cochlea (2) or…



Coronal T1 C+ FS MR demonstrates a large GTP filling the middle ear cavity. The floor is intact, separating the tumor from the jugular bulb below.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1064.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

# Cholesteatoma of the Middle Ear

## IMAGING FINDINGS

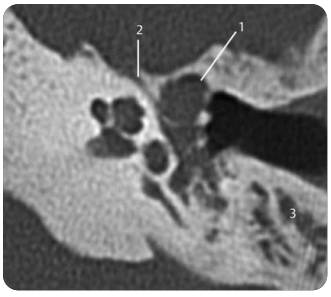
- Soft tissue in Prussak space (attic) or the posterior mesotympanum
- Blunted scutum; eroded, medially displaced ossicles
- Bright on non-EPI diffusion MRI; no enhancement

## TOP DIFFERENTIALS

- Granulation tissue or effusion (no bone erosion)
- Cholesterol granuloma (bright on T1)
- Glomus tympanicum (enhancing, on the promontory)

## CHECKLIST

- Five questions: tegmen, lateral canal fistula, facial canal, sinus tympani, ossicles
- Non-EPI DWI for residual or recurrent disease




CT, axial. The most frequently encountered mass in the middle ear is cholesteatoma, and therefore it is (also) shown in this chapter. In this…



Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

## Labyrinthine Fistula due to Cholesteatoma

### IMAGING FINDINGS

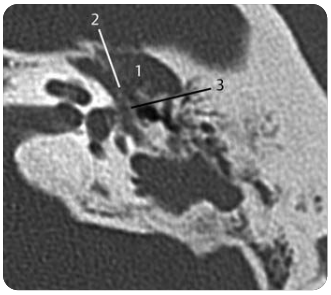
- Loss of the bony cover of the lateral semicircular canal next to cholesteatoma
- Best seen on coronal and Pöschl or Stenvers reformats
- Labyrinthine fluid on T2 may be reduced if labyrinthitis follows

### TOP DIFFERENTIALS

- Partial volume of thin normal bone
- Superior canal dehiscence
- Post-surgical defect

### CHECKLIST

- Report fistula site and size before surgery
- Confirm on thin slices in two planes




CT, axial. Note the smooth-bordered opacification suggestive of cholesteatoma in the anterior part of the middle ear (1), at the level of the…



Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

MIDDLE EAR & MASTOID · CONGENITAL

## Congenital Cholesteatoma

### IMAGING FINDINGS

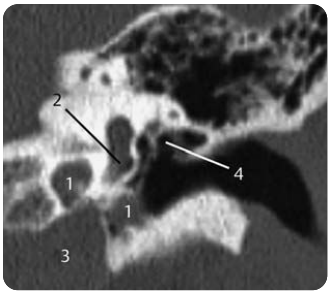
- Round, well-defined pearl behind an intact drum, usually anterosuperior mesotympanum
- Ossicles displaced or eroded, especially the incus long process and stapes
- Bright on non-EPI DWI; no enhancement

### TOP DIFFERENTIALS

- Acquired cholesteatoma
- Glomus tympanicum
- Facial nerve schwannoma of the tympanic segment

### CHECKLIST

- Intact drum and no history of otitis
- Look for mastoid and petrous apex extension




CT, coronal. CT excluded deeper invagination, but revealed other pathology. Coincidentally, an expansive infralabyrinthine…



Coronal temporal bone CT of the right ear demonstrates a large congenital cholesteatoma filling the middle ear cavity with subtle long process of…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1026.

For personal study. Follow local protocols and senior advice for patients.

TEMPORAL BONE & SKULL BASE · TRAUMA

# Skull Base Fractures

## IMAGING FINDINGS

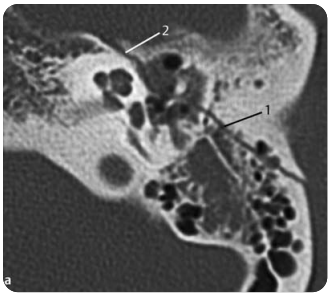
- Fracture line through the temporal bone or skull base; air in the middle ear or cranium
- Otic capsule sparing (most) or violating
- Fluid in the middle ear and mastoid

## TOP DIFFERENTIALS

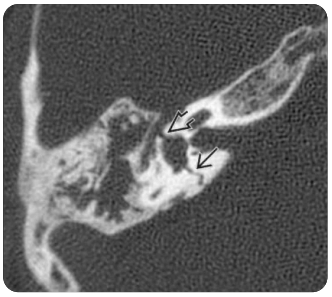
- Sutures, fissures and small canals (pseudofractures)
- Vascular grooves

## CHECKLIST

- Ossicles, facial canal (geniculate), tegmen and CSF leak
- Carotid canal involved: CT angiography




CT, axial. Patient with head trauma showing a fracture line through the mastoid (1) extending just anterior to the cochlea (2). Luxation of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

INNER EAR & IAC · CONGENITAL

# Malformations of the Inner Ear

## IMAGING FINDINGS

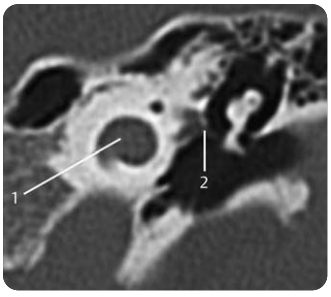
- Abnormal cochlea, vestibule or canals on CT and high-resolution T2
- IP-II: cystic apex, normal basal turn, large vestibular aqueduct
- Common cavity, cochlear hypoplasia, IP-I and IP-III by appearance

## TOP DIFFERENTIALS

- Labyrinthitis ossificans
- Normal variant of canal size

## CHECKLIST

- Cochlear nerve present on oblique sagittal T2
- Cochlear nerve canal width; gusher risk (IP-I, IP-III)




CT, coronal. This coronal view of the cochlea does not show any bony modiolus (1). This empty cochlea is suggestive of an incomplete partition…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1102.

INNER EAR & IAC · CONGENITAL

# Vestibular Aqueduct, Enlargements

## IMAGING FINDINGS

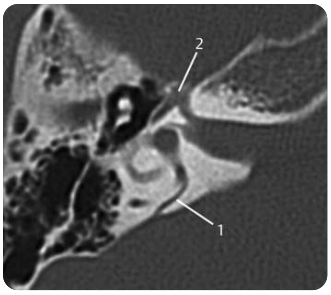
- Vestibular aqueduct wider than about 1 mm at the midpoint or 2 mm at the operculum
- Large endolymphatic sac on T2
- Often with IP-II cochlear deficiency

## TOP DIFFERENTIALS

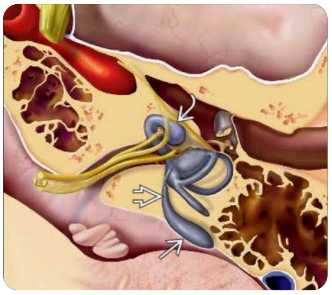
- Normal aqueduct (compare with the posterior canal)
- IP-I cystic cochleovestibular malformation

## CHECKLIST

- Look at both ears: often bilateral
- Pendred syndrome association




CT, axial. The normal vestibular aqueduct (1) in its passage from the vestibule to its junction with sinuses in the region of the sigmoid sinus. Ther…



In left inner ear, large endolymphatic sac epidural and intraosseous components are shown. Cochlea is malformed, with absent septation…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1104.

INNER EAR & IAC · BONE DISEASE

# Labyrinthitis Ossificans (Ossification after Meningitis)

## IMAGING FINDINGS

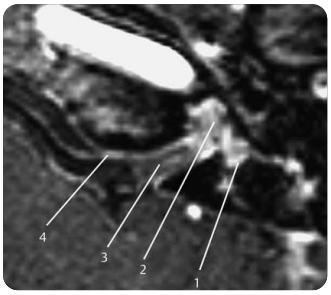
- Loss of the bright fluid signal of the cochlea on high-resolution T2
- Later: bone in the basal turn and canals on CT
- Early: enhancement of the labyrinth

## TOP DIFFERENTIALS

- Labyrinthine or cochlear aplasia
- Cochlear otosclerosis
- Intralabyrinthine schwannoma

## CHECKLIST

- After meningitis: urgent MRI and cochlear implantation
- Report patency of the basal turn




MRI, T1-weighted with gadolinium enhancement, axial. There is enhancement of the vestibule (1), cochlear contents (2), and the…



Axial bone CT in a 2-year-old boy with bilateral sensorineural hearing loss

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116.

TEMPORAL BONE & SKULL BASE · TUMOURS

# Pseudotumors in the Apex of the Petrous Bone

## IMAGING FINDINGS

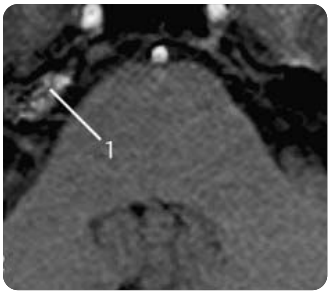
- Asymmetric marrow: bright on T1, suppresses with fat saturation, not expanded
- Trapped fluid: bright on T2, intact septa
- Cholesterol granuloma: bright on T1 and T2, expansile

## TOP DIFFERENTIALS

- Cholesteatoma or epidermoid (DWI bright)
- Petrous apicitis (destruction, enhancement)
- Carotid aneurysm

## CHECKLIST

- Compare with the other apex
- CT for expansion and cortex




MRI, T1-weighted, axial. T1-weighted image shows that the lesion (1) has high signal intensity compared with brain tissue (2). The contralateral…



Axial bone CT reveals an expansile cholesterol granuloma in the left PA with marginal bone dehiscence present. The right PA is well…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1152.

INNER EAR & IAC · TUMOURS

# Vestibular Schwannoma

## IMAGING FINDINGS

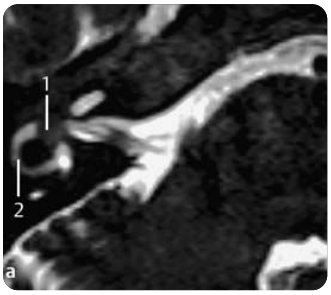
- Enhancing mass centred on the porus, "ice cream on a cone"
- Dark filling defect in bright CSF on high-resolution T2; fundal cap of fluid
- Acute angle with the petrous bone; widened porus; cystic change in larger tumours

## TOP DIFFERENTIALS

- Meningioma (broad dural base, dural tail)
- Epidermoid cyst (DWI bright)
- Facial nerve schwannoma (labyrinthine segment)

## CHECKLIST

- Size in canal and cistern; brainstem and fourth ventricle (Koos)
- Fundal cap and cochlear nerve canal for hearing preservation




MRI, T2-weighted, axial.A partial lack of fluid in the vestibule and horizontal semicircular canals is evident (1). A mass lesion is present in the vestibul…



Axial T2WI MR reveals a small intracanalicular vestibular schwannoma visualized as a soft tissue intensity mass surrounded by…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222.

INNER EAR & IAC · INFECTION & INFLAMMATION

# (Epi)dermoid Cyst

## IMAGING FINDINGS

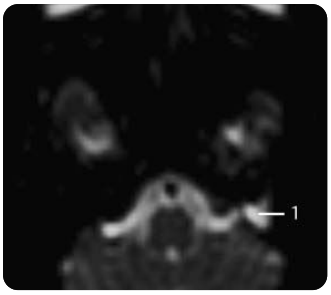
- CSF-like lesion that insinuates around nerves and vessels
- Incomplete suppression on FLAIR
- Bright on diffusion; no enhancement

## TOP DIFFERENTIALS

- Arachnoid cyst (follows CSF on every sequence, DWI dark)
- Cystic schwannoma

## CHECKLIST

- Encasement of cranial nerves and vessels
- Extent across cisterns before surgery




Patient with progressive and asymmetrical perceptive hearing loss.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.9 a



Axial CECT shows a large CPA epidermoid cyst. Note that this non-enhancing low-density lesion appears to invade the left cerebellar…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1202.

INNER EAR & IAC · TUMOURS

# Meningioma of the Cerebellopontine Angle

## IMAGING FINDINGS

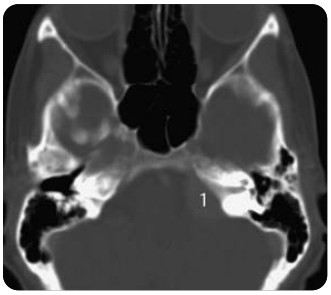
- Broad-based dural mass, obtuse angle with the petrous bone
- Eccentric to the porus; dural tail
- Calcification and hyperostosis; uniform enhancement

## TOP DIFFERENTIALS

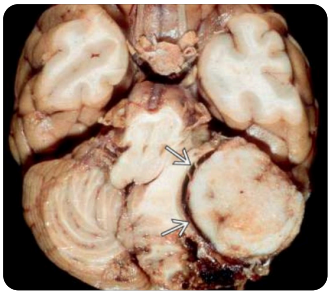
- Vestibular schwannoma
- Epidermoid cyst
- Dural metastasis

## CHECKLIST

- Extension into the IAC, jugular foramen and Meckel cave
- Venous sinus involvement




CT, axial. Just before the birth of her child, screening with CT without contrast enhancement suggested a lesion in the left CPA (1). The size of…



Gross pathologic section viewed from below shows a large CPA meningioma with a broad dural base compressing the cerebellum. The specimen…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1228.

TEMPORAL BONE & SKULL BASE · TUMOURS

# Glomus Jugulare Tumor (Paraganglioma)

## IMAGING FINDINGS

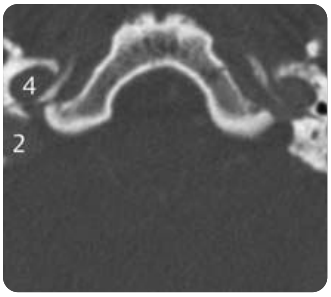
- Jugular foramen mass with permeative, moth-eaten bone erosion
- "Salt and pepper" on T1 with intense enhancement
- Spread superolaterally into the middle ear

## TOP DIFFERENTIALS

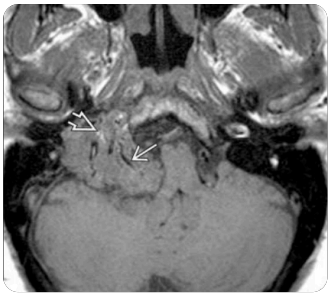
- Jugular foramen schwannoma (smooth scalloping)
- Meningioma (sclerosis, dural tail)
- Dehiscent jugular bulb or metastasis

## CHECKLIST

- Carotid, jugular vein and lower cranial nerves
- Multicentric paragangliomas: scan the neck




Patient with pulsating tinnitus, without any pathology behind the eardrum.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.20 a



Axial T1WI MR shows a large mass arising from the jugular foramen with multiple areas of salt & pepper. The "salt" represents blood products or…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 892.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

# Dural Sinus Thrombosis

## IMAGING FINDINGS

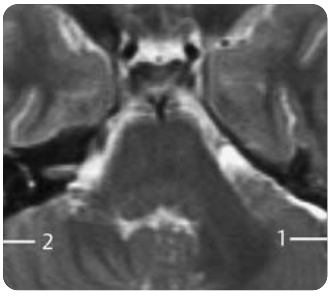
- Filling defect in the sigmoid or transverse sinus on contrast CT or MR venography
- Empty delta sign; dense sinus on non-contrast CT
- Adjacent mastoiditis with bone erosion over the sinus

## TOP DIFFERENTIALS

- Arachnoid granulation (round, CSF signal)
- Hypoplastic transverse sinus
- Slow flow artefact

## CHECKLIST

- Extension to the jugular vein and other sinuses
- Venous infarction and raised intracranial pressure




MRI, T2-weighted, axial. Otomastoiditis can spread intracranially, resulting in dural sinus thrombophlebitis, which may compromise the sigmoid and…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

# Otomastoiditis and its complications

## IMAGING FINDINGS

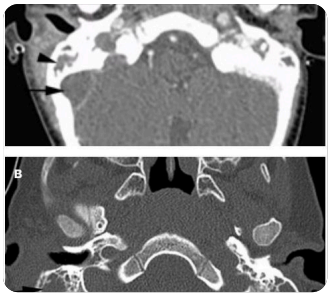
- Opacified middle ear and mastoid
- Coalescence: loss of the mastoid septa and cortical erosion
- Complications: subperiosteal abscess, sigmoid sinus thrombosis, epidural or brain abscess, labyrinthitis

## TOP DIFFERENTIALS

- Uncomplicated effusion (septa intact)
- Cholesteatoma
- Langerhans cell histiocytosis or rhabdomyosarcoma in a child

## CHECKLIST

- Contrast CT or MRI with venography when complications are suspected
- Check the sigmoid sinus, tegmen, labyrinth and petrous apex



Post-contrast CT shows (a) right coalescent mastoiditis (black arrowhead) and ipsilateral sigmoid sinus thrombosis (black arrow) – bot…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.